

Algebra 4

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1 A Brief Introduction to Category Theory

§1.1 Basic Definitions

§1.1.i Categories

Definition 1.1.1. A **category** $\mathcal{C}$ consists of:

- a collection of objects, denoted as $\text{ob}(\mathcal{C})$
- for any two objects $A, B \in \text{ob}(\mathcal{C})$, set of morphisms $\text{Hom}_{\mathcal{C}}(A, B)$
- A composition operation $\circ: \text{Hom}_{\mathcal{C}}(A, B) \times \text{Hom}_{\mathcal{C}}(B, C) \rightarrow \text{Hom}_{\mathcal{C}}(A, C)$ defined by $(f, g) \mapsto g \circ f$
- An element $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$ for each $A \in \text{ob}(\mathcal{C})$, called the identity.

such that the following hold:

- associativity: $h \circ (g \circ f) = (h \circ g) \circ f$
- identity: $\text{id}_B \circ f = f = f \circ \text{id}_A$,

where $f \in \text{Hom}_{\mathcal{C}}(A, B), g \in \text{Hom}_{\mathcal{C}}(B, C), h \in \text{Hom}_{\mathcal{C}}(C, D)$.

Remark 1.1.2 — Note that the composition operation is a part of the definition of a category, so when formally verifying that a category is indeed a category, then we include this. We often just don't include this because in many cases, it is just *usual composition*.

Definition 1.1.3. Let $f: A \rightarrow B$ be a morphism in $\mathcal{C}$ (that is, $A, B \in \text{ob}(\mathcal{C})$ and $f \in \text{Hom}_{\mathcal{C}}(A, B)$). Then, f is an **isomorphism** if there exists $g: B \rightarrow A$ such that $f \circ g = \text{id}_B$ and $g \circ f = \text{id}_A$.

Example 1.1.4

The category of sets, **Set**, has objects as sets and morphisms as any function of sets. The composition operation here is just composition of functions. In the category **Set**, isomorphisms and bijections are equivalent.

Example 1.1.5

The category of groups, **Group**, has objects as groups and morphisms and group homomorphisms. The composition here is just usual composition of homomorphisms.

Example 1.1.6

The category of (left) R -modules, where R is a ring, is denoted $R\text{-mod}$. Here, morphisms are homomorphisms of R -modules.

Example 1.1.7

The category **Top** of topological spaces has objects as topological spaces and morphisms as continuous maps. Isomorphisms here are homeomorphisms

Definition 1.1.8. A **subcategory** $\mathcal{D}$ of $\mathcal{C}$ is a sub-collection of objects $\text{ob}(\mathcal{D}) \subseteq \text{ob}(\mathcal{C})$ and for each $A, B \in \text{ob}(\mathcal{D})$, a subset $\text{Hom}_{\mathcal{D}}(A, B) \subseteq \text{Hom}_{\mathcal{C}}(A, B)$. It should contain the identity morphism and closed under composition. In other words, $\mathcal{D}$ should be a category in its own right.

Definition 1.1.9. A subcategory $\mathcal{D} \subseteq \mathcal{C}$ is a **full** subcategory if $\text{Hom}_{\mathcal{D}}(A, B) = \text{Hom}_{\mathcal{C}}(A, B)$, for all $A, B \in \text{ob}(\mathcal{D})$.

Example 1.1.10

We see that the category of abelian groups **Ab** is a full subcategory of **Group**.

Example 1.1.11

Every R -module can be considered as an abelian group. But this does not mean the category of R -modules is a subcategory of the category of abelian groups. There is no way to look at an abelian group and say “that one is an R -module”. Being an R -module involves extra structure. (Although, if $R = \mathbb{Q}$ this is not really true.)

Example 1.1.12

We can view **Set** as a full subcategory of **Top**, by viewing sets as discrete topological spaces. That is, every set is open and closed.

Example 1.1.13

Let $\mathcal{C}$ be any category. Then, consider $\mathcal{C}^{\text{id}} \subseteq \mathcal{C}$, where $\mathcal{C}^{\text{id}}$ has only the identity morphism. This is not full unless $\mathcal{C}$ only had identity morphisms.

Remark 1.1.14 — Categories where all the morphisms are isomorphisms are called groupoids, and the construction above is not very unusual, as we may restrict to this.

Definition 1.1.15. A morphism $f: B \rightarrow C$ in a category $\mathcal{C}$ is **monic** (or **monomorphism**) if for any two morphisms $e_1, e_2: A \rightarrow B$, if $f \circ e_1 = f \circ e_2 \implies e_1 = e_2$. That is, we can cancel f on the left.

Definition 1.1.16. A morphism $f: B \rightarrow C$ in a category $\mathcal{C}$ is **epic** (or **epimorphism**) if for any two morphisms $g_1, g_2: C \rightarrow D$, if $g_1 \circ f = g_2 \circ f \implies g_1 = g_2$. That is, we can cancel f on the right.

Exercise 1.1.17. In the category **Set**, we have that monic maps are equivalent to injective functions. Further, epic maps are equivalent to surjective functions.

Solution: Consider the situation in the definition above. Let $x \in A$. Then, suppose that $f(e_1(x)) = f(e_2(x))$. By definition, this implies then that $e_1(x) = e_2(x)$. This is exactly the definition of f being injective.

Consider the situation in the definition above. Suppose that f is not surjective, so let $c \in C$ such that $f(b) \neq c$, for all $b \in B$. Now, define

$$g_1(d) = \{d\}, \quad g_2(d) = \{d\} \cap f(B), \quad d \in C.$$

For every $b \in B$, we note that $g_1(f(b)) = g_2(f(b))$. Thus, $g_1 = g_2$, since f is epic. However, $\{c\} = g_1(c) \neq g_2(c) = \emptyset$, since $\{c\} \cap f(B) = \emptyset$. This is a contradiction, so f must have been surjective. $\square$

Exercise 1.1.18. Let $\mathcal{C}$ be a category. A morphism $f: A \rightarrow B$ in $\mathcal{C}$ is called a **split monomorphism** (respectively **split epimorphism**) if there exists a morphism $g: B \rightarrow A$ in $\mathcal{C}$ such that $gf = \text{id}_A$ (respectively $fg = \text{id}_B$). Show that the following are equivalent:

- (i) f is an isomorphism.
- (ii) f is split monic and split epic.
- (iii) f is monic and split epic.
- (iv) f is split monic and epic.

Solution: The equivalence (i) $\iff$ (ii) is clear. Suppose (i). Let $e_1, e_2: A' \rightarrow A$, and suppose $f \circ e_1 = f \circ e_2$. Then, since f is an isomorphism, we can apply f^{-1} to the left and deduce that $e_1 = e_2$. Thus, (i) $\implies$ (iii). Similarly, we can do the same for (i) $\implies$ (iv). Now assume (iii), so f is monic and split epic. Split epic implies that f has a right inverse, say g , so $fg = \text{id}_B$. Then, $fgf = f$. Since f is monic, then $gf = \text{id}_A$, so we get that g is a left inverse as well. The case assuming (iv) $\implies$ (i) follows analogously. $\square$

Example 1.1.19

In **Ring** (the category of rings and ring homomorphisms), we note that the above results for **Set** is not true. For instance, the inclusion map $\mathbb{Z} \hookrightarrow \mathbb{Q}$, the inclusion is epic and monic but not an isomorphism.

Exercise 1.1.20. Show that the inclusion $\iota: \mathbb{Z} \hookrightarrow \mathbb{Q}$ is epic in the category of rings.

Solution: Let $g_1, g_2: \mathbb{Q} \rightarrow R$ be ring homomorphisms, where R is some ring. Assume that $g_1(\iota(n)) = g_2(\iota(n))$, for some $n \in \mathbb{Z}$. In other words, we're assuming that $g_1(n) = g_2(n)$. In fact, this implies that $g_1 = g_2$ on $\mathbb{Q}$ due to the following. Let $\frac{p}{q} \in \mathbb{Q}$ (so $p, q \in \mathbb{Z}$ and $q \neq 0$); it follows that

$$g_1\left(\frac{p}{q}\right) = g_1(p)g_1(q)^{-1} = g_2(p)g_2(q)^{-1} = g_2\left(\frac{p}{q}\right).$$

Note that the equality $g_1\left(\frac{p}{q}\right) = g_1(p)g_1(q)^{-1}$ follows from $\frac{p}{q} = p \cdot \frac{1}{q} = p \cdot q^{-1}$. $\square$

Exercise 1.1.21. Show that the inclusion map $\iota: \mathbb{Q} \hookrightarrow \mathbb{R}$ in the category **Haus** of Hausdorff topological spaces is epic. Note that **Haus** is a full subcategory of **Top**.

Solution: Let $g_1, g_2: \mathbb{R} \rightarrow X$ be continuous maps, where X is a Hausdorff topological space. Assume that $g_1(\iota(r)) = g_2(\iota(r))$, for all $r \in \mathbb{Q}$. That is, $g_1(r) = g_2(r)$, for all $r \in \mathbb{Q}$. By the density of rationals, this is enough to deduce that $g_1 = g_2$ on all of $\mathbb{R}$. Take some arbitrary $a \in \mathbb{R}$, and take a sequence $\{a_i\}_i$ with each $a_i \in \mathbb{Q}$ which converges to a . Then,

$$g_1(a) = \lim_{n \rightarrow \infty} g_1(a_i) = \lim_{n \rightarrow \infty} g_2(a_i) = g_2(a).$$

Note that their limits are the same because in Hausdorff spaces, convergent sequences have unique limits. To be specific, we have that

$$g_1(a) = g_1\left(\lim_{n \rightarrow \infty} a_i\right) = \lim_{n \rightarrow \infty} g_1(a_i),$$

where the second equality comes from the fact that continuity implies sequential continuity in topological spaces $\square$

Exercise 1.1.22. Describe a category with more than one object such that every map is monic and epic but any two distinct objects are not isomorphic.

Solution: Consider the category of posets, $(P, \leq)$. Here, the objects are the elements of P , and $\leq$ is the partial order. In posets, there is a unique morphism $x \rightarrow y$ if and only if $x \leq y$. Now, let $f: x \rightarrow y$ such that for all $g_1, g_2: y \rightarrow z$, we have $g_1 f = g_2 f$. Then, $g_1 = g_2$ is trivially true, so f is a epic.

Now consider $f: y \rightarrow z$ and $g_1, g_2: x \rightarrow y$. Then, $f g_1 = f g_2$ implying that $g_1 = g_2$ is trivially true again, so f is also monic.

Therefore, f is both epic and monic, but clearly, two distinct objects are not isomorphic, since there is no arrow going from the *bigger* object to the *smaller* one. $\square$

Definition 1.1.23. Any category $\mathcal{C}$ has an **opposite category** $\mathcal{C}^{\text{op}}$ with $\text{ob}(\mathcal{C}^{\text{op}}) = \text{ob}(\mathcal{C})$ with $\text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) = \text{Hom}_{\mathcal{C}}(B, A)$.

Remark 1.1.24 — There is a natural composition rule $f \circ g = g \circ f$ (where the left is in $\mathcal{C}^{\text{op}}$ and the right is in $\mathcal{C}$).

Example 1.1.25

If G is a group, there is a category BG of one object $\{e\} = \text{ob}(BG)$, $\text{Hom}_{BG}(e, e) = G$. This is a category, since we have the identity inside. The composition rule here is as follows: let $g: e \rightarrow e$ and $f: e \rightarrow e$; then, $g \circ f = g \cdot f$, where the right $\cdot$ is the group operation.

Remark 1.1.26 — We note that morphisms from an object to itself are called endomorphisms.

Remark 1.1.27 — We note that $(BG)^{\text{op}} = B(G^{\text{op}})$, where G^{op} is the opposite group.

§1.1.ii Functors

Definition 1.1.28. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A **(covariant) functor** $F: \mathcal{C} \rightarrow \mathcal{D}$ is a rule that associates to every object C of $\mathcal{C}$ an object $F(C)$ of $\mathcal{D}$, and to every morphism $f: C_1 \rightarrow C_2$ in $\mathcal{C}$ a morphism $F(f): F(C_1) \rightarrow F(C_2)$ in $\mathcal{D}$, such that the following conditions hold:

- for every object $A \in \mathcal{C}$, $F(\text{id}_A) = \text{id}_{F(A)}$
- for every $f \in \text{Hom}_{\mathcal{C}}(A, B)$ and $g \in \text{Hom}_{\mathcal{C}}(B, C)$, $F(g \circ f) = F(g) \circ F(f)$.

Definition 1.1.29. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A **contravariant functor** $F: \mathcal{C} \rightarrow \mathcal{D}$ is a covariant functor $F: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$.

Remark 1.1.30 — To unravel this definition, we note that the only real difference is that $F(f \circ g) = F(g) \circ F(f)$. Verify this.

Proof. We verify the remark above. Let $f \in A \rightarrow B$, and $f^{\text{op}}: B \rightarrow A$ be the reversed arrow in $\mathcal{C}^{\text{op}}$. Same with $g: B \rightarrow C$ and $g^{\text{op}}: C \rightarrow B$. Then,

$$F((g \circ f)^{\text{op}}) = F(f^{\text{op}} \circ g^{\text{op}}) = F(f^{\text{op}}) \circ F(g^{\text{op}}).$$

Translating this back to F , this says $F(g \circ f) = F(f) \circ F(g)$. □

Example 1.1.31

$\mathbf{Vect}_k$ is the category of vector spaces over a field k with morphisms as linear maps. Let G be a group. A functor $F: BG \rightarrow \mathbf{Vect}_k$ is *the same as* a representation of G . That is, $F(e) = V$, for some vector space V , and for all $g: e \rightarrow e$ (so $g \in G$), a linear map $F(g): V \rightarrow V$ which satisfies $F(gh) = F(g) \circ F(h) \in \text{Hom}_k(V, V)$.

Example 1.1.32

For any category $\mathcal{C}$ and any $A \in \text{ob}(\mathcal{C})$, there is a contravariant functor $\mathcal{C} \rightarrow \mathbf{Set}$, which is $\text{Hom}_{\mathcal{C}}(-, A)$. That is, $\text{Hom}_{\mathcal{C}}(B, A)$, a set (for an object B of $\mathcal{C}$). For any $f: B \rightarrow C$ in $\mathcal{C}$, this functor maps f to pre-composition with f , $- \circ f: \text{Hom}_{\mathcal{C}}(C, A) \rightarrow \text{Hom}_{\mathcal{C}}(B, A)$.

Example 1.1.33

If R is a ring, then we noted that $R\text{-mod}$ is not a subcategory of $\mathbf{Ab}$. Instead, there is a functor $F: R\text{-mod} \rightarrow \mathbf{Ab}$ taking an R -module to its underlying abelian group, and taking a morphism of R -modules to the same morphism of abelian groups.

§1.1.iii Natural Transformations

Definition 1.1.34. Given categories $\mathcal{C}$ and $\mathcal{D}$, and functors $F, G: \mathcal{C} \rightarrow \mathcal{D}$, a **natural transformation** $\eta: F \Rightarrow G$, is the following. For each object A in $\mathcal{C}$, $\eta_A: F(A) \rightarrow G(A)$ is a morphism in $\mathcal{D}$ such that for all $f: A \rightarrow B$ in $\mathcal{C}$, the following diagram commutes:

$$\begin{array}{ccc} F(A) & \xrightarrow{\eta_A} & G(A) \\ \downarrow F(f) & & \downarrow G(f) \\ F(B) & \xrightarrow{\eta_B} & G(B) \end{array}$$

That is, $G(f) \circ \eta_A = \eta_B \circ F(f)$. This is called the **naturality square**. If η_A is an isomorphism for every $A \in \text{ob}(\mathcal{C})$, then η is called a **natural isomorphism**, often written $\eta: F \cong G$. The set of natural transformations from F to G is written $\text{Nat}(F, G)$.

Example 1.1.35

There are at least two (obvious) functors $F, G: \mathbf{Set} \rightarrow \mathbf{Top}$. The discrete topology functor $D: \mathbf{Set} \rightarrow \mathbf{Top}$ assigns a set S with the discrete topology. The trivial (or indiscrete) topology functor $T: \mathbf{Set} \rightarrow \mathbf{Top}$ assigns a set S with the trivial topology, with only $\emptyset$ and S open. Note that the identity function provides a natural transformation $\eta: F \Rightarrow G$.

Example 1.1.36

Take F to be the identity functor from $\mathbf{Set}$ to $\mathbf{Set}$ and $G: \mathbf{Set} \rightarrow \mathbf{Set}$, where $G(S) = S \times \{\emptyset\}$. Verify that F is naturally isomorphic to G . That is, there exists $\eta: F \Rightarrow G$ that is naturally isomorphic.

Exercise 1.1.37. For a finite dimensional k -vector space V there is a linear map $V \rightarrow V^{**}$ given by $v \mapsto (f \mapsto f(v))$. Show that this defines a natural transformation from the identity to the double dualization functor on $\mathbf{Vect}_k$. When V is finite dimensional, show that η_V is a natural isomorphism.

Solution: Let $V \in \text{ob}(\mathbf{Vect}_k)$. Let $\eta: \text{id} \Rightarrow (-)^{**}$ defined by $\eta_V: V \rightarrow V^{**}$, $v \mapsto \text{eval}_v$. Here, $v \in V$ and $\text{eval}_v: V^* \rightarrow k$ is the ‘evaluation at v ’ map, where if $f \in V^*$, $\text{eval}_v(f) = f(v)$. We want to show that the following square commutes. Let V, W be finite dimensional vector spaces and $f: V \rightarrow W$ be a linear map.

$$\begin{array}{ccc} \text{id}(V) & \xrightarrow{\eta_V} & V^{**} \\ \downarrow \text{id}(f) & & \downarrow (f)^{**} \\ \text{id}(W) & \xrightarrow{\eta_W} & W^{**} \end{array}$$

Let $v \in V$. Then, going right then down gives

$$(f)^{**} \circ \eta_V(v) = (f)^{**}(\text{eval}_v) = \text{eval}_v \circ f^*$$

Going down then right gives

$$\eta_W \circ \text{id}(f)(v) = \eta_W(f(v)) = \text{eval}_{f(v)}$$

Now we need to see how these act on $g \in V^*$. The top acts as

$$\text{eval}_v \circ f^*(g) = \text{eval}_v(g(f)) = g(f(v)).$$

The bottom acts as $\text{eval}_{f(v)}(g) = g(f(v))$. This was arbitrary for g , so this square commutes.

Next, we show each η_V is an isomorphism when V is finite dimensional. Note that $\dim V = \dim V^{**}$, so showing injectivity or surjectivity is enough. We show injectivity. Suppose that $\eta_v(f) = 0$, for all $f \in V^*$. If $v \neq 0$, we can extend this to a basis of V . Then, defining $f(v) = 1$, and 0 for other basis elements, we reach a contradiction. Thus, $v = 0$. $\square$

§1.1.iv Equivalence of Categories

Definition 1.1.38. A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is called an **equivalence of categories** if there is another functor $G: \mathcal{D} \rightarrow \mathcal{C}$ such that $F \circ G: \mathcal{D} \rightarrow \mathcal{D}$ is naturally isomorphic to $\text{id}_{\mathcal{D}}$ and $G \circ F: \mathcal{C} \rightarrow \mathcal{C}$ is naturally isomorphic to $\text{id}_{\mathcal{C}}$. Here, we denote this as $F \circ G \cong \text{id}_{\mathcal{D}}$ and $G \circ F \cong \text{id}_{\mathcal{C}}$.

Example 1.1.39

The category **Mat** of matrices has objects $n \in \mathbb{N}_{\geq 0}$ with $\text{Hom}_{\mathbf{Mat}_k}(n, m) = M_{m \times n}(k)$.

Exercise 1.1.40. Consider the category **fdVect** $_k$ of finite dimensional k -vector spaces. There is a functor $F: \mathbf{Mat}_k \rightarrow \mathbf{fdVect}_k$ defined by $n \mapsto F(n) = k^n$ and $A \mapsto A \cdot -: k^n \rightarrow k^m$, where $A \in M_{m \times n}(k)$. Show that this is an equivalence of categories.

Solution: We need to define a functor $G: \mathbf{fdVect}_k \rightarrow \mathbf{Mat}_k$ such that $F \circ G: \mathbf{fdVect}_k \rightarrow \mathbf{fdVect}_k$ is naturally isomorphic to $\text{id}_{\mathbf{fdVect}_k}$ and $G \circ F: \mathbf{Mat}_k \rightarrow \mathbf{Mat}_k$ is naturally isomorphic to $\text{id}_{\mathbf{Mat}_k}$. Let $G: k^n \mapsto n$ and $T \mapsto [T]_B$, where B is a basis that is chosen. This is, indeed, a functor, and composition of linear maps correspond to multiplication of matrices through the functor. Let $n \leq m$ be positive integers now, and consider the following diagram:

$$\begin{array}{ccc} GF(n) & \xrightarrow{\eta_n} & \text{id}_{\mathbf{Mat}_k}(n) \\ \downarrow GF(A) & & \downarrow \text{id}_{\mathbf{Mat}_k}(A) \\ GF(m) & \xrightarrow{\eta_m} & \text{id}_{\mathbf{Mat}_k}(m) \end{array}$$

Here, the vertical maps A are matrices. Clearly, this square commutes and the η_n and η_m are just identity maps, which map $n \mapsto n$ and $m \mapsto m$, so they are isomorphisms. Next, consider two k -vector spaces V, W , with respective dimensions n, m .

$$\begin{array}{ccc} FG(V) & \xrightarrow{\eta_V} & \text{id}_{\mathbf{fdVect}_k}(V) \\ \downarrow FG(T) & & \downarrow \text{id}_{\mathbf{fdVect}_k}(T) \\ FG(W) & \xrightarrow{\eta_W} & \text{id}_{\mathbf{fdVect}_k}(W) \end{array}$$

Again, this square clearly commutes. Here, η_V is a linear map between the n -dimensional vector spaces $FG(V)$ to V . Thus, η_V is an isomorphism. (same for η_W) Thus, FG is naturally isomorphic to the identity, and so we have that **Mat** $_k$ and **fdVect** $_k$ is an equivalence of categories. $\square$

Remark 1.1.41 — $\mathbf{Mat}_k$ is called a **skeleton** of $\mathbf{Vect}_k$.

Definition 1.1.42. A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is called **faithful** if $F: \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B))$ is injective, for all $A, B \in \text{ob}(\mathcal{C})$.

Definition 1.1.43. A functor $F: \mathcal{C} \rightarrow \mathcal{D}$ is called **full** if $F: \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B))$ is surjective, for all $A, B \in \text{ob}(\mathcal{C})$.

Definition 1.1.44. A functor F is **fully faithful** if it is both.

Definition 1.1.45. A **concrete category** is a category $\mathcal{C}$ together with a faithful functor to $\mathbf{Set}$.

Example 1.1.46

If $\mathcal{D} \subseteq \mathcal{C}$ is a full subcategory, then the inclusion $\mathcal{D} \hookrightarrow \mathcal{C}$ is fully faithful.

Example 1.1.47

For example, $\mathbf{Ab} \hookrightarrow \mathbf{Group}$ is a fully faithful functor.

Exercise 1.1.48. Define a category $\mathbf{Cat}$ whose objects are categories.

Solution: The category $\mathbf{Cat}$ has objects as categories. The morphisms are (covariant) functors. Composition is as usual, and the identity morphism is just the identity functor. Associativity and identity are clear from definition. $\square$

Exercise 1.1.49. For two categories $\mathcal{C}$ and $\mathcal{D}$ define the category of functors, denoted by $\text{Fun}(\mathcal{C}, \mathcal{D})$ (or $[\mathcal{C}, \mathcal{D}]$), whose objects are functors from $\mathcal{C}$ to $\mathcal{D}$.

Solution: Objects here are functors, and the morphisms between them are just natural transformations. Composition is just composition of natural transformations. Note that there is also the identity natural transformation which is just the identity morphism in $\mathcal{D}$. Associativity holds from the category $\mathcal{D}$ and the naturality square. Identity is also clear. $\square$

Exercise 1.1.50. Let R be a ring. Let $\mathbf{Ar}_R$ be the category of arrows over R : objects in this category are maps $M \xrightarrow{f} N$ of R -modules, and morphisms $f \rightarrow f'$ in this category are commutative diagrams

$$\begin{array}{ccc} M & \xrightarrow{f} & N \\ g \downarrow & & \downarrow h \\ M' & \xrightarrow{f'} & N' \end{array}$$

Now let $U_2(R)$ be the ring of 2×2 upper-triangular matrices over R . Prove that $\mathbf{Ar}_R$ is equivalent to $\mathbf{mod}\text{-}U_2(R)$.

Solution: Given an arrow $M \xrightarrow{f} N$, we define a right $U_2(R)$ -module structure on $M \oplus N$ via

$$(m, n) \cdot \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = (ma, f(m)b + nc).$$

On the arrows, we define $g \oplus h: M \oplus N \rightarrow M' \oplus N'$ via

$$(g \oplus h)(m, n) = (g(m), h(n)),$$

where $f'g = hf$. Then, this is $U_2(R)$ -linear:

$$\begin{aligned} (g \oplus h) \left((m, n) \cdot \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right) &= (g(ma), h(f(m)b + nc)) \\ &= (g(m)a, f'g(m)b + h(n)c) = (g(m), h(n)) \cdot \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \end{aligned}$$

Thus, we have a functor from $F: \mathbf{Ar}_R \rightarrow U_2(R)\text{-mod}$.

We now define a functor going the other way. Let X be a $U_2(R)$ -module. Let

$$e_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then, $X = Xe_1 \oplus Xe_2$, and we define $f_X: Xe_1 \rightarrow Xe_2$ as

$$f_X(x) = x \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

This lands in Xe_2 because

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} e_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Now, if $\varphi: X \rightarrow Y$ is a $U_2(R)$ -module map, then it restricts to maps $\varphi_1: Xe_1 \rightarrow Ye_1$ and $\varphi_2: Xe_2 \rightarrow Ye_2$. In particular,

$$\varphi_2(f_X(x)) = \varphi \left(x \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right) = \varphi(x) \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = f_Y(\varphi_1(x)).$$

Therefore, we have a functor $G: U_2(R)\text{-mod} \rightarrow \mathbf{Ar}_R$. Moreover, these functors are inverses to each other, and are naturally isomorphic to the identities. $\square$

§1.1.v The Yoneda Lemma

Proposition 1.1.51 (Yoneda lemma)

Let $F: \mathcal{C} \rightarrow \mathbf{Set}$ be a contravariant functor, and let A be an object of $\mathcal{C}$. Note that $\text{Hom}_{\mathcal{C}}(-, A): \mathcal{C} \rightarrow \mathbf{Set}$ is also a contravariant functor. There is a bijection of sets

$$\text{Nat}(\text{Hom}_{\mathcal{C}}(-, A), F) \cong F(A).$$

Now let A and B be objects of $\mathcal{C}$. Particularly, there is a natural isomorphism of functors

$$\text{Hom}_{\mathcal{C}}(-, A) \cong \text{Hom}_{\mathcal{C}}(-, B)$$

if and only if $A \cong B$.

Proof. The relevant map to consider is the following:

$$\begin{aligned}\Phi: \text{Nat}(\text{Hom}_{\mathcal{C}}(-, A), F) &\rightarrow F(A) \\ \alpha &\mapsto \alpha_A(\text{id}_A).\end{aligned}$$

Firstly, note that this map is well-defined, since a natural transformation of α at the object $A \in \text{ob}(\mathcal{C})$ is a function between sets $\alpha_A: \text{Hom}_{\mathcal{C}}(A, A) \rightarrow F(A)$. It therefore makes sense to apply α_A to the identity id_A . We prove that this is a bijection.

We first show injectivity. Suppose that We have two maps $\alpha, \beta \in \text{Nat}(\text{Hom}_{\mathcal{C}}(-, A), F)$ and that $\Phi(\alpha) = \Phi(\beta)$. Then, we have $\beta_A(\text{id}_A) = \alpha_A(\text{id}_A)$. We want to show that $\beta = \alpha$, which is equivalent to show that for every object in $B \in \text{ob}(\mathcal{C})$, $\alpha_B = \beta_B$. Let $B \in \text{ob}(\mathcal{C})$, and let $f \in \text{Hom}_{\mathcal{C}}(B, A)$. Then, the naturality of α says that

$$\begin{array}{ccc}\text{Hom}_{\mathcal{C}}(A, A) & \xrightarrow{\alpha_A} & F(A) \\ \downarrow - \circ f & & \downarrow F(f) \\ \text{Hom}_{\mathcal{C}}(B, A) & \xrightarrow{\alpha_B} & F(B)\end{array}$$

We know that id_A always exists, so applying the diagram above to $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$ (and noting that $\text{id}_A \circ f = f$, we have

$$\alpha_B(f) = F(f)\alpha_A(\text{id}_A).$$

Analogously, we have that $\beta_B(f) = F(f)\beta_A(\text{id}_A)$. By assumption, we assumed that $\alpha_A(\text{id}_A) = \beta_A(\text{id}_A)$, so we attain that $\alpha_B(f) = \beta_B(f)$, proving injectivity.

Now for surjectivity. Take some $x \in F(A)$, which is an element in a set. From here, I want to find a natural map $\alpha: \text{Hom}_{\mathcal{C}}(-, A) \Rightarrow F$. Given an object $B \in \mathcal{C}$ and $f \in \text{Hom}_{\mathcal{C}}(B, A)$, I will take $\alpha_B(f) = F(f)(x)$. In particular, by taking $B = A$ and $f = \text{id}_A$, we get that

$$\alpha_A(\text{id}_A) = F(\text{id}_A)(x) = \text{id}_{F(A)}(x) = x,$$

since F is a (contravariant) functor. We just need to show that this map α is natural.

Let B, B' be any objects in $\mathcal{C}$ and $g: B' \rightarrow B$. By definition, we must show the commutativity of the following square:

$$\begin{array}{ccc}\text{Hom}_{\mathcal{C}}(B, A) & \xrightarrow{\alpha_B} & F(B) \\ \downarrow - \circ g & & \downarrow F(g) \\ \text{Hom}_{\mathcal{C}}(B', A) & \xrightarrow{\alpha_{B'}} & F(B')\end{array}$$

Let $f \in \text{Hom}_{\mathcal{C}}(B, A)$ and $x \in F(A)$. Then, we have that

$$F(g) \circ \alpha_B(f) = F(g)F(f)(x).$$

On the other hand, we have that

$$\alpha_{B'} \circ (f \circ g) = F(f \circ g)(x).$$

Indeed, since F is contravariant, we have that the two expressions are equivalent, concluding the proof. We've shown that this is a bijection, so we are done.

As for the second statement, this can be deduced from the first part by setting $F = \text{Hom}_{\mathcal{C}}(-, B)$. Yoneda then says that there is some natural isomorphism $\eta: \text{Hom}_{\mathcal{C}}(-, A) \Rightarrow \text{Hom}_{\mathcal{C}}(-, B)$ which corresponds to some $f \in \text{Hom}_{\mathcal{C}}(A, B)$. Conversely, the inverse η^{-1} corresponds to some $g \in \text{Hom}_{\mathcal{C}}(B, A)$. Then, we use the naturality of η to get the following:

$$\begin{aligned} g \circ f &= \eta_B^{-1}(\text{id}_B) \circ f = \eta_A^{-1} \circ (\text{id}_B \circ f) = \eta_A^{-1} \circ \eta_A(\text{id}_A) = \text{id}_A \\ f \circ g &= \eta_A(\text{id}_A) \circ g = \eta_B(\text{id}_A \circ g) = \eta_B \circ \eta_B^{-1}(\text{id}_B) = \text{id}_B \end{aligned}$$

Thus, f, g are inverses to each other and so $A \cong B$. The converse to this statement is immediate. $\square$

Remark 1.1.52 — The second part says that if you know how all the morphisms into an object A work, then you actually know the object A perfectly.

Exercise 1.1.53 (Yoneda Embedding). Let $\mathcal{C}$ be a category and $A \in \text{ob}(\mathcal{C})$. Then, we have a contravariant functor $F_A = \text{Hom}_{\mathcal{C}}(-, A): \mathcal{C} \rightarrow \mathbf{Set}$, which induces a covariant functor $\Psi: \mathcal{C} \rightarrow \text{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set})$. Moreover, prove that Ψ is fully faithful. This is called the **Yoneda embedding**.

Solution: Let $\Psi: A \mapsto F_A$. Note that F_A is indeed a functor from $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$. Let $A, B \in \text{ob}(\mathcal{C})$, and let $F = F_B$ in the Yoneda lemma. This gives us the bijection of sets

$$\text{Nat}(\text{Hom}_{\mathcal{C}}(-, A), \text{Hom}_{\mathcal{C}}(-, B)) \cong \text{Hom}_{\mathcal{C}}(B, A).$$

Of course, we can also prove this directly. We note that if $\varphi \in \text{Nat}(\text{Hom}_{\mathcal{C}}(-, A), \text{Hom}_{\mathcal{C}}(-, B))$, then φ we know that $\varphi_A(\text{id}_A): A \rightarrow B$ from the proof of the Yoneda lemma. Therefore, every natural transformation has exactly one corresponding morphism in $\text{Hom}_{\mathcal{C}}(A, B)$. Covariance follows essentially by definition: $\Psi(f)(g) := f \circ g$.

On morphisms, if $f: A \rightarrow B$ in $\mathcal{C}$, then we define $\Psi(f)$ as the natural transformation such that for each $X \in \text{ob}(\mathcal{C})$, we get $\Psi(f)_X: F_A(X) \rightarrow F_B(X)$ is defined by $g \mapsto f \circ g$; that is, post-composition. We want to show the following diagram commutes.

$$\begin{array}{ccc} F_A(X) & \xrightarrow{\Psi(f)_X} & F_B(X) \\ \downarrow F_A(g) & & \downarrow F_B(g) \\ F_A(Y) & \xrightarrow{\Psi(f)_Y} & F_B(Y) \end{array}$$

Here, $f: A \rightarrow B$ and $g: Y \rightarrow X$. Now, let $h \in F_A(X) = \text{Hom}_{\mathcal{C}}(X, A)$. First going down then right, we have that

$$\Psi(f)_Y \circ F_A(g)(h) = \Psi(f)_Y(h \circ g) = f \circ (h \circ g).$$

Conversely, going right then down, we have that

$$F_B(g) \circ \Psi(f)_X(h) = (f \circ h) = (f \circ h) \circ g.$$

Clearly, these are equal by associativity. Thus, the diagram commutes.

Now, we need to show that the functor Ψ is fully faithful. First, we show that Ψ is faithful. Let $A, B \in \text{ob}(\mathcal{C})$ be arbitrary. We need to show that

$$\Psi: \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\text{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set})}(\Psi(A), \Psi(B))$$

is injective. Let $f, g \in \text{Hom}_{\mathcal{C}}(A, B)$, and suppose that $\Psi(f) = \Psi(g)$ as natural transformations. We may get that $\Psi(f)_A = \Psi(g)_A$, and $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$, so we have that

$$f = f \circ \text{id}_A = \Psi(f)_A(\text{id}_A) = \Psi(g)_A(\text{id}_A) = g \circ \text{id}_A = g.$$

Thus, injective.

Next, we show fullness. We now need to show that Ψ is surjective. Let $\eta \in \text{Hom}_{\text{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set})}(\Psi(A), \Psi(B))$. This is a natural transformation from $\text{Hom}_{\mathcal{C}}(-, A) \Rightarrow \text{Hom}_{\mathcal{C}}(-, B)$. Let $f = \eta(\text{id}_A) \in \text{Hom}_{\mathcal{C}}(A, B)$. Then, we claim that $\eta = \Psi(f)$. Let $X \in \text{ob}(\mathcal{C})$, $g: X \rightarrow A$, and consider the following naturality square:

$$\begin{array}{ccc} F_A(A) & \xrightarrow{\eta_A} & F_A(B) \\ \downarrow - \circ g & & \downarrow - \circ g \\ F_A(X) & \xrightarrow{\eta_X} & F_B(X) \end{array}$$

Evaluating at id_A , down then right gives

$$\eta_X(\text{id}_A \circ g) = \eta_X(g) = f \circ g.$$

This is exactly $\Psi(f)_X(g)$. Thus, $\eta_X = \Psi(f)_X$, for all X , so $\eta = \Psi(f)$, and we get surjectivity. This concludes the proof. $\square$

For completeness, I will just list the (covariant) version of the Yoneda lemma:

Corollary 1.1.54 ((Covariant Yoneda Lemma))

Let $F: \mathcal{C} \rightarrow \mathbf{Set}$ be a covariant functor, and $A \in \text{ob}(\mathcal{C})$. Then, there is a bijection of sets

$$\text{Nat}(\text{Hom}_{\mathcal{C}}(A, -), F) \cong F(A).$$

Furthermore, let $A, B \in \text{ob}(\mathcal{C})$. Particularly, we have a natural isomorphism

$$\text{Hom}_{\mathcal{C}}(A, -) \cong \text{Hom}_{\mathcal{C}}(B, -)$$

if and only if $A \cong B$.

§1.2 Adjunctions

Definition 1.2.1. A pair of functors $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{D} \rightarrow \mathcal{C}$ are called **adjoint** (or is an **adjunction**) if there is a bijection of sets for all $C \in \text{ob}(\mathcal{C})$ and $D \in \text{ob}(\mathcal{D})$

$$\tau_{C,D}: \text{Hom}_{\mathcal{D}}(L(C), D) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(C, R(D))$$

which is natural in C and D . That is, the following diagrams commute: for all $f: C \rightarrow C'$ in $\mathcal{C}$ and $g: D \rightarrow D'$ in $\mathcal{D}$,

$$\begin{array}{ccccc} \text{Hom}_{\mathcal{D}}(L(C'), D) & \xrightarrow{- \circ L(f)} & \text{Hom}_{\mathcal{D}}(L(C), D) & \xrightarrow{g \circ -} & \text{Hom}_{\mathcal{D}}(L(C), D') \\ \downarrow \tau_{C', D} & & \downarrow \tau_{C, D} & & \downarrow \tau_{C, D'} \\ \text{Hom}_{\mathcal{C}}(C', R(D)) & \xrightarrow{- \circ f} & \text{Hom}_{\mathcal{C}}(C, R(D)) & \xrightarrow{R(g) \circ -} & \text{Hom}_{\mathcal{C}}(C, R(D')) \end{array}$$

In this case, we say that L is **left adjoint** to R , or R is **right adjoint** to L .

Exercise 1.2.2. Let $U: \mathbf{Top} \rightarrow \mathbf{Set}$ be the underlying set functor. Determine (and prove) whether there are adjunctions involving U and the discrete/trivial topology functors D and T above.

Solution: We have that $\mathrm{Hom}_{\mathbf{Top}}(D(Y), X) \cong \mathrm{Hom}_{\mathbf{Set}}(Y, U(X))$, where Y is a set and X is a topological space. Particularly, any continuous map $D(Y) \rightarrow X$ is firstly a map of sets from Y to $U(X)$, and any set-theoretic map $Y \rightarrow U(X)$ is continuous from $D(Y) \rightarrow X$. In fact, the natural isomorphism is the identity.

We also have that $\mathrm{Hom}_{\mathbf{Top}}(X, T(Y)) \cong \mathrm{Hom}_{\mathbf{Set}}(U(X), Y)$, with X a topological space and Y a set. Any continuous map $X \rightarrow T(Y)$ maps to the empty set or Y entirely, so it's a set-theoretic function. Conversely, any set-theoretic function from $U(X) \rightarrow Y$ is continuous, since the preimage of Y is just X and the preimage of the $\emptyset$ is $\emptyset$.

These are natural... □

Exercise 1.2.3. Show that right adjoints are unique in the sense if R_1, R_2 are both right adjoints of a functor L , then there is a natural isomorphism $R_1 \cong R_2$. The same goes for left adjoints. *Hint: Use the Yoneda Lemma*

Solution: Since we have two right adjoints, we have that there is a natural bijection of sets

$$\mathrm{Hom}_{\mathcal{C}}(-, R_1(D)) \cong \mathrm{Hom}_{\mathcal{D}}(L(-), D) \cong \mathrm{Hom}_{\mathcal{C}}(-, R_2(D)),$$

where $D \in \mathcal{C}$. Thus, $\mathrm{Hom}_{\mathcal{C}}(-, R_1(D))$ and $\mathrm{Hom}_{\mathcal{C}}(-, R_2(D))$ are naturally isomorphic. The Yoneda lemma tells us that $R_1(D) \cong R_2(D)$ naturally. This is the natural isomorphism we were searching for, so we are done. For left adjoints, we require the covariant Yoneda lemma. □

Exercise 1.2.4.

1. Let $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{D} \rightarrow \mathcal{C}$ be an adjoint pair of functors. Show that there are two natural transformations

$$\eta: \mathrm{id}_{\mathcal{C}} \Rightarrow RL \quad \text{and} \quad \varepsilon: LR \Rightarrow \mathrm{id}_{\mathcal{D}}$$

such that the compositions

$$R\varepsilon \circ \eta R = \mathrm{id}_R \quad \text{and} \quad \varepsilon L \circ L\eta = \mathrm{id}_L.$$

The natural transformation η is called the *unit* of the adjunction, and ε is called the *co-unit*.

2. Let L and R be functors as above, and let η and ε be natural transformations satisfying

$$R\varepsilon \circ \eta R = \mathrm{id}_R \quad \text{and} \quad \varepsilon L \circ L\eta = \mathrm{id}_L.$$

Show that L and R are (left and right) adjoint.

Solution: 1. We construct the desired natural transformation η and ε . First, since L, R is an adjoint pair, we have a bijection of sets $\Phi_{C,D}: \mathrm{Hom}_{\mathcal{D}}(L(C), D) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(C, R(D))$

which is natural in C and D . This gives us the following commutative square:

$$\begin{array}{ccccc} \mathrm{Hom}_{\mathcal{D}}(L(C'), D) & \xrightarrow{-\circ L(f)} & \mathrm{Hom}_{\mathcal{D}}(L(C), D) & \xrightarrow{g \circ -} & \mathrm{Hom}_{\mathcal{D}}(L(C), D') \\ \downarrow \Phi_{C', D} & & \downarrow \Phi_{C, D} & & \downarrow \Phi_{C, D'} \\ \mathrm{Hom}_{\mathcal{C}}(C', R(D)) & \xrightarrow{-\circ f} & \mathrm{Hom}_{\mathcal{C}}(C, R(D)) & \xrightarrow{R(g) \circ -} & \mathrm{Hom}_{\mathcal{C}}(C, R(D')) \end{array}$$

Here, $f: C \rightarrow C'$ and $g: D \rightarrow D'$.

Define $\eta: \mathrm{id}_{\mathcal{C}} \Rightarrow RL$ as follows. If $C \in \mathrm{ob}(\mathcal{C})$, then $C \mapsto \Phi_{C, L(C)}(\mathrm{id}_{L(C)})$. Conversely, if $\varepsilon: L(R) \Rightarrow \mathrm{id}_{\mathcal{D}}$ is defined as follows. If $D \in \mathrm{ob}(\mathcal{C})$, then $D \mapsto \Phi_{R(R), D}^{-1}(\mathrm{id}_{R(D)})$. We verify that these are natural transformations; that is, we want to show that

$$RL(f) \circ \eta_C = \eta_{C'} \circ \mathrm{id}_C(f) = \eta_{C'} \circ f.$$

Here, the second equality comes from the fact that $\mathrm{id}_C(f) = f$. In the commutative diagram above, let $L(C') = D$, so that $\mathrm{id}_D \in \mathrm{Hom}_{\mathcal{D}}(D, D)$. Then, commutativity of the left square gives

$$\Phi_{C', D}(\mathrm{id}_D) \circ f = \Phi_{C, D}(L(f)),$$

but notice that $\Phi_{C', D}(\mathrm{id}_D) = \eta_{C'}$, since $L(C') = D$. Thus, $\eta_{C'} \circ f = \Phi_{C, D}(L(f))$. Notice that this is the RHS of the commutativity we want, so we now consider what $\Phi_{C, D}(L(f))$ is. For this, note that $L(f) = L(f) \circ \mathrm{id}_{L(C)}$. Further, we have that $L(f): D \rightarrow D'$ (since $L: C \mapsto D, C' \mapsto D'$). Then the naturality in D tells us that

$$\Phi(L(f) \circ \mathrm{id}_{L(C)}) = R(L(f)) \circ \Phi_{C, D}(\mathrm{id}_{L(C)}) = R(L(f)) \circ \eta_C.$$

Therefore, we get the desired naturality:

$$RL(f) \circ \eta_C = \Phi(L(f)) = \eta_{C'} \circ f.$$

The naturality for ε follows analogously. Now, we just need to verify the composition rules. Again, we just verify this for the first rule, since the second will follow analogously.

We want to show $R\varepsilon \circ \eta R = \mathrm{id}_R$. First, note that $\eta(R(D)) = \Phi_{R(D), LR(D)}(\mathrm{id}_{LR(D)})$. We also have, by naturality from the square on the adjunction pair L and R , that $\Phi_{R(D), D}(\varepsilon_D) = R(\varepsilon_D) \circ \Phi_{RD, LR(D)}(\mathrm{id}_{LR(D)})$. Therefore, we get that

$$R\varepsilon \circ \eta R = \Phi_{R(D), D}(\varepsilon_D).$$

However, by definition, this last expression is just $\mathrm{id}_{R(D)}$. Thus, we've shown this, concluding the proof.

2. Let $R\varepsilon \circ \eta R = \mathrm{id}_R$ and $\varepsilon L \circ L\eta = \mathrm{id}_L$. We define $\Phi_{C, D}: \mathrm{Hom}_{\mathcal{D}}(L(C), D) \rightarrow \mathrm{Hom}_{\mathcal{C}}(C, R(D))$: $\Psi_{C, D}$ as follows:

$$\begin{aligned} \Phi_{C, D}(f) &= R(f) \circ \eta_C \\ \Psi_{C, D}(g) &= \varepsilon_D \circ L(g) \end{aligned}$$

Here, $f \in \mathrm{Hom}_{\mathcal{D}}(L(C), D)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(C, R(D))$. These are inverses to each other. We will check for one direction. We have that

$$\begin{aligned} \Psi_{C, D} \circ \Phi_{C, D}(f) &= \Psi_{C, D}(R(f) \circ \eta_C) \\ &= \varepsilon_D \circ L(R(f) \circ \eta_C) \\ &= \varepsilon_D \circ L(R(f)) \circ L(\eta_C). \end{aligned}$$

Now, we use the following naturality square for ϵ on $L(C)$ and D .

$$\begin{array}{ccc} LRL(C) & \xrightarrow{LR(f)} & LR(D) \\ \downarrow \epsilon_{L(C)} & & \downarrow \epsilon_D \\ L(C) & \xrightarrow{f} & D \end{array}$$

Therefore, we have that

$$\Psi_{C,D} \circ \Phi_{C,D}(f) = f \circ \epsilon_{L(C)} \circ L(\eta_C) = f \circ \text{id}_{L(C)} = f.$$

This show that Ψ and Φ are inverses to each other, so these are isomorphisms. This gives a bijection of sets as desired. The naturality is also clear from definition. If $g: D \rightarrow D'$, then we have that

$$\Phi_{C,D'}(g \circ f) = R(g) \circ R(f) \circ \eta_C = R(g) \circ \Phi_{C,D}(f).$$

□

Remark 1.2.5 — This provides another characterization of adjunctions.

§1.2.i An Example: Group and Ab

Let **Ab** be the category of abelian groups which has morphisms as group homomorphisms. There is a **forgetful** functor

$$F: \mathbf{Ab} \rightarrow \mathbf{Group}$$

which forgets that an abelian group is abelian (F is just the identity on objects and morphisms). It is obviously fully faithful.

There is also a functor in the other direction

$$(-)^{\text{ab}}: \mathbf{Group} \rightarrow \mathbf{Ab}$$

which sends G to its abelianization

$$G^{\text{ab}} := G/[G, G].$$

Here $[G, G]$ is the commutator subgroup; the subgroup of G generated by all commutators $aba^{-1}b^{-1}$.

Claim 1.2.6. The forgetful functor and abelianization map form an adjunction for **Group** and **Ab**.

To prove this, we will show that $(-)^{\text{ab}}$ is left adjoint to F . We need to show that for every group G and every abelian group A , the map

$$\begin{aligned} \text{Hom}_{\mathbf{Ab}}(G^{\text{ab}}, A) &\rightarrow \text{Hom}_{\mathbf{Group}}(G, A) \\ (f: G^{\text{ab}} \rightarrow A) &\mapsto (\bar{f}: G \rightarrow G^{\text{ab}} \xrightarrow{f} A) \end{aligned} \tag{*}$$

is a bijection, and natural in (G, A) .

To see that the map above is injective, consider two group homomorphisms $f, g: G^{\text{ab}} \rightarrow A$ such that $\bar{f} = \bar{g}$. Then $f \circ s = g \circ s$ where s is the quotient map $s: G \twoheadrightarrow G^{\text{ab}}$. But s is a surjective group homomorphism, so s is epic in **Group**. Hence $f = g$ and $(*)$ is injective.

To see surjectivity, let $f: G \rightarrow A$ be a group homomorphism. Then f kills commutators because A is abelian:

$$\begin{aligned} f(aba^{-1}b^{-1}) &= f(a)f(b)f(a)^{-1}f(b)^{-1} \\ &= f(a)f(a)^{-1}f(b)f(b)^{-1} \\ &= e. \end{aligned}$$

Thus, $[G, G] \subset \ker(f)$. This implies that f factors through the quotient $G/[G, G] = G^{\text{ab}}$

$$\begin{array}{ccc} G & \xrightarrow{f} & A \\ \downarrow s & \searrow \exists! g & \\ G^{\text{ab}} & & \end{array}$$

Clearly $g \mapsto f$ under $(*)$, so $(*)$ is also a surjection.

It remains to show that this is naturality, stare at the following diagram until it commutes.

$$\begin{array}{ccccc} \text{Hom}_{\mathbf{Ab}}((G')^{\text{ab}}, A) & \xrightarrow{- \circ (f)^{\text{ab}}} & \text{Hom}_{\mathbf{Ab}}(G^{\text{ab}}, A) & \xrightarrow{g \circ -} & \text{Hom}_{\mathbf{Ab}}(G^{\text{ab}}, A') \\ \downarrow & & \downarrow & & \downarrow \\ \text{Hom}_{\mathbf{Group}}(G', A) & \xrightarrow{- \circ f} & \text{Hom}_{\mathbf{Group}}(G, A) & \xrightarrow{g \circ -} & \text{Hom}_{\mathbf{Group}}(G, A') \end{array}$$

Exercise 1.2.7. Show that the above diagram commutes.

Exercise 1.2.8. In the category **Group**, epimorphisms are precisely surjections.

§1.3 Products and Coproducts

Definition 1.3.1. Let $\mathcal{C}$ be a category, and I be an index set, not necessarily finite. Let $\{C_i: i \in I\}$ be a collection of objects in $\mathcal{C}$. A **product** $C := \prod_{i \in I} C_i$ is an object C of $\mathcal{C}$ and morphisms $\pi_i: C \rightarrow C_i$, for all $i \in I$ such that for every object $D \in \text{ob } \mathcal{C}$ and every family of morphisms $f_i: D \rightarrow C_i$, there exists a unique map $g: D \rightarrow C$ such that the following diagram commutes for all i :

$$\begin{array}{ccc} D & & \\ \exists! g \downarrow & \searrow f_i & \\ C & \xrightarrow{\pi_i} & C_i \end{array}$$

Example 1.3.2

Let $\mathcal{C} = \mathbf{fdVect}_k$, and $C_i = k$ for all i . Then, the product doesn't exist in $\mathcal{C}$ if I is infinite. If I is finite, then $\prod_{i \in I} C_i = k^n$, where $n = |I|$. The projection maps are $\pi_i: k^n \rightarrow k$, the projection to the i -th factor. It suffices to just check that the appropriate diagram commutes. Given $f_i: V \rightarrow k$ for all i , we shall define the unique map $g: V \rightarrow k^n$ as

$$g(v) = \begin{pmatrix} f_1(v) \\ \vdots \\ f_n(v) \end{pmatrix}$$

However, if $\mathcal{C}$ is $\mathbf{Vect}_k$, then $\prod_{i \in I} C_i$ always exists. For any indexed set of vector spaces C_i , $i \in I$, we take

$$\prod_{i \in I} C_i := \{ \text{functions } I \rightarrow \cup_{i \in I} F(C_i), f(i) \in C_i \},$$

where $F(C_i)$ is the underlying set of C_i .

Example 1.3.3

In $\mathcal{C} = \mathbf{Set}$, products are *Cartesian products*. For instance, let $I = \{1, 2\}$. Then,

$$\prod_{i \in I} X_i = X_1 \times X_2.$$

More generally, we have

$$\prod_{i \in I} X_i = \{ \text{functions } f: I \rightarrow \cup_{i \in I} X_i, f(i) \in X_i \}.$$

Example 1.3.4

A product in $\mathbf{Group}$ is usual product of groups.

Example 1.3.5

The product in the category of R -modules, where R is a ring, is the direct product.

Example 1.3.6

The product in the category of topological spaces is the Cartesian product of underlying sets, equipped with the product topology.

Definition 1.3.7. Let $\mathcal{C}$ be a category, and I be an index set, not necessarily finite. Let $\{C_i: i \in I\}$ be a collection of objects in $\mathcal{C}$. A **coproduct** $\coprod_{i \in I} C_i$ is an object C of $\mathcal{C}$ and morphisms $\iota_i: C_i \rightarrow C$ for all $i \in I$ such that for every object $D \in \text{ob } \mathcal{C}$ and every family of morphisms $f_i: C_i \rightarrow D$, there exist a unique map $g: C \rightarrow D$ such that the following

diagram commutes for all $i \in I$:

$$\begin{array}{ccc} C_i & \xrightarrow{\iota_i} & C \\ & \searrow f_i & \downarrow \exists! g \\ & & D \end{array}$$

Definition 1.3.8. Let R be a ring and M_i be a collection of left R -modules. Then, we define the **direct sum** $\oplus_{i \in I} M_i$ to be the set

$$\{(x_1, x_2, \dots) : x_i \in M_i, \text{ all but finitely many are nonzero}\}.$$

Example 1.3.9

Let $\mathcal{C} = \mathbf{Vect}_k$. Coproducts in $\mathbf{Vect}_k$ always exist, and are often called *direct sums*. Let V_i be vector spaces indexed by some set I .

$$\begin{aligned} \coprod_{i \in I} V_i &= \bigoplus_{i \in I} V_i \\ &= \{ \text{functions } f: I \rightarrow \cup_{i \in I} V_i \text{ such that } f(i) \in V_i \\ &\quad \text{and } f(i) = 0 \text{ for all but finitely many } i \in I \}. \end{aligned}$$

The maps here are the inclusions $\iota_i: V_i \rightarrow \bigoplus_{i \in I} V_i$ defined by

$$\iota_i(v) = \text{the function } f: I \rightarrow \cup_{i \in I} V_i \text{ given by } f(j) = \begin{cases} 0 & i \neq j \\ v & i = j \end{cases}.$$

Here is a sketch proof of the above. If we are given a vector space W and maps $g_i: V_i \rightarrow W$ for all i , then we can define the map $\bigoplus_{i \in I} V_i \rightarrow W$ as described above. If $f \in \bigoplus_{i \in I} V_i$, this gets mapped to $\sum_{i \in I} g_i(f(i))$.

In the case that I is finite and $|I| = n$, then $\bigoplus_{i \in I} V_i = V_1 \oplus \dots \oplus V_n$, and

$$\begin{aligned} k^n &\rightarrow W \\ \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} &\mapsto g(v_1) + \dots + g(v_n) \end{aligned}$$

We shall note that this sum is well-defined. This is because all but finitely many of the $f(i)$ are zero.

Remark 1.3.10 — In infinite dimensional cases, coproducts are smaller than direct sums.

Example 1.3.11

In **Set**, coproducts are disjoint unions.

Example 1.3.12

In **Group**, products are just cartesian products. Coproducts in **Group** is the free product. Thus, if $I = \{1, 2\}$, then

$$\prod_{i \in I} G_i = G_1 \times G_2, \quad \coprod_{i \in I} G_i = G_1 * G_2.$$

Exercise 1.3.13 (Important). Let R be a ring and M_i be a collection of modules. Then,

$$\prod_{i \in I} M_i \cong \bigoplus_{i \in I} M_i.$$

Thus, in the category of left modules $R\text{-mod}$, we have the coproducts and products coincide.

Solution: We do this by showing that $\bigoplus_{i \in I} M_i$ has the universal property defined by the coproduct. That is, given some other R -module X and R -module maps $f_i: M_i \rightarrow X$, there exists a unique R -module map $g: \bigoplus_{i \in I} M_i \rightarrow X$ such that for all i and every $m_i \in M_i$, we have

$$g(\iota_i(m_i)) = f_i(m_i).$$

We show the existence and uniqueness of g . The map g we construct is

$$g((m_i)_{i \in I}) := \sum_{i \in I} f_i(m_i).$$

Note that this is well-defined because all but finitely many terms m_i are nonzero. Further, this is indeed an R -module map because it respects addition on both sides and because the R -action is preserved. We now show that g is the unique map defined by the universal property. Let $(m_i)_{i \in I} \in \bigoplus_{i \in I} M_i$ be an arbitrary element. Then, we can write

$$(m_i) = \sum_{i \in I} \iota_i(m_i).$$

Then, we have that

$$g((m_i)_{i \in I}) = g\left(\sum_{i \in I} \iota_i(m_i)\right) = \sum_{i \in I} g(\iota_i(m_i)) = \sum_{i \in I} f_i(m_i),$$

so we are done. $\square$

Exercise 1.3.14. Show that products in $\mathcal{C}$ are ‘the same’ as coproducts in $\mathcal{C}^{\text{op}}$.

Solution: Essentially flipping the arrows in each definition shows that products and coproducts are dual notions. $\square$

Proposition 1.3.15

If the product (or coproduct) $\prod_{i \in I} C_i$ exists in $\mathcal{C}$, then it is unique up to unique isomorphism.

Proof. We just do this for products. Suppose we have two products, C and C' with respective projections. Then, by the universal property, there exist unique maps $f: C \rightarrow$

C' and $g: C' \rightarrow C$. Thus, we have a unique map $f \circ g: C' \rightarrow C'$. Then, $\text{id}_{C'}$ is another map which respects the commutativity with the projections, so $f \circ g = \text{id}_{C'}$ by uniqueness. Similarly, $g \circ f = \text{id}_C$, so we have that f and g were isomorphisms. The dualized argument may be used for the coproduct. $\square$

2 Complex of R -modules, Projective, Injective, Flat Modules

§2.1 Definitions and Examples

Let R be an associative ring with unit. Recall that a left R -module is an abelian group with a bilinear associative operation $R \times M \rightarrow M$ defined by

$$r, m \mapsto r \cdot m,$$

and $1 \cdot m = m$, for all $m \in M$. Conversely, a right R -module is an abelian group but with a right action defined in the analogous way. Unless explicitly mentioned, all our modules are left modules.

Example 2.1.1

0 is an R -module.

Example 2.1.2

Take $M = R^n = \left\{ \begin{pmatrix} r_1 \\ \vdots \\ r_n \end{pmatrix} : r_i \in R \right\}$ is an R -module, with the action defined by

$$r \cdot \begin{pmatrix} r_1 \\ \vdots \\ r_n \end{pmatrix} := \begin{pmatrix} rr_1 \\ \vdots \\ rr_n \end{pmatrix}.$$

Example 2.1.3

Any (left) ideal $I \subseteq R$ is an R -module.

Example 2.1.4

Given a linear transformation $L: V \rightarrow V$ of a k -vector space V , one turns V into a $k[x]$ -module by making x act by L .

Definition 2.1.5. A **homomorphism** of (left) R -modules is a map $f: M \rightarrow N$ of abelian groups that preserves the action of R . That is, $f(r \cdot m) = r \cdot f(m)$, where $r \in R$ and $m \in M$. An isomorphism of R -modules is a bijective map.

Notation 2.1.6. We define $R\text{-mod}$ to be the category of left R -modules, and $\text{mod-}R$ to be the category of right R -modules.

Remark 2.1.7 — Note that if R is commutative, then the left and right versions of associativity coincide. A formal way to say this would be to say that there is an equivalence of categories from $R - \mathbf{mod}$ to $\mathbf{mod} - R$.

Remark 2.1.8 — Given a ring R , we define the opposite ring R^{op} by replacing multiplication of xy with yx . A right R -module is therefore equivalent to a left R^{op} -module, and if R is commutative, R and R^{op} coincide.

Example 2.1.9

We note that $k - \mathbf{mod}$ and $\mathbf{Vect}_k$ are isomorphic, for where k is a field.

Example 2.1.10

$\mathbb{Z} - \mathbf{mod}$ are isomorphic to $\mathbf{Ab}$. We note that that $\mathbb{Z}$ -action $\mathbb{Z} \times M \rightarrow M$ is forced.

Example 2.1.11

Now let G be a group and k be a field. Let

$$R = \bigoplus_{g \in G} k \cdot g,$$

where the direct sum is as an abelian group. let $\alpha, \beta \in k$ and $g \in G$. Then the product $(\alpha \cdot g) \cdot (\beta \cdot h)$ is defined by

$$(\alpha \cdot g) \cdot (\beta \cdot h) := (\alpha\beta)(g \cdot h).$$

This makes a group into a ring, called the **group ring** $k[G]$. We note that $k[G] - \mathbf{mod}$ are isomorphic to the category of representations of a group G over a field k .

Example 2.1.12

Let S be a set. Suppose we are given a left R -module M_s for each $s \in S$. The **direct product** $\prod_{s \in S} M_s$ of modules M_s as the product of sets M_s indexed by S , so that its elements are written as $(m_s)_{s \in S}$, where $m_s \in M_s$. The addition is defined coordinate wise, so the action of $r \in R$ multiplies each coordinate by r . For a fixed element $s_0 \in S$ the surjective map

$$\prod_{s \in S} M_s \rightarrow M_{s_0}$$

that forgets all coordinates other than the s_0 -coordinate is a map of R -modules. If all M_s are isomorphic to a given module M , we denote the direct product by M^S , because this is the same as the set of all functions $S \rightarrow M$.

Example 2.1.13

The **direct sum** $\bigoplus_{s \in S} M_s$ is the submodule of $\prod_{s \in S} M_s$ given by the condition that all but finitely many coordinates are zero. For a fixed element $s_0 \in S$ the injective map

$$M_{s_0} \rightarrow \bigoplus_{s \in S} M_s$$

that sends $m \in M_{s_0}$ to $(m_s)_{s \in S}$, where $m_s = 0$ if $s \neq s_0$ and $m_{s_0} = m$, is a map of R -modules.

Remark 2.1.14 — If S is finite, then there is no difference between the direct sum and the direct product.

Remark 2.1.15 — Recall that in the category of R -modules, the direct sum just defined is the coproduct in the category of R -modules.

Definition 2.1.16. A **free module** is any module isomorphic to

$$\bigoplus_{i \in I} R,$$

for some I .

Remark 2.1.17 — Here, I can be of any cardinality (finite, countable, uncountable).

Exercise 2.1.18. Show that there is an adjunction between **Set** and $R\text{-mod}$. We have

$$F: \mathbf{Set} \leftrightarrow R\text{-mod}: U,$$

where U maps an R -module to its underlying set and F maps a set to its free module. That is, if I is a set, then $F(I) = \bigoplus_{i \in I} R$. Here, F is the left adjoint and U is the right adjoint.

Exercise 2.1.19. Let k be a field. Show that every k -module is free.

Solution: k -modules are just vector spaces. Vector spaces are just direct sums of k . $\square$

§2.2 Complexes of R -modules

Definition 2.2.1. Let R be a ring. A sequence of (finite or infinite) R -modules and maps is a **complex** if, in the diagram,

$$\cdots \rightarrow M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \xrightarrow{f_{i+1}} M_{i+2} \rightarrow \cdots,$$

each M_i is an R -module and each $f_i: M_i \rightarrow M_{i+1}$ is a map of R -modules (for all i) such that $f_{i+1}f_i = 0$, for all i .

Notation 2.2.2. We may write a complex totally as $M_\bullet$, or just M (when lazy).

Remark 2.2.3 — Note that the sequence above can be finite or infinite (indexed by $\mathbb{Z}$).

Definition 2.2.4. Complex possess a **homology**. Fix i and look at

$$H_i(M_*) = \ker(f_i)/\text{Im}(f_{i-1}).$$

This is called the i -th **homology** of M_* .

Remark 2.2.5 — We note that $\text{Im}(f_{i+1}) \subseteq \ker(f_i)$, by the condition $f_{i+1}f_i = 0$.

Remark 2.2.6 — When $H_i(M_*) \cong 0$, we may say that there is *no homology*.

Definition 2.2.7. A complex $M_\bullet$ is called **exact** if $\ker(f_i) = \text{Im}(f_{i-1})$, for all i . In other words, if $H_i(M_*) \cong 0$ for all i .

Notation 2.2.8. A more common notation of this would be to just write

$$\dots \xrightarrow{\partial} M_i \xrightarrow{\partial} M_{i-1} \xrightarrow{\partial} \dots$$

We can then just write $\partial^2 = 0$, which is what we will continue with.

Definition 2.2.9. An exact complex of the following form

$$0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 0$$

is called a **short exact sequence** (s.e.s).

Remark 2.2.10 — Note that exactness at A tells us that $\ker(\alpha) = 0$; this is equivalent to α being injective. Exactness at C tells us that $\text{Im}(\beta) = \ker(C \rightarrow 0)$; this is equivalent to β being surjective. Of course, exactness at B tells us that $\ker(\beta) = \text{Im}(\alpha)$. Ultimately, all of this tells us that

$$C \cong B/A.$$

Remark 2.2.11 — We may write $0 \rightarrow A \rightarrow B$ just to mean that we have an injection from A to B .

Exercise 2.2.12. Show that a map of R -modules $\alpha: A \rightarrow B$ is:

1. injective if and only if $\ker \alpha = 0$ if and only if α is a monomorphism.
2. surjective if and only if $\text{coker } \alpha = 0$ if and only if α is an epimorphism.

Solution: 1. Suppose α is injective. Suppose $\alpha(x) = 0$ and $x \neq 0$. Write $x = y - z$, for some $y \neq z$. Then, $\alpha(y - z) = \alpha(y) - \alpha(z) = 0$, so $y = z$, a contradiction. Now suppose $\ker \alpha = 0$. First, we have that if $\alpha(x) = \alpha(y)$, then $\alpha(x - y) = 0$, so $x - y = 0$, proving injectivity. We have the first equivalence. Now, suppose that there are $e_1, e_2: C \rightarrow A$ such that $\alpha e_1 = \alpha e_2$. We want to show that $e_1 = e_2$. Well, we see that

$$\alpha(e_1 - e_2)(x) = 0,$$

so $e_1(x) - e_2(x) = 0$ for all x . Thus, monomorphism. Now suppose α is a monomorphism. Now, let $e_1, e_2: \ker(\alpha) \rightarrow A$, where e_1 is the inclusion map and e_2 is the trivial map. Then, we have that $\alpha e_1 = \alpha e_2$. Since α is a monomorphism, we have that $e_1 = e_2$, so $0 = \text{Im} e_1 = \ker(\alpha)$. Thus, we're done.

2. Let α be surjective. Then, $B = \text{Im} \alpha$, so $\text{coker } \alpha = B/\text{Im} \alpha = 0$. Now suppose $\text{coker } \alpha = 0$. Let $y \in B$. $B/\text{Im} \alpha = 0$, so $y \in \text{Im} \alpha$. We have the first equivalence. Thus, α is surjective. Now let $f_1, f_2: B \rightarrow C$ such that $f_1 \alpha = f_2 \alpha$. By surjectivity of α , we have that $f_1(x) = f_2(x)$ for all $x \in B$, since there exists some $y \in A$ such that $\alpha(y) = x$. Now suppose α is an epimorphism. Let $f_1, f_2: B \rightarrow \text{coker } \alpha$ with f_1 be the natural projection and f_2 be the trivial map. Then, we have $f_1 \alpha = f_2 \alpha$, so $f_1 = f_2$ since α is epi. Therefore, $0 = \text{Im} f_2 = \text{Im} f_1 = \text{coker } \alpha$. $\square$

Definition 2.2.13. Two short exact sequences are **isomorphic** if there is a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{\alpha} & B & \xrightarrow{\beta} & C & \longrightarrow & 0 \\ & & \downarrow f & & \downarrow g & & \downarrow h & & \\ 0 & \longrightarrow & A' & \xrightarrow{\alpha'} & B' & \xrightarrow{\beta'} & C & \longrightarrow & 0 \end{array}$$

such that f, g, h are isomorphisms.

Definition 2.2.14. A short exact sequence isomorphic to the following form is called a **split** exact sequence:

$$0 \rightarrow A \xrightarrow{\alpha} A \oplus C \xrightarrow{\beta} C \rightarrow 0,$$

where $\alpha(x) = (x, 0)$ and $\beta((x, y)) = y$.

Example 2.2.15

Take $R = \mathbb{Z}/4\mathbb{Z}$, and consider the sequence

$$0 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0.$$

This is split.

Example 2.2.16

Take $R = \mathbb{Z}/4\mathbb{Z}$ again. Consider the sequence

$$0 \rightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$$

This is non-split s.e.s. of R -modules.

Proposition 2.2.17

A short exact sequence of left R -modules

$$0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 0$$

is split if and only if one of the following equivalent conditions hold:

- (i) there exists a map $\sigma: C \rightarrow B$ such that $\beta\sigma = \text{id}_C$. Such a ‘right inverse’ is called a **section** of β .
- (ii) there exists a retraction $r: B \rightarrow A$ such that $r\alpha = \text{id}_A$. Such a ‘left inverse’ is called a **retraction** of α .

Proof. First suppose that the sequence splits. Thus, we can write

$$0 \rightarrow A \rightarrow A \oplus B \rightarrow B \rightarrow 0.$$

Then, we can let r be the projection map and σ be the inclusion map. These clearly work and are a retraction and section, respectively.

Now assume that the sequence admits a section. We want to show that the sequence splits. We shall construct an isomorphism $f: B \rightarrow A \oplus C$ that makes the following diagram commute.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{\alpha} & B & \xrightarrow[\beta]{\sigma} & C & \longrightarrow & 0 \\ & & \text{id}_A \downarrow = & & f \downarrow \cong? & & \text{id}_C \downarrow = & & \\ 0 & \longrightarrow & A & \xrightarrow{\iota} & A \oplus C & \xrightarrow{\pi} & C & \longrightarrow & 0 \end{array}$$

We define this as

$$f(b) = (\alpha^{-1}(b - \sigma\beta(b)), \beta(b)).$$

Then,

$$\beta(b - \sigma\beta(b)) = \beta(b) - \beta\sigma\beta(b) = \beta(b) - \beta(b) = 0.$$

That is, $b - \sigma\beta(b) \in \ker(\beta) = \text{Im}(\alpha)$. α is injective, so there exists a unique $\alpha^{-1}(b - \sigma\beta(b))$. We shall now verify that this diagram commutes and that f is an isomorphism. We get that

$$f\alpha(a) = (\alpha^{-1}(\alpha(a) - \sigma\beta\alpha(a)), \beta\alpha(a)) = (a, 0) = \iota(a).$$

Next, we have

$$\pi f(b) = \pi(\alpha^{-1}(b - \sigma\beta(b)), \beta(b)) = \beta(b).$$

Thus, the diagram commutes. What remains is to show that f is an isomorphism. If $f(b) = 0$, then

$$(\alpha^{-1}(b - \sigma\beta(b)), \beta(b)) = 0 = (0, 0).$$

In particular, $\beta(b) = 0$, so $b \in \ker(\beta) = \text{Im}(\alpha)$, so there exists $a \in A$ such that $b = \alpha(a)$. However,

$$f(\alpha(a)) = \iota(a) = (a, 0) = 0,$$

as assumed. Thus, $a = 0$, so $b = 0$, giving injectivity. Let $a \in A$, $c \in C$. We compute $f(\sigma(c) + \alpha(a))$.

$$\begin{aligned} f(\sigma(c) + \alpha(a)) &= (\alpha^{-1}(\alpha(a) - \sigma\beta\alpha(a)), \beta\alpha(a)) + (\alpha^{-1}(\sigma(c) - \sigma\beta\sigma(c)), \beta\sigma(c)) \\ &= (a, 0) + (0, c). \end{aligned}$$

This is an arbitrary element of $A \oplus C$, so f is surjective.

The final part for (ii) implies the splitness is an exercise. $\square$

Remark 2.2.18 — We note that the final part is formally *dual*.

Exercise 2.2.19. Show that the s.e.s.

$$0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 0$$

is split when C is free.

Solution: We will see this later, but C is projective, so we get a map from $s: C \rightarrow B$ such that $\pi s = \text{id}_C$. $\square$

Exercise 2.2.20. Consider the following commutative diagram of R -modules

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A_1 & \longrightarrow & B_1 & \longrightarrow & C_1 \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A_2 & \longrightarrow & B_2 & \longrightarrow & C_2 \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A_3 & \longrightarrow & B_3 & \longrightarrow & C_3 \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

Suppose the rows and the middle column are exact. Show that the left column is exact if and only if the right column is exact. *Hint: Turn the diagram 90° .*

Solution: This can be done by a big diagram chase. First, assume that the left column is exact. We start with surjectivity of $C_2 \rightarrow C_3$. Given an element $c \in C_3$, we can find some $b \in B_3$ by surjectivity. Then, by surjectivity of $B_2 \rightarrow B_3$, we can find some $b' \in B_2$. Then, there exists some element $c' \in C_2$ mapped under $B_2 \rightarrow C_2$ by b' , which is equal by commutativity.

Next is injectivity. Suppose that $c \in C_1$ maps to zero under $C_1 \rightarrow C_2$. Then, by surjectivity of $B_1 \rightarrow C_1$, there exists some element $b \in B_1$ which maps to c . Note that by commutativity, b mapped under $B_1 \rightarrow B_2 \rightarrow C_2$ maps to zero as well. Note that the $\ker(B_2 \rightarrow C_2) = \text{Im}(A_2 \rightarrow B_2)$, so there exists some $a' \in A_2$ which maps to the image of b under $B_1 \rightarrow B_2$ then. Now, note that the image of a' under $A_2 \rightarrow A_3 \rightarrow B_3$ is equal to its image under $A_2 \rightarrow B_2 \rightarrow B_3$ by commutativity. But, the image of a' under $A_2 \rightarrow B_2$ is equal to the image of b under $B_1 \rightarrow B_2$, so the image of a' under $A_2 \rightarrow A_3 \rightarrow B_3$ is equal to the image of b under $B_1 \rightarrow B_2 \rightarrow B_3$, which is just zero. Then, $A_3 \rightarrow B_3$ is injective, so a' under $A_2 \rightarrow A_3$ maps to zero. Thus, there exists a preimage of a' , say $a \in A_1$ under the map $A_1 \rightarrow A_2$. Now, the image of b under $B_1 \rightarrow B_2$ and the image of a under $A_1 \rightarrow A_2 \rightarrow B_2$ are equal by commutativity at the top left. Thus, b is equal to the image of a under $A_1 \rightarrow B_1$ by injectivity. It follows then that

Now for exactness. By commutativity and exactness of the middle column with the right, it immediately follows that $C_1 \rightarrow C_2 \rightarrow C_3$ is the zero map, so we just need to show that $\ker(C_2 \rightarrow C_3) \subseteq \text{Im}(C_1 \rightarrow C_2)$. Let $c \in C_2$ map to zero under $C_2 \rightarrow C_3$. By surjectivity, we find $b \in B_2$ which maps to c through surjectivity at the middle. However, by commutativity, b goes to zero through $B_2 \rightarrow B_3 \rightarrow C_3$, so there exists some $a' \in A_3$ which maps to the same image of b through $B_2 \rightarrow B_3$. By surjectivity, this lifts to some $a \in A_2$. Now, we note that the difference of a under $A_2 \rightarrow B_2$ and b under $B_2 \rightarrow B_3$ map to zero by commutativity, so there is some $b' \in B_1$ which maps to this difference. Then, it suffices to just find c through $B_2 \rightarrow C_2$ then. c is equal to $B_2 \rightarrow C_2$ applied to a under $A_2 \rightarrow B_2$ plus b' under $B_1 \rightarrow B_2$. The image under a disappears by exactness, and the part with b' gives an element in C_1 due to commutativity. We are therefore done, since we have found some element in C_1 which maps to c .

We've only shown one implication, but we can finish the other by a similar manner... $\square$

Definition 2.3.1. An R -module M is called a **direct summand** of an R -module N if there exists some R -module M' such that $N \cong M \oplus M'$.

$$\begin{array}{ccc} & P & \\ \swarrow \exists h & \downarrow g & \\ M & \xrightarrow{f} & N \end{array}$$

Here, we say that f **lifts** q .

$$0 \longrightarrow A \longrightarrow M \overset{f}{\underset{h}{\longrightarrow}} P \longrightarrow 0$$

That is if P is projective, then any s.e.s. ending with P splits.

A direct sum $\oplus_{s \in S} M_s$ of R -modules is projective if and only if each M_s is projective.

$$g := \sum_{s \in S} g_s$$

is a well-defined function $M \rightarrow B$. It is clear that $\beta g = f$.

Conversely, assume that M is projective. A map $f_s: M_s \rightarrow C$ for one element $s \in S$ extends trivially to a map $f: M \rightarrow C$ whose restriction to any other summand is zero. It lifts to a map $g: M \rightarrow B$. Restricting to M_s we get a lifting of f_s . $\square$

Example 2.3.5

Free R -modules are projective.

Example 2.3.6

If $R = \mathbb{Z}/6\mathbb{Z}$, then $R \cong \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, an isomorphism of rings. This induces an isomorphism of R -modules. Thus, both $\mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$ are projective.

Proposition 2.3.7

Let R be a ring. An R -module P is projective if and only if P is a direct summand of a free module.

Proof. We note that P is a direct summand of a free module means that there is some R -module Q such that $P \oplus Q \cong \bigoplus_{i \in I} R$. By the equivalent characterization of ‘splitting’, this is then equivalent to the following statement: there exists a free module F and $p: F \rightarrow P$ and $s: P \rightarrow F$ such that $ps = \text{id}_P$; that is, there is a section to p .

Now, assume that P is projective. Pick a set of generators $x_i \in P$ indexed by some $i \in I$ (some set). For instance, take every element of P . Define the map

$$f: \bigoplus_{i \in I} R \rightarrow P$$

$$p_i \mapsto x_i.$$

where p_i is the ‘ i -th basis vector’, where

$$(p_i)_j = \begin{cases} 0 & i \neq j, \\ 1 & i = j. \end{cases}$$

Note that $f: \bigoplus_{i \in I} R \rightarrow P$ is surjective since $p_i \mapsto x_i$, and all the x_i are all in the image of f (and the $\text{Im}(f)$ is an R -submodule). Thus, $P = \text{Im}(f)$. We note that projectivity is not used yet. Now consider the following diagram:

$$\begin{array}{ccc} & & P \\ & \nearrow \exists h & \downarrow \text{id} \\ F := \bigoplus_{i \in I} R & \xrightarrow{f} & P \end{array}$$

Note that the existence of h exists by projectivity. However, this means that we have a map $f: F \rightarrow P$ and a map $h: P \rightarrow F$ such that $fh = \text{id}_P$. Thus, P is a direct summand by the remark at the beginning of the proof.

Now for the converse. Assume that P is a direct summand of a free module, so $P = \oplus_{i \in I} R$. This is equivalent to the fact that all summands are projective by the previous proposition. Therefore, R is projective, so $\oplus_{i \in I} R$ is projective, implying that P is projective, since it is a summand. $\square$

Remark 2.3.8 — We note that in the above proof for the forward implication, we used that every module is a quotient of a free R -module. In particular, the free R -module F is the freely generated R -module by elements of M .

Example 2.3.9

We note that projective modules are not necessarily free. Let $R = \mathbb{Z}/6\mathbb{Z}$ and M be the principal ideal of R generated by 3, so $M \cong \mathbb{Z}/2\mathbb{Z}$. The abelian group $\mathbb{Z}/6\mathbb{Z}$ is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, which gives an isomorphism of $\mathbb{Z}/6\mathbb{Z}$ -modules $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$. Thus, by the previous proposition $\mathbb{Z}/2\mathbb{Z}$ is a projective $\mathbb{Z}/6\mathbb{Z}$ -module but because 2 is not a power of 6, it’s certainly not free.

Exercise 2.3.10. Construct infinitely many pairwise non-isomorphic projective $\mathbb{Z}/12\mathbb{Z}$ -modules, none of which are free.

Solution: First, note that $\mathbb{Z}/12\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ by the Chinese remainder theorem. Of course, $\mathbb{Z}/12\mathbb{Z}$ is a free $\mathbb{Z}/12\mathbb{Z}$ -module, and $\mathbb{Z}/3\mathbb{Z}$ and $\mathbb{Z}/4\mathbb{Z}$ are direct summands of it, so these are free. Thus, the following are non-isomorphic projective $\mathbb{Z}/12\mathbb{Z}$ -modules:

$$(\mathbb{Z}/3\mathbb{Z})^n \oplus (\mathbb{Z}/4\mathbb{Z})^m, \quad n, m \geq 0.$$

$\square$

Definition 2.3.11. An R -module I is called **injective** if for any injective R -module map $f: N \hookrightarrow M$ and any map $g: N \rightarrow I$, there exists a map $h: M \rightarrow I$ such that $g = h \circ f$.

$$\begin{array}{ccc} & & I \\ & \nearrow \exists h & \uparrow g \\ M & \xleftarrow{f} & N \end{array}$$

Here, we say g **extends** f .

Lemma 2.3.12

An R -module $M = \prod_{i \in I} M_i$ is injective if and only if each M_i is injective.

Proof. Let $M = \prod_{s \in S} M_s$ be a direct product of R -modules and let $f: A \rightarrow M$ be a map of R -modules. If M_s is injective for each $s \in S$, then the s -coordinate of f , which is a map $f_s: A \rightarrow M_s$, is such that there exists $g_s: B \rightarrow M_s$ with the property $g_s \alpha = f_s$. The product (which can be infinite)

$$g := \prod_{s \in S} g_s$$

is a well-defined function $B \rightarrow M$. It is clear that $g \alpha = f$.

Conversely, assume that M is injective. A map $f_s: A \rightarrow M_s$ for one element $s \in S$ extends trivially to a map $f: A \rightarrow M$ (all coordinates zero except s). Then there is a map $g: B \rightarrow M$ such that $g\alpha = f$. The s -coordinate g_s of g has the desired property $g_s\alpha = f_s$. $\square$

Theorem 2.3.13 (Baer's Criterion of Injectivity)

Let M be a (left) R -module. Then, the following conditions are equivalent:

- (i) M is injective.
- (ii) for any (left) ideal I of R , any map of (left) R -modules $f: I \rightarrow M$ extends to a (left) R -module map $g: R \rightarrow M$ such that $g|_I = f$.
- (iii) any map of (left) R -modules $f: I \rightarrow M$, where I is a (left) ideal, is of the form $f(x) = xb$, for some fixed $b \in M$.

Proof. To see the equivalence of (ii) and (iii), we note that to give a map of left R -modules $R \rightarrow M$ is the same as to specify the image of $1 \in R$. So if $f(x) = xm$ for any $x \in I$, then f can be extended to R by sending 1 to m . Conversely, any map $I \rightarrow M$ which is a restriction of some map $F: R \rightarrow M$ has the form $F(x) = xF(1)$.

We now prove the theorem by proving (i) $\iff$ (iii). If M is injective, then any $f: I \rightarrow M$ extends to a map of R -modules $R \rightarrow M$, so one direction is clear.

The proof of the converse requires Zorn's lemma. Let us recall its statement. Let S be a non-empty set with a partial order $\leq$. A totally ordered subset $C \subset S$ is called a *chain*. An upper bound of a subset of S is an element $t \in S$ such that $t \geq x$ for any x in this subset. Zorn's lemma states that if any chain in S has an upper bound, then S has a maximal element, that is, an element $m \in S$ such that $m \leq x$ for $x \in S$ implies $m = x$.

Let $A \subset B$ be left R -modules and let $g: A \rightarrow M$ be a map of left R -modules. To show that M is injective we need to show that g extends to a map of R -modules $B \rightarrow M$. Let S be the set of pairs (A', g') , where A' is a R -submodule of B containing A and $g': A' \rightarrow M$ is a map of R -modules whose restriction to A is g . We make S a partially ordered set:

$$(A', g') \leq (A'', g'') \quad \text{if } A' \subset A'' \text{ and the restriction of } g'' \text{ to } A' \text{ is } g'.$$

S is nonempty, because (A, g) is in S , and each chain in S has an upper bound: take the union of the submodules A' in this chain; this is a submodule of B together with a map of R -modules to M which extends $g: A \rightarrow M$. By Zorn's lemma S has a maximal element. Call it (A_0, g_0) .

Let us prove that $A_0 = B$. This will finish the proof, since then we have the desired diagram:

$$\begin{array}{ccc} & & M \\ & \nearrow \exists g_0 & \uparrow g \\ B & \longleftarrow & A \end{array}$$

By contradiction, suppose $A_0 \neq B$, then there exists $\beta \in B \setminus A_0$. Now, let $A_1 = A_0 + R\beta$; that is, the set of all sums $a + r\beta$, where $a \in A_0$ and $r \in R$. This is the R -submodule

generated by A_0 and β , so properly contains A_0 . To deduce a contradiction it is enough to extend g_0 to $g_1: A_1 \rightarrow M$, for which we need to define the image $m = g_1(\beta) \in M$. This would contradict the maximality of (A_0, g_0) .

Now let $I = \{r \in R: r\beta \in A_0\}$. This is a proper ideal of R ($1 \notin I$). Consider the map $h: I \rightarrow M$ defined by $h(r) = g_0(r\beta)$ (this map can be thought of as $I \rightarrow A_0 \cap (R\beta) \rightarrow M$, where $r \mapsto r\beta \mapsto g_0(r\beta)$). By the hypothesis (iii), there is a $b \in M$ such that $h(r) = rb$. Now define $g_1: A_1 \rightarrow M$ as follows:

$$g_1(a + r\beta) = g_0(a) + rb.$$

This is clearly R -linear, so we just check that this map is well-defined. That is, if $a + r\beta = a' + r'\beta$ for $a, a' \in A_0$ and $r, r' \in R$, then

$$g_0(a) + rb = g_0(a') + r'b.$$

This implies that $a - a' = (r' - r)\beta$, hence $r - r' \in I$ and in this case

$$g_0(a) - g_0(a') = g_0(a - a') = g_0((r' - r)\beta) = h(r' - r) = (r' - r)b.$$

This provides

$$g_0(a) + rb = g_0(a') + r'b,$$

so g_1 is well defined. We now have $(A_0, g_0) < (A_1, g_1)$, which is a contradiction to the maximality. Note that $g_1|_{A_0} = g_0$ is clear from how we define g_1 . Thus, $A_0 = B$, so we get that M is injective. $\square$

Example 2.3.14

Let k be a field, $R = k[[X]]$, and $M = k[X^{-1}]$, polynomials in X^{-1} . Let $x \in R$ and $x^{-i} \in M$, then

$$x \cdot x^{-i} := \begin{cases} x^{-i+1} & \text{if } i > 0, \\ 0 & \text{if } i = 0. \end{cases}$$

We can extend this to the following

$$x^j \cdot x^{-i} := \begin{cases} x^{-i+j} & \text{if } i > j, \\ 0 & \text{if } i \leq j. \end{cases}$$

More generally, we extend this to any power series $f(x) \in k[[X]] = R$. Let $f(x) = \sum a_j x^j$. Then,

$$f(x) \cdot x^{-i} := \sum a_j x^j \cdot x^{-i}.$$

Note that the sum above has finitely many terms by the definitions of the R -action on M . Note that M is not finitely generated, because we require arbitrarily small powers of x . This module M is injective.

Proof. R is a PID, so every ideal of R is of the form (x^i) , for some i (further, a DVR). We need to check that the following diagram commutes:

$$\begin{array}{ccc} & & M \\ & \nearrow \exists h? & \uparrow \text{any } g \\ B & \longleftarrow & A \end{array}$$

Say $g(X^i) = m \in M$, where $m = f(X^{-1})$, a polynomial in X^{-1} . We extend g to h by setting $h(1) = x^{-i}f(x^{-1})$. This works (check), and then we apply Baer's. $\square$

Example 2.3.15

Let $R = \mathbb{Z}/n\mathbb{Z}$, and $M = \mathbb{Z}/n\mathbb{Z}$. Then, M is injective. In this case, R is called *self-injective*.

Proof. We want to apply Baer again. Note that $\mathbb{Z}/n\mathbb{Z}$ is a principal ideal ring. Say $I \subseteq \mathbb{Z}/n\mathbb{Z}$ is our ideal. Then, $I = a\mathbb{Z}/n\mathbb{Z}$. Then, for a map $f: I \rightarrow \mathbb{Z}/n\mathbb{Z}$, we're interested in $f(a)$. Now, note that $\frac{n}{a}$ kills $f(a)$, and I is precisely the set of elements killed by $\frac{n}{a}$. Therefore, $f(a) \in a\mathbb{Z}/n\mathbb{Z}$, and so $f(a) = a \cdot m$, for some $m \in \mathbb{Z}/n\mathbb{Z}$. Then, if we define $F: \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ as $F(1) = m$, then we see that $F(a) = a \cdot m = f(a)$, so we are done, by Baer's criterion. $\square$

Exercise 2.3.16 (Dual Basis Criterion for Projectivity). Let R be a ring. Prove the following statement. P is a projective R -module if and only if there is a set I and elements $\{e_i\}_{i \in I}$ of P and elements $\{f_i\}_{i \in I}$ of $\text{Hom}_R(P, R)$ such that $p = \sum_i e_i f_i(p)$ for all $p \in P$ (and the sum is always finite).

Solution. First suppose that P is projective. Then, pick a set of generators $e_i \in P$ which are indexed by some set $i \in I$, and defined the surjection

$$\beta: \bigoplus_{i \in I} R \rightarrow P$$

$$p_i \mapsto e_i,$$

where p_i is the i -th basis vector. We then have the following exact sequence then:

$$0 \rightarrow \ker \beta \rightarrow F \xrightarrow{\beta} P \rightarrow 0$$

Note that since this ends with a projective module, this sequence splits, giving us a section of β , say $\sigma: P \rightarrow F$. Now let $p \in P$. Then, we must have that

$$\sigma(p) = \sum_{i \in I} r_i p_i.$$

We define the maps $f_i \in \text{Hom}_R(P, R)$ to be the map $f_i(p) = r_i$; these are the coordinate functions. Then, we have that

$$p = \beta\sigma(p) = \beta\left(\sum_{i \in I} r_i p_i\right) = \sum_{i \in I} r_i \beta(p_i) = \sum_{i \in I} f_i(p) e_i.$$

Note that this sum is finite because the expression $\sum_{i \in I} r_i p_i$ is finite.

Now for the converse. We will show that P is a direct summand of a free R -module. First, since e_i clearly generates P , we can define a surjection β analogous to before:

$$\beta: \bigoplus_{i \in I} R \rightarrow P$$

$$p_i \mapsto e_i,$$

We show that P is a direct summand now by showing that β admits a section $\sigma: P \rightarrow F$. Let $p \in P$. Then, by the property given we can write $p = \sum_{i \in I} e_i f_i(p)$. Then, we define σ as

$$\sigma(p) = \sum_{i \in I} f_i(p) p_i.$$

Then, since the sum involving p is finite, the sum in $\sigma(p)$ is finite as well, so this is well-defined. Further, this is R -linear. We just need to check that this defines a section as claimed.

$$\beta\sigma(p) = \beta\left(\sum_{i \in I} f_i(p) p_i\right) = \sum_{i \in I} f_i(p) \beta(p_i) = \sum_{i \in I} f_i(p) e_i = p,$$

so we are done. Thus, P is a direct summand of F as claimed. $\square$

Exercise 2.3.17. Let $R = C[0, 1]$ be the ring of continuous real-valued functions on $[0, 1]$. Let $P = C_{(0)}[0, 1]$ be the ideal in R of those functions f such that $f[0, \epsilon] = 0$ for some $\epsilon > 0$. Show that P is a projective R -module. Do the same for smooth functions as well.

§2.4 Hom

Let A and B be left R -modules. We denote by $\text{Hom}_R(A, B)$ the set of maps of R -modules $A \rightarrow B$. It is an abelian group under addition of maps. If R is commutative then $\text{Hom}_R(A, B)$ is an R -module: if $f \in \text{Hom}_R(A, B)$ and $r \in R$ then rf is the map $(rf)(a) = r \cdot f(a) = f(ra)$. This doesn't work if R is non-commutative (check this).

Proposition 2.4.1

Let $\{A_i\}$, A , $\{B_i\}$, B be R -modules. There are canonical (natural) isomorphisms

$$\text{Hom}_R\left(\bigoplus_{i \in I} A_i, B\right) \cong \prod_{i \in I} \text{Hom}_R(A_i, B)$$

and

$$\text{Hom}_R\left(A, \prod_{i \in I} B_i\right) \cong \prod_{i \in I} \text{Hom}_R(A, B_i).$$

Proof. This follows immediately from the universal property of products and coproducts. Given a map $g: \bigoplus_{i \in I} A_i \rightarrow B$, the inclusions $\iota_j: A_j \rightarrow \bigoplus_{i \in I} A_i$ allow us to define the maps $g_j = g \circ \iota_j$ such that the following diagram commutes for all $j \in I$:

$$\begin{array}{ccc} A_j & & \\ \downarrow \iota_j & \searrow g_j & \\ \bigoplus_{i \in I} A_i & \xrightarrow{g} & Y \end{array}$$

The first isomorphism is given by $f \mapsto (f \circ \iota_i)_{i \in I}$. The inverse sends $(f_i)_{i \in I}$ to the unique R -module map $f: \bigoplus_{i \in I} A_i \rightarrow B$ making the diagrams

$$\begin{array}{ccc} A_j & & \\ \downarrow \iota_j & \searrow f_j & \\ \bigoplus_{i \in I} A_i & \xrightarrow{f} & B \end{array}$$

commute, which exists due to the universal property. This first isomorphism therefore follows via the universal property of coproducts. The argument for the second isomorphism is similar. $\square$

Corollary 2.4.2

There are canonical isomorphisms

$$\mathrm{Hom}_R(A_1 \oplus A_2, B) \cong \mathrm{Hom}_R(A_1, B) \oplus \mathrm{Hom}_R(A_2, B)$$

and

$$\mathrm{Hom}_R(A, B_1 \oplus B_2) \cong \mathrm{Hom}_R(A, B_1) \oplus \mathrm{Hom}_R(A, B_2).$$

Proof. Finite products are the same as finite coproducts in the category of R -modules and the category of abelian groups. $\square$

Remark 2.4.3 — There is a map

$$\bigoplus_{i \in I} \mathrm{Hom}_R(A, B_i) \longrightarrow \mathrm{Hom}_R\left(A, \bigoplus_{i \in I} B_i\right),$$

but this may not be an isomorphism if I is infinite. Consider the case $A = \bigoplus_{i \in I} B_i$ where each B_i is non-zero. Then the identity map on A is not in the image of the above map if I is infinite. Modules A such that the above map is an isomorphism for all families $\{B_i\}$ are sometimes called *compact*.

Similarly,

$$\bigoplus_{i \in I} \mathrm{Hom}_R(A_i, B) \longrightarrow \mathrm{Hom}_R\left(\prod_{i \in I} A_i, B\right)$$

need not be an isomorphism.

Remark 2.4.4 — Note that the group $\mathrm{Hom}_R(R, M)$ is canonically isomorphic to M via the map sending $f: R \rightarrow M$ to $f(1)$.

Let

$$0 \longrightarrow A_1 \longrightarrow A_2 \longrightarrow A_3 \longrightarrow 0 \quad (2.1)$$

be an exact sequence of left R -modules.

Remark 2.4.5 — Let M be an R -module. We shall note that $\mathrm{Hom}_R(-, M)$ is a contravariant functor. That is, taking $f: A_1 \rightarrow A_2$, then $\mathrm{Hom}_R(A_2, M) \rightarrow \mathrm{Hom}_R(A_1, M)$ can be defined by $g \mapsto gf$, where $g: A_2 \rightarrow M$.

Conversely, one can check similarly that $\mathrm{Hom}_R(M, -)$ is a (covariant) functor.

Given the remark above, we get the sequence

$$0 \leftarrow \mathrm{Hom}(A_1, M) \leftarrow \mathrm{Hom}(A_2, M) \leftarrow \mathrm{Hom}(A_3, M) \leftarrow 0.$$

We now ask when this short sequence of Hom is exact.

Remark 2.4.6 — In the case of a split exact sequence, the previous corollary gives that

$$0 \rightarrow A_1 \rightarrow A_1 \oplus A_2 \rightarrow A_2 \rightarrow 0,$$

then we can apply $\text{Hom}(-, B)$ to get a split sequence.

$$0 \leftarrow \text{Hom}(A_1, B) \leftarrow \text{Hom}(A_1 \oplus A_2, B) \leftarrow \text{Hom}(A_2, B) \leftarrow 0.$$

Lemma 2.4.7

Suppose we have (2.1).

(i) The sequence

$$0 \rightarrow \text{Hom}_R(A_3, B) \rightarrow \text{Hom}_R(A_2, B) \rightarrow \text{Hom}_R(A_1, B)$$

is exact. Note that we're missing surjectivity of the last map for s.e.s. Here, we say that $\text{Hom}_R(-, B)$ is **left exact**.

(ii) B is injective if and only if

$$0 \rightarrow \text{Hom}_R(A_3, B) \rightarrow \text{Hom}_R(A_2, B) \rightarrow \text{Hom}_R(A_1, B) \rightarrow 0$$

is exact for every short exact sequence (2.1). That is, $\text{Hom}_R(-, B)$ into an injective preserves s.e.s.

Proof. Let $\alpha: A_1 \rightarrow A_2$ and $\beta: A_2 \rightarrow A_3$ in the original s.e.s. There are two things to check. If $f: A_2 \rightarrow B$ such that $f\alpha = 0$, we need to show that there is $g: A_3 \rightarrow B$ such that $g\beta = f$. We note that

$$A_3 = \text{Im}\beta \cong A_2 / \ker \beta = A_2 / \text{Im}\alpha = \text{coker}(\alpha) = A_2 / A_1.$$

Let φ denote this isomorphism. Then, consider the map $g: A_2/A_1 \rightarrow B$ defined by $g(a + \text{Im}\alpha) = f(a)$. Here, we note that

$$g(\pi(a)) = f(a).$$

The diagram we have here is as follows.

$$\begin{array}{ccccc} A_1 & \xrightarrow{\alpha} & A_2 & \xrightarrow{\beta} & A_3 \\ \parallel & & \parallel & & \downarrow \wr \varphi \\ A_1 & \longrightarrow & A_2 & \xrightarrow{\pi} & A_2/A_1 \\ & & \downarrow f & \nearrow g & \\ & & B & & \end{array}$$

Then, we claim that $g \circ \varphi: A_3 \rightarrow B$ is a map such that $f = (g \circ \varphi) \circ \beta$. We can calculate this directly now:

$$f(a) = g(\pi(a)) = g(\varphi(\beta(a))).$$

We also need to check exactness at $\text{Hom}_R(A_2, B)$. Given $g: A_3 \rightarrow B$, such that $g\beta = 0$, then $g = 0$. This holds because β is surjective (epimorphism).

Now for the second. Suppose that B is injective. We need to check exactness at $\text{Hom}_R(A_1, B)$. That is, for all $f: A_1 \rightarrow B$, there exists $g: A_2 \rightarrow B$ such that $f = g\alpha$ by definition. Since α is injective, injectivity of B ensures that we have a map g such that $f = g\alpha$, so we're done. Time for the converse. Suppose we have exactness; that is, for all $f: A_1 \rightarrow B$, there exists $g: A_2 \rightarrow B$ such that $f = g\alpha$. This is essentially the definition of injectivity. $\square$

Lemma 2.4.8

Suppose we have (2.1).

(i) The sequence

$$0 \longrightarrow \text{Hom}_R(B, A_1) \longrightarrow \text{Hom}_R(B, A_2) \longrightarrow \text{Hom}_R(B, A_3)$$

is exact. We say that $\text{Hom}_R(B, -)$ is a **left-exact functor**.

(ii) B is projective if and only if

$$0 \longrightarrow \text{Hom}_R(B, A_1) \longrightarrow \text{Hom}_R(B, A_2) \longrightarrow \text{Hom}_R(B, A_3) \longrightarrow 0$$

is exact for every short exact sequence (2.1).

Proof. Once again can directly check that (i) always works, and (ii) follows from the definition of projective. $\square$

Remark 2.4.9 — To finish off, we note that $\text{Hom}_R(M, -)$ always preserves kernels. Further, $\text{Hom}_R(M, -)$ commutes with products. The reason why we mention this is because kernels and products are examples of (categorical) limits.

$\text{Hom}(-, M)$ takes cokernels to kernels and takes coproducts to products. Cokernels and coproducts are examples of colimits.

§2.5 Tensor Products

Let A be a right R -module and B a left R -module. If R is a commutative ring then we don't need the above left or right; we just take two R -modules. We define $A \otimes_R B$ as follows.

Let F be the free abelian group generated by pairs (a, b) with $a \in A$, $b \in B$: $F = \bigoplus_{(a,b) \in A \times B} \mathbb{Z}$.

Remark 2.5.1 — Note that F is not the abelian group $A \oplus B$ consisting of pairs (a, b) . It is freely generated by them, so it contains things like $2(a, b) - 5(a', b')$. We note that $(0, 0) \neq 0$, or $2(m, n) \neq (2m, 2n)$. Typically F will have infinitely many generators.

Now, let $S \subset F$ be the $\mathbb{Z}$ -submodule generated by all elements of the form

$$(a + a', b) - (a, b) - (a', b), \quad (a, b + b') - (a, b) - (a, b'),$$

and also by the elements

$$(ar, b) - (a, rb),$$

for all $a \in A$, $b \in B$, and $r \in R$. Note that we can attain $n(a, b) = (na, b) = (a, nb)$, where $n \in \mathbb{Z}$. Given this, we define

$$A \otimes_R B := F/S.$$

The image of the generator (a, b) in $A \otimes_R B$ is denoted by $a \otimes b$. From the definition we obtain the relations

$$\begin{aligned} (a + a') \otimes b &= a \otimes b + a' \otimes b, \\ a \otimes (b + b') &= a \otimes b + a \otimes b', \\ ar \otimes b &= a \otimes rb, \end{aligned}$$

for all $a, a' \in A$, $b, b' \in B$, and $r \in R$. We note that the first two are essentially distributivity and the last is bilinearity.

Remark 2.5.2 — We note that if R is not commutative, $A \otimes_R B$ is only an abelian group. Only when R is commutative do we get that $A \otimes_R B$ is an R -module in its own right.

Example 2.5.3

Let us calculate $R \otimes_R M$, where R is a right module over itself, and M is a left R -module. The abelian group $R \otimes_R M$ is generated by elements $r \otimes m = 1 \otimes rm$, so the map

$$\begin{aligned} f: M &\rightarrow R \otimes_R M \\ m &\mapsto 1 \otimes m \end{aligned}$$

is surjective. Let us prove that f is injective, and hence an isomorphism (of R -modules).

Proof. Let F be the free abelian group with generators (r, m) , $r \in R$, $m \in M$. Define a (inverse) map $g: F \rightarrow M$ by mapping the generator (r, m) to rm (this makes sense because F is a free abelian group). We then check that the following elements of F go to 0:

$$(r + r', m) - (r, m) - (r', m), \quad (r, m + m') - (r, m) - (r, m'), \quad (rr', m) - (r, r'm),$$

for any $r, r' \in R$ and any $m, m' \in M$. Then g sends the subgroup $S \subset F$ generated by these elements to 0, hence g descends to a homomorphism $F/S = R \otimes_R M \rightarrow M$. The composition $m \mapsto 1 \otimes m \mapsto m$ is id_M , thus we have an isomorphism

$$f: M \xrightarrow{\sim} R \otimes_R M.$$

□

Remark 2.5.4 — By the example, one could say that R is the *unit* of $-\otimes_R -$.

Lemma 2.5.5

In any tensor product $M \otimes_R N$, we have that $0 \otimes n = 0$ and $m \otimes 0 = 0$.

Proof. We note that $0 \otimes n + 0 \otimes n = (0 + 0) \otimes n = 0 \otimes n$, so $0 \otimes n = 0$. The other-side is clear. $\square$

Example 2.5.6

Take $R = \mathbb{Z}$. In this case, we will often drop the subscript $\mathbb{Z}$. Here, we obtain the tensor product of abelian groups G_1 and G_2 , denoted by $G_1 \otimes_{\mathbb{Z}} G_2$. For instance, we obtain a canonical isomorphism $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z} \cong \mathbb{Z}$. Similarly, $\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}/2 \cong \mathbb{Z}/2$. Sometimes, this can be counterintuitive, as in the following example.

$\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}/3 = 0$. This follows from the relations $2(a \otimes b) = 2a \otimes b = 0 \otimes b = 0$ and $3(a \otimes b) = a \otimes 3b = a \otimes 0 = 0$, hence $a \otimes b = 0$ for all $a \in A$ and $b \in B$.

Exercise 2.5.7. Generalize the example above. That is, prove that $\mathbb{Z}/n\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/m\mathbb{Z} = \mathbb{Z}/\gcd(n, m)\mathbb{Z}$.

Proof. Let $d = \gcd(n, m) = an + bm$, for some integers $a, b \in \mathbb{Z}$. Consider $\mathbb{Z} \mapsto \mathbb{Z}/n\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/m\mathbb{Z}$ defined by $z \mapsto z(1 \otimes 1)$. We note that:

$$d(x \otimes y) = (an + bm)(x \otimes y) = an(x \otimes y) + bm(x \otimes y) = a(nx \otimes y) + b(x \otimes my) = 0.$$

Thus, the kernel of this map is $d\mathbb{Z}$ is in the kernel. Conversely, if some integer was in the kernel, then it better be divisible by both n and m . Thus, the kernel is contained in $d\mathbb{Z}$, proving the isomorphism. Furthermore, this is clearly surjective. $\square$

Exercise 2.5.8. Generalize the above even further. Let R be a commutative ring. What should $R/I \otimes_R R/J$ be? *Hint: It should be isomorphic to $R/(I + J)$.*

Proof. The desired isomorphism is $R/I \otimes_R R/J \cong R/(I + J)$. Consider the map $\phi: R/(I + J) \rightarrow R/I \otimes_R R/J$ defined by $r \mapsto (r + I) \otimes (1 + J)$. We also consider the map $\psi: R/I \otimes_R R/J \rightarrow R/(I + J)$ defined by $(r + I) \otimes (s + J) \mapsto rs + (I + J)$. We note that $\psi\phi$ and $\phi\psi$ are both the identity, so they are inverses, proving the isomorphism. $\square$

Proposition 2.5.9

The map

$$\begin{aligned}\varphi: M \times N &\rightarrow M \otimes_R N \\ (m, n) &\mapsto m \otimes n\end{aligned}$$

is a covariant functor in each argument.. That is, if $f: M \rightarrow M'$ is a map of right R -modules, then

$$\begin{aligned}f \otimes_R \text{id}_N: M \otimes_R N &\rightarrow M' \otimes_R N \\ m \otimes n &\mapsto f(m) \otimes n\end{aligned}$$

is a map of abelian groups. Similarly, $g: N \rightarrow N'$ a map of left R -modules gives

$$\begin{aligned}\text{id}_M \otimes_R g: M \otimes_R N &\rightarrow M \otimes_R N' \\ m \otimes n &\mapsto m \otimes g(n)\end{aligned}$$

Proof. We first check that $f \otimes_R N$ is well defined. Let F be the free module from before. Then, we need to see that the map

$$\begin{aligned}F &\rightarrow M \otimes_R N \\ (m, n) &\mapsto f(m) \otimes n\end{aligned}$$

vanishes on S . This is an easy check. We note that (mr, n) maps to $f(mr) \otimes n$, which is equal to $f(m)r \otimes n$, which is equal to $f(m) \otimes rn$. However, this last term is the image of (m, rn) . The same conclusion holds for the other generators. Therefore, we get an induced map

$$M \otimes_R N = F/S \rightarrow M' \otimes_R N.$$

□

Remark 2.5.10 — This tells us that the tensor product is a covariant functor in each argument. We may denote this as $\mathbf{mod}\text{-}R \times R\text{-}\mathbf{mod} \rightarrow \mathbf{Ab}$.

Proposition 2.5.11 (The Universal Property of Tensor Products)

Let R be a ring, and M be a right R -module and N be a left R -module, and G be an abelian group. Then, for every bilinear map $f: M \times N \rightarrow G$ which satisfies $f(mr, n) = f(m, rn)$ (for all $m \in M, n \in N, r \in R$), then f can be uniquely written as $f = g \circ \varphi$,

$$g: A \otimes_R B \rightarrow G$$

is a (group) homomorphism and $\varphi: A \times B \rightarrow A \otimes_R B$.

Remark 2.5.12 — If G is an R -module, the map g is a map of R -modules, of course.

Remark 2.5.13 — We will not provide a proof, for the proof is analogous to the calculations above. The point is that the quotient by F gives the bilinearity conditions.

Remark 2.5.14 — We emphasize that $M \otimes_R N$ is just an abelian group; explicitly, not an R -module. In the case that R is commutative, $M \otimes_R N$ is an R -module by defining

$$r \cdot (m \otimes n) = mr \otimes n = m \otimes rn.$$

However, we note that $M \otimes_R N$ is a module over the center of R ,

$$Z(R) = \{r \in R : rs = sr, \forall s \in R\}.$$

Explicitly, $\mathbb{Z}$ is in the center $Z(R)$, so we can ‘move’ these around.

Example 2.5.15

Let k be a field, and V be a nonzero finite dimensional vector space. Let $R = \text{End}_k(V)$, the endomorphism ring of V over k . This indeed forms a ring by multiplication. Let U, W be two more finite dimensional vector space. Then, $N = \text{Hom}_k(U, V)$ is a left R -module. Given $f \in R = \text{End}_k(V)$, then $g: U \rightarrow V$, and we define a left R -module structure by $f \cdot g = f \circ g$. We emphasize that $\text{End}_k(V)$ is noncommutative, so switching the order of multiplication would not make sense.

Likewise, let $M = \text{Hom}_k(V, W)$. This is a right R -module by the following. Let $g \in M$ and $f \in R$. Then, we define $g \cdot f = g \circ f$.

We now claim that

$$M \otimes_R N = \text{Hom}_k(V, W) \otimes_{\text{End}_k(V, V)} \text{Hom}_k(U, V) \cong \text{Hom}_k(U, W)$$

defined by $g \otimes h \mapsto g \circ h$. The remarkable aspect is that the resulting isomorphism has nothing to do with V ; thus, we can’t endow this with an R -module structure.

Exercise 2.5.16. Prove the claim above.

Solution: We want to show that $\text{Hom}_k(V, W) \otimes_{\text{End}_k(V, V)} \text{Hom}_k(U, V) \cong \text{Hom}_k(U, W)$ defined by $g \otimes h \mapsto g \circ h$ is an isomorphism. To show this, we note that these are both k -module (so k -vector spaces) and use finite dimensionality; particularly, the fact that the dimensions are equal). Then, showing either injectivity or surjectivity is enough. The way we show surjectivity is by first realizing that $\text{Hom}_k(U, W) \cong W \otimes_k U^*$ (thanks to finite dimensionality). The map here to consider is

$$\begin{aligned} W \otimes_k U^* &\rightarrow \text{Hom}_k(U, W) \\ w \otimes \varphi &\mapsto \varphi(\cdot)w. \end{aligned}$$

This is an isomorphism because this map is a surjection between spaces of equal finite dimension. We prove surjectivity here. Let $T \in \text{Hom}_k(U, W)$. Now, write φ_i as the usual dual basis of U^* and $T(u_i) = w_i \in W$, for each basis element $u_i \in U$; that is, $\varphi_i(u_j) = \delta_{ij}$.

Then, we claim that T is the image of $\sum_{i=1}^n w_i \otimes \varphi_i$. We have that this element maps to

$$\sum_{i=1}^n \varphi_i(\cdot) w_i,$$

so evaluating this at each basis element u_i , we get that

$$T(v_i) = \left(\sum_{i=1}^n \varphi_i(\cdot) w_i \right) (u_i) = w_i.$$

Thus, we have the isomorphism. Now, we just need to show that

$$(W \otimes_k V^*) \otimes_{\text{End}(V)} (V \otimes U^* \xrightarrow{\sim} W \otimes U^*$$

is indeed an isomorphism. Our map $g \otimes h \mapsto g \circ h$ translates to

$$(w \otimes \varphi) \otimes (v \otimes \psi) \mapsto \varphi(v)(w \otimes \psi).$$

We want to show this map is surjective. Take any $w \otimes \psi \in W \otimes U^*$. Choose any element $v \in V$ and $\varphi \in V^*$ such that $\varphi(v) = 1$. Then, we have that

$$(w \otimes \varphi) \otimes (v \otimes \psi) = \varphi(v)(w \otimes \psi) = w \otimes \psi,$$

so we have proven surjectivity. Thus, we have the desired isomorphism. $\square$

Theorem 2.5.17 (Tensor-Hom Adjunction)

Let R be a commutative ring and A, B, C be R -**mod**. There is a natural isomorphism

$$\text{Hom}_R(A \otimes_R B, C) \cong \text{Hom}_R(B, \text{Hom}_R(A, C)).$$

Proof. First, note that $\text{Hom}_R(A, B)$ is an R -module in its own right. We first show that there is an isomorphism of R -modules

$$\text{Bilin}_R(A \times B, C) \cong \text{Hom}_R(B, \text{Hom}_R(A, C)). \quad (2.2)$$

Here, $\text{Bilin}_R(A \times B, C)$ denotes the set of R -bilinear maps $f: A \times B \rightarrow C$ with the structure of an R -module by addition of maps $(f, g) \mapsto f + g$ and the ring action $(r, f) \mapsto rf$. Now, let $f \in \text{Bilin}_R(A \times B, C)$. Consider the map $\Phi: f \mapsto (b \mapsto f(\cdot, b))$. Note that $f(\cdot, b): A \rightarrow C$ for $b \in B$, so this lands in the correct target. First, note that

$$\Phi(rf) \mapsto (b \mapsto rf(\cdot, b)) = r(b \mapsto f(\cdot, b)) = r\Phi(f),$$

so R -linearity is clear ($b \in B$). Additivity is also clear, so Φ is a map of R -modules. We now show that this is an isomorphism. We define the inverse $\Psi: F \mapsto f$, where $f(a, b) = F(b)(a)$ for $a \in A$ and $b \in B$. Note that f is bilinear because F is linear in both A and B . R -linearity is also clear and additivity is preserved as well, so Ψ is a R -module map. We now show that these maps are inverses. Let $f \in \text{Bilin}_R(A \times B, C)$.

$$\Psi\Phi(f) = \Psi(* \mapsto f(\cdot, *)) = f.$$

Conversely, let $F \in \text{Hom}_R(B, \text{Hom}_R(A, C))$.

$$\Phi\Psi(F) = \Phi(f) = (* \mapsto f(\cdot,)),$$

but then this last expression is just F . Thus, we have the isomorphism.

By commutativity of R , $f(ar, b) = rf(a, b) = f(a, rb)$. Therefore, the universal property of the tensor product provides the R -module isomorphism

$$\text{Bilin}_R(A \times B, C) \cong \text{Hom}_R(A \otimes_R B, C)$$

defined by

$$f = g \circ \varphi \mapsto g.$$

This is clearly an isomorphism due to the uniqueness of g as given in the universal property.

Combining the two isomorphisms, we have that

$$\text{Hom}_R(A \otimes_R B, C) \cong \text{Hom}_R(B, \text{Hom}_R(A, C)).$$

These were all R -module isomorphisms, and natural in B and C , so we are done. $\square$

Remark 2.5.18 — This says that $A \otimes_R -$ (left adjoint) and $\text{Hom}_R(A, -)$ (right adjoint) are an adjunction.

Corollary 2.5.19

There is a natural isomorphism

$$A \otimes_R (B \oplus C) \cong (A \otimes_R B) \oplus (A \otimes_R C).$$

More generally, we have the following lemma.

Lemma 2.5.20

Let A be a right R -module and $M_i, i \in I$, are left R -modules, then there is a natural isomorphism

$$A \otimes_R \left(\bigoplus_{i \in I} M_i \right) \cong \bigoplus_{i \in I} (A \otimes_R M_i).$$

Proof. On the one hand, the natural map $A \times (\bigoplus_{i \in I} M_i) \rightarrow \bigoplus_{i \in I} (A \otimes_R M_i)$ defined by

$$(a, (m_i)_{i \in I}) \mapsto (a \otimes m_i)_{i \in I}$$

is bilinear and satisfies $f(ar, (m_i)_{i \in I}) = f(a, r(m_i)_{i \in I})$, so uniquely factors through the tensor product, providing one map

$$\mu: A \otimes_R \left(\bigoplus_{i \in I} M_i \right) \rightarrow \bigoplus_{i \in I} (A \otimes_R M_i)$$

Conversely, the natural map $A \otimes_R M_i \rightarrow A \otimes_R (\bigoplus_{i \in I} M_i)$ gives rise a to a map $\bigoplus_{i \in I} (A \otimes_R M_i) \rightarrow A \otimes_R (\bigoplus_{i \in I} M_i)$ via the universal property of coproducts; we note that this is bilinear, so is well-defined.

$$\nu: \bigoplus_{i \in I} (A \otimes_R M_i) \rightarrow A \otimes_R (\bigoplus_{i \in I} M_i)$$

We now just check that μ and ν compose to give identities. $\square$

Remark 2.5.21 — We note that tensor products don't work well with respect to infinite products.

Remark 2.5.22 — We note that if we have a split exact sequence

$$0 \rightarrow A \rightarrow A \oplus B \rightarrow B \rightarrow 0,$$

then the proposition above says that

$$0 \rightarrow A \otimes_R M \rightarrow (A \oplus B) \otimes_R M \rightarrow B \otimes_R M \rightarrow 0$$

is split exact.

Lemma 2.5.23 (Tensor by $- \otimes_R M$ is Right Exact)

Let

$$0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 0 \quad (2.3)$$

be an exact sequence of right R -modules. For any left R -module M , the sequence of abelian groups

$$A \otimes_R M \xrightarrow{\alpha \otimes \text{id}} B \otimes_R M \xrightarrow{\beta \otimes \text{id}} C \otimes_R M \rightarrow 0$$

is exact.

Proof. Let Q be the quotient of $B \otimes_R M$ by the image of $\alpha \otimes \text{id}$. We have a natural map

$$u: Q \rightarrow C \otimes_R M$$

induced by $\beta \otimes \text{id}$. It is enough to show that u is an isomorphism. We do this by constructing a left and right inverse, but for this, we use a set-theoretic section of β .

Since β is surjective, we can choose a set-theoretic section $s: C \rightarrow B$, i.e. a function (not necessarily a homomorphism and not necessarily respecting the R -action) such that $\beta \circ s = \text{id}_C$. Now define a set-theoretic map $v: C \otimes_R M \rightarrow Q$ defined by $a \otimes m \mapsto s(c) \otimes m$ (where $a \in C$ and $m \in M$), which is contained in $B \otimes_R M / (\text{Im}(\alpha \otimes \text{id}))$. This map does not depend on the choice of s : choose some other choice s' , so that

$$s(c) \otimes m - s'(c) \otimes m = (s(c) - s'(c)) \otimes m.$$

Since s, s' are both sections of β , they map to the same preimage of c . $\beta(s(c) - s'(c)) = 0$, so there exists some $a \in A$ such that $\alpha(a) = s(c) - s'(c)$, so $\alpha(a) \otimes m$ is in the image of $\alpha \otimes \text{id}$. Therefore, $s(c) \otimes m - s'(c) \otimes m = 0$ in Q . Similar calculations show that our function $v: C \otimes_R M \rightarrow Q$ is bilinear and satisfies $v(cr, b) = v(c, rb)$. Thus, v is a well-defined R -linear map. Thus, we have that

$$uv(c \otimes m) = u(s(c) \otimes m) = \beta(s(c)) \otimes m = c \otimes m,$$

so we get that u is surjective.

We now show injectivity. First, we note that if $b \in B$, then

$$\beta(b - s\beta(b)) = \beta(b) - \beta s\beta(b) = \beta(b) - \beta(b) = 0,$$

so $b - s\beta(b) \in \ker(\beta) = \text{Im}\alpha$. We compute that

$$vu(b \otimes m) = v(\beta(b) \otimes m) = s\beta(b) \otimes m = b \otimes m.$$

Thus, vu is the identity, so we get that u is injective. Moreover, note that we've constructed an inverse to u . $\square$

Exercise 2.5.24. Let I be a right ideal of a ring R and M be a left R -module. Prove that $R/I \otimes_R M \cong M/IM$.

Solution: Note that given the sequence

$$0 \rightarrow I \xrightarrow{\alpha} R \xrightarrow{\beta} R/I \rightarrow 0$$

is exact, so applying $- \otimes_R M$, we attain the exact sequence

$$I \otimes_R M \xrightarrow{\alpha \otimes \text{id}} R \otimes_R M \xrightarrow{\beta \otimes \text{id}} R/I \otimes_R M \rightarrow 0.$$

We note that $R \otimes_R M \cong M$ under the map ϕ which map $r \otimes m \mapsto rm$. Since $\ker \beta = \text{Im}\alpha$, if we can show that $\text{Im}\alpha \cong IM$, then we are done. Now, note that $\text{Im}(\alpha \otimes \text{id}) \cong \phi(\text{Im}(\alpha \otimes \text{id})) = IM$ since ϕ is an isomorphism. Thus, we have the desired isomorphism:

$$M/IM \cong R/I \otimes_R M.$$

$\square$

Exercise 2.5.25. Let M be a (left) R -module and I be a (left) ideal of R . Prove that $\text{Hom}_R(R/I, M) \cong \{m \in M : Im = 0\}$.

Solution: For the forward direction, we note that $f(1 + I)$ is determined by some element, say m . Then, $f(r + I) = r \cdot f(1 + I) = r \cdot m$. For this to be well-defined, we have that if $r - r' \in I$, then $(r - r') \cdot m = 0$. Thus, mapping $f \mapsto m$ is the desired morphism.

For the inverse, we define a morphism f coming from m via $m \mapsto f$ such that $f(r + I) = r \cdot m$. This is well defined, since if $r \in I$, then $f(r + I) = 0$ because $Im = 0$. R -linearity is also clear.

These also clearly form inverses! $\square$

Let A be an abelian group and let n be a positive integer. Define $A/nA = A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$. By Lemma 2.5.23 the s.e.s.

$$0 \rightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$$

(where the second arrow is multiplication by n). Applying $- \otimes_{\mathbb{Z}} A$, this gives rise to the exact sequence

$$A[n] \rightarrow A \rightarrow A \rightarrow A/nA \rightarrow 0, \quad (2.4)$$

where $A[n] = \{a \in A : na = 0\}$. We shall note that we had to alter the very left entry, and can't simply copy down the 0.

Exercise 2.5.26. Prove that $\mathbb{Z}/n\mathbb{Z} \otimes_{\mathbb{Z}} A \cong A/nA$.

Proof. Define a map $i \otimes a \mapsto ia$. This is bilinear, so is a well-defined map. Conversely, send $a \mapsto 1 \otimes a$. This is well-defined because $na \mapsto 1 \otimes na = n \otimes a = 0$. So this is well-defined as well. One can check that these are inverses. $\square$

Tensoring (2.3) by $\mathbb{Z}/n$ gives an exact sequence

$$A_1/n \rightarrow A_2/n \rightarrow A_3/n \rightarrow 0.$$

Definition 2.5.27. A left R -module M is called **flat** if $- \otimes_R M$ preserves short exact sequences of right R -modules.

Remark 2.5.28 — Given a s.e.s. as in (2.3), Lemma 2.5.23, which induces the exact tells us that flat-ness depends on the exactness at $A \otimes_R M$ in the following induced exact sequence:

$$A \otimes_R M \xrightarrow{\alpha \otimes \text{id}} B \otimes_R M \xrightarrow{\beta \otimes \text{id}} C \otimes_R M \rightarrow 0$$

In other words, flatness is equivalent to showing that the map $\alpha \otimes \text{id}$ is a monomorphism (injection). We may say that flatness is equivalent to preserving injectivity of $- \otimes_R M$.

Example 2.5.29

R is a flat R -module, since $A \otimes_R R \cong A$. In other words, $- \otimes_R R$ is naturally isomorphic to the identity functor.

Example 2.5.30

We see from our example above that $\mathbb{Z}/n\mathbb{Z}$ is not flat as an abelian group.

Lemma 2.5.31

Let $M \cong \bigoplus_{i \in I} M_i$ be a direct sum of (left) R -modules. Then, M is flat if and only if all M_i are flat.

Proof. M is flat if and only if for any s.e.s.

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

the induced sequence is exact:

$$0 \rightarrow A \otimes_R \left(\bigoplus_{i \in I} M_i \right) \rightarrow B \otimes_R \left(\bigoplus_{i \in I} M_i \right) \rightarrow C \otimes_R \left(\bigoplus_{i \in I} M_i \right) \rightarrow 0$$

The sequence above is isomorphic to the following sequence

$$0 \rightarrow \bigoplus_{i \in I} (A \otimes_R M_i) \rightarrow \bigoplus_{i \in I} (B \otimes_R M_i) \rightarrow \bigoplus_{i \in I} (C \otimes_R M_i) \rightarrow 0.$$

This sequence is exact if and only if the component parts

$$0 \rightarrow A \otimes_R M_i \rightarrow B \otimes_R M_i \rightarrow C \otimes_R M_i \rightarrow 0.$$

are exact, which is equivalent to M_i all being flat. $\square$

Exercise 2.5.32. Convince yourself that the direct sum of the sequences is exact if and only if the component parts are exact.

Solution. This is true because if we compare the component wise maps and the overall map, we have that f_i is injective for all i if and only if $\oplus_{i \in I} (f_i)_i$ is injective. $\square$

Proposition 2.5.33

Projective modules are flat.

Proof. Lemma 2.5.31 implies that a direct sum of modules is flat if and only if each summand is flat. We know that R is a flat R -module, so Lemma 2.5.31 implies that all free modules are flat. Now, Proposition 2.3.4 tells us that projective modules are direct summands of a free module, so they must be flat in their own right. $\square$

Remark 2.5.34 — We will see that the converse is not true: $\mathbb{Q}$ is flat as an abelian group (by a later theorem) and is manifestly not a direct summand of a power of $\mathbb{Z}$ (since it is divisible), hence not projective.

Definition 2.5.35. An element $a \in R$ is a **right nonzero divisor** if $ax = 0$ implies that $x = 0$, i.e. the right R -module map $R \mapsto R$ defined by $x \mapsto ax$ is injective.

Lemma 2.5.36

If M is a flat left R -module, then for any right non-zero divisor $a \in R$ and non-zero $m \in M$ we have $am \neq 0$. That is, a is a non-zero divisor on M .

Proof. Consider the injective map $R \rightarrow R$ of right R -modules sending 1 to a . If M is flat, then the induced map

$$R \otimes_R M \rightarrow R \otimes_R M, \quad (r \otimes m) \mapsto (ar) \otimes m$$

is also injective. Note that $R \otimes_R M$ is canonically isomorphic to M . The identification $m \mapsto 1 \otimes m$ of M with $R \otimes_R M$ allows us to rewrite this map as a map $M \rightarrow M$ defined by m to am . $\square$

Remark 2.5.37 — Submodules of projective or injective or flat modules need not inherit the same property.

Example 2.5.38 (Projective module with non-projective submodule)

Let $R = \mathbb{Z}/4\mathbb{Z}$ and $I = 2 \cdot \mathbb{Z}/4\mathbb{Z}$. Then if we consider R as a module over itself, it is projective. However, I is not a projective R -module since it is not a direct summand of a free R -module.

Example 2.5.39 (Injective module with non-injective submodule)

Let $R = \mathbb{Z}$. Then, we will see that $\mathbb{Q}$ is injective because it is divisible. However, $\mathbb{Z}$ as a module over itself is not flat because it is not divisible. More generally, we will see that any R -module is isomorphic to a submodule of an injective module, so we can just choose any non-injective module we wish.

Example 2.5.40 (Flat module with non-flat submodule)

Let $R = \mathbb{Z}/4\mathbb{Z}$ and $I = 2 \cdot \mathbb{Z}/4\mathbb{Z}$. Then if we consider R as a module over itself, it is projective, hence flat. However, I is a R -submodule which is not flat. To see this, consider the injection $I \hookrightarrow R$. Then, we apply $- \otimes_R I$. We get as follows:

$$I \otimes_R I \rightarrow I$$

If I were injective, this would continue to be an injection, but this is not the case. This is clearly the trivial map, since this map sends $a \otimes b \mapsto ab$. However, we note that

$$I \otimes_R I \cong 2\mathbb{Z}/4\mathbb{Z} \otimes_{\mathbb{Z}/4\mathbb{Z}} 2\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z},$$

so the kernel is nonzero.

Example 2.5.41

Let A be an abelian group and let n be a positive integer. There is an exact sequence

$$0 \longrightarrow A[n] \longrightarrow A \xrightarrow{n} A \longrightarrow A/nA \longrightarrow 0,$$

where

$$A[n] = \{a \in A \mid na = 0\}.$$

This need not be trivial, for example the case $A = \mathbb{Z}/n$.

We can view this as a tensor product of A with the short exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}/n \longrightarrow 0.$$

That is, we can check that $A/nA \cong A \otimes_{\mathbb{Z}} \mathbb{Z}/n$, and tensor the short exact sequence with A to get

$$0 \longrightarrow A[n] \longrightarrow A \otimes_{\mathbb{Z}} \mathbb{Z} \longrightarrow A \otimes_{\mathbb{Z}} \mathbb{Z} \longrightarrow A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

We will later come back to where this extra term $A[n]$ comes from, in terms of the tensor product. This is applying $\mathrm{Tor}_{\bullet}^{\mathbb{Z}}(A, -)$ to the s.e.s. above!

3 Modules over Integral Domains

§3.1 Torsion and Divisible Submodules

Let R be an integral domain, that is, a commutative ring with unit such that $ab = 0$ implies $a = 0$ or $b = 0$. If $1 \neq 0$ and R is commutative, then the multiplication by $a \neq 0$ in R is a nonzero divisor if and only if the multiplication by a map is injective (the side doesn't matter by commutativity).

Definition 3.1.1. Let M be an R -modules, and $m \in M$. m is a **torsion element** if $rm = 0$ for some nonzero $r \in R$. Define M_{tors} as the submodule of torsion elements in M .

Definition 3.1.2. If $M_{\text{tors}} = 0$, then M is called **torsion-free**. If $M = M_{\text{tors}}$, then M is called a **torsion module**.

Example 3.1.3

Let $R = \mathbb{Z}$, so R -modules are abelian groups. Abelian groups $\mathbb{Z}$ and $\mathbb{Q}$ are torsion-free, whereas $\mathbb{Z}/n\mathbb{Z}$ (for $n \neq 0$) and $\mathbb{Q}/\mathbb{Z}$ are torsion groups.

Exercise 3.1.4.

1. Let M be the quotient of $\mathbb{Z}^2$ by the subgroup generated by (m, n) , where $m, n \in \mathbb{Z}$. Determine M_{tors} .
2. What is $(\mathbb{R}/\mathbb{Z})_{\text{tors}}$?

Solution: 1. $\mathbb{Z}^2/(m, n) \cong \mathbb{Z} \oplus \mathbb{Z}/d\mathbb{Z}$, where $d = \gcd(m, n)$. Therefore, $M_{\text{tors}} \cong \mathbb{Z}/d\mathbb{Z}$, and is generated by elements like $(\frac{m}{d}, \frac{n}{d})$.

2. $(\mathbb{R}/\mathbb{Z})_{\text{tors}}$ contains elements which are an integer, after some integer multiple. This is just $\mathbb{Q}/\mathbb{Z}$. $\square$

Exercise 3.1.5. Let M be a torsion $\mathbb{Z}$ -module and N be a divisible $\mathbb{Z}$ -module. Then, show that $M \otimes_{\mathbb{Z}} N = 0$.

Solution: Let $m \otimes n \in M \otimes_{\mathbb{Z}} N$. Suppose $am = 0$, where $a \in \mathbb{Z}$. Then, there exists $l \in N$ such that $al = n$. Thus,

$$m \otimes n = m \otimes al = am \otimes l = 0.$$

$\square$

Exercise 3.1.6. Let R be any ring, and A be a torsion R -module and B be a torsion free R -module. Then, $\text{Hom}_R(A, B) = 0$.

Solution: Let $a \in A$. If $r \cdot a = 0$ for some $r \in R$, then

$$rf(a) = f(ra) = 0.$$

However, since B is torsion free, it follows that $f(a) = 0$. Since a was arbitrary, $f = 0$. $\square$

Example 3.1.7
 $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/6\mathbb{Z}, \mathbb{Z}) = 0.$
Lemma 3.1.8

Flat modules (and thus projective modules) over integral domains are torsion free.

Proof. Let $a \in R$ be nonzero, so that $a \cdot : R \rightarrow R$ is injective. Then, applying $F \otimes_R -$, the map $a \cdot : F \rightarrow F$ is also injective since $F \otimes_R -$ preserves injections. Thus, F is torsion free. $\square$

Definition 3.1.9. An element $m \in M$ is called **divisible** (or **infinitely divisible**) if for every nonzero $a \in R$, there exists an element $n \in M$ such that $an = m$. We denote by $M_{\text{div}} = \{m \in M : m \text{ is divisible}\}$. M is **divisible** (or **infinitely divisible**) if $M = M_{\text{div}}$.

Example 3.1.10

Let $R = \mathbb{Z}$. Then, $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are divisible. $\mathbb{Z}$ is not divisible.

Example 3.1.11

$\mathbb{Z}/n\mathbb{Z}$ is never divisible as an abelian group, i.e., a $\mathbb{Z}$ -module. For instance, take $1 \in \mathbb{Z}/n\mathbb{Z}$, where $n \in \mathbb{Z}$. Then, there is no x such that $nx = 1$, since $nx = 0$ (even if n is prime). However, $\mathbb{Z}/p\mathbb{Z}$ is divisible as a $\mathbb{Z}/p\mathbb{Z}$ -module.

Lemma 3.1.12

Injective modules over integral domains are divisible.

Proof. Let M be injective. Then, let $m \in M$ and $a \neq 0$ be in R . Then, aR is a principal ideal, and R being an integral domain implies that $R \xrightarrow{a} aR$ is an isomorphism. Due to this isomorphism, the map $R \rightarrow M$ defined by $r \mapsto rm$ can be precomposed by the inverse of $R \rightarrow aR$ to get a map $f: aR \rightarrow M$. Here, we note that $f(ax) = xm$. Since aR can be included into R , Baer's criterion tells us that f extends to a map of R -modules $F: R \rightarrow M$. In particular, if $n = F(1)$, then $m = f(a) = F(a \cdot 1) = aF(1) = an$. Thus, M is divisible. $\square$

Exercise 3.1.13. Show that the coproduct $\bigoplus_{n \in \mathbb{N}} \mathbb{Z}/n\mathbb{Z}$ is a torsion $\mathbb{Z}$ -module, but the product $\prod_{n \in \mathbb{N}} \mathbb{Z}/n\mathbb{Z}$ is not.

Solution: Recall that the direct sum contains elements of tuples, of which all but finitely many entries are 0. Suppose $a := (a_{i_1}, \dots, a_{i_k}) \in \bigoplus_{n \in \mathbb{N}} \mathbb{Z}/n\mathbb{Z}$. Then, letting $d = \gcd(a_{i_1}, \dots, a_{i_k})$, we have that $da = 0$.

Now, the product contains tuples, where there may be infinitely many nonzero entries. It is then clear that the product is not torsion. For instance, consider the tuple $a := (0, 1, 2, 3, \dots)$. There is clearly no integer $n \in \mathbb{Z}$ such that $na = 0$, so we are done. $\square$

Exercise 3.1.14. Let R be an integral domain and M be a finitely generated torsion-free R -module. Show that M is isomorphic to a submodule of a free R -module of finite rank.

Solution: Let $K = \text{Frac}(R)$. We first realize that $M \hookrightarrow M \otimes_R K$ defined by $m \mapsto m \otimes 1$ is an injection. Indeed, if $m \otimes 1 = 0$, then $rm \otimes 1 = 0$, for any $r \in R$. However, M is torsion free, so this forces $m = 0$. Now, since M is finitely generated, we note that we get an isomorphism

$$M \otimes_R K \cong K^d.$$

Let v_i be a set of vector which span K^d as a K -vector space. Now, if $\{m_i\}$ is a finite set of generators of M over R , then write

$$m_i = \sum_{j=1}^d a_{ij} v_j.$$

Note that the $a_j \in K$, so let a be the least common multiple of the denominators of the a_{ij} . Then, we have that am_i has all coefficients in R . Therefore, it follows that

$$aM \subseteq R^d \subseteq K^d.$$

To conclude, we construct an isomorphism from $M \rightarrow aM$ given by $\iota: M \hookrightarrow M^d$. We do this simply by sending an element by multiplication by a . This is injective because R is an integral domain, and it's clearly surjective. $\square$

§3.2 Modules over PIDs

Recall that a ring R is a PID if R is commutative (with $1 \neq 0$), R is a domain, and all ideals are principal. Some recurring examples are $\mathbb{Z}, k[x], k[[x]]$. In this case, we may consider a diagram: Let I be an ideal of R represented by an element a .

$$\begin{array}{ccc} & & M \\ & \nearrow \exists f & \uparrow \cdot m \\ R & \xleftarrow{a \cdot} & R \\ \parallel & & \downarrow \sim a \cdot \\ R & \hookrightarrow & I = (a) \end{array}$$

Proposition 3.2.1

Let M be a module over a PID R . Then, M is injective if and only if M is divisible.

Proof. Note that the forward implication is already proven (by the previous lemma). The converse is what we work on now. We use Baer's criterion. Let $I = (a) \subseteq R$. Since R is a PID, $(a) \cong R$. Let $f: I \rightarrow M$ be a R -module map. M is divisible, so let $m = f(a)$ and $n \in M$ such that $m = an$. Then, $f: I \rightarrow M$ extends to a R -module map $F: R \rightarrow M$ which sends 1 to n that restricts to f . Explicitly, f takes the form $f(x) = xn$. By Baer's criterion, we're done. $\square$

Remark 3.2.2 — The crux is that we can essentially reverse the argument because now, every ideal is principal.

Example 3.2.3

$\mathbb{Q}$ is an injective $\mathbb{Z}$ -module. $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$ -module.

Theorem 3.2.4 (Submodules of Free Modules are Free)

Let R be a PID. Then, every submodule of a free module is free. In particular, every submodule of R^n is isomorphic to R^m for some $m \leq n$.

Proof. Let M be a submodule of a free module $M \subseteq F = \bigoplus_{i \in S} R$, where S is an indexing set. We want to show that M is free. Given $T \subseteq S$, we shall write $M_T := M \cap \bigoplus_{i \in T} R$. For instance, $M_S = M$ and $M_\emptyset = 0$. Consider the set of pairs (T, B) where $T \subseteq S$ and $B \subseteq M$ such that M_T is a free module on the basis B . That is, we have the isomorphism

$$\bigoplus_{b \in B} R \xrightarrow{\sim} M_T$$

$$(r_b)_{b \in B} \mapsto \sum r_b b.$$

We partially order this set by $(T, B) \leq (T', B')$ if $T \subseteq T'$ and $B \subseteq B'$. This partially ordered set is nonempty, since $(\emptyset, \emptyset)$ is in it. This is because $M_\emptyset = 0$ is a free module on the basis $\emptyset$.

We wish to apply Zorn's lemma, so we check that all chains have an upper bound. Therefore, let $\{(T_i, B_i)\}_{i \in I}$ be a chain in this partial order. Then, we claim that $(T, B) = (\bigcup T_i, \bigcup B_i)$ is an upper bound in the chain. Setting $T = \bigcup T_i$, the module M_T is free on the basis $B = \bigcup B_i$. First, B spans M_T since any element of M_T belongs to M_{T_i} for some i , so it is spanned by B_i . Secondly, if there is a finite sum $\sum r_j b_j = 0$ in M_T for some $r_j \in R$, $b_j \in B$, then since the sum is finite there is a B_i containing all of the b_j , and so $\sum r_j b_j = 0$ in M_{T_i} . Since B_i is a basis, then all the r_j must be zero. Thus, (T, B) is an upper bound on this chain, so by Zorn's lemma, there exists a maximal element (T, B) (not necessarily same as before) such that M_T is free on a basis B .

The goal is now to show that $T = S$, for then this implies that $M_T = M_S = M$ is free. Assume for the sake of contradiction that there is some $t \in S \setminus T$. Consider the map $\pi_t: M_{T \cup \{t\}} \rightarrow R$, which comes from noting

$$M_{T \cup \{t\}} = M \cap \bigoplus_{T \cup \{t\}} R \subseteq \bigoplus_{T \cup \{t\}} R,$$

and that $\bigoplus_{T \cup \{t\}} R$ can project onto R at the t -factor. This map π_t need not be surjective, and is just the restriction of $\bigoplus_{T \cup \{t\}} R \twoheadrightarrow R$ to $M_{T \cup \{t\}}$. $\text{Im}(\pi_t) \subseteq R$ is an R -submodule of R , so is just an ideal. Since R is a PID, $\text{Im}(\pi_t)$ is principal, so let $\text{Im}(\pi_t) = (a_t) \subseteq R$. In particular, this is a free module because R is a domain. If $a_t = 0$, then $\text{Im}(\pi_t) = 0$ and if $a_t \neq 0$, then $\text{Im}(\pi_t) \cong R$ (multiplication by a_t map). Now, consider the kernel of π_t :

$$\ker(\pi_t) = \{m = (r_i)_{i \in T \cup \{t\}} \in M_{T \cup \{t\}} : r_t = 0\} = M_T.$$

Therefore, we have a s.e.s. as follows:

$$0 \rightarrow M_T \rightarrow M_{T \cup \{t\}} \xrightarrow{\pi_t} (a_t) \rightarrow 0.$$

Since (a_t) is free, this sequence is actually split. Therefore, $M_{T \cup \{t\}} \cong M_T \oplus (a_t)$, which is a direct sum of free modules; thus, free itself. It is free on a basis $B \cup \{b_t\}$, where $b_t \in M_{T \cup \{t\}}$ such that $\pi_t(b_t) = a_t$. If $a_t = 0$, then $t \in T$, which is a contradiction. Note that if $a_t \neq 0$, then we reach a contradiction, for we have found a pair $(T \cup \{t\}, B \cup \{b\}) > (T, B)$ (strictly containment). This contradicts the maximality of (T, B) . Therefore, we must have $T = S$, and we have shown what is desired.

To finish, we just need to show that the rank is indeed less than or equal to n . Let S be the indexing set of a free submodule $M \subseteq R^n$. Let K be the field of fractions of R , and Then,

$$M \otimes_R K \cong \left(\bigoplus_S R \right) \otimes_R K \cong \bigoplus_S (R \otimes_R K) \cong \bigoplus_S K,$$

which is a $|S|$ -dimensional K -vector space. Further, $M \subseteq R^n$, so $M \otimes_R K \subseteq R^n \otimes_R K \cong K^n$. Therefore, $M \otimes_R K$ is a K -vector subspace, so $\dim_K(M \otimes_R K) = |S| \leq n$. Note that K is a flat R -module, which justifies the subset. $\square$

Remark 3.2.5 — Note that this means every ideal $I \subseteq R$ is a free module.

Remark 3.2.6 — The point of including the data of the bases into the partial ordering sets is because unions of free modules are not necessarily free. In particular, consider $\frac{1}{n}\mathbb{Z} \subseteq \mathbb{Q}$. The union of such submodules is not free, but we want them to be.

Remark 3.2.7 — We note that this is something special regarding PIDs. The *natural* extension of this would be Hilbert's syzygy theorem.

Example 3.2.8

Consider $R = k[x, y]$, the polynomial ring over a field k in two variables. Consider $M = I = (x, y)$, which we claim is not free. We have $y \cdot x = x \cdot y$, which implies that the module is not free. Thus, R is not a PID.

Corollary 3.2.9

If R is a PID, then all projective modules are free. Thus, they are equivalent over PIDs.

Proof. If M is projective, it is a direct summand of a free module, so a submodule of a free module. Thus, free. $\square$

Theorem 3.2.10

If R is a PID and M is an R -module, then M is flat if and only if M is torsion free.

Proof. Note that all flat modules are torsion free over integral domains. Now, assume that M is torsion free. Let $f: A \hookrightarrow B$ be an injective R -module homomorphism. The goal is to show that $f \otimes_R M: A \otimes_R M \rightarrow B \otimes_R M$ is injective, since M be then be flat.

Step 1: We show that it's enough to prove the claim when B is free. We know that every R -module can be written as a quotient of a free R -module. That is, there exists a free module F such that we have a s.e.s.

$$0 \rightarrow K \rightarrow F \xrightarrow{\pi} B \rightarrow 0.$$

Here, K is the kernel of the map from $F \twoheadrightarrow B$. Define $A' \subseteq F$ to be

$$A' = \{x \in F: \pi(x) \in \text{Im}(f)\}.$$

By assumption, the induced map $g \otimes_R M: A' \otimes_R M \rightarrow F \otimes_R M$ is injective. In a diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & K & \xrightarrow{t} & A' & \xrightarrow{\pi_A} & A \longrightarrow 0 \\ & & \parallel & & \downarrow g & & \downarrow \\ 0 & \longrightarrow & K & \xrightarrow{h} & F & \xrightarrow{\pi} & B \longrightarrow 0 \end{array}$$

Note that $\ker(A' \rightarrow A) = K$. $K \subseteq \ker(A' \rightarrow A)$ because $\pi(x) = 0 \in \text{Im}(f)$, where $x \in K$. Conversely, $K \supseteq \ker(A' \rightarrow A)$, since if $x \in A'$ and $\pi_A(x) = 0$, then certainly $\pi(x) = 0$. Thus, $x \in K$. This gives rise to two right exact sequences:

$$\begin{array}{ccccccc} K \otimes_R M & \longrightarrow & A' \otimes_R M & \longrightarrow & A \otimes_R M & \longrightarrow & 0 \\ & & \downarrow \cap & & \downarrow & & \\ K \otimes_R M & \longrightarrow & F \otimes_R M & \longrightarrow & B \otimes_R M & \longrightarrow & 0 \end{array}$$

Note the middle vertical arrow is injective by our assumption that it's enough to prove for free modules. We now want to show the right arrow is actually injective. Suppose that there is some $y \in A \otimes_R M$ which maps to 0; we will show that $y = 0$. We now diagram chase. We have a surjection at the top, so there is some y' such that $(\pi_A \otimes_R M)(y') = y$. Then, look at $(g \otimes_R M)(y') = z \in F \otimes_R M$. This maps to 0, so there exists some w such that $(h \otimes_R M)(w) = z$. By equality, w is in the top $K \otimes_R M$ as well. Now, we have that

$$(g \otimes_R M)(t \otimes_R M)(w) = z = (g \otimes_R M)(y'),$$

so $y' = (t \otimes_R M)(w)$ by injectivity of $g \otimes_R M$. By exactness, we have that

$$y = (\pi_A \otimes_R M)(y') = (\pi_A \otimes_R M)(t \otimes_R M)(w) = 0.$$

Thus, $y = 0$. We have shown that this is a valid simplification. We now need to prove that B is free.

Step 2: Now, we show that it is sufficient to do this when $B = R$; that is, that if $B = R$, then the result holds for free modules. By the first step, it's enough to consider an inclusion $A \subset F$, where F is a free module. Given any $x \in A \otimes_R M$, it can be written as $\sum_i a_i \otimes m_i$, some finite sum. Defining $A' \subset A$ to be the submodule generated by $a_1, \dots, a_n$, we have that $A' \subset R^n$. To be precise, A' is generated under the map $f: A \hookrightarrow B = F$. Therefore, $F = R^n \oplus F'$, where F' is some other free R -module

generated by the generators of F not contained in the first factor R^n . Note that we have the isomorphism $F \otimes_R M \cong (R^n \otimes_R M) \oplus (F' \otimes_R M)$ and the following diagram.

$$\begin{array}{ccc} A' \otimes_R M & \xrightarrow{\quad} & R^n \otimes_R M \\ \downarrow \cap & & \downarrow \cap \\ A \otimes_R M & \longrightarrow & F \otimes_R M \cong (R^n \otimes_R M) \oplus (F' \otimes_R M) \end{array}$$

We want to show the injectivity of the bottom arrow. What we've done right now is taken some $x \in A \otimes_R M$ and built this top part of the diagram. Viewing x as an element in $A' \otimes_R M$, x then maps to some y in $F' \otimes_R M$, which maps to the first component of $F' \otimes_R M \oplus F'' \otimes_R M$; that is, to $(y, 0)$. We assume that this is zero (since we want to show injectivity), so we assume that $y = 0$, and we want to show $x = 0$. Therefore, it is sufficient to show that the map $A' \otimes_R M \rightarrow R^n \otimes_R M$ is injective. This is what the next paragraph is about.

To prove this, we proceed by induction. For $n = 1$, this is our assumption. So assume the result is true p to $n - 1$. Write $R = R \oplus R^{n-1}$, $A_1 = A' \cap R$ (the first component), and A_2 be the image of A' in R^{n-1} (the second component) under the projection map. That is,

$$A_2 = \text{Im}(A \hookrightarrow R^n \twoheadrightarrow R^{n-1}).$$

We get short exact sequences and a commutative diagram as below:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A_1 & \longrightarrow & A' & \longrightarrow & A_2 \longrightarrow 0 \\ & & \downarrow \cap & & \downarrow \cap & & \downarrow \cap \\ 0 & \longrightarrow & R & \longrightarrow & R^n & \longrightarrow & R^{n-1} \longrightarrow 0 \end{array}$$

It is worth noting that at the bottom, we have a split sequence. Applying $- \otimes_R M$ provides the following diagram.

$$\begin{array}{ccccccc} A_1 \otimes_R M & \longrightarrow & A' \otimes_R M & \longrightarrow & A_2 \otimes_R M & \longrightarrow & 0 \\ \downarrow \cap & & \downarrow & & \downarrow \cap & & \\ R \otimes_R M & \longrightarrow & R^n \otimes_R M & \longrightarrow & R^{n-1} \otimes_R M & \longrightarrow & 0 \end{array}$$

However, note that $R^n = R \oplus R^{n-1}$ and that tensor products commute with direct sums. Thus, have a split sequence at the bottom, so we can attach a zero at the left.

$$\begin{array}{ccccccc} A_1 \otimes_R M & \longrightarrow & A' \otimes_R M & \longrightarrow & A_2 \otimes_R M & \longrightarrow & 0 \\ \downarrow \cap & & \downarrow & & \downarrow \cap & & \\ 0 & \longrightarrow & R \otimes_R M & \hookrightarrow & R^n \otimes_R M & \longrightarrow & R^{n-1} \otimes_R M \longrightarrow 0 \end{array}$$

We want to show the middle vertical map is injective; this can be proven by diagram chasing. Suppose $x \in A' \otimes_R M$ maps to 0 in $R^n \otimes_R M$. Then commutativity on the right says that x maps to 0 in $A_2 \otimes_R M$. Then, x is in the kernel, so x is in the image of some element in $A_1 \otimes_R M$. However, this element maps to 0 by commutativity on the left, so this element was 0 by composing two injections on the left. Therefore, x is in the image of 0, so $x = 0$. Thus, the middle map is injective and the induction is complete.

Step 3: We prove this for $B = R$. Thus, assume we have an injection $f: A \hookrightarrow R$. $\text{Im}(f)$ is a R -submodule of R , so an ideal, and because R is a PID (first time using this!), this ideal is principal. Thus, $\text{Im}(f) = aR$. In particular, $A \cong (a)$. Thus, we want to prove that $f \otimes_R M: (a) \otimes_R M \rightarrow R \otimes_R M$ is injective. Note that f induces $\beta := f \otimes_R M: R \otimes_R M \rightarrow aR \otimes_R M$, which is surjective. Note that this map is just multiplication by a in the first entry. The composition $\alpha\beta: R \otimes_R M \rightarrow R \otimes_R M$ is identified with the map $m \mapsto am$. As M is torsion free, this map is injective. Since $\alpha\beta$ is injective and β is surjective, it must be that α is injective, so we are done. $\square$

Example 3.2.11

Now we can justify the example of a module that is flat but not projective. $\mathbb{Q}$ is torsion free, so flat. However, $\mathbb{Q}$ is divisible, so it’s not a direct summand of a free abelian group; thus, $\mathbb{Q}$ is not projective. Thus, $\mathbb{Q}$ is a flat $\mathbb{Z}$ -module which isn’t free.

Example 3.2.12

$\mathbb{Q}/\mathbb{Z}$ is a divisible $\mathbb{Z}$ -module. Therefore, it is injective.

Remark 3.2.13 — For every R , there is a unique map $\mathbb{Z} \rightarrow R$, so any R -module is a $\mathbb{Z}$ -module. We use this basic observation to leverage a lot of information to arbitrary rings.

Exercise 3.2.14. Give an example of a submodule of a free module which is not free.

Solution: Let $R = \mathbb{Z}/pq\mathbb{Z}$ (for p, q prime). Then, $\mathbb{Z}/pq\mathbb{Z}$ is clearly a free $\mathbb{Z}/pq\mathbb{Z}$ -module. Consider then the submodule $\mathbb{Z}/p\mathbb{Z}$. This is not free. $\square$

Exercise 3.2.15. Let R be a PID and M be a R -module which is free of finite rank. Then, every submodule of M is free (of finite rank $\leq n$).

Solution: This essentially follows by the structure theorem for finitely generated modules over PIDs. If M is a free R -module of finite rank, then in particular, it must be finitely generated, so the theorem applies. Thus,

$$M \cong \bigoplus_{i=1}^n R.$$

If N is a submodule of M , then it follows that N must be free of rank less than n . $\square$

Exercise 3.2.16. Let M be an R -module. Prove that there is a canonical isomorphism of abelian groups

$$\text{Hom}_R(M, \text{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})) \cong \text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z}).$$

Here, we consider $\text{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})$ with the left R -module structure given in the lectures.

Solution: We can construct an explicit map.

$$\begin{aligned} \Phi: \text{Hom}_R(M, \text{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})) &\rightarrow \text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z}) \\ f &\mapsto f(\cdot)(1) \end{aligned}$$

We first note that it is clear that Φ is a map between abelian groups. We define the inverse Ψ . Let $g \in \text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$. We define $\Psi(g)(m)(x) = g(xm)$. We just check that these are inverses. We have that

$$\Psi\Phi(f(m)) = \Psi(f(m)(1)) = f(m).$$

Conversely,

$$\Phi(\Psi(g)(m)(x)) = \Phi(g(xm)) = g(m)(x).$$

Thus, we are done. This is also natural in R . $\square$

Remark 3.2.17 — This can also be seen through the tensor-Hom adjunction.

§3.3 Existence of Enough Injectives

This is analogous to the fact that every R -module is a quotient of a free (hence projective) module.

Theorem 3.3.1

Let R be a (not necessarily commutative) ring. Any left R -module can be embedded into an injective R -module, i.e. for any module M , there exists an injective R -module homomorphism from any $M \hookrightarrow I$, where I is an injective module. That is, every left R -module is isomorphic to a submodule of an injective module.

Proof. Step 1: We first show that for any abelian group G and any nonzero element $x \in G$, there is a homomorphism $f: G \rightarrow \mathbb{Q}/\mathbb{Z}$ such that $f(x) \neq 0$. Let $C = x \cdot \mathbb{Z} \subseteq G$, a cyclic subgroups generated by x . We first construct a map $C \rightarrow \mathbb{Q}/\mathbb{Z}$ such that x maps to something not equal 0. If the order of x is n , i.e. $C = \mathbb{Z}/n\mathbb{Z}$, then $x \mapsto \frac{1}{n}$. If the order of x is infinite, i.e. $C \cong \mathbb{Z}$, then $x \mapsto \frac{1}{2}$ (indeed, this x can be sent to anything nonzero). Since $\mathbb{Q}/\mathbb{Z}$ is injective (by Baer's criterion), we have the following diagram:

$$\begin{array}{ccc} C & \longrightarrow & \mathbb{Q}/\mathbb{Z} \\ \downarrow & \nearrow \exists f & \\ G & & \end{array}$$

Here, if $x \in C$, then we have constructed $f(x)$ such that $f(x) \neq 0$.

Before moving on to the next step, define $S := \text{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})$. We show that S is a left R -module by using the right R -module structure on the free module R . Here, we define

$$(r \cdot f)(x) = f(x \cdot r).$$

We shall check associativity here:

$$(s \cdot (r \cdot f))(x) = (r \cdot f)(x \cdot s) = f(x \cdot s \cdot r) = f(x \cdot (sr)) = ((sr) \cdot f)(x)$$

The other conditions can be checked clearly.

Step 2: Now, we show that for every R -module M , and every nonzero $m \in M$, there exists an R -module homomorphism $f: M \rightarrow S$ such that $f(m) \neq 0$. We note that

$$\text{Hom}_R(M, S) = \text{Hom}_R(M, \text{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})) \cong \text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$$

by the tensor-Hom adjunction. In particular, this last isomorphism is described as follows:

$$\begin{aligned} \text{Hom}_R(M, \text{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})) &\cong \text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z}) \\ f &\mapsto g, \quad g(m) = f(m)(1) \\ f, \quad f(m) &= (r \mapsto g(rm)) \leftarrow g \end{aligned}$$

One should check that $f \leftrightarrow g$ are bijections. We know that there exists $g: M \rightarrow \mathbb{Q}/\mathbb{Z}$, a map of abelian groups such that $g(m) \neq 0$ (by the previous step). The corresponding f is defined by $f(m) = (r \mapsto g(rm))$. Thus, in particular, $f(m)(1) = g(m) \neq 0$. This means that we have a $f: M \rightarrow S = \text{Hom}_{\mathbb{Z}}(R, \mathbb{Q}/\mathbb{Z})$ with $f(m) \neq 0$. This concludes this step.

Step 3: We show that S is injective. For this, we use the isomorphism in step 2 again. We have

$$\text{Hom}_R(-, S) \cong \text{Hom}_{\mathbb{Z}}(-, \mathbb{Q}/\mathbb{Z}),$$

which is a natural isomorphism of functors. Indeed, one should check that given $h: N \rightarrow M$, then the following square commutes (and the stated isomorphisms are actually isomorphisms).

$$\begin{array}{ccc} \text{Hom}_R(M, S) & \xrightarrow{\sim} & \text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z}) \\ \downarrow - \circ h & & \downarrow - \circ h \\ \text{Hom}_R(N, S) & \xrightarrow{\sim} & \text{Hom}_{\mathbb{Z}}(N, \mathbb{Q}/\mathbb{Z}) \end{array}.$$

Indeed, we have that

$$\begin{array}{ccc} f & \xrightarrow{\quad} & g \\ \downarrow & & \downarrow \\ fh & \xrightarrow{\quad} & g' = gh \end{array},$$

where the maps g, g' are defined accordingly. We emphasize this natural isomorphism because we have a characterization of injective modules. That is, I is injective if and only if $\text{Hom}_R(-, I)$ preserves short exact sequences. Since $\mathbb{Q}/\mathbb{Z}$ is injective, it follows that $\text{Hom}_{\mathbb{Z}}(-, \mathbb{Q}/\mathbb{Z})$ preserves s.e.s., so $\text{Hom}_R(-, S)$ preserves s.e.s., implying that S is injective.

Final Step: Finally, we find an injection $M \hookrightarrow I$, where I is some injective R -module I . Let

$$I = \prod_{f \in \text{Hom}_R(M, S)} S.$$

This is injective since products of injective modules are injective. Now define

$$\begin{aligned} F: M &\rightarrow I \\ m &\mapsto (f(m))_{f \in \text{Hom}_R(M, S)} \end{aligned}$$

rm is mapped to $(r \cdot f(m))_{f \in \text{Hom}_R(M, S)}$ because each f is an R -module homomorphism, so this is a map of left R -modules. We know that if $m \neq 0$, then there is at least one component of this indexed family such that $f(m) \neq 0$. This implies that $\ker(F) = 0$, so F is injective. $\square$

Example 3.3.2

$R = k[x]/(x^n)$ is a self-injective ring, i.e. the left R -module R is injective (and free).

4 Resolutions and Derived Functors

Definition 4.0.1. Let R be a ring. A **chain complex** of R -modules is a sequence of R -modules

$$\dots \xrightarrow{d} C_{n+1} \xrightarrow{d} C_n \xrightarrow{d} C_{n-1} \rightarrow \dots,$$

such that the maps d are R -module map such that $d^2 = 0$; that is, $\ker d = \operatorname{Im} d$. The entire sequence is often denoted $C_\bullet$.

Definition 4.0.2. A **cochain complex** is that same but using the notation

$$\dots \xrightarrow{d} C_{n-1} \xrightarrow{d} C_n \xrightarrow{d} C_{n+1} \dots$$

Definition 4.0.3. The maps d in the above are called **differentials**.

Remark 4.0.4 — Any chain complex $C_\bullet$ has a sequence of **homology groups**:

$$H_n(C_\bullet) = \ker(d: C_n \rightarrow C_{n-1}) / \operatorname{Im}(d: C_{n+1} \rightarrow C_n).$$

We note that $C_\bullet$ is exact if and only if $H_n(C) = 0$ for all n . **Cohomology groups** refer to the same successive quotient, but with $C_\bullet$ being a cochain complex.

Definition 4.0.5. If $C_\bullet, D_\bullet$ are two (chain) complexes, a **chain map** $f_\bullet: C_\bullet \rightarrow D_\bullet$ is a collection of R -module maps $f_m: C_m \rightarrow D_m$ such that $df_m = f_{m-1}d$, for all m . That is, all the squares commute:

$$\begin{array}{ccccccc} \dots & \longrightarrow & C_{n+1} & \xrightarrow{d} & C_n & \xrightarrow{d} & C_{n-1} \longrightarrow \dots \\ & & \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} \\ \dots & \longrightarrow & C_{n+1} & \xrightarrow{d} & C_n & \xrightarrow{d} & C_{n-1} \xrightarrow{d} \dots \end{array}$$

A map of cochain complexes is the same but for cochain complexes.

Remark 4.0.6 — A chain map $f_\bullet: C_\bullet \rightarrow D_\bullet$ induces maps $H_n(f_\bullet): H_n(C_\bullet) \rightarrow H_n(D_\bullet)$, for all n . To show this, we must check that the kernel maps to the kernel, and the image maps to the image.

Proof. Indeed, let $a \in \ker(A_n \rightarrow A_{n-1})$, i.e. $d(a) = 0$. So $df_n(a) = f_{n+1}d(a) = 0$ (where we have used that $f_\bullet$ is a map of chain complexes). Thus, $f_n(a) \in \ker(B_n \rightarrow B_{n-1})$. Now let $a \in \operatorname{Im}(A_{n+1} \rightarrow A_n)$, i.e. $a = d(a')$ for some $a' \in A_{n+1}$. Then $f_n(a) = f_nd(a') = df_{n+1}(a')$, so $f_n(a) \in \operatorname{Im}(B_{n+1} \rightarrow B_n)$. Hence we get an induced map

$$H_n(A_\bullet) \rightarrow H_n(B_\bullet).$$

□

Remark 4.0.7 — Another way of phrasing the remark above is that homology, $H_n(-): \text{ch}(R\text{-mod}) \rightarrow R\text{-mod}$ is a functor (where $\text{ch}(R\text{-mod})$ is the category of chain complexes (of left R -modules) and chain maps; if we have right $\text{mod-}R$, then it respects this).

§4.1 Projective and Injective Resolutions

Definition 4.1.1. Let M be an R -module. A **left resolution** of M is a (right bounded) chain complex of the form

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0$$

(a complex concentrated in nonnegative degrees) with a map $P_0 \rightarrow M$ such that

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

is an exact sequence. We denote this as $P_\bullet \rightarrow M$. If every P_n is a projective R -module, then $P_\bullet$ is called a **projective resolution**.

Remark 4.1.2 — We note that the right bounded chain complex $P_\bullet$, terminating at P_0 , is necessarily exact.

Example 4.1.3 (Trivial Resolution)

$P_0 = M$ and $P_{\geq 1} = 0$ would work.

$$\cdots \rightarrow 0 \rightarrow 0 \rightarrow M \xrightarrow{\text{id}} M \rightarrow 0$$

Definition 4.1.4. A map $f_\bullet: C_\bullet \rightarrow D_\bullet$ of complexes is called a **quasi-isomorphism** if $H_n(f_\bullet): H_n(C) \rightarrow H_n(D)$ is an isomorphism, for all n .

Remark 4.1.5 — It should be clear that a chain map isomorphism is a quasi-isomorphism, but the converse is not necessarily true.

Remark 4.1.6 — Note that to have a chain complex $P_\bullet$ with a homomorphism $P_0 \rightarrow M$ is precisely the same data as a morphism of complexes

$$\begin{array}{ccccccc} \cdots & \xrightarrow{d} & P_3 & \xrightarrow{d} & P_2 & \xrightarrow{d} & P_1 & \xrightarrow{d} & P_0 & \longrightarrow & 0 \\ & & \downarrow f_3 & & \downarrow f_2 & & \downarrow f_1 & & \downarrow f_0 & & \\ \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

where M is in degree zero. To ask for the complex $P_\bullet$ to be exact precisely means that $H_n(P_\bullet) = 0$ for all $n \neq 0$ and

$$\begin{aligned} H_0(P_\bullet) &= P_0 / \text{Im}(P_1 \xrightarrow{d} P_0) \cong P_0 / \ker(P_0 \xrightarrow{f_0} M) \cong \text{Im}(f_0) \\ &= \ker(M \rightarrow 0) \cong M = H_0(M). \end{aligned}$$

So a left resolution is precisely a complex concentrated in nonnegative degrees which is quasi-isomorphic to the complex with M concentrated in degree 0.

Remark 4.1.7 — What we're really saying in the above is that a projective resolution is equivalent to a quasi-isomorphism, meaning that we have one diagram if and only if we have the other. In particular, that one diagram implies that other and vice versa.

Definition 4.1.8. Let M be an R -module. A **right resolution** of M is a (left bounded) cochain complex

$$I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

with a map $M \rightarrow I^0$ such that

$$0 \rightarrow M \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \dots$$

is exact. This is denoted $M \rightarrow I^\bullet$. This is an **injective resolution** if each of the I^i is injective.

Remark 4.1.9 — This is equivalent to a quasi-isomorphism:

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \longrightarrow & 0 & \longrightarrow & 0 \longrightarrow \dots \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & I^0 & \longrightarrow & I^1 & \longrightarrow & I^2 \longrightarrow \dots \end{array}$$

Example 4.1.10

Let $R = \mathbb{Z}$ and $M = \mathbb{Z}/n\mathbb{Z}$. Let us write down a projective resolution of M .

$$0 \rightarrow 0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$$

Remark 4.1.11 — We say that the length of a projective resolution is equal to the maximal n such that $P_n \neq 0$.

Example 4.1.12

Let $R = \mathbb{Z}$ and $M = \mathbb{Z}$. We find an injective resolution of M . Since R is a PID, injective modules are equivalent to divisible ones.

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$$

We note that $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are divisible modules. This is an injective resolution of length 1.

Example 4.1.13

The same s.e.s.

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$$

is a left resolution for $\mathbb{Q}/\mathbb{Z}$. Note that $\mathbb{Q}$ is not projective, so this is not a projective resolution. However, since $\mathbb{Q}$ is torsion-free, this is a flat resolution of $\mathbb{Q}/\mathbb{Z}$.

Example 4.1.14

Let $R = k[x]/(x^n)$, for some positive integer n (for a field k). This can be thought of as a k -vector space $k \oplus kx \oplus \cdots \oplus kx^{n-1}$. Take $M = R/(x) \cong k$. This is called the *residue field* of R . Let us write a projective resolution of M .

$$\cdots \xrightarrow{x^{n-1}} R \xrightarrow{\cdot x} R \xrightarrow{x^{n-1}} R \xrightarrow{\cdot x} R \longrightarrow k \longrightarrow 0$$

This is an infinite projective resolution.

One can generalize this slightly to say that if instead $M = R/(x^i)$, then we have the repeated maps $\cdot x^i$ and $\cdot x^{n-i}$. If $M = R/(f)$, then we know that at least can make

$$? \cdots R \xrightarrow{f} R \rightarrow R/(f) \rightarrow 0.$$

Theorem 4.1.15

Every R -module M has a projective and an injective resolution.

Proof. We start with the projective resolution. We know that for any M there is a surjective map $\epsilon: P_0 \rightarrow M$, where P_0 is free, hence projective. Let $M_1 = \ker(P_0 \rightarrow M)$. again, there is a surjection $P_1 \rightarrow M_1$, where P_1 is projective. We then have

$$\begin{array}{ccccccc} P_1 & \longrightarrow & P_0 & \longrightarrow & M & \longrightarrow & 0 \\ & \searrow & \uparrow & & & & \\ & & M_1 & & & & \end{array}$$

If we let d be the composition $P_1 \rightarrow M_1 \rightarrow P_0$, then it has image M_1 . Thus, $\text{Im}d = \ker(P_0 \rightarrow M)$. We continue this. Set $M_n := \ker(P_{n-1} \rightarrow M_{n-1})$. Then, pick $P_n \rightarrow M_n$, and build

$$\begin{array}{ccccccccccc} P_n & \xrightarrow{d} & P_{n-1} & \xrightarrow{d} & P_{n-2} & \xrightarrow{d} & \cdots & \longrightarrow & P_0 & \longrightarrow & M & \longrightarrow & 0 \\ & \searrow & \uparrow & & & & & & & & & & \\ & & M_n & & & & & & & & & & \end{array}$$

The injective case is done similarly. By the existence of enough injectives, we know that M can inject into any I^0 . Set $M^1 = \text{coker}(M \rightarrow I^0)$. Then, let M^1 inject into I^1 . Set d as the composition $I^0 \rightarrow M^1 \rightarrow I^1$. Then, the kernel of d is the image of $M \rightarrow I^0$. Thus, $\text{Im}(M \rightarrow I^0) = \ker(d)$. $\square$

Proposition 4.1.16

Suppose $P_\bullet \rightarrow M$ is a projective resolution and $f: M \rightarrow N$ is a R -module homomorphism. Then, for any left resolution $Q_\bullet \rightarrow N$, there exists a chain map $f_\bullet: P_\bullet \rightarrow Q_\bullet$ inducing f as $H_0(f_\bullet) = f$ such that the following diagram commutes:

$$\begin{array}{ccccccccc} \dots & \longrightarrow & P_2 & \xrightarrow{d} & P_1 & \xrightarrow{d} & P_0 & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow f_2 & & \downarrow f_1 & & \downarrow f_0 & & \downarrow f & & \\ \dots & \longrightarrow & Q_2 & \xrightarrow{d} & Q_1 & \xrightarrow{d} & Q_0 & \longrightarrow & N & \longrightarrow & 0 \end{array}$$

Moreover, if $g_\bullet: P_\bullet \rightarrow Q_\bullet$ is another such chain map (that the above diagram commutes), then there exists maps $s_\bullet: P_\bullet \rightarrow Q_{\bullet+1}$ such that

$$f_n - g_n = s_{n-1}d + ds_n, \quad f_0 - g_0 = ds_0$$

Proof. We proceed by induction. First is the base case. We have

$$\begin{array}{ccc} P_0 & \longrightarrow & M \\ \exists f_0 \downarrow & \searrow & \downarrow f \\ Q_0 & \longrightarrow & N \end{array}$$

We note that the map f_0 exists, since P_0 is projective. Thus, we've found f_0 and so our base case is complete. Now suppose we've constructed up to f_{n-1} .

$$\begin{array}{ccccccccccc} \dots & \longrightarrow & P_n & \xrightarrow{d_n^P} & P_{n-1} & \xrightarrow{d_{n-1}^P} & \dots & \longrightarrow & P_0 & \xrightarrow{d} & M & \longrightarrow & 0 \\ & & & & \downarrow f_{n-1} & & & & \downarrow f_0 & & \downarrow f & & \\ \dots & \longrightarrow & Q_n & \xrightarrow{d} & Q_{n-1} & \xrightarrow{d_{n-1}^Q} & \dots & \longrightarrow & Q_0 & \xrightarrow{d} & N & \longrightarrow & 0 \end{array}$$

We note that $f_{n-1}d_n^P$ lands inside $\ker(d_{n-1}^Q)$ (by exactness of $P_\bullet$). Because P_n is projective, the map $P_n \rightarrow \ker(d_{n-1}^Q)$ must lift to $f_n: P_n \rightarrow Q_n$. Note that this makes the square commute; that is, $f_{n-1}d_n^P = d_n^Q f_n$. This completes the induction.

For the second part, we consider $f_n - g_n$. Since both f_n and g_n allow the diagram to commute, we have

$$\begin{array}{ccccccc} \dots & \longrightarrow & P_2 & \xrightarrow{d} & P_1 & \xrightarrow{d} & P_0 & \xrightarrow{d} & M \\ & & \downarrow f_2 - g_2 & & \downarrow f_1 - g_1 & & \downarrow f_0 - g_0 & & \downarrow 0 \\ \dots & \longrightarrow & Q_2 & \xrightarrow{d} & Q_1 & \xrightarrow{d} & Q_0 & \xrightarrow{d} & N \end{array}$$

We inductively construct s in a similar fashion. We will show the first step here. First, we want to find $s_0: P_0 \rightarrow Q_1$ such that $ds_0 = f_0 - g_0$ (since $s_{-1} = 0$). For this, we note that $f_0 - g_0: P_0 \rightarrow \ker(Q_0 \rightarrow N)$. This is by the commutativity of the very right square. Therefore, we have a picture:

$$\begin{array}{ccc} & P_0 & \\ \exists s_0 \swarrow & \downarrow f_0 - g_0 & \\ Q_1 & \xrightarrow{d} & \ker(Q_0 \rightarrow N) \end{array}$$

Note that the existence of s_0 follows from P_0 being projective. The rest follows by induction again (exercise). $\square$

Example 4.1.17

Let $R = \mathbb{Z}$ and $n \mid m$, and $M = \mathbb{Z}/m\mathbb{Z}$, $N = \mathbb{Z}/n\mathbb{Z}$. We can construct maps as follows.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\cdot m} & \mathbb{Z} & \longrightarrow & \mathbb{Z}/m\mathbb{Z} \longrightarrow 0 \\ & & \downarrow \cdot & & \downarrow \text{id} & & \downarrow \\ 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{\cdot n} & \mathbb{Z} & \longrightarrow & \mathbb{Z}/n\mathbb{Z} \longrightarrow 0 \end{array}$$

Example 4.1.18

Let $R = k[x]/(x^n)$, and $N = R/(x)$ and $M = R/(x^i)$, for some $i < n$.

$$\begin{array}{ccccccc} \dots & \longrightarrow & R & \xrightarrow{x^{n-i}} & R & \xrightarrow{x^i} & R \longrightarrow M \longrightarrow 0 \\ & & \downarrow \text{id} & & \downarrow x^{i-1} & & \downarrow \text{id} \\ \dots & \longrightarrow & R & \xrightarrow{x^{n-1}} & R & \xrightarrow{x^{n-1}} & R \longrightarrow N \longrightarrow 0 \end{array}$$

This then repeats to the left.

Definition 4.1.19. Let $C_\bullet, D_\bullet$ be two complexes of R -modules. Let $f_\bullet, g_\bullet: C_\bullet \rightrightarrows D_\bullet$ be two chain maps. A **chain homotopy** $f_\bullet \Rightarrow g_\bullet$ is $s_\bullet$ given by $s_n: C_n \rightarrow D_{n+1}$ such that $s_{n-1}d + ds_n = f_n - g_n$. Here, $f_\bullet$ and $g_\bullet$ are called **chain homotopic** if there exist a chain homotopy $s_\bullet: f_\bullet \Rightarrow g_\bullet$.

Lemma 4.1.20

If $f_\bullet, g_\bullet: C_\bullet \rightrightarrows D_\bullet$ are chain homotopic chain maps, then $H_n(f_\bullet), H_n(g_\bullet): H_n(C_\bullet) \rightarrow H_n(D_\bullet)$ are equal, i.e. $H_n(f_\bullet) = H_n(g_\bullet)$.

Proof. Let $s_{n-1}d + ds_n = f_n - g_n$, for all n . $f_n = g_n$ in homology. Let $[c] \in H_n(C_\bullet)$. We will be a bit precise with our notation for clarity. Then,

$$(s_{n-1}d_n^C + d_{n+1}^D s_n)[c] = d_{n+1}^D s_n([c])$$

to begin with, since $c \in \ker d_n^C$. In $H_n(D_\bullet)$, the remaining term vanishes, since we quotient by $\text{Im} d_{n+1}^D$. Therefore, $H_n(f_n) = H_n(g_n)$. $\square$

Proposition 4.1.21

Let R be a ring, M be an R -module. If $P_\bullet$ and $Q_\bullet$ are two projective resolutions of M , then they are **homotopy equivalent**, i.e. there exist chain maps $f_\bullet: P_\bullet \rightarrow Q_\bullet$ and $g_\bullet: Q_\bullet \rightarrow P_\bullet$ and homotopies $s_\bullet: P_\bullet \rightarrow P_{\bullet+1}$, $t_\bullet: Q_\bullet \rightarrow Q_{\bullet+1}$ such that

$$\begin{aligned} gf &= \text{id}_P + ds + sd, \\ fg &= \text{id}_Q + dt + td. \end{aligned}$$

In other words, there exist chain homotopies $s_\bullet: \text{id}_P \Rightarrow gf$ and $t_\bullet: \text{id}_Q \Rightarrow fg$.

Proof. Consider the following:

$$\begin{array}{ccc}
 \vdots & & \vdots \\
 \downarrow & & \downarrow \\
 P_1 & \xrightarrow{\exists f_1} & Q_1 \\
 \downarrow & & \downarrow \\
 P_0 & \xrightarrow{\exists f_0} & Q_0 \\
 \downarrow & & \downarrow \\
 M & \xrightarrow{\text{id}} & M
 \end{array}$$

The previous proposition tells us that f_* exists. Similarly, we look at $M \xleftarrow{\text{id}} M$ to attain $g_*: Q_\bullet \rightarrow P_\bullet$. Now, look at the following:

$$\begin{array}{ccc}
 \vdots & & \vdots \\
 \downarrow & & \downarrow \\
 P_1 & \xrightarrow[g_1 f_1]{\text{id}} & P_1 \\
 \downarrow & & \downarrow \\
 P_0 & \xrightarrow[g_0 f_0]{\text{id}} & P_0 \\
 \downarrow & & \downarrow \\
 M & \xleftarrow{\text{id}} & M
 \end{array}$$

Both $\text{id}_{P_\bullet}$ and gf are chain maps which map this diagram commute. The previous proposition tells us that there exists $s: P_\bullet \rightarrow P_{\bullet+1}$ such that $gf - \text{id} = d^P s + s d^Q$. Similarly, such a t exists, and we are done. $\square$

Exercise 4.1.22. Let M, N be R -modules and $f: M \rightarrow N$ be a map of R -modules. Let

$$M \rightarrow J^0 \rightarrow J^1 \rightarrow \dots$$

be a right resolution, and

$$N \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$$

be an injective right resolution. Then, state and prove the corresponding proposition about existence of chain maps.

Solution: We can just dualize the argument by viewing these as projective resolutions in $R\text{-mod}^{\text{op}}$. $\square$

§4.2 The Snake Lemma and More

Theorem 4.2.1 (Snake Lemma)

Let

$$\begin{array}{ccccccc} L & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & N & \longrightarrow & 0 \\ \downarrow f & & \downarrow g & & \downarrow h & & \\ 0 & \longrightarrow & L' & \xrightarrow{\alpha'} & M' & \xrightarrow{\beta'} & N' \end{array}$$

be a diagram with exact rows. Then, there is an induced exact sequence as follows:

$$\ker(f) \xrightarrow{\alpha} \ker(g) \xrightarrow{\beta} \ker(h) \xrightarrow{\delta} \operatorname{coker}(f) \xrightarrow{\alpha'} \operatorname{coker}(g) \xrightarrow{\beta'} \operatorname{coker}(h),$$

where $\delta(n) = \ell'$, where

$$\alpha'(\ell') = m' = g(m),$$

with $\beta(m) = n$.

Proof. Exactness at $\ker(g)$ and $\operatorname{coker}(g)$ are immediate from the original, so we just check exactness at $\ker(h)$ and $\operatorname{coker}(f)$. First, $\ker(h)$. We show

$$\operatorname{Im}(\beta : \ker g \rightarrow \ker h) = \ker(\delta).$$

Let $n \in \ker h$. By definition of δ , choose $m \in M$ and $\ell' \in L'$ such that

$$\beta(m) = n, \quad \alpha'(\ell') = g(m).$$

Then $\delta(n) = [\ell'] \in \operatorname{coker} f$. We have $\delta(n) = 0$ if and only if $\ell' \in \operatorname{Im} f$, say $\ell' = f(\ell)$ for some $\ell \in L$. Then

$$g(m) = \alpha'(\ell') = \alpha'(f(\ell)) = g(\alpha(\ell)),$$

so

$$g(m - \alpha(\ell)) = 0,$$

hence $m - \alpha(\ell) \in \ker g$. Moreover,

$$\beta(m - \alpha(\ell)) = \beta(m) - \beta(\alpha(\ell)) = n - 0 = n.$$

Thus $n \in \operatorname{Im}(\ker g \rightarrow \ker h)$. Conversely, if $n = \beta(m)$ with $m \in \ker g$, then $g(m) = 0$. In the construction of $\delta(n)$ we may take $\ell' = 0$, so $\delta(n) = 0$. Hence exactness holds at $\ker h$.

Now for exactness at $\operatorname{coker}(f)$.

$$\operatorname{Im}(\delta) = \ker(\alpha' : \operatorname{coker} f \rightarrow \operatorname{coker} g).$$

Let $n \in \ker h$ and let $\delta(n) = [\ell'] \in \operatorname{coker} f$. Then for some $m \in M$,

$$\beta(m) = n, \quad \alpha'(\ell') = g(m).$$

Thus $\alpha'(\ell') \in \operatorname{Im} g$, so $[\alpha'(\ell')] = 0$ in $\operatorname{coker} g$. Hence

$$\operatorname{Im} \delta \subseteq \ker \alpha'.$$

Conversely, let $[\ell'] \in \operatorname{coker} f$ lie in $\ker \alpha'$. Then

$$\alpha'(\ell') \in \operatorname{Im} g,$$

so there exists $m \in M$ with

$$g(m) = \alpha'(\ell').$$

Let $n = \beta(m)$. Then

$$h(n) = h(\beta(m)) = \beta'(g(m)) = \beta'(\alpha'(\ell')) = 0,$$

so $n \in \ker h$. By construction,

$$\delta(n) = [\ell'].$$

Thus $\ker \alpha' \subseteq \text{Im} \delta$. Therefore exactness holds at $\text{coker } f$. $\square$

Remark 4.2.2 — If our original sequence was

$$\begin{array}{ccccccc} 0 & \longrightarrow & L & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & N \longrightarrow 0 \\ & & \downarrow f & & \downarrow g & & \downarrow h \\ 0 & \longrightarrow & L' & \xrightarrow{\alpha'} & M' & \xrightarrow{\beta'} & N' \longrightarrow 0 \end{array},$$

then the induced sequence would be

$$0 \rightarrow \ker(f) \rightarrow \ker(g) \rightarrow \ker(h) \rightarrow \text{coker}(f) \rightarrow \text{coker}(g) \rightarrow \text{coker}(h) \rightarrow 0.$$

This is immediate, since if α is monic and β' is epic, these properties descend to the kernels/ cokernels.

Lemma 4.2.3 (Long Exact Sequence)

Let

$$0 \rightarrow A_\bullet \xrightarrow{\alpha} B_\bullet \xrightarrow{\beta} C_\bullet \rightarrow 0$$

be a s.e.s. of complexes of R -modules. That is, each

$$0 \rightarrow A_n \xrightarrow{\alpha_n} B_n \xrightarrow{\beta_n} C_n \rightarrow 0$$

is a s.e.s. for each $n \geq 0$. Then, there exists a long exact sequence on homology as follows:

$$\begin{array}{ccccccc} \dots & \longrightarrow & H_{n+1}(A_\bullet) & \longrightarrow & H_{n+1}(B_\bullet) & \longrightarrow & H_{n+1}(C_\bullet) \\ & & & & \searrow \delta & & \\ & & H_n(A_\bullet) & \longrightarrow & H_n(B_\bullet) & \longrightarrow & H_n(C_\bullet) \\ & & & & \searrow \delta & & \\ & & \dots & \longrightarrow & H_0(A_\bullet) & \longrightarrow & H_0(B_\bullet) \longrightarrow H_0(C_\bullet) \longrightarrow 0 \end{array}$$

Proof. The proof is to employ the snake lemma to the following diagram:

$$\begin{array}{ccccccc} A_n/d(A_{n+1}) & \xrightarrow{\alpha_n} & B_n/d(B_{n+1}) & \xrightarrow{\beta_n} & C_n/d(C_{n+1}) & \longrightarrow & 0 \\ \downarrow d & & \downarrow & & \downarrow & & \\ 0 \longrightarrow \ker(d|_{A_{n-1}}) & \xrightarrow{\alpha_{n-1}} & \ker(d|_{B_{n-1}}) & \xrightarrow{\beta_{n-1}} & \ker(d|_{C_{n-1}}) & & \end{array}$$

We note that the vertical maps d are well defined. In the above, $\ker d|_{A_{n-1}} := \ker(d: A_{n-1} \rightarrow A_{n-2})$ and analogously for the others. Further, note that the horizontal sequences are exact. Then, we can apply the snake lemma. Then, we get an exact sequence as follows:

$$\begin{array}{ccccccc} H_n(A) & \xrightarrow{H_n(\alpha)} & H_n(B) & \xrightarrow{H_n(\beta)} & H_n(C) & \xrightarrow{\delta} & \\ & \searrow & & & & \nearrow & \\ & & H_{n-1}(A) & \xrightarrow{H_{n-1}(\alpha)} & H_{n-1}(B) & \xrightarrow{H_{n-1}(\beta)} & H_{n-1}(C) \end{array}$$

Here, we note that

$$H_n(A) = \ker(d: A_n/d(A_{n+1}) \rightarrow \ker(d|_{A_{n-1}})),$$

and so forth. We can continue and apply for all n , and note that the exact sequences glue together into a long exact sequence. $\square$

Corollary 4.2.4 (Two-out-of-Three Property/ Five Lemma)

If two of $A_\bullet, B_\bullet, C_\bullet$ are exact ($H_n = 0$ for all n), then the other one is exact as well.

Proof. We won't draw out the diagram but, in the above, if two out of the three homology groups are 0, then the third one must be zero as well, at every index. $\square$

Theorem 4.2.5 (Horse-Shoe Lemma)

Let

$$0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 0$$

be a s.e.s. of R -modules. Suppose we have projective resolutions $P_\bullet \rightarrow A$ and $R_\bullet \rightarrow C$. Then, define $Q_n := P_n \oplus R_n$, for all $n \geq 0$. Then, we can define maps $Q_n \rightarrow Q_{n-1}$ and a map $Q_0 \rightarrow B$ to make $Q_\bullet$ a projective resolution of B in such a way that we get chain maps

$$\begin{array}{ccccccc} 0 & \longrightarrow & P_\bullet & \xrightarrow{\tilde{\alpha}} & Q_\bullet & \xrightarrow{\tilde{\beta}} & R_\bullet \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \end{array}$$

that form a s.e.s. of chain complexes. That is, for all n ,

$$0 \longrightarrow P_n \xrightarrow{\tilde{\alpha}_n} Q_n \xrightarrow{\tilde{\beta}_n} R_n \longrightarrow 0$$

is exact (note that this last sequence splits, since these are projective modules).

Proof. Let $P_\bullet, R_\bullet$ be (any) resolutions. We have the following diagram:

$$\begin{array}{ccccccc}
 & & P_2 & & R_2 & & \\
 & & \downarrow & & \downarrow & & \\
 & & P_1 & & R_1 & & \\
 & & \downarrow & & \downarrow & & \\
 & & P_0 & & R_0 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0
 \end{array}$$

We want to ‘build the middle’ of the ‘horseshoe’. We set $Q_0 = P_0 \oplus R_0$. We can define $\tilde{\alpha}_0 = \iota_1$ (inclusion) and $\tilde{\beta}_0 = \pi_2$ (projection). We also want a map from $P_0 \oplus R_0 \rightarrow B$. On P_0 , this is just $P_0 \rightarrow A \rightarrow B$. For R_0 , we use the lift given from R_0 being projective. Thus, both new squares commute. We can then fill it out more:

$$\begin{array}{ccccccc}
 & & \vdots & & \vdots & & \\
 & & \downarrow & & \downarrow & & \\
 & & P_2 & & R_2 & & \\
 & & \downarrow & & \downarrow & & \\
 & & P_1 & & R_1 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & P_0 & \xrightarrow{\iota_1} & P_0 \oplus R_0 & \xrightarrow{\pi_2} & R_0 \longrightarrow 0 \\
 & & \downarrow & & \downarrow d & & \downarrow \\
 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0
 \end{array}$$

We continue this inductively now. We will just do one more step. Define $Q_1 := P_1 \oplus R_1$, $\tilde{\alpha}_1 = \iota_1$ (inclusion), and $\tilde{\beta}_1 = \pi_2$ (projection). Again, we must fill in the boundary map $d: P_1 \oplus R_1 \rightarrow P_0 \oplus R_0$. On P_1 , we are really constrained by commutativity of the square. Thus,

$$d^Q|_{P_1} = \tilde{\alpha}_0 \circ d^P.$$

On R_1 , we apply the snake lemma to the diagram

$$\begin{array}{ccccccc}
 & & P_1 & & & & R_1 \\
 & & \downarrow d & & & & \downarrow d \\
 & & \ker(f) & \longrightarrow & \ker(g) & \longrightarrow & \ker(h) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & P_0 & \longrightarrow & Q_0 & \longrightarrow & R_0 \longrightarrow 0 \\
 & & \downarrow f & & \downarrow g & & \downarrow h \\
 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & \text{coker}(f) = 0 & \longrightarrow & \text{coker}(g) = 0 & \longrightarrow & \text{coker}(h) = 0 \longrightarrow 0
 \end{array}$$

$\swarrow \exists \gamma$ (dashed arrow from $\ker(h)$ to $\ker(g)$)
 $\searrow \delta$ (curved arrow from $\ker(h)$ to A)

Note that the cokernel of f, g, h are all zero because $P_\bullet, Q_\bullet$ are projective and Q_0 was constructed so that it forms a projective resolution. Thus, we define

$$d^Q|_{R_1} := \gamma,$$

where γ comes from the diagram above. Indeed, this is contained in $P_0 \oplus R_0 = Q_0$ then.

Now, we continue inductively. Then, the exactness of $Q_\bullet$ is the only thing that is left to prove. This follows by the long exact sequence on homology. Viewing the homology, we have that $H_n(P_\bullet) = 0 = H_n(R_\bullet)$, so $H_n(Q_\bullet) = 0$. $\square$

Remark 4.2.6 — It is maybe not worth mentioning this, but to construct the resolution $Q_\bullet$, we only required that $R_\bullet$ was projective. So, where did we use that $P_\bullet$ was projective? This was used to deduce that $Q_i = P_i \oplus R_i$ is projective, since each of the P_i, R_i are projective.

Exercise 4.2.7 (Five Lemma). Consider the following commutative diagram of R -modules

$$\begin{array}{ccccccccc} A_1 & \longrightarrow & A_2 & \longrightarrow & A_3 & \longrightarrow & A_4 & \longrightarrow & A_5 \\ f_1 \downarrow & & f_2 \downarrow & & f_3 \downarrow & & f_4 \downarrow & & f_5 \downarrow \\ B_1 & \longrightarrow & B_2 & \longrightarrow & B_3 & \longrightarrow & B_4 & \longrightarrow & B_5 \end{array}$$

where the top and bottom row are exact. Show the following:

1. If f_1 is surjective and f_2, f_4 are injective, then f_3 is injective.
2. If f_2, f_4 are surjective and f_5 are injective, then f_3 is surjective.

Solution: Done via diagram chase. $\square$

Exercise 4.2.8 (Short Five Lemma). Consider the following commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_1 & \longrightarrow & B_1 & \longrightarrow & C_1 & \longrightarrow & 0 \\ & & f \downarrow & & g \downarrow & & h \downarrow & & \\ 0 & \longrightarrow & A_0 & \longrightarrow & B_0 & \longrightarrow & C & \longrightarrow & 0 \end{array}$$

where the rows are exact and f, h are isomorphisms. Then, show that g is an isomorphism.

Solution: We could just apply the 2-out-of-3 lemma. To expand this, we can induce the LES of homology as follows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & H_1(A) & \longrightarrow & H_1(B) & \longrightarrow & H_1(C) \\ & & & & & & \searrow \delta \\ & & & & & & \nearrow \\ & & & & H_0(A) & \longrightarrow & H_0(B) & \longrightarrow & H_0(C) & \longrightarrow & 0 \end{array}$$

Then, since f and h are isomorphisms, we note that $\ker(f) = \operatorname{coker}(f) = \ker(h) = \operatorname{coker}(h) = 0$. Therefore, we get that $H_1(B) = H_0(B) = 0$, so the middle is exact. Therefore, g is an isomorphism as well. $\square$

Exercise 4.2.9 (Short Five Lemma). Consider the following commutative diagram of R -modules

Exercise 4.2.10 (Nine Lemma). Consider the following commutative diagram of R -modules

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A_1 & \longrightarrow & B_1 & \longrightarrow & C_1 \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A_2 & \longrightarrow & B_2 & \longrightarrow & C_2 \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A_3 & \longrightarrow & B_3 & \longrightarrow & C_3 \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

Suppose the rows and the middle column are exact. Show that the left column is exact if and only if the right column is exact.

Solution: We've done this before via diagram chasing, but we now use homology as it is more powerful. Suppose that the left is exact to begin with. We shall show that $A_\bullet$ and $C_\bullet$ are actually complexes. In particular, we just need to show that $C_1 \rightarrow C_2 \rightarrow C_3$ is zero, since the others are 0, and the same for $A_\bullet$. For C , this is immediate. We want to calculate $d_1^C d_2^C(c)$. However, by surjectivity at the top, $c = g_1(b)$, where g_1 is the map $B_1 \rightarrow C_1$. Then,

$$d_1^C d_2^C(c) = d_1^C d_2^C(g_1(b)) = g_3(d_2^B(d_1^B(b))) = 0.$$

Now for A , this follows similarly. Since $A_3 \rightarrow B_3$ is injective, we have that

$$f_3(d_2^A(d_1^A(a))) = d_2^B(d_1^B(f_1(a))) = 0.$$

Therefore, $A_\bullet$ and $C_\bullet$ form a complex. Thus, we now apply the 2-out-of-3 property and we are done. $\square$

Exercise 4.2.11. Let $f^\bullet: A^\bullet \rightarrow B^\bullet$ be a morphism of cochain complexes. Define a cochain complex $C(f)$ such that $f^\bullet$ is a quasi-isomorphism if and only if $C(f)$ is exact. Try to do this yourself, but if you're not sure where to start (or confused about what's being asked) look up the 'cone' of a chain map (and prove for yourself that it works).

Solution: We note that $f^\bullet$ is a quasi-isomorphism if and only if $H^n(A^\bullet) \cong H^n(B^\bullet)$. Therefore, we want to find a cochain complex $C(f)$ such that it fits into a s.e.s. of cochain complexes with $A^\bullet$ and $B^\bullet$. The right $C(f)$ will be as follows: $C(f)^n := B^n \oplus A^{n+1}$, with differential $d^n(b, a) := (d_B^n b + f^{n+1}(a), -d_A^{n+1} a)$. This is the cone of a chain map, as stated in the hint. We first check that it is a cochain complex; that is, that $d^2 = 0$. Let

$(b, a) \in B^n \oplus A^{n+1}$. Then, we have as follows:

$$\begin{aligned} d^2(b, a) &= d(d_B^n b + f^{n+1}(a), -d_A^{n+1}a) \\ &= (d_B^{n+1}(d_B^n b + f^{n+1}(a)) + f^{n+2}(-d_A^{n+1}a), -d_A^{n+2}(-d_A^{n+1}a)) \\ &= (d_B^{n+1}(f^{n+1}(a))) + f^{n+2}(-d_A^{n+1}a), 0) \\ &= (0, 0). \end{aligned}$$

Here, we’ve used that $d_B^2 = d_A^2 = 0$, and that $d_B f = f d_A$, because f is a cochain map. Thus, this is a cochain complex. Now, this induces the ses of cochain complexes

$$0 \rightarrow B^\bullet \rightarrow C(f)^\bullet \rightarrow A^{\bullet+1} \rightarrow 0,$$

since at each $\bullet = n$, we get the split sequence

$$0 \rightarrow B^n \rightarrow B^n \oplus A^{n+1} \rightarrow A^{n+1} \rightarrow 0.$$

It follows that there exists a long exact sequence on homology, by the lemma on the long exact sequence:

$$\begin{array}{ccccccc} \dots & \longrightarrow & H^n(B) & \longrightarrow & H^n(C(f)) & \longrightarrow & H^{n+1}(A) \\ & & & & & & \searrow \delta \\ & & & & & & H^{n+1}(B) \longrightarrow H^{n+1}(C(f)) \longrightarrow H^{n+2}(A) \longrightarrow \dots \end{array}$$

The claim is now that the connecting morphism δ is actually $H^{n+1}(f)$. Then, it would follow that $H^{n+1}(f)$ is an isomorphism (for all n) if and only if $H^\bullet(C(f)) = 0$, which is exactly the desired claim. It now suffices to show that δ is indeed $H(f)$. Let $[a] \in H^{n+1}(A)$. Then, $d_A(a) = 0$. If we view a as an element in A^{n+1} and lift it to $(0, a) \in C(f)^n$, then it follows that $d(0, a) = (f^{n+1}(a), 0)$ because $d_A(a) = 0$. Now, going back to the snake lemma, we see that $\delta([a]) = [\ell]$, where $\iota_{n+1}([\ell]) = d^n[(0, a)] = [(f^{n+1}(a), 0)]$; however, ι is just the inclusion, so $[(\ell, 0)] = [(f^{n+1}(a), 0)]$, so $\delta([a]) = [f^{n+1}(a)]$. $\square$

§4.3 Derived Functors

Let R be a ring, and let $F: R\text{-}\mathbf{mod} \rightarrow \mathbf{Ab}$ be a functor. More generally, let $F: C \rightarrow D$ be a functor between two abelian categories (most results here hold for abelian categories; check Section 3.5 in [1]).

Definition 4.3.1. F is called **additive** if

$$F: \text{Hom}_R(M, N) \rightarrow \text{Hom}_{\mathbf{Ab}}(F(M), F(N))$$

is a map of abelian groups. That is, $F(f + g) = F(f) + F(g)$, where $f, g: M \rightarrow N$ is a map of R -modules.

Lemma 4.3.2

If F is an additive functor, then

$$F(M \oplus N) \cong F(M) \oplus F(N).$$

That is, F preserves split exact sequences.

Proof. The key fact is that the direct sum of M and N is specified by the following data: X is an R -module with maps

$$\begin{array}{ccccc} M & \xrightarrow{i_1} & X & \xrightarrow{\pi_1} & M \\ & \searrow i_2 & & \searrow \pi_2 & \\ N & & & & N \end{array}$$

such that $\pi_1 i_1 = \text{id}_M$, $\pi_2 i_2 = \text{id}_N$, $\pi_2 i_1 = 0$, $\pi_1 i_2 = 0$. If X has this data, then the map

$$M \oplus N \xrightarrow{(i_1, i_2)} X$$

and

$$X \xrightarrow{\begin{pmatrix} \pi_1 \\ \pi_2 \end{pmatrix}} M \oplus N$$

are isomorphisms, by the splitting lemma. Now, if F is an additive functor, then we get that

$$\begin{array}{ccccc} F(M) & \xrightarrow{F(i_1)} & F(X) & \xrightarrow{F(\pi_1)} & F(M) \\ & \searrow F(i_2) & & \searrow F(\pi_2) & \\ F(N) & & & & F(N) \end{array}$$

such that $F(\pi_1)F(i_1) = \text{id}_{F(M)}$, $F(\pi_2)F(i_2) = \text{id}_{F(N)}$ (by functoriality), and $F(\pi_2)F(i_1) = 0$, $F(\pi_1)F(i_2) = 0$ (since any additive functor preserves 0). Therefore,

$$F(X) \cong F(M) \oplus F(N).$$

□

The definitions of injective and projective objects in arbitrary abelian categories are the same as in the category of R -modules.

Definition 4.3.3. We say that an abelian category $\mathcal{C}$ has **enough projective objects** if every object is a surjective image of a projective object. Similarly, $\mathcal{C}$ has **enough injective objects** if for every object there is an injective morphism into an injective object. We have already seen that the category of R -modules has enough projectives and enough injectives.

Definition 4.3.4. a **right exact additive functor** F is an additive functor that preserves exactness at the right of a sequence

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0.$$

Remark 4.3.5 — We note that we usually just refer to right exact additive functors as right exact functors.

Definition 4.3.6. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a right exact additive functor between abelian categories such that $\mathcal{C}$ has enough projectives. Then, the i -th **left derived functor** of F is $L_i F(M) := H_i(F(P_\bullet))$, where $P_\bullet \rightarrow M$ is a projective resolution.

Remark 4.3.7 — Here, note that $F(P_\bullet)$ is the complex

$$\cdots \rightarrow F(P_2) \xrightarrow{F(d)} F(P_1) \xrightarrow{F(d)} F(P_0)$$

and $F(d) \circ F(d) = F(d \circ d) = F(0) = 0$, so we may take the homology.

Remark 4.3.8 — Note that in the case of $\mathcal{C} = R\text{-mod}$, we have proven that there are enough projectives.

Theorem 4.3.9

Let $F: R\text{-mod} \rightarrow \mathbf{Ab}$ (or between abelian categories $F: \mathcal{C} \rightarrow \mathcal{D}$, with $\mathcal{C}$ has enough projectives) be a right exact additive functor. Then, we have the following:

- (i) $L_0 F = F$.
- (ii) $L_i F(M)$ doesn't depend on the choice of $P_\bullet$ (i.e. a different choice of resolution of all modules lead to naturally isomorphic derived functors $L_i F$ s).
- (iii) $L_i F$ is an additive functor for all i .
- (iv) Any s.e.s. of R -modules

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

gives rise to a long exact sequence

$$\begin{array}{ccccccc} \cdots & \longrightarrow & L_2 F(A) & \xrightarrow{L_2 F(\alpha)} & L_2 F(B) & \xrightarrow{L_2 F(\beta)} & L_2 F(C) \\ & & & & & & \searrow \delta \\ & & & & L_1 F(A) & \xrightarrow{L_1 F(\alpha)} & L_1 F(B) & \xrightarrow{L_1 F(\beta)} & L_1 F(C) \\ & & & & & & \searrow \delta \\ & & & & & & F(A) & \xrightarrow{F(\alpha)} & F(B) & \xrightarrow{F(\beta)} & F(C) & \longrightarrow & 0 \end{array}$$

Proof. (i). Take our projective resolution $P_\bullet \rightarrow M$ and take the H_0 of

$$\cdots \rightarrow F(P_1) \rightarrow F(P_0) \rightarrow 0.$$

Then, $H_0(F(P_\bullet)) = \text{coker}(F(P_1) \rightarrow F(P_0))$. Since F is right exact, the sequence

$$\cdots \rightarrow F(P_1) \rightarrow F(P_0) \rightarrow F(M) \rightarrow 0.$$

is exact. Therefore, $F(M) \cong \text{coker}(F(P_1) \rightarrow F(P_0))$, so we are done.

(ii). Let $Q_\bullet \rightarrow M$ be another projective resolution. By uniqueness of homotopy, we have

$$\begin{array}{ccc} Q_\bullet & \begin{array}{c} \xleftarrow{f} \\ \xrightarrow{g} \end{array} & P_\bullet \\ & \searrow \quad \swarrow & \\ & M & \end{array}$$

Then, there exist chain maps f, g and homotopies $s: Q_\bullet \rightarrow Q_{\bullet+1}$, $t: P_\bullet \rightarrow P_{\bullet+1}$ such that

$$fg - \text{id}_Q = ds + sd, \quad gf - \text{id}_P = dt + td.$$

Then, we get that $F(f): F(Q_\bullet) \leftarrow F(P_\bullet): F(g)$ and $F(s), F(t)$, such that $F(s), F(t)$, such that

$$F(f)F(g) - \text{id}_{F(Q)} = dF(s) + F(s)f$$

and

$$F(g)F(f) - \text{id}_{F(P)} = dF(t) + F(t)d,$$

since F is an additive functor. Consequently, when we take homology, the $dF(s) + F(s)d$ and $dF(t) + F(t)d$ vanish. Thus, we get

$$H_i(F(f)): H_i(F(Q_\bullet)) \leftarrow H_i(F(P_\bullet)): H_i(F(g))$$

such that

$$H_i(F(f)) \circ H_i(F(g)) = \text{id}_{H_i(F(Q_\bullet))}, \quad H_i(F(g)) \circ H_i(F(f)) = \text{id}_{H_i(F(P_\bullet))}.$$

Thus, $H_i(F(f))$ and $H_i(F(g))$ are inverse isomorphisms. Thus, $L_i F(M) \cong L_i F(M)$, where the LHS is defined by $Q_\bullet$ and the RHS is defined by $P_\bullet$.

(iii): Given a map of R -modules $\alpha: M \rightarrow N$, take projective resolutions

$$\begin{array}{ccc} P_\bullet & \xrightarrow{\exists \tilde{\alpha}} & Q_\bullet \\ \downarrow & & \downarrow \\ M & \xrightarrow{\alpha} & N \end{array}$$

We note that such a chain map $\tilde{\alpha}$ exists by lifting α , which is unique up to homotopy. Therefore, we get a map of complexes $F(\tilde{\alpha}): F(P_\bullet) \rightarrow F(Q_\bullet)$, since F is an additive functor. From this, we attain that $H_i(F(\tilde{\alpha})): H_i(F(P_\bullet)) \rightarrow H_i(F(Q_\bullet))$. Note that this is independent of the choice of $\tilde{\alpha}$, since it is unique up to homotopy. One can then check that

$$M \xrightarrow{\alpha} N \xrightarrow{\beta} U$$

are two maps and define

$$\begin{array}{ccccc} L_i F(M) & \xrightarrow{L_i F(\alpha)} & L_i F(N) & \xrightarrow{L_i F(\beta)} & L_i F(U) \\ & \searrow & & \nearrow & \\ & & L_i F(\beta\alpha) & & \end{array}$$

Thus, $L_i F$ is a functor. Further, if we have $\alpha_1, \alpha_2: M \rightarrow N$, then $\alpha_1 + \alpha_2: M \rightarrow N$, and

$$\begin{array}{ccc} P_\bullet & \xrightarrow{\tilde{\alpha}_1 + \tilde{\alpha}_2} & Q_\bullet \\ \downarrow & & \downarrow \\ M & \xrightarrow{\alpha_1 + \alpha_2} & N \end{array}$$

Then, $\tilde{\alpha}_1 + \tilde{\alpha}_2$ is a lift of $\alpha_1 + \alpha_2$. This implies that

$$L_i F(\alpha + \beta) = L_i F(\alpha) + L_i F(\beta),$$

so $L_i F$ is an additive functor.

(iv). Let

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

be an exact sequence of R -modules, and let $P_\bullet \rightarrow A$ and $R_\bullet \rightarrow C$ be projective resolutions. Then, we use the horse shoe lemma to say that there exists a resolution $Q_\bullet \rightarrow B$ such that

$$\begin{array}{ccccccc} 0 & \longrightarrow & P_\bullet & \longrightarrow & Q_\bullet & \longrightarrow & R_\bullet \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \end{array}$$

is an exact sequence of chain complexes. By independence of the choice of projective resolution, we can compute $L_i F$ by using these resolutions. Since R_i are projective for all i , the sequence

$$0 \rightarrow P_i \rightarrow Q_i \rightarrow R_i \rightarrow 0$$

is split exact, i.e. $Q_i \cong P_i \oplus R_i$. Since F is an additive functor, it preserves the split exact sequence:

$$0 \rightarrow F(P_i) \rightarrow F(Q_i) \rightarrow F(R_i) \rightarrow 0$$

Therefore,

$$0 \rightarrow F(P_\bullet) \rightarrow F(Q_\bullet) \rightarrow F(R_\bullet) \rightarrow 0$$

is an exact sequence of complexes (note that we don’t have split here, but we don’t require it!). Now, by the snake lemma/ long exact sequence lemma, we are done. $\square$

Remark 4.3.10 — In the setting of abelian categories, ‘enough projectives’ just mean that we can create a projective resolution for any object in C .

Definition 4.3.11. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a left exact additive functor between abelian categories, where C has enough injectives. Let $M \rightarrow I^\bullet$ be an injective resolution. Then, the i -th right derived functor is defined as $R^i F(M) := H^i(F(I^\bullet))$.

Exercise 4.3.12. Prove that the above definition is well-defined and has all the analogous properties of left derived functors. Further, reprove the details of the theorem above.

Solution: This is as below. We dualize everything and attain the desired result. $\square$

Theorem 4.3.13

Let $F: R\text{-mod} \rightarrow \mathbf{Ab}$ (or between abelian categories $F: \mathcal{C} \rightarrow \mathcal{D}$, with $\mathcal{C}$ has enough injectives) be a left exact additive functor. Then, we have the following:

- (i) $R^0 F = F$.
- (ii) $R^i F(M)$ doesn't depend on the choice of $I^\bullet$ (i.e. a different choice of resolution of all modules lead to naturally isomorphic derived functors $R^i F$ s).
- (iii) $R^i F$ is an additive functor for all i .
- (iv) Any s.e.s. of R -modules

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

gives rise to a long exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & F(A) & \xrightarrow{F(\alpha)} & F(B) & \xrightarrow{F(\beta)} & F(C) \\ & & & & & & \searrow \delta \\ & & R^1 F(A) & \xrightarrow{R^1 F(\alpha)} & R^1 F(B) & \xrightarrow{R^1 F(\beta)} & R^1 F(C) \\ & & & & & & \searrow \delta \\ & & R^2 F(A) & \xrightarrow{R^2 F(\alpha)} & R^2 F(B) & \xrightarrow{R^2 F(\beta)} & R^2 F(C) \longrightarrow \dots \end{array}$$

Remark 4.3.14 — One advantage of working in abelian categories is that we can just take opposite category, and the above claim becomes trivial.

§4.4 Ext and Tor

Exercise 4.4.1. Let R be a ring and M be a left R -module. Prove that $- \otimes_R M$ is an additive functor.

Proof. Let $f, g: A \rightarrow B$ be a map of right R -modules. Then,

$$((f + g) \otimes_R \text{id}_M)(a \otimes m) = (f + g)(a) \otimes m = f(a) \otimes m + g(a) \otimes m,$$

by the bilinearity of $\otimes$. Therefore, $(f + g) \otimes_R \text{id}_M = f \otimes_R \text{id}_M + g \otimes_R \text{id}_M$. $\square$

Remark 4.4.2 — $\text{Hom}_R(M, -)$ is clearly an additive functor.

Definition 4.4.3. Let R be a ring and M be a left R -module. Let $F(-) = - \otimes_R M: \text{mod-}R \rightarrow \mathbf{Ab}$. Then, we define $\text{Tor}_i^R(-, M) := L_i F(-) = L_i(- \otimes_R M) = H_i(P_\bullet \otimes_R M)$, where $P_\bullet \rightarrow (-)$ is a projective resolution of $(-)$.

Definition 4.4.4. Let R be a ring and M be a right R -module. Let $F = \text{Hom}_R(M, -): R\text{-mod} \rightarrow \mathbf{Ab}$. Then, we define $\text{Ext}_R^i(M, -) = R^i F = R^i(\text{Hom}_R(M, -))$.

Remark 4.4.5 — The analogous definition could be made for right R -modules. The point is that $\text{Ext}_R^i(M, N)$ should have M and N both being right or left R -modules.

Exercise 4.4.6. Calculate that $\text{Tor}_0^R(M, N) = M \otimes_R N$, where R is a ring, N is a right-module, and M is a right-module.

Solution: We begin with a projective resolution

$$\dots \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} M \rightarrow 0.$$

Then, we tensor it with N so that we have the following complex.

$$\dots \xrightarrow{d_2 \otimes \text{id}_N} P_1 \otimes_R N \xrightarrow{d_1 \otimes \text{id}_N} P_0 \otimes_R N \rightarrow 0.$$

We have that

$$\text{Tor}_0^R(M, N) = P_0 \otimes_R N / \text{Im}(d_1 \otimes \text{id}_N),$$

so we want to find what $\text{Im}(d_1 \otimes \text{id}_N)$ is. However, note that $- \otimes_R N$ is right exact, so we also have the exact sequence

$$P_1 \otimes_R N \xrightarrow{d_1 \otimes \text{id}_N} P_0 \otimes_R N \xrightarrow{d_0 \otimes \text{id}_N} M \otimes_R N \rightarrow 0.$$

Thus, $\text{Im}(d_1 \otimes \text{id}_N) = \ker(d_0 \otimes \text{id}_N)$ and we have that

$$P_0 \otimes_R N / \ker(d_0 \otimes \text{id}_N) \cong M \otimes_R N$$

by exactness. It follows then that

$$\text{Tor}_0^R(M, N) \cong M \otimes_R N.$$

□

Exercise 4.4.7. Prove the analogous result for the exercise above. That is, for a right R -modules M and N , $\text{Ext}_R^0(M, N) = \text{Hom}_R(M, N)$.

Solution: Let $N \rightarrow I^\bullet$ be an injective resolution for N . Then, we apply $\text{Hom}_R(M, -)$ to the sequence $I^\bullet$. We get that

$$\text{Ext}_R^0(M, N) = \ker(\text{Hom}_R(M, I^0) \rightarrow \text{Hom}_R(M, I^1)).$$

However, because $\text{Hom}_R(M, -)$ is left exact, we have that the kernel above is equal to the image $\text{Im}(\text{Hom}_R(M, N) \rightarrow \text{Hom}_R(M, I^0))$. However, this map is injective (by definition of an injective resolution), so the image is isomorphic to $\text{Hom}_R(M, N)$. Thus, we are done. □

Remark 4.4.8 — The above two exercises reflect what we expect from the theorem on L_i and R^i at their 0th spot.

Example 4.4.9

Compute $\text{Tor}^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A)$.

Solution: First, we note that $\mathrm{Tor}^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) = L_i(\mathbb{Z}/n\mathbb{Z} \otimes_{\mathbb{Z}} A)$. We need a projective resolution of $\mathbb{Z}/n\mathbb{Z}$ now.

$$0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$$

We apply the functor $L_i(- \otimes_{\mathbb{Z}} A)$ now. This gives

$$0 \rightarrow A \xrightarrow{n} A \rightarrow 0.$$

Now, we just take the homology of this new sequence. Thus,

$$\mathrm{Tor}_i^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) = \begin{cases} A/nA & i = 0, \\ A[n] & i = 1, \\ 0 & i \geq 2. \end{cases}$$

Here, $A[n] = \{a \in A : na = 0\}$ is the n -torsion subgroup. □

Theorem 4.4.10 (Balanced-ness of Ext and Tor)

- (i) Tor is balanced, i.e. for a left R -module N and a right R -module M ,

$$L_i(- \otimes_R N)(M) = L_i(M \otimes_R -)(N).$$

That is, if we have a projective resolution $P_{\bullet} \rightarrow M$ or $Q_{\bullet} \rightarrow N$, then,

$$H_i(P_{\bullet} \otimes_R N) \cong H_i(M \otimes_R Q_{\bullet}).$$

- (ii) Ext is balanced. Let M and N be right modules. Then,

$$R^i \mathrm{Hom}_R(-, N)(M) \cong R^i \mathrm{Hom}_R(M, -)(N).$$

That is, we can either use a projective resolution $P_{\bullet} \rightarrow M$ or an injective resolution $N \rightarrow I^{\bullet}$ due to the contravariance of $\mathrm{Hom}_R(-, N)$:

$$H^i \mathrm{Hom}_R(P_{\bullet}, N) \cong H^i \mathrm{Hom}_R(M, I^{\bullet})$$

Proof. Not proven in this course; check Weibel (but can use in exercises). □

Remark 4.4.11 — Note that the above tells us that for Tor, one can resolve either component. For Ext, we need to be more careful, and so this tells us that we can take a projective resolution of N (in the second equality).

Exercise 4.4.12. Show that $\mathrm{Hom}_R(A, B) \cong A^* \otimes_R B$, if A is finitely generated and projective.

Exercise 4.4.13. Compute $\mathrm{Ext}_{\mathbb{Z}}^i(\mathbb{Z}/n\mathbb{Z}, A)$. *Hint: Use the previous theorem.*

Solution: We note that $\mathrm{Ext}_{\mathbb{Z}}^i(\mathbb{Z}/n\mathbb{Z}, -) = R^i(\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, -))$, so by balanced-ness, we can similarly just calculate $R^i(\mathrm{Hom}_{\mathbb{Z}}(-, A))$. We note that

$$0 \rightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \xrightarrow{\mathrm{mod } n} \mathbb{Z}/n\mathbb{Z} \rightarrow 0$$

is a projective resolution of $\mathbb{Z}/n\mathbb{Z}$. We apply $\text{Hom}_{\mathbb{Z}}(-, A)$, so we get the sequence

$$0 \leftarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, A) \xleftarrow{\times n} \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, A) \leftarrow 0.$$

However, $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}, A) \cong A$ because any map $\varphi: \mathbb{Z} \rightarrow A$ is determined by the image of 1, which can be any element of A . Thus, the above sequence is now as follows:

$$0 \leftarrow A \xleftarrow{\times n} A \leftarrow 0$$

Thus,

$$\text{Ext}_{\mathbb{Z}}^k(\mathbb{Z}/n\mathbb{Z}, A) = \begin{cases} A[n] & k = 0, \\ A/nA & k = 1, \\ 0 & k > 1. \end{cases}$$

□

Example 4.4.14

Let $R = k[X]/(X^n)$, where k is a field. Let $i < n$, and define $M = R/(X^i) = k[X]/(X^i)$. A projective resolution of M is

$$\cdots \rightarrow R \xrightarrow{x^i} R \xrightarrow{x^{n-i}} R \xrightarrow{x^i} R \rightarrow M \rightarrow 0$$

Let $j < n$ as well, and set $N = R/(X^j)$. We can compute

$$\text{Tor}_t^R(M, N) = \text{Tor}_t^{k[X]/(X^n)}(k[X]/(X^i), k[X]/(X^j)).$$

Then, if the above projective resolution is $P_{\bullet}$, then we must take $P_{\bullet} \otimes_R N$. Then, since $R \otimes_R N \cong N$, this is as follows:

$$\cdots \rightarrow N \xrightarrow{x^i} N \xrightarrow{x^{n-i}} N \xrightarrow{x^i} N \rightarrow 0$$

We now just calculate the homology of the sequence. We get that

$$\text{Tor}_1^R(M, N) = \ker(x^i: N \rightarrow N) / \text{Im}(x^{n-i}: N \rightarrow N) = \frac{(x^{j-\min(j,i)})}{(x^{\min(j,n-i)})}.$$

We can also calculate the homology as

$$\text{Tor}_2^R(M, N) = \ker(x^{n-i}: N \rightarrow N) / \text{Im}(x^i: N \rightarrow N) = \frac{(x^{j-\min(j,n-i)})}{(x^{\min(j,i)})}.$$

The remaining odd and even terms agree with these.

Remark 4.4.15 — Note that when $R = k[X]/(X^n)$, it is both injective and projective over itself.

Example 4.4.16 (Koszul Complex)

Let $R = k[X, Y]$, where k is a field. Let $M = R/(X, Y) \cong k$. Compute $\text{Tor}_*^R(k, k) = \text{Tor}_*^R(M, M)$.

Solution: We first write a projective resolution.

$$0 \rightarrow R \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} R^2 \xrightarrow{(x, y)} R \rightarrow M \rightarrow 0$$

Note that the kernel of the map (x, y) (which sends a basis of R^2 to x and y) is

$$\left\{ \begin{pmatrix} f \\ g \end{pmatrix} = h \begin{pmatrix} -y \\ x \end{pmatrix} : h \in R \right\},$$

and the map $\begin{pmatrix} -y \\ x \end{pmatrix}$ is injective, giving us the 0 at the end. To compute $\text{Tor}_n^R(k, k)$, we calculate the n -th homology of the projective resolution above tensored with k :

$$0 \rightarrow R/(x, y) \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} R^2/(x, y) \xrightarrow{(x, y)} R/(x, y) \rightarrow 0$$

However, this is just

$$0 \rightarrow k \xrightarrow{0} k^2 \xrightarrow{0} k \rightarrow 0.$$

Thus, we have that

$$\text{Tor}_n^R(k, k) = \begin{cases} k & n = 0, \\ k^2 & n = 1, \\ k & n = 2. \end{cases}$$

□

Remark 4.4.17 — We note that when the differential maps are zero, then it is easy to compute the homology; it is just the space itself.

Remark 4.4.18 — Note that if R is a PID, we know that $\text{Tor}_n^R(M, N) = 0$ and $\text{Ext}_R^n(M, N) = 0$ for $n \geq 2$. For this, we use the fact that submodules of free modules over PIDs are free, and projectives and free modules are equivalent. For this, let M be a R -module. Then, take any P which is projective which surjects:

$$P \rightarrow M \rightarrow 0$$

Thus, P is free, and the kernel of $P \rightarrow M$ is also free. Setting $F = \ker(P \rightarrow M)$, we can extend this to the following projective resolution:

$$0 \rightarrow F \rightarrow P \rightarrow M \rightarrow 0.$$

Note that this very left map is injective.

Remark 4.4.19 — When $R = k[X, Y]$, there is an analogous statement of the fact above stating that $\text{Tor}_n^R(M, N) = 0$ and $\text{Ext}_R^n(M, N) = 0$ for $n \geq 3$. This is because every module has a projective resolution of length ≤ 2 .

Remark 4.4.20 — Define the **global dimension** of R to be the supremum over all modules M taken over the smallest length of a projective resolution of M . Then, Hilbert's syzygy theorem states that the global dimension of $k[X_1, \dots, X_n] = n$.

Proposition 4.4.21

If P is a projective right R -module, then $\text{Tor}_i^R(P, B) = 0$ for any left R -module B and all $i \geq 1$. If Q is a projective right R -module then $\text{Ext}_R^i(Q, A) = 0$ for any right R -module A and all $i \geq 1$.

Proof. The 1-term complex P together with the identity map $P = P$ is a projective resolution of P as an R -module. $\square$

Proposition 4.4.22

Let R be a ring.

- (i) A right R -module A is projective if and only if $\text{Ext}_R^1(A, B) = 0$ for any right R -module B (and then $\text{Ext}_R^n(A, B) = 0$ for any B and any $n \geq 1$).
- (ii) A right R -module B is injective if and only if $\text{Ext}_R^1(A, B) = 0$ for any right R -module A (and then $\text{Ext}_R^n(A, B) = 0$ for any A and any $n \geq 1$).
- (iii) A left R -module B is flat if and only if $\text{Tor}_1^R(A, B) = 0$ for any right R -module A (and then $\text{Tor}_n^R(A, B) = 0$ for any A and any $n \geq 1$).

Proof. (iii). Assume C is flat. Let $P_\bullet \rightarrow A$ be a projective resolution. Then,

$$\text{Tor}_1^R(A, C) = H_1(P_\bullet \otimes_R C).$$

However, note that

$$P_2 \rightarrow P_1 \rightarrow P_0$$

is exact at P_1 . Tensoring $-\otimes_R C$ is an exact functor by definition of a flat module. Thus,

$$P_2 \otimes_R C \rightarrow P_1 \otimes_R C \rightarrow P_0 \otimes_R C$$

is exact, so $H_1(P_\bullet \otimes_R C) = 0$. Now suppose the converse holds. Let

$$0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$$

be a s.e.s. We want to show that $-\otimes_R C$ preserves exactness. We attain a long exact sequence of $\text{Tor}_\bullet^R(-, C)$.

$$\begin{array}{ccccccc} \dots & \longrightarrow & \text{Tor}_1^R(X, C) & \longrightarrow & \text{Tor}_1^R(Y, C) & \longrightarrow & \text{Tor}_1^R(Z, C) \\ & & & & & \searrow & \\ & & & & & & X \otimes_R C \longrightarrow Y \otimes_R C \longrightarrow Z \otimes_R C \longrightarrow 0 \end{array}$$

The above is zero (by assumption), so the bottom (horizontal line) is a short exact sequence. Thus, $-\otimes_R C$ preserves exactness, so C is flat. The proof for (i) and (ii) are similar. Essentially the crux for the others is that

- (i) A being projective is equivalent to $\text{Hom}_R(A, -)$ being exact.
- (ii) B being injective is equivalent to $\text{Hom}_R(-, B)$ being exact.

$\square$

§4.4.i Ext and Tor for Abelian Groups

We now consider Ext and Tor for $\mathbb{Z}$ -modules.

Proposition 4.4.23

For all $n \geq 2$, and A, B abelian groups,

$$\operatorname{Ext}_{\mathbb{Z}}^n(A, B) = 0, \quad \operatorname{Tor}_{\mathbb{Z}}^n(A, B) = 0.$$

Proof. We have seen many times that there exists a free abelian group F and a surjective homomorphism

$$F \rightarrow B.$$

Let F' be the kernel of $F \rightarrow B$. Since $\mathbb{Z}$ is a PID, the abelian group F' is also free. Hence

$$0 \rightarrow F' \rightarrow F$$

is a projective resolution of B . Since it has only two non-zero terms, all left derived functors L_i vanish for $i > 1$.

Let I be an injective abelian group such that there is an injective homomorphism

$$B \rightarrow I.$$

Since $\mathbb{Z}$ is a PID, being injective and being divisible are the same thing for abelian groups. Thus I is divisible, but then I/B is divisible too. Hence I/B is injective, and

$$I \rightarrow I/B \rightarrow 0$$

is an injective resolution of B . Therefore, all right derived functors R^i vanish for $i > 1$. $\square$

Remark 4.4.24 — The same result holds with $\mathbb{Z}$ replaced with any PID R , as we can see from the proof.

Remark 4.4.25 — We know that for any $n \geq 1$ we have

$$\operatorname{Tor}_n^{\mathbb{Z}}(\mathbb{Z}, B) = \operatorname{Tor}_n^{\mathbb{Z}}(A, \mathbb{Z}) = \operatorname{Tor}_n^{\mathbb{Z}}(\mathbb{Q}, B) = \operatorname{Tor}_n^{\mathbb{Z}}(A, \mathbb{Q}) = 0,$$

because $\mathbb{Z}$ is free and $\mathbb{Q}$ is flat.

We understand Ext^0 and Tor_0 , so we try to understand Ext^1 and Tor_1 .

Proposition 4.4.26

For any abelian group B ,

$$\operatorname{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B) = B_{\text{tors}} = \{b \in B : \exists n \geq 1 \text{ s.t. } nb = 0\}.$$

Proof. First, we look at the s.e.s.

$$0 \rightarrow B_{\text{tors}} \rightarrow B \rightarrow B/B_{\text{tors}} \rightarrow 0.$$

Note that B/B_{tors} is torsion free. Therefore, we apply $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} -$ to get a long exact sequence as follows.

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \text{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B_{\text{tors}}) & \longrightarrow & \text{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B) & \longrightarrow & \text{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B/B_{\text{tors}}) \\
 & & & & & & \searrow \\
 & & & & & & \mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} B_{\text{tors}} \longrightarrow \mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} B \longrightarrow \mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} B/B_{\text{tors}} \longrightarrow 0
 \end{array}$$

The zero above comes from $\text{Tor}_2^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B_{\text{tors}}) = 0$. Note that $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} B_{\text{tors}}$ is zero because we are tensoring a torsion module with a divisible module, so this is zero. Further, $\text{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B/B_{\text{tors}}) = 0$ because B/B_{tors} is flat. Therefore, we may prove the result for B_{tors} . We use the exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0,$$

to get the left exact sequence from $- \otimes_{\mathbb{Z}} B$ as follows:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \text{Tor}_1(\mathbb{Z}, B_{\text{tors}}) & \longrightarrow & \text{Tor}_1(\mathbb{Q}, B_{\text{tors}}) & \longrightarrow & \text{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B_{\text{tors}}) \\
 & & & & & & \searrow \\
 & & & & & & \mathbb{Z} \otimes_{\mathbb{Z}} B_{\text{tors}} \longrightarrow \mathbb{Q} \otimes_{\mathbb{Z}} B_{\text{tors}} \longrightarrow \mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} B_{\text{tors}} \longrightarrow 0
 \end{array}$$

In the above, we have that $\mathbb{Z} \otimes_{\mathbb{Z}} B = B$, and that everything besides $\text{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B)$ is zero. Thus, the map from $\text{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B)$ to B is an isomorphism, so we are done. $\square$

Remark 4.4.27 — If we know about colimits, this is easy:

$$\text{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B) = \text{Tor}_1^{\mathbb{Z}}\left(\varinjlim_n \mathbb{Z}/n\mathbb{Z}, B\right) \cong \varinjlim_n \text{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, B) \cong \varinjlim_n B[n] = B_{\text{tors}}.$$

Exercise 4.4.28. Show that $(\mathbb{Z}/a\mathbb{Z})[b] \cong \mathbb{Z}/d\mathbb{Z}$, where $d = \gcd(a, b)$.

Solution: We just construct a map of abelian groups. Consider $\varphi: \mathbb{Z} \rightarrow (\mathbb{Z}/a\mathbb{Z})[b]$ defined by $n \mapsto n \bmod a$. Then, it is clear that $nb \equiv 0 \pmod{a}$ if and only if $d \mid n$. Therefore, the kernel of this map is $d\mathbb{Z}$ and so we are done. $\square$

§4.5 Ext and Extensions

Definition 4.5.1. Let A, B be R -modules. An **extension** of A by B is a s.e.s. of R -modules

$$0 \rightarrow B \rightarrow M \rightarrow A \rightarrow 0.$$

Definition 4.5.2. Two extensions (of A by B), say ξ and ξ' are **equivalent** if there is a commutative diagram

$$\begin{array}{ccccccc}
 \xi & & 0 & \longrightarrow & B & \longrightarrow & M & \longrightarrow & A & \longrightarrow & 0 \\
 & & & & \parallel & & \downarrow & & \parallel & & \\
 \xi' & & 0 & \longrightarrow & B & \longrightarrow & M' & \longrightarrow & A & \longrightarrow & 0
 \end{array}$$

Remark 4.5.3 — We note that the middle arrow is actually an isomorphism, as one can show. For this, we use the short five lemma. The equality maps are, a priori, isomorphisms, so indeed this is true.

There is a connection with Ext^1 , using long exact sequences of a s.e.s. Given a sequence ξ

$$0 \rightarrow B \rightarrow M \rightarrow A \rightarrow 0,$$

and a R module N , we get a left exact sequence

$$0 \rightarrow \text{Hom}(N, B) \rightarrow \text{Hom}(N, M) \rightarrow \text{Hom}(N, A) \xrightarrow{\delta} \text{Ext}^1(N, B) \rightarrow \dots$$

Here, δ comes from the connecting homomorphisms. If we take $N = A$, we have a map $\text{id}: A \rightarrow A$. This gives

$$\begin{aligned} \text{Hom}(A, A) &\longrightarrow \text{Ext}_R^1(A, B) \\ \text{id} &\longmapsto \delta(\text{id}) = [\xi] \end{aligned}$$

In the above, $[\xi] \in \text{Ext}_R^1$ is called the **class** of the extension.

Remark 4.5.4 — Given two s.e.s. in a commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & L & \longrightarrow & M & \longrightarrow & N & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & L' & \longrightarrow & M' & \longrightarrow & N' & \longrightarrow & 0 \end{array}$$

we have a commutative diagram

$$\begin{array}{ccc} \text{Hom}_R(A, N) & \xrightarrow{\delta} & \text{Ext}_R^1(A, L) \\ \downarrow & & \downarrow \\ \text{Hom}_R(A, N') & \xrightarrow{\delta} & \text{Ext}_R^1(A, L') \end{array}$$

For this, check [1], Theorem 2.4.6

§4.5.i Baer Sums

Definition 4.5.5. Given two extensions ξ, ξ' of C by A ,

$$\xi \quad 0 \longrightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \longrightarrow 0$$

$$\xi' \quad 0 \longrightarrow A \xrightarrow{\alpha'} B' \xrightarrow{\beta'} C \longrightarrow 0$$

The **Baer sum** $\xi + \xi'$ is the extension

$$0 \rightarrow A \rightarrow X \rightarrow C \rightarrow 0$$

of C by A , where X is the homology in the middle of

$$A \xrightarrow{(\alpha, -\alpha')} B \oplus B' \xrightarrow{\beta - \beta'} C$$

That is, $X = \{(b, b') \in B \oplus B' : \beta b = \beta' b'\} / \{(\alpha a, -\alpha' a) : a \in A\}$. Further, the map $A \rightarrow X$ is obtained from $(\alpha, 0)$, and the map $X \rightarrow C$ is obtained from β (applied to the first component).

Remark 4.5.6 — Note that

$$A \xrightarrow{(\alpha, \alpha')} B \oplus B' \xrightarrow{\beta + \beta'} C$$

is actually a complex, as one can verify. Further, the Baer sum is a short exact sequence.

Proposition 4.5.7

The set $\{\text{all extensions of } C \text{ by } A\} / \sim$, where $\sim$ denotes the equivalence of such extensions, is an abelian group under the Baer sum. The zero element of this abelian group is the class of any split sequence. That is, that class

$$[0 \rightarrow A \rightarrow A \oplus C \rightarrow C \rightarrow 0].$$

Proof. This follows from the theorem identifying the $\{\text{extensions}\} / \sim \leftrightarrow \text{Ext}^1$. $\square$

Example 4.5.8

Let $R = \mathbb{Z}$. Consider the non-split sequence

$$\xi: 0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0.$$

We compute $\xi + \xi$. In this case,

$$\begin{aligned} X &= \{(x, y) \in \mathbb{Z}^2 : x \equiv y \pmod{2}\} / \{(2x, -2x) : x \in \mathbb{Z}\} \\ &\cong \mathbb{Z}(1, -1) \oplus \mathbb{Z}(2, 0) / \mathbb{Z}(2, -2) \\ &\cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}. \end{aligned}$$

We have

$$0 \rightarrow \mathbb{Z} \xrightarrow{(n, \bar{n})} \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \xrightarrow{(a, \bar{b}) \mapsto \overline{a+b}} \mathbb{Z}/2\mathbb{Z} \rightarrow 0.$$

This is a split exact sequence, if we examine where elements get sent to. The first map is a canonical inclusion and the second is the canonical projection. Therefore,

$$[\xi] + [\xi] = 2[\xi] = 0,$$

the class of the split exact sequence.

Theorem 4.5.9

Taking the class of an extension of C by A induces a bijection

$$\begin{aligned} \{\text{extensions of } C \text{ by } A\} / \sim &\xrightarrow{\sim} \text{Ext}_R^1(C, A) \\ \xi &\mapsto [\xi] \end{aligned}$$

Under this bijection, the taking Baer sum corresponds to addition in $\text{Ext}_R^1(C, A)$, providing an isomorphism of abelian groups.

Proof. We first need to show that equivalent extensions yield the same class. This follows immediately from the naturality of δ . We first show this map is surjective. We do this by providing a map $\text{Ext}_R^1(C, A) \rightarrow \{\text{extensions of } C \text{ by } A\} / \sim$. First, choose an injective map $A \hookrightarrow I$ into an injective R -module I . Write down the s.e.s.

$$0 \rightarrow A \rightarrow I \rightarrow M \rightarrow 0,$$

where $M = I/A$. This induces an exact sequence of $\text{Hom}_R(C, -)$:

$$0 \rightarrow \text{Hom}_R(C, A) \rightarrow \text{Hom}_R(C, I) \rightarrow \text{Hom}_R(C, M) \rightarrow \text{Ext}_R^1(C, A) \rightarrow \underbrace{\text{Ext}_R^1(C, I)}_{=0}$$

Thus, $\text{Hom}_R(C, M) \rightarrow \text{Ext}_R^1(C, A)$ is surjective. Given $x \in \text{Ext}_R^1(C, A)$, we lift to $\phi \in \text{Hom}_R(C, M)$. We construct the corresponding extension as a pullback.

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \xhookrightarrow{\iota} & I & \xrightarrow{\pi} & M \longrightarrow 0 \\ & & \parallel & & \uparrow & & \uparrow \phi \\ 0 & \longrightarrow & A & \longrightarrow & X & \longrightarrow & C \longrightarrow 0 \end{array}$$

Here, $X := \{(i, c) \in I \oplus C : \pi(i) = \phi(c)\}$. Check that the kernel of $X \rightarrow C$ is isomorphic to A and the natural projection $X \rightarrow C$ is surjective; that is, the horizontal sequence is exact. We define ξ_x to be

$$0 \rightarrow A \rightarrow X \rightarrow C \rightarrow 0.$$

We need to check that $\xi \mapsto [\xi] \mapsto \xi_{[\xi]} = \xi$. First, consider the naturality of δ :

$$\begin{array}{ccc} \text{Hom}_R(C, M) & \xrightarrow{\delta} & \text{Ext}_R^1(C, A) \\ \phi_* \uparrow & & \parallel \\ \text{Hom}_R(C, C) & \xrightarrow{\delta} & \text{Ext}_R^1(C, A) \end{array}$$

Note that $x \in \text{Ext}_R^1(C, A)$ and we constructed a class $[\xi_x]$ so that $[\xi_x] = x$. Consider $\text{id}_C \in \text{Hom}_R(C, C)$. This maps to ϕ vertically and to $[\xi_x]$ horizontally. The class of $[\xi_x]$ is

$$[\xi_x] = \delta(\phi)(\text{id}_C) = \delta(\phi) = x,$$

where the last equality is by the definition of ϕ . Therefore, we've shown surjectivity.

We want to check that this construction is independent of our choice of ϕ . If ϕ' is another lift $\text{Hom}_R(C, M)$ of x , then $\delta(\phi - \phi') = \delta(\phi) - \delta(\phi') = x - x = 0$. Thus, there exists $\rho: C \rightarrow I$ such that $\pi\rho = \phi - \phi'$, by exactness. We now show that the extension with ϕ and the extension with ϕ' are equivalent.

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & X & \longrightarrow & C \longrightarrow 0 \\ & & \parallel & & \downarrow & & \parallel \\ 0 & \longrightarrow & A & \longrightarrow & X' & \longrightarrow & C \longrightarrow 0 \end{array}$$

Note that $X = \{(i, c) \in I \oplus C : \pi i = \phi c\}$ and $X' = \{(i, c) \in I \oplus C : \pi i = \phi' c\}$. Here, the vertical map $X \rightarrow X'$ is defined by $(i, c) \mapsto (i + \rho c, c)$. First, this map is well defined. If $(i, c) \in X$, then $\pi i = \phi c$. Then,

$$\pi(i - \rho c) = \pi i - \pi \rho c = \pi i - (\phi - \phi')c = \phi' c$$

We must also check this diagram commutes, but that shall be left as an exercise.

We've shown that our map is well-defined and is surjective, so we just need to show that the other composition provides the identity. That is, $\xi \mapsto [\xi] \mapsto \xi_{[\xi]} = \xi$. Given ξ

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

and

$$0 \longrightarrow A \longrightarrow I \longrightarrow M \longrightarrow 0$$

as before. Note that since I is injective, there exists a map $B \rightarrow I$ such that

$$\begin{array}{ccc} & I & \\ \uparrow & \nearrow \psi & \\ A & \hookrightarrow & B \end{array}$$

Then, we can write a diagram as follows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & I & \longrightarrow & M & \longrightarrow & 0 \\ & & \parallel & & \uparrow \psi & & \uparrow \bar{\psi} & & \\ 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \end{array}$$

Here, the map $\bar{\psi}$ is induced by ψ , which comes from the functoriality of cokernels. The final thing to note is that B is isomorphic to the pullback $X = \{(i, c) \in I \oplus C : \pi i = \psi c\}$. One can prove this using the snake lemma (exercise). This means that the original ξ

$$0 \longrightarrow A \longleftarrow B \longleftarrow C \longleftarrow 0$$

agrees with the pullback construction of $\xi_{[\xi]}$. This establishes the bijection.

It remains to show that the Baer sum maps to addition in Ext_R^1 . Given $x_1, x_2 \in \text{Ext}_R^1(C, A)$, say we have $\xi_1 \mapsto x$ and $\xi_2 \mapsto x_2$ under the bijection just established, where these extensions are constructed as pullbacks using $\phi_1, \phi_2: C \rightarrow M$. We can consider the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & I & \xrightarrow{\pi} & M & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow \phi_1 + \phi_2 & & \\ 0 & \longrightarrow & A \oplus A & \longrightarrow & B \oplus B' & \longrightarrow & C \oplus C & \longrightarrow & 0 \end{array}$$

Here, the vertical map $A \oplus A \rightarrow A$ is defined by $(a, a') \mapsto a - a'$ and the vertical map $B \oplus B' \rightarrow I$ is defined by $(i, c) + (i', c') \mapsto i + i'$. Here, the kernel of the map from $A \oplus A \rightarrow A$ is

$$\{(a, a) \in A \oplus A : a \in A\} \cong A.$$

This induces a diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & I & \longrightarrow & M & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & (A \oplus A)/\{(a, a)\} \cong A & \longrightarrow & (B \oplus B')/A & \longrightarrow & (C \oplus C) & \longrightarrow & 0 \end{array}$$

Consider $C \rightarrow C \oplus C$ defined by $c \mapsto (c, c)$. Let X be the preimage of

$$\{(c, 0) : c \in C\}$$

under the map $(B \oplus B')/A \rightarrow C$. It is worth noting that X is isomorphic to the homology of

$$A \rightarrow B \oplus B' \rightarrow C.$$

This gives rise to a diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & I & \longrightarrow & M & \longrightarrow & 0 \\ & & \uparrow & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & (A \oplus A)/A \cong A & \longrightarrow & X & \longrightarrow & C & \longrightarrow & 0 \end{array}$$

One can check that the bottom horizontal sequence is exact. Note the bottom is the Baer of ξ and ξ' . Here, the map from $A \rightarrow A$ is the identity. Once again, one can show that this X is the pullback of

$$\begin{array}{ccc} I & \longrightarrow & M \\ \uparrow & \phi_1 + \phi_2 \uparrow & \\ X & \dashrightarrow & C \end{array}$$

using the snake lemma. This show that the Baer sum $\xi + \xi'$ corresponds to $\phi_1 + \phi_2$, applying the connecting homomorphism δ to $\phi_1 + \phi_2$ yields $x_1 + x_2$. This concludes the proof. $\square$

Exercise 4.5.10. Write down an extension whose equivalence class is inverse to the class of a given extension (hint: change the sign of one of the maps) and show that the Baer sum of these extensions is split.

Solution: One can check that the inverse of the Baer sum of a sequence ξ

$$0 \rightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \rightarrow 0$$

can be given by

$$0 \rightarrow A \xrightarrow{-\alpha} B \xrightarrow{\beta} C \rightarrow 0$$

We now claim that the Baer sum of these forms a split extension. Let

$$X = \{(a, b) \in B \oplus B : \beta(a) = \beta(b)\} / \{\alpha(a), \alpha(a) : a \in A\}.$$

Then, we have

$$0 \rightarrow A \xrightarrow{\alpha} X \xrightarrow{\beta} C \rightarrow 0.$$

We show this is a split extension by forming a section to β . We define $\sigma: C \rightarrow B$ as $\sigma(c) = [(b, b)]$, where $\beta(b) = c$. We check well-defined ness. First, $\beta\sigma(c) = c$, so this is good. If b' is another lift of c , then we want to show that $[(b', b')] = [(b, b)]$, for $\beta(b) = \beta(b')$. Well, $b - b' \in \ker \beta = \text{Im} \alpha$, so $\alpha(a) + b' = b$. Therefore, $[(b, b)] = [(b', b')]$ in X . Thus, this splits. $\square$

Remark 4.5.11 — Further, if R is a commutative ring, then we may even give it an R -module structure.

Exercise 4.5.12. Let

$$\xi_1 : \quad 0 \longrightarrow \mathbb{Z} \xrightarrow{i_1} \mathbb{Z} \xrightarrow{p_1} \mathbb{Z}/4 \longrightarrow 0$$

be the short exact sequence where

$$i_1(n) = 4n, \quad p_1(m) = \overline{m}.$$

Let

$$\xi_2 : 0 \longrightarrow \mathbb{Z} \xrightarrow{i_2} \mathbb{Z} \oplus \mathbb{Z}/2 \xrightarrow{p_2} \mathbb{Z}/4 \longrightarrow 0$$

be the short exact sequence where

$$i_2(n) = (2n, \overline{n}), \quad p_2(a, \overline{b}) = \overline{a + 2b}.$$

Compute the Baer sum

$$[\xi_1] + [\xi_2] \in \text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/4, \mathbb{Z}),$$

and identify the resulting extension class under the canonical isomorphism

$$\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/4, \mathbb{Z}) \cong \mathbb{Z}/4.$$

Solution: We first calculate $X = \ker(p_1 - p_2)/(i_1, -i_2)$. Call $\ker(p_1 - p_2) = F$ and $(i_1, -i_2) = D$. Then,

$$F = \{(a, (b, \overline{c})) \in \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} : a \equiv b + 2c \pmod{4}\}.$$

First, assume $c \equiv 0 \pmod{2}$. Then, we get that $a \equiv b \pmod{4}$, so this is generated by $(4, (0, \overline{0}))$ and $(1, (1, \overline{0}))$. Now, let $\overline{c} = 1 \pmod{2}$. Then, $a \equiv b + 2 \pmod{4}$, so we have that $a = b + 2 + 4k$. Thus, write

$$(b + 2 + 4k, (b, \overline{1})) = b(1, (1, \overline{0})) + (2, (0, \overline{1})) + k(4, (0, \overline{0})).$$

Therefore, we get a new generator $(2, (0, \overline{1}))$. However, note that $2(2, (0, \overline{1})) = (4, (0, \overline{0}))$, so we have that $z := (2, (0, \overline{1}))$ and $x := (1, (1, \overline{0}))$ generate F .

Next, we have that D is generated by $(4, (-2, \overline{-1}))$. This can be written as follows:

$$(4, (-2, \overline{-1})) = 3z - 2x.$$

Therefore,

$$X = (\mathbb{Z}x \oplus \mathbb{Z}z)/\mathbb{Z}(3z - 2x).$$

We now calculate the maps. First, the one from $\mathbb{Z} \rightarrow X$ is induced by i_1 , so $1 \mapsto (4, (0, \overline{0})) = 2z$. Next the map $X \rightarrow \mathbb{Z}/4\mathbb{Z}$ is induced by p_1 , so $z \mapsto 2 \pmod{4}$ and $x \mapsto 1 \pmod{4}$ (these are equal). Let us try to simplify this a bit. Consider the map $X \rightarrow \mathbb{Z}$ defined by $x \mapsto 3t$, $z \mapsto 2t$. Then, $X \cong \mathbb{Z}t$. Further, the maps alter as follows:

$$i : \mathbb{Z} \rightarrow \mathbb{Z}t, \quad 1 \mapsto 4t$$

and

$$q : \mathbb{Z}t \rightarrow \mathbb{Z}/4\mathbb{Z}, \quad t \mapsto 3 \pmod{4}.$$

Note that this second map is because $3t \mapsto 1 \pmod{4}$, so $t \mapsto 3 \pmod{4}$. Therefore, the Baer extension is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{1 \mapsto 4t} \mathbb{Z} \xrightarrow{t \mapsto 3 \pmod{4}} \mathbb{Z}/4\mathbb{Z} \longrightarrow 0.$$

Noting that $H_{\mathbb{Z}}^1(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}) = \mathbb{Z}/4\mathbb{Z}$. Notice that this is not the trivial class. However, we can calculate that $[\xi_2] + [\xi_2] = 0$ in $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z})$. Thus, let us calculate the Baer sum of ξ_2 with itself. We use similar notation to before.

$$F = \{((a, \overline{b}), (c, \overline{d})) \in (\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}) \oplus (\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}) : a + 2b \equiv c + 2d \pmod{4}\}.$$

We find generators now. Let $\bar{b} = \bar{c} = 0$. This gives $((0, \bar{0}), (4, \bar{0}))$ and $((1, \bar{0}), (1, \bar{0}))$. Next let $\bar{b} = 1, \bar{c} = 0$. Then, $c = a + 2 + 4k$, so we can express this as follows:

$$((a, \bar{1}), (a + 2 + 4k, \bar{0})) = a((1, \bar{0}), (1, \bar{0})) + ((0, \bar{1}), (2, \bar{0})) + k((0, \bar{0}), (4, \bar{0})).$$

Thus, we get $((0, \bar{1}), (2, \bar{0}))$, and again, $2((0, \bar{1}), (2, \bar{0})) = ((0, \bar{0}), (4, \bar{0}))$, so we currently have the generators $y := ((0, \bar{1}), (2, \bar{0}))$ and $x := ((1, \bar{0}), (1, \bar{0}))$. If $\bar{b} = 0$ and $\bar{c} = 1$, we get the symmetric situation, which pops out $z := ((2, \bar{0}), (0, \bar{1}))$ and $((4, \bar{0}), (0, \bar{0}))$. However,

$$((4, \bar{0}), (0, \bar{0})) = 4x - 2y.$$

z , on the other hand, is not expressible with x and y . If $\bar{b} = \bar{c} = 1$, then we just get $a \equiv c \pmod{4}$, so this doesn't produce anything new. Thus, we have a set of generators, which are x, y, z . However, we shall note that x, y, z are subject to the following relation:

$$4x - 2y - 2z = 0.$$

Next, F is generated by $((2, \bar{1}), (-2, -\bar{1})) = z - y$. Therefore,

$$X = (\mathbb{Z}x \oplus \mathbb{Z}y \oplus \mathbb{Z}z) / (\mathbb{Z}(4x - 2y - 2z) \oplus \mathbb{Z}(z - y))$$

The map $i: \mathbb{Z} \rightarrow X$ is $1 \mapsto ((2, \bar{1}), (0, \bar{0})) = 2z - 2x + y$. Also, we have the map $p: X \rightarrow \mathbb{Z}/4\mathbb{Z}$ as $x \mapsto 1 \pmod{4}$, $y \mapsto 2 \pmod{4}$, $z \mapsto 2 \pmod{4}$.

Let's try to find a neater presentation for X . We note that X can be simplified to the following:

$$X = (\mathbb{Z}x \oplus \mathbb{Z}y \oplus \mathbb{Z}z) / (\mathbb{Z}(4x - 2y - 2z) \oplus \mathbb{Z}(z - y)) \cong (\mathbb{Z}x \oplus \mathbb{Z}y) / \mathbb{Z}(4(x - y)),$$

so if we set $u = x$, $v = x - y$, then we find that

$$X\mathbb{Z}u \oplus (\mathbb{Z}/4\mathbb{Z})v.$$

Thus, $[\xi_2] + [\xi_2]$ is split. It follows that $[\xi_1] + [\xi_2]$ is the equivalence class corresponding to $[3]$ in $\mathbb{Z}/4\mathbb{Z}$. $\square$

5 Group Cohomology

Let G be a group unless specified otherwise.

§5.1 The Group Ring

Definition 5.1.1. $\mathbb{Z}[G]$ (or $\mathbb{Z}[G]$) is the free $\mathbb{Z}$ -module on elements $g \in G$. That is, an element in $\mathbb{Z}[G]$ looks like a finite sum $\sum_i n_i g_i$, $n_i \in \mathbb{Z}$ and $g_i \in G$. This is called the **group ring**.

Definition 5.1.2. $\mathbb{Z}[G]$ -modules are called **G -modules**.

We note that $\mathbb{Z}[G]$ has a product defined as one would expect:

$$\left(\sum_i n_i g_i \right) \left(\sum_j n_j g'_j \right) := \sum_{i,j} n_i n_j g_i g'_j$$

Remark 5.1.3 — The generator attached to the unit element $e \in G$ is the multiplicative unit of $\mathbb{Z}[G]$, so we shall denote it by 1. Associativity of G implies that $\mathbb{Z}[G]$ is an associative ring. We shall call the generators attached to $g \in G$ the **canonical (standard) generators** or the canonical $\mathbb{Z}$ -basis of $\mathbb{Z}[G]$.

Remark 5.1.4 — Note that we can do $R[G]$, for any commutative ring R to form the group ring. In this case, we call this a **group algebra**. This is especially important for when $R = k$, a field. This is a functor from **Group** to **Ring** defined by $G \mapsto R[G]$. This is left adjoint to **Ring** $\rightarrow$ **Group** defined by $R \mapsto R^\times$.

Exercise 5.1.5. Verify that the group ring functor is indeed adjoint to the functor taking a ring to its multiplicative group.

Exercise 5.1.6. Verify that $\mathbb{Z}[G]$ is commutative if and only if G is abelian.

Solution: It suffices to show this on a generating element, since the rest are just $\mathbb{Z}$ -linear combination of such elements. Thus, we have that $g \cdot h = h \cdot g$ if and only if G is abelian. $\square$

Exercise 5.1.7. Verify that a G -module M is the same as an abelian group M , and for each $g \in G$, an action $g: M \rightarrow M$ is compatible with the group operation.

Exercise 5.1.8. Let $G = C_n = \langle s \rangle$, the cyclic group of order n with generator s . Then, show that

$$\mathbb{Z}[G] = \mathbb{Z} \oplus \mathbb{Z}s \oplus \mathbb{Z}s^2 \oplus \cdots \oplus \mathbb{Z}s^{n-1}.$$

Show that

$$s \cdot \sum_{i=0}^{n-1} a_i s^i = a_{n-1} + \sum_{i=1}^{n-1} a_{i-1} s^i.$$

Example 5.1.9

Let $G = \mathbb{Z} = \langle t \rangle$. Then, consider the group ring $\mathbb{Z}[\mathbb{Z}]$. Then, we have the isomorphism of rings

$$\begin{aligned} \mathbb{Z}[\mathbb{Z}] &\xrightarrow{\sim} \mathbb{Z}[t, t^{-1}] \\ n &\mapsto t^n. \end{aligned}$$

This is clear if we see that the LHS is isomorphic to $\oplus \mathbb{Z}n$ and the RHS is isomorphic to $\oplus \mathbb{Z}t^n$.

Abuse of Notation 5.1.10. Let M, N be G -modules. We shall write $\text{Hom}_G(M, N)$ for $\text{Hom}_{\mathbb{Z}[G]}(M, N)$. Further, we will just write $M \otimes_G N$ for $M \otimes_{\mathbb{Z}[G]} N$. That is, we often just drop the ‘ $\mathbb{Z}$ ’.

Definition 5.1.11. Given a group G and a G -module M , the set of **G -invariants** of M as $M^G := \{m \in M : g \cdot m = m, \forall g \in G\}$.

Exercise 5.1.12. Given a group G and a G -module M , verify that M^G is an abelian group.

Definition 5.1.13. Let M be a G -module. We define the **coinvariants** of M as $M_G := M / \{m - g \cdot m : g \in G, m \in M\}$.

Definition 5.1.14. A G -module N is called **trivial** if $g \cdot m = m$, for all $m \in N$.

Remark 5.1.15 — It is easy to see that M^G and M_G are both trivial G -modules. More precisely, $M^G \subseteq M$ is a trivial G -submodule and $M \twoheadrightarrow M_G$ is a trivial G -quotient module.

Exercise 5.1.16 ($(-)_G \dashv \text{triv} \dashv (-)^G$). Prove that the invariants and coinvariants $(-)^G$ and $(-)_G$ are functors from G -modules **$G\text{-mod}$** to abelian groups **$\mathbf{Ab}$** . There is also a functor

$$\text{triv} : \mathbf{Ab} \rightarrow G\text{-mod}$$

that takes an abelian group and gives it the trivial G -module structure. (Note that this functor is exact, because it does not change the underlying abelian group.) Prove that $(-)^G$ and $(-)_G$ are right and left adjoints to triv .

Solution: Let $L \xrightarrow{f} M \xrightarrow{h} N$ be G -modules. First, $(-)^G$ maps a G -module to M^G . $(-)^G$ also maps f to $f^G : L^G \rightarrow M^G$ by restriction; thus, if $\ell \in L^G$, then $f^G(\ell) = f(\ell)$. This is well defined, because $g \cdot f^G(\ell) = f^G(g \cdot \ell) = f^G(\ell)$. Since f^G is just the restriction of f , this clearly satisfies the unit and composition conditions: $\text{id}^G = \text{id}_{M^G}$, $(h \circ f)^G = h \circ f = h^G \circ f^G$. Thus, $(-)^G$ is a functor. We check that $(-)^G$ is a right adjoint to triv . Let $A \in \mathbf{Ab}$ and M be a $G\text{-mod}$. We define the maps $\tau_{A,M} : \text{Hom}_G(\text{triv}(A), M) \rightarrow \text{Hom}_{\mathbf{Ab}}(A, (M)^G)$ as $f \mapsto f$, where the latter just forgets about the G -module structure.

This is well defined because $g \cdot f(a) = f(g \cdot a) = f(g)$, so $f(a) \in M^G$. Let B be another abelian group, and $f: A \rightarrow B$ be a map of abelian groups. Let $h \in \text{Hom}_G(\text{triv}(B), M)$. Then, we want $h \circ \text{triv}(f) = h \circ f$, but this is clearly true because $\text{triv}(f)$ just gives the map f the trivial action. Now let $h \in \text{Hom}_G(\text{triv}(A), M)$ and $f: M \rightarrow N$ be a G -module map. We want to show that $f \circ h = f^G \circ h$. This is also clearly true because when $f \circ h$ gets mapped from $\text{Hom}_G(\text{triv}(A), N)$ to $\text{Hom}_{\mathbf{Ab}}(A, M^G)$, it forgets about its G -action. Thus we have a natural transformation. To show that it is an isomorphism, we define an inverse. If $f \in \text{Hom}_{\mathbf{Ab}}(A, M^G)$, we map it to f in $\text{Hom}_G(\text{triv}(A), M)$. This is also well defined because $g \cdot f(a) = f(a)$, so $f \in \text{Hom}_{\mathbf{Ab}}(A, M^G)$. The other case is left for later. $\square$

Remark 5.1.17 — Any group A can be considered as a trivial G -module. We do this by just setting $g \cdot a = a$, for all g and all a .

Proposition 5.1.18

Let G be a group.

- (i) Define $I := \{\sum a_i g_i : \sum_i a_i = 0\} \subseteq \mathbb{Z}[G]$, called the **augmentation ideal**. Then, $I = \ker \phi$, where

$$\begin{aligned} \phi: \mathbb{Z}[G] &\rightarrow \mathbb{Z} \\ g &\mapsto 1. \end{aligned}$$

Then, the set $\{1 - g : g \in G\}$ is a basis for I , and $\mathbb{Z}[G]/I \cong \mathbb{Z}$, where $\mathbb{Z}$ here is the trivial G -module.

- (ii) Assume G is finite. Define $N = \sum_{g \in G} g \in \mathbb{Z}[G]$, called the **norm element** of $\mathbb{Z}[G]$. $N \in Z(\mathbb{Z}[G])$, and $(\mathbb{Z}[G])^G = \mathbb{Z}N \cong \mathbb{Z}$.

Proof. (i). Note that the map ϕ as shown is surjective G -module map. Now, $\phi(\sum_i a_i g_i) \mapsto \sum a_i$. Thus, $\ker \phi = I$ by definition. Note that $(1 - g : g \in G) \subseteq I$ by definition and we have that $\mathbb{Z}[G]/(1 - g : g \in G) \cong \mathbb{Z}$. This implies that $I = (1 - g : g \in G)$, and since g forms a basis of $\mathbb{Z}[G]$, it follows that $\{1 - g : g \in G\}$ forms a basis of I .

(ii). Note that $gN = N$. In other words, $N \in (\mathbb{Z}[G])^G$. Furthermore, $gNg^{-1} = N$ for any $g \in G$, so N commutes with every element in G . Thus, $N \in Z(\mathbb{Z}[G])$. We need to show that reverse containment now. Let $\sum_i a_i g_i \in (\mathbb{Z}[G])^G$. Then, for all $g \in G$,

$$\sum a_i g g_i = g \sum a_i g_i = \sum a_i g_i.$$

We focus on both sides of the equality now. Since the action of G on itself is transitive, it follows that because $\{g_i\}$ forms a basis for $\mathbb{Z}[G]$, $a_i = a_j$ for all i, j . Thus, $\sum a_i g_i = a_0 N$. Therefore,

$$(\mathbb{Z}[G])^G = \mathbb{Z}N \cong \mathbb{Z}.$$

$\square$

Exercise 5.1.19. Let S be a set and let F be the group freely generated by S . Consider the kernel I of the augmentation map $\sigma: \mathbb{Z}[F] \rightarrow \mathbb{Z}$. Prove that I is a free $\mathbb{Z}[F]$ -module.

Solution: Let σ map $\sum_{g \in F} a_g g$ to $\sum_{g \in F} a_g$. Let $I = \ker \sigma$. We claim that I is free $\mathbb{Z}[F]$ -module generated by $\{1 - s : s \in S\}$. We first show that if $g, h \in F$, then

$$gh - 1 = g(h - 1) + (g - 1).$$

Also note that $s^{-1} - 1 = -s^{-1}(1 - s)$. Therefore, for any $g \in F$, $g - 1$ lies in the set generated by $\{s - 1 : s \in S\}$. Using this, we can decompose any word of the form $s_1^{\pm 1} \dots s_k^{\pm 1}$ into a linear combination of elements which look like $1 - s$, for $s \in S$. Now, as an abelian group, I is generated by such elements as well. Particularly, if $\sum_{g \in F} a_g g \in I$, then the coefficients sum to zero, so

$$\sum_{g \in F} a_g g = \sum_{g \in F} a_g g - \sum_{g \in F} a_g = \sum_{g \in F} a_g (g - 1).$$

It follows that we have a surjective map

$$\Phi: \bigoplus_{s \in S} \mathbb{Z}[F]e_s \mapsto I$$

defined by mapping $e_s \mapsto s - 1$. We show this is also injective.

$$\sum_{s \in S} a_s (s - 1) = 0.$$

a_s is in $\mathbb{Z}[F]$, so write each a_s as a sum $a_s = \sum_{g \in F} n_{g,s} g$. Here, $n_{g,s}$ is a finite collection of integers. Then, the above sum becomes

$$\sum_{s \in S, g \in F} n_{g,s} g (s - 1) = 0.$$

Here, we have a finite number of terms, so amongst all words g , choose the one which has maximal length such that $n_{g,s} \neq 0$. If this word does not end with s^{-1} , then there is no word that kills this in the subtracted part, so this is a contradiction. If such a maximal lengthed word does end in s^{-1} , then $g = us^{-1}$, for some $u \in S$ and so we get $g(s - 1) = u - g$. Noting that $-g$ is of maximal length, if it can be killed by some other word, then it is some element of the form $g's'$, where $g' \in F$ and $s' \in S$. However, then g' has length one less than g , so by uniqueness of the representation of g , we get that $g' = u$ and $s' = s^{-1}$, which contradicts that $s' \in S$. $\square$

§5.1.i Definition of Group Homology and Cohomology

Definition 5.1.20. Let M be a G -module. We define the n -th group homology with coefficients in M as

$$H_n(G, M) := \text{Tor}_n^{\mathbb{Z}[G]}(\mathbb{Z}, M),$$

where the $\mathbb{Z}$ appearing on the right is the trivial G -module $\mathbb{Z}$.

Definition 5.1.21. Let M be a G -module. We define the n -th group cohomology with coefficients in M as

$$H^n(G, M) := \text{Ext}_{\mathbb{Z}[G]}^n(\mathbb{Z}, M),$$

where (again) the $\mathbb{Z}$ appearing on the right is the trivial G -module $\mathbb{Z}$.

Example 5.1.22

First consider when $n = 0$. Then,

$$\begin{aligned}
 H_0(G, M) &= \operatorname{Tor}_0^{\mathbb{Z}[G]}(\mathbb{Z}, M) \cong \mathbb{Z} \otimes_G M \\
 &\cong \mathbb{Z}[G]/I \otimes_G M \\
 &\cong M/IM \\
 &\cong M/\{(1-g)m : g \in G, m \in M\} \\
 &\cong M/\{m - gm : g \in G, m \in M\} \\
 &\cong M_G.
 \end{aligned}$$

Here, we note that we've used that $\mathbb{Z} \cong \mathbb{Z}[G]/I$ in the second line, and the proposition in the fourth. Therefore, we see that $H_0(G, M) \cong M_G$, so $H_n(G, -)$ is the n -th left derived functor of coinvariants.

Example 5.1.23

Analogously, we can see that the $H^0(G, M) \cong \operatorname{Hom}_G(\mathbb{Z}, M) \cong M^G$, and so $H^n(G, -)$ is the n -th right derived functor of invariants. In particular, the crux of this is that

$$\begin{aligned}
 \operatorname{Hom}_G(\mathbb{Z}, M) &\cong \operatorname{Hom}_G(\mathbb{Z}[G]/I, M) \\
 &\cong \{m \in M : Im = 0\} \\
 &= \{m \in M : (1-g)m = 0, \forall g \in G\} \\
 &= M^G
 \end{aligned}$$

Exercise 5.1.24. In the above, we have the general isomorphism as follows. Let R be a ring and I be a (left) ideal, M be a (left) R -module. Then,

$$\operatorname{Hom}_R(R/I, M) \cong \operatorname{Ann}_I(M),$$

where the RHS is the annihilator of M in I .

Solution: The isomorphism $\operatorname{Hom}_R(R/I, M) \rightarrow \operatorname{Ann}_I(M)$ is defined by $f \mapsto f(1 + I)$. Note that if $i \in I$, then $i \cdot f(1 + I) = f(i(1 + I)) = f(0) = 0$, so this is well-defined. As for the reverse map, we define $m \mapsto f$, where $f: r + I \mapsto rm$. This is well define because if $r \in I$, then f is the zero map. It is clear that these are R -module maps, and that these maps are inverses to each other. $\square$

Example 5.1.25 (The Infinite Cyclic Group)

Let $G = \mathbb{Z} = \langle t \rangle$. Note that in this case, we've shown that $\mathbb{Z}[G] \cong \mathbb{Z}[t, t^{-1}]$. Let us compute $H^\bullet(G, M) = \operatorname{Ext}_{\mathbb{Z}[G]}^\bullet(\mathbb{Z}, M)$. Here, M is a $\mathbb{Z}$ -module and $\mathbb{Z}$ here is a trivial G -module as usual.

Solution: We have that $H^\bullet(G, M) = \operatorname{Ext}_{\mathbb{Z}[G]}^\bullet(\mathbb{Z}, M) \cong \operatorname{Ext}_{\mathbb{Z}[t, t^{-1}]}^\bullet(\mathbb{Z}, M)$. Here, M is a module over $\mathbb{Z}[G] \cong \mathbb{Z}[t, t^{-1}]$. We first require a projective resolution of $\mathbb{Z}$ (or an injective resolution of M , but this is harder). Note that $\mathbb{Z} \cong \mathbb{Z}[G]/I$, where $I = \{1 - g : g \in G\}$ is the augmentation ideal, so our augmentation ideal is now $I = (1 - t^n : n \in \mathbb{Z}) = (1 - t)$,

due to the identity $(1 - t^n) = (1 - t)(1 + \dots + t^{n-1})$. We have as follows:

$$0 \rightarrow \underbrace{\mathbb{Z}[t, t^{-1}]}_{=: P_1} \xrightarrow{1-t} \underbrace{\mathbb{Z}[t, t^{-1}]}_{=: P_0} \xrightarrow{\text{mod } I} \mathbb{Z} \rightarrow 0$$

Our projective resolution $P_\bullet$ is then the two left terms. We apply $\text{Hom}_{\mathbb{Z}[t, t^{-1}]}(-, M)$ to $P_\bullet$. We have

$$0 \rightarrow \text{Hom}_{\mathbb{Z}[t, t^{-1}]}(P_0, M) \xrightarrow{1-t} \text{Hom}_{\mathbb{Z}[t, t^{-1}]}(P_1, M) \rightarrow 0.$$

Since $P_1 = \mathbb{Z}[t, t^{-1}]$, using the more general isomorphism $\text{Hom}_R(R, M) \cong M$, we get that the sequence above is

$$0 \rightarrow M \xrightarrow{1-t} M \rightarrow 0$$

Therefore,

$$H^0(G, M) = \{m \in M : (1 - t)m = 0\} = \{m \in M : m = tm\} = M^G.$$

This follows exactly what we said above. Next, we calculate $H^1(G, M)$. We have

$$H^1(G, M) = M/(1 - t)M = M/\{m - tm : m \in M\} = M_G = H_0(G, M).$$

For $n \geq 2$, it is clear that $H^n(G, M) = 0$. This fact is obvious also from the fact that $\mathbb{Z}[t, t^{-1}]$ is a PID. Therefore,

$$H^0(\mathbb{Z}, M) = M^{\mathbb{Z}}, \quad H^1(\mathbb{Z}, M) = M_{\mathbb{Z}}, \quad H^n(\mathbb{Z}, M) = 0 \text{ for } n \geq 2.$$

□

Remark 5.1.26 — We note that it is unusual to see that homology and cohomology groups agree.

Example 5.1.27

We can, analogously, calculate that

$$H_0(\mathbb{Z}, M) = M_{\mathbb{Z}}, \quad H_1(\mathbb{Z}, M) = M^{\mathbb{Z}}, \quad H_n(\mathbb{Z}, M) = 0 \text{ for } n \geq 2.$$

Example 5.1.28 (Finite Cyclic Groups)

Let $G = C_m = \langle s \rangle$, the cyclic group of order m . As before, $\mathbb{Z}[G] \cong \mathbb{Z}[s]/(s^m - 1)$. We calculate $H^\bullet(G, M)$ now.

Solution: We first need a projective resolution of the trivial module $\mathbb{Z}$. We have that

$$\mathbb{Z} \cong \mathbb{Z}[G]/I \cong \mathbb{Z}[s]/(s^m - 1, s - 1) \cong \mathbb{Z},$$

where $I = (1 - s)$ is the augmentation ideal. We have the following projective resolution of $\mathbb{Z}$. Let $N = \sum_{g \in G} g \in \mathbb{Z}[G]$ denote the norm element. Note that in this case,

$$N = 1 + s + \dots + s^{n-1}.$$

It is worth noting that $N(1 - s) = 0$. Then, we proved that $\mathbb{Z}N = (\mathbb{Z}[G])^G = \ker(\mathbb{Z}[G] \xrightarrow{1-s} \mathbb{Z}[G])$.

$$\dots \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{1-s} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{1-s} \mathbb{Z}[G] \xrightarrow{\text{mod } I} \mathbb{Z} \rightarrow 0$$

Note that the discussion right above the sequence tells us that this is exact. Let $P_\bullet$ be the projective resolution. We apply $\text{Hom}_{\mathbb{Z}[G]}(-, M)$ to $P_\bullet$. We note that every module appearing is $\mathbb{Z}[G]$ itself, so $\text{Hom}_{\mathbb{Z}[G]}(P_i, M) \cong M$ for all i . We have our co-chain complex then:

$$0 \rightarrow M \xrightarrow{1-s} M \xrightarrow{N} M \xrightarrow{1-s} M \xrightarrow{N} \dots$$

We have as follows:

$$H^n(C_m, M) = \begin{cases} M^G & n = 0, \\ \ker(M \xrightarrow{N} M)/(1-s)M & n > 0 \text{ odd}, \\ M^s/MN & n > 0 \text{ even}. \end{cases}$$

Note that we could write the odd ones as $\ker(M \xrightarrow{N} M)_G$. □

Example 5.1.29

Analogously, we can calculate that

$$H_n(C_m, M) = \begin{cases} M_G & n = 0, \\ M^s/NM & n > 0 \text{ odd}, \\ \ker(M \xrightarrow{N} M)/(1-s)M & n > 0 \text{ even}. \end{cases}$$

§5.2 The Standard Resolution

We compute the general case $H^*(G, M) = \text{Ext}_{\mathbb{Z}[G]}(\mathbb{Z}, M)$, the *derived functors of invariants*. We can do this using a projective resolution of the trivial $\mathbb{Z}[G]$ -module $\mathbb{Z}$. Here, we talk about a canonical resolution of $\mathbb{Z}$ that always works, essentially independent of G . This is an amazing fact.

Notation 5.2.1. Define $P_n := \mathbb{Z}[G^{n+1}] = \mathbb{Z}[G \times \dots \times G]$, the group ring of $G \times \dots \times G$ $n+1$ times.

P_n is a G -module with diagonal action

$$g \cdot \underbrace{(g_1, \dots, g_{n+1})}_{\in P_n} = (gg_1, \dots, gg_{n+1}).$$

We note that P_n is a free abelian group.

Proposition 5.2.2

This diagonal action makes P_n into a free $\mathbb{Z}[G]$ -module. We have $P_n = \mathbb{Z}[G] \otimes_{\mathbb{Z}} Q_n$, where Q_n is a free abelian group generated by

$$(e, a_1, a_1a_2, a_1a_2a_3, \dots, a_1 \dots a_n),$$

for all $(a_1, \dots, a_n) \in G^n$.

Proof. For each $a = (a_1, a_2, \dots, a_n) \in G^n$ define $P_n(a)$ as a G -submodule of P_n generated as an abelian group by $(g, ga_1, ga_1a_2, \dots, ga_1a_2 \dots a_n)$ for all $g \in G$:

$$P_n(a) = \bigoplus_{g \in G} \mathbb{Z}(g, ga_1, ga_1a_2, \dots, ga_1a_2 \dots a_n) = \mathbb{Z}[G](e, a_1, a_1a_2, \dots, a_1a_2 \dots a_n),$$

so $P_n(a) \cong \mathbb{Z}[G]$. It is clear that P_n is the direct sum of its submodules $P_n(a)$ for $a \in G^n$:

$$P_n = \bigoplus_{a \in G^n} \mathbb{Z}[G](e, a_1, a_1 a_2, \dots, a_1 a_2 \cdots a_n).$$

Hence P_n is a free G -module and $P_n = \mathbb{Z}[G] \otimes_{\mathbb{Z}} Q_n$. $\square$

We now want to make $P_\bullet$ into a complex. In other words, we want a differential $d: P_n \rightarrow P_{n-1}$ such that $d^2 = 0$. Define

$$d((g_0, \dots, g_n)) := \sum_{i=0}^n (-1)^i (g_0, \dots, \hat{g}_i, \dots, g_n),$$

where the $\hat{g}_i$ means the i -th coordinate is deleted. By the proposition below, $(P_\bullet, d)$ is a complex.

Proposition 5.2.3

$(P_\bullet, d)$ is a complex, and the complex

$$\dots \xrightarrow{d} P_2 \xrightarrow{d} P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} \mathbb{Z} \rightarrow 0$$

is exact, where $\epsilon: P_0 \rightarrow \mathbb{Z}$ sends every element of G to 1; that is, the sum of coefficients. Here, note that $P_0 \cong \mathbb{Z}[G]$.

Proof. We need to verify that $d^2 = 0$. We have that

$$\begin{aligned} d^2((g_0, \dots, g_n)) &= \sum_i \sum_j \left((-1)^i (-1)^{j-1} + (-1)^j (-1)^i \right) (g_0, \dots, \hat{g}_i, \dots, \hat{g}_j, \dots, g_n) \\ &= 0 \end{aligned}$$

We note that we got the expansion on the right by noting that the term $(g_0, \dots, \hat{g}_i, \dots, \hat{g}_j, \dots, g_n)$ for $0 \leq i < j \leq n$ appears twice in $d^2(g_0, \dots, g_n)$. That is, we can delete i then j or delete j then delete i . Thus, $d^2 = 0$. We now show exactness. For this, we use a homotopy. Let us call $\tilde{P}$ to be P including $\mathbb{Z}$ ($\mathbb{Z}$ being the -1 index); that is,

$$\dots \xrightarrow{d} P_2 \xrightarrow{d} P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} \mathbb{Z}.$$

If we can find $s_\bullet: \tilde{P}_\bullet \rightarrow \tilde{P}_{\bullet+1}$ such that $d_{n+1}s_n + s_{n-1}d_n = \text{id}_n$, then id and 0 are homotopic. In other words, id and 0 induce the same map in homology. That is,

$$H_n(\tilde{P}_\bullet) \xrightarrow{\text{id}=0} H_n(\tilde{P}_\bullet).$$

In other words, $H_n(\tilde{P}_\bullet) = 0$ for all n . Now, we just need to construct s such that

$$\begin{array}{ccccccc} & & \xleftarrow{s} & \xleftarrow{s} & \xleftarrow{s} & \xleftarrow{s} & \\ \dots & \xrightarrow{\quad} & P_2 & \xrightarrow{d} & P_1 & \xrightarrow{d} & P_0 \xrightarrow{\epsilon} \mathbb{Z} \end{array}$$

Fix $h \in G$, and define

$$s_n(g_0, \dots, g_n) := (h, g_0, \dots, g_n).$$

This is a map from $P_n \rightarrow P_{n+1}$ and $\mathbb{Z} \rightarrow P_0$. We now check this works. We have

$$\begin{aligned} d_{n+1}s_n(g_0, \dots, g_n) &= (g_0, \dots, g_n) - (h, g_1, \dots, g_n) + (h, g_0, \dots, g_n) \cdots \pm (h, g_0, \dots, g_{n-1}) \end{aligned}$$

whereas

$$s_{n-1}d_n(g_0, \dots, g_n) = (h, g_1, \dots, g_n) - (h, g_0, \dots, g_n) \cdots \mp (h, g_0, \dots, g_{n-1})$$

Thus, everything cancels except for $(g_0, \dots, g_n)$. In other words,

$$d_{n+1}s_n + s_{n-1}d_n = \text{id}_n.$$

This finishes the proof, because the homology is zero, so the sequence is exact. $\square$

Remark 5.2.4 — We note that the maps $s_n: \tilde{P}_n \rightarrow \tilde{P}_{n+1}$ is not a map of G -modules. For instance, $s_{-1}(n) = nh$, but this is not compatible with the G -action. Applying some g action to n , we have that $n \xrightarrow{g} n$, but $ngh = g \cdot s_{-1}(n) \neq s_{-1}(gn) = nh$. However, it is a map of abelian groups. Therefore, we've shown that $\tilde{P}_\bullet$ is exact as a complex of abelian groups, but this is the same as being exact as a complex of G -modules when G is finite (this is crucial). It is possible to make s into a map of G -modules by summing over all h s:

$$\begin{aligned} S_n: P_n &\rightarrow P_{n+1} \\ (g_0, \dots, g_n) &\mapsto \sum_{h \in G} (h, g_0, \dots, g_n) \end{aligned}$$

We verify this just at the -1 -th index. Recall that we have

$$\begin{array}{ccccccc} & \overset{S}{\curvearrowright} & & \overset{S}{\curvearrowright} & & \overset{S}{\curvearrowright} & \overset{N}{\curvearrowright} \\ \cdots & \longrightarrow & P_2 & \xrightarrow{d} & P_1 & \xrightarrow{d} & P_0 \xrightarrow{d} \mathbb{Z} \end{array}$$

We note that the first map $S: \mathbb{Z} \rightarrow P_0$ is defined by multiplication by the norm map N . This works because N is preserved under the G -action.

Proposition 5.2.5

Let G be a finite group. The map $S_\bullet: \tilde{P}_\bullet \rightarrow \tilde{P}_{\bullet+1}$ as defined above is a map of complexes of G -modules, and satisfies

$$dS + Sd = |G| \cdot \text{id},$$

where $|G|$ is the cardinality of G .

Proof. This is clear, since the map S just summed over all $h \in G$. $\square$

Remark 5.2.6 — For the worrier: why then does the above proof show that the complex $P_\bullet$ is exact? When defining a chain homotopy for a complex of R -modules, I wanted the homotopy maps s_n to be R -module maps. Here we are dealing with $\mathbb{Z}[G]$ -modules but the s_n are only $\mathbb{Z}$ -module maps. Well, the point is that for any ring R , the forgetful functor from $R\text{-mod}$ to $\mathbf{Ab}$ is exact and faithful, so we are okay just to check exactness as a complex of abelian groups.

We now return to computing group cohomology. Let M be a G -module. Let $P_\bullet$ be the standard resolution. Then, $H^n(G, M) = H^n(\text{Hom}_G(P_\bullet, M))$. Since S is a G -module map for all n , we get a map

$$S_n^M := \text{Hom}_G(S_n, M) : \text{Hom}(P_n, M) \rightarrow \text{Hom}_G(P_{n+1}, M).$$

Using the previous proposition, $S_\bullet^M$ satisfies the same formula

$$d_{n+1}S_n^M + S_{n-1}^M d_n = |G| \cdot \text{id}.$$

This says that the multiplication by $|G|$ is equal to the zero map up to homotopy of complexes of G -modules. We've proven the following:

Proposition 5.2.7

If G is a finite group, then for each $n \geq 1$ the abelian group $H^n(G, M)$ is annihilated by $|G|$.

Let R be a ring considered as a left R -module, any abelian group A , and any left R -module M we have

$$\text{Hom}_R(R \otimes_{\mathbb{Z}} A, M) = \text{Hom}_{\mathbb{Z}}(A, M),$$

say by the tensor-hom adjunction. Indeed, a homomorphism of R -modules $R \otimes_{\mathbb{Z}} A \rightarrow M$ is uniquely determined by its values at $1 \otimes a$, $a \in A$. Conversely, any homomorphism of abelian groups $f : A \rightarrow M$ naturally extends to a homomorphism of R -modules $R \otimes_{\mathbb{Z}} A \rightarrow M$ sending $r \otimes a$ to $rf(a)$. In particular,

$$\text{Hom}_G(P_n, M) = \text{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G] \otimes_{\mathbb{Z}} Q_n, M) = \text{Hom}_{\mathbb{Z}}(Q_n, M).$$

Thus $H^n(G, M)$ are the cohomology groups of the cochain complex

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}_{\mathbb{Z}}(Q_0, M) & \xrightarrow{d} & \text{Hom}_{\mathbb{Z}}(Q_1, M) & \xrightarrow{d} & \dots \xrightarrow{d} \text{Hom}_{\mathbb{Z}}(Q_n, M) \xrightarrow{d} \dots \\ & & \sim \downarrow & & & & \\ & & M & & & & \end{array}$$

Corollary 5.2.8

Let G be a finite group of order $|G| = m$. Let M be a G -module such that one of the following conditions hold:

- $aM = 0$ for some $a \in \mathbb{Z}$ with $\gcd(a, m) = 1$
- M is a $\mathbb{Q}$ -vector space

Then $H^n(G, M) = 0$, for all $n \geq 1$.

Proof. By the discussion above, $|G| = m$ is zero on $H^n(G, M)$ for $n > 0$. However, $a \cdot H^n(G, M) = 0$ for $n > 0$ by definition. Therefore, $H^n(G, M)$ is annihilated by $\gcd(a, m) \subseteq \mathbb{Z}$, but $\gcd(a, m) = 1$. Therefore, $H^n(G, M) = 0$.

Likewise, if M is a $\mathbb{Q}$ -vector space, then so is $H^n(G, M)$ for all n . However, $m \cdot H^n(G, M) = 0$, but $m \neq 0$, so it must be that $H^n(G, M) = 0$. $\square$

Example 5.2.9

For example, let $G = \mathbb{Z}/2\mathbb{Z}$ and $M = \mathbb{Z}/3\mathbb{Z}$ with the trivial G -action. Then, $H^n(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}) = 0$ for $n \geq 1$.

Example 5.2.10

Let K be a number field and $G = (K/\mathbb{Q})$, with the G -action being the natural Galois action on K . Then, $H^0(G, K) = \mathbb{Q}$ and $H^n(G, K) = 0$ for $n \geq 1$.

Using the standard resolution, we compute the group cohomology. We have that

$$H^n(G, M) = H^n(\text{Hom}_G(P_\bullet, M)),$$

where $P_\bullet$ is the standard resolution. We have that

$$\begin{aligned} \text{Hom}_G(P_n, M) &= \text{Hom}_G \left(\bigoplus_{(a_1, \dots, a_n) \in G^n} (e, a_1, \dots, a_1 \dots a_n), M \right) \\ &\cong \text{Fun}(\{(e, a_1, \dots, a_1 \dots a_n), \text{ for } (a_1, \dots, a_1 \dots a_n) \in G^n\}, M) \\ &\cong \text{Fun}(G^n, M), \end{aligned}$$

where Fun denotes the space of functions from the first entry to the second. This follows from the fact that G -module morphisms out of a free module are uniquely determined by what they do on basis elements. Note that $\text{Fun}(G^0, M) \cong M$. Therefore, we get a cochain complex.

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \xrightarrow{d} & \text{Fun}(G, M) & \xrightarrow{d} & \text{Fun}(G \times G, M) \longrightarrow \dots \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ & & \text{Hom}_G(P_0, M) & \xrightarrow{d} & \text{Hom}_G(P_1, M) & \xrightarrow{d} & \text{Hom}_G(P_1, M) \xrightarrow{d} \dots \end{array}$$

Lemma 5.2.11

The induced differential $d: \text{Fun}(G^n, M) \rightarrow \text{Fun}(G^{n+1}, M)$ is as follows:

- $n = 0$: $d(m)(g) = gm - m$
- $n = 1$: $d(f)(g_1, g_2) = g_1 f(g_2) - f(g_1 g_2) + f(g_1)$
- $n = 2$: $d(f)(g_1, g_2, g_3) = g_1 f(g_2 g_3) - f(g_1 g_2, g_3) + f(g_1, g_2 g_3) - f(g_1, g_2)$
- For general n , the formula for $d(f)(g_1, \dots, g_{n+1})$ is:

$$\begin{aligned} &g_1 f(g_2, \dots, g_{n+1}) - f(g_1 g_2, g_3, \dots, g_{n+1}) + f(g_1, g_2 g_3, \dots, g_{n+1}) - \dots \\ &\quad + \dots (-1)^n f(g_1, \dots, g_n g_{n+1}) + (-1)^{n+1} f(g_1, \dots, g_n) \end{aligned}$$

Proof. We do induction. First $n = 0$: Note that

$$\begin{array}{ccc} M & \xrightarrow{d} & \text{Fun}(G, M) \\ \downarrow \sim & & \downarrow \sim \\ \text{Hom}_G(\mathbb{Z}[G], M) & \xrightarrow{d} & \text{Hom}_G(\mathbb{Z}[G \times G], M) \end{array}$$

We compute this by going below. The map below sends $m \mapsto (g \mapsto gm)$, and then horizontal, we get that

$$\begin{array}{ccc} m & \dashrightarrow & ((1, g) \mapsto gm - m) \\ \downarrow & & \uparrow \\ (g \mapsto gm) & \mapsto & ((g, h) \mapsto hm - gm) \end{array}$$

Note that the vertical arrow going up on the right is justified by taking $(1, g)$ as a basis for $\mathbb{Z}[G \times G]$. Thus, true.

We do $n = 1$, and call it a day, since the calculation is essentially the same. We use the following diagram now.

$$\begin{array}{ccc} \text{Fun}(G, M) & \xrightarrow{d} & \text{Fun}(G \times G, M) \\ \downarrow \sim & & \downarrow \sim \\ \text{Hom}_G(\mathbb{Z}[G \times G], M) & \xrightarrow{d} & \text{Hom}_G(\mathbb{Z}[G \times G \times G], M) \end{array}$$

We have the following.

$$\begin{array}{ccc} f & \dashrightarrow & ((g_1, g_2) \mapsto g_1 f(g_2) - f(g_1 g_2) + f(g_1)) \\ \downarrow & & \uparrow \\ ((g_1, g_2) = g_1(1, g_1^{-1} g_2) \mapsto g_1 f(g_1^{-1} g_2)) & \mapsto & ((g_1, g_2, g_3) \mapsto g_2 f(g_2^{-1} g_3) - g_1 f(g_1^{-1} g_3) + g_1 f(g_1^{-1} g_2)) \end{array}$$

Note that in the vertical right arrow, we use the basis $(1, g_1, g_1 g_2)$ as a basis for $\mathbb{Z}[G \times G \times G]$, where $(g_1, g_2) \in G \times G$. Thus, true. $\square$

Let us summarize what we have done above in the following theorem.

Theorem 5.2.12

Let G be a group and M be a G -module. Then, $H^n(G, M)$ is the n -th cohomology of the complex

$$M \longrightarrow \text{Fun}(G, M) \xrightarrow{d} \text{Fun}(G \times G, M) \xrightarrow{d} \dots$$

where if $f \in \text{Fun}(G^n, M)$, then

$$\begin{aligned} d(f)(g_1, \dots, g_{n+1}) &= g_1 f(g_2, \dots, g_{n+1}) - f(g_1 g_2, g_3, \dots, g_{n+1}) + \dots \\ &\quad + (-1)^n f(g_1, \dots, g_n g_{n+1}) + (-1)^{n+1} f(g_1, \dots, g_n). \end{aligned}$$

Definition 5.2.13. Elements of the set $\{ \text{functions } f: G \rightarrow M: f(g_1 g_2) = g_1 f(g_2) + f(g_1) \}$ are called the **1-cocycles**.

Definition 5.2.14. Elements of the set $\{ \text{functions } f: G \rightarrow M \text{ of the form } g \mapsto gm - m \}$ are called **trivial 1-cocycles**.

Definition 5.2.15. More generally, we call elements of f which are in the k -cocycles as elements which lie in the kernel of d_k , and those in the image of d_{k+1} as trivial k -cocycles.

Theorem 5.2.16 (Low Homology Groups)

Let G be a group and M be a G -module. Then, we have the following:

1. $n = 0$: $H^0(G, M) = M^G$
2. $n = 1$: $H^1(G, M)$ is the quotient of the group

$$\{f \in \text{Fun}(G, M) : f(g_1 g_2) = g_1 f(g_2) + f(g_1)\}$$

by

$$\{f \in \text{Fun}(G, M) \text{ of the form } g \mapsto gm - m\}.$$

That is, the set of 1-cocycles quotiented by the trivial 1-cocycles.

3. $n = 2$: $H^2(G, M)$ is the quotient of the group

$$\{f \in \text{Fun}(G^2, M) : g_1 f(g_2, g_3) - f(g_1 g_2, g_3) + f(g_1, g_2 g_3) - f(g_1, g_2) = 0\}$$

by

$$\{f \in \text{Fun}(G^2, M) \text{ of the form } (g_1, g_2) \mapsto g_1 f(g_2) - f(g_1 g_2) + f(g_1)\}.$$

That is, the quotient of 2-cocycles by the trivial 2-cocycles.

Remark 5.2.17 — 1-cocycles are also called *crossed homomorphisms* from G to M . Note that if the G action on M is trivial, then 1-cocycles are the same as group homomorphisms. Further, the trivial 1-cocycle conditions, so we just have that $H^1(G, M) = \text{Hom}(G, M)$.

Example 5.2.18

If G is finite, then $H^1(G, \mathbb{Z}) = 0$. This is because $\mathbb{Z}$ is a trivial $\mathbb{Z}[G]$ -module, and so $f(G) \subseteq \mathbb{Z}$ is a finite subgroup. $\mathbb{Z}$ has no nontrivial finite subgroups, so must be trivial.

Example 5.2.19

If G is trivial and k is a field, then $H^1(G, K^\times)$ provide the 1-dimensional characters of G .

Exercise 5.2.20. Let S be a set and let F be the group freely generated by S . Calculate $H^n(F, M)$ and $H_n(F, M)$ for all $n \geq 0$, where M is a trivial $\mathbb{Z}[F]$ -module.

Solution: We use Exercise 5.1.19. We know that the augmentation ideal is $I \cong \bigoplus_{s \in S} \mathbb{Z}[F](1 - s)$, so we can construct the following s.e.s.:

$$0 \rightarrow I \xrightarrow{\iota} \mathbb{Z}[F] \xrightarrow{\sigma} \mathbb{Z} \rightarrow 0$$

Here, σ is the map from Exercise 5.1.19, and ι is the canonical inclusion, since I is the kernel of the map; therefore, this is exact. Let us first calculate homology. For this, we

apply $-\otimes_F M$ to the projective resolution part to get

$$0 \rightarrow \bigoplus_{s \in S} M \rightarrow M \rightarrow 0$$

Particularly, $H_n(F, M) = 0$, for $n \geq 2$. We claim that the second map becomes the zero map. Let $(\sum_s g_s(1-s)) \otimes m \in I \otimes M$, where $g_s \in \mathbb{Z}[F]$. Then, $\iota \otimes \text{id}_M$ maps this to

$$(\sum_s g_s(1-s)) \otimes m = 1 \otimes (\sum_s g_s(1-s))m = \sum g_s \otimes 0 = 0.$$

This is because M is a trivial $\mathbb{Z}[F]$ -module. Therefore, the second map is the zero map, giving $H_1(F, M) = \bigoplus_{s \in S} M$. Further, we now see that $H_0(F, M) = M$. As for cohomology, we do the same, but apply $\text{Hom}_{\mathbb{Z}[F]}(-, M)$. Because direct sums factor through the first entry, we get that

$$0 \rightarrow M \rightarrow \prod_{s \in S} M \rightarrow 0.$$

Similarly, the second map is the zero map because if $f \in \text{Hom}_{\mathbb{Z}[F]}(\mathbb{Z}[F], M)$ with $f(1) = m$, then $\iota^*(f) = f \circ \iota$ applied to some element $\sum_s g_s(1-s)$ can be computed as

$$f \circ \iota \left(\sum_s g_s(1-s) \right) = f \left(\sum_s g_s(1-s) \right) = \left(\sum_s g_s(1-s) \right) \cdot f(1) = \left(\sum_s g_s(1-s) \right) m = 0.$$

Thus, $H^1(F, M) = \prod_{s \in S} M$, $H^0(F, M) = M$ and $H^n(F, M) = 0$ for $n \geq 2$. Ultimately, we have as follows:

$$H^n(F, M) = \begin{cases} M & n = 0, \\ \prod_{s \in S} M & n = 1, \\ 0 & n \geq 2 \end{cases}, \quad H_n(F, M) = \begin{cases} M & n = 0, \\ \bigoplus_{s \in S} M & n = 1, \\ 0 & n \geq 2 \end{cases}$$

□

§5.3 Change of Group

We already know that $H^\bullet(G, -)$ is a functor. That is, we can change the ‘coefficients’ M . One way to compare groups is via restriction maps.

Definition 5.3.1. Let $\varphi: H \rightarrow G$ be a homomorphism of groups and M be a G -module. Then, there is a **restriction map**

$$\text{res}_H^G: \text{Fun}(G^n, M) \rightarrow \text{Fun}(H^n, M) \\ f \mapsto f \circ (\varphi^n).$$

Here, φ^n means φ applied in each entry or coordinate of H . This induces a chain map (which one should check)

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \longrightarrow & \text{Fun}(G, M) & \xrightarrow{d} & \text{Fun}(G \times G, M) \xrightarrow{d} \dots \\ & & \parallel & & \downarrow \text{res}_1 & & \downarrow \text{res}_2 \\ 0 & \longrightarrow & M & \longrightarrow & \text{Fun}(H, M) & \xrightarrow{d} & \text{Fun}(H \times H, M) \xrightarrow{d} \dots \end{array}$$

It follows that there is a map $\text{res}_H^G: H^n(G, M) \rightarrow H^n(H, M)$.

Remark 5.3.2 ($\mathbb{Z}[G]$ is a free $\mathbb{Z}[H]$ -module) — If $H \subseteq G$, then $\text{res}(f) = f|_{H^n}$. In fact, if $H \subseteq G$, we can think of this another way. Let $P_\bullet \rightarrow \mathbb{Z}$ be any projective resolution of the trivial $\mathbb{Z}[G]$ -module. Note that $\mathbb{Z}[H] \subseteq \mathbb{Z}[G] = \bigoplus_{g \in G} \mathbb{Z}g$. Therefore, we may think of $\mathbb{Z}[G]$ as follows

$$\mathbb{Z}[G] = \bigoplus_{g \in G} \mathbb{Z}g = \bigoplus_{\substack{h \in H \\ g \in G/H}} \mathbb{Z}hg$$

Thus, we see that

$$\mathbb{Z}[H] \subseteq \mathbb{Z}[G] \cong \bigoplus_{g \in G/H} \mathbb{Z}[H] \cdot g \cong \mathbb{Z}[H]^{\oplus [G:H]}.$$

In other words, $\mathbb{Z}[G]$ is a free (left) $\mathbb{Z}[H]$ -module. We note that the isomorphism of the above is of left $\mathbb{Z}[H]$ -modules.

Remark 5.3.3 — A useful fact about this is that $\mathbb{Z}[G]$ is a projective $\mathbb{Z}[H]$ module. Therefore, if $P_\bullet \rightarrow \mathbb{Z}$ is a projective resolution of $\mathbb{Z}$ as a trivial $\mathbb{Z}[G]$ -module, then it's also a projective resolution of $\mathbb{Z}$ as a trivial $\mathbb{Z}[H]$ -module (because projective modules over $\mathbb{Z}[H]$ are precisely direct summands). Note that this does not imply that $H^k(H, M) \cong H^k(G, M)$, but $P_\bullet$ be used to calculate $H^n(H, M)$ as the cohomology groups of $\text{Hom}_{\mathbb{Z}[H]}(P_\bullet, M)$. That is, we get a map

$$\text{res}_H^G: H^n(G, M) = H^n(\text{Hom}_{\mathbb{Z}[G]}(P_\bullet, M)) \rightarrow H^n(\text{Hom}_{\mathbb{Z}[H]}(P_\bullet, M)) = H^n(H, M)$$

The key here is that any $\mathbb{Z}[G]$ -linear map is also $\mathbb{Z}[H]$ -linear, since H is a subgroup of G .

Remark 5.3.4 — When $H \subseteq G$, something similar can be said for injective resolutions too. Let $\mathbb{Z} \rightarrow I^\bullet$ be an injective resolution of the trivial $\mathbb{Z}[G]$ -module. We shall show that this is an injective resolution of $\mathbb{Z}$, as a trivial $\mathbb{Z}[H]$ -module as well.

Proof. What we really need to show is that if I is an injective $\mathbb{Z}[G]$ -module, then it is also an injective $\mathbb{Z}[H]$ -module. Since $\mathbb{Z}[G]$ is free as a $\mathbb{Z}[H]$ -module, it is flat, so $\mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} -$ is exact. $f: A \rightarrow I$ be any map of $\mathbb{Z}[H]$ -modules, and $A \hookrightarrow B$ be any injection (between $\mathbb{Z}[H]$ -modules). Then, define an R -linear map $F: \mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} A \rightarrow I$ via

$$F(g \otimes a) = g \cdot f(a).$$

This is taking over $\mathbb{Z}[G]$ now, so by injectivity of I (over $\mathbb{Z}[G]$), we may extend this to a map $\tilde{F}: \mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} B \rightarrow I$ by Baer's criterion. Now, we descend this to an extension $\tilde{f}: B \rightarrow I$ by defining

$$\tilde{f}(b) := \tilde{F}(1 \otimes b).$$

Then, $\tilde{f}$ is $\mathbb{Z}[H]$ -linear and for all $a \in A$, $\tilde{f}(a) = \tilde{F}(1 \otimes a) = f(a)$. $\square$

We next question how $H^n(G/H, -)$ relates with $H^n(G, -)$, where $H \subseteq G$ is a normal subgroup.

Remark 5.3.5 — Note that G -modules are not G/H -modules unless H acts trivially. However, there is a functor $G\mathbf{mod} \rightarrow G/H\mathbf{mod}$ which maps M to M^H . This has a well-defined G/H -action, where if $\bar{g} \in G/H$, then

$$\bar{g} \cdot m := g \cdot m.$$

The well-definedness comes from $m \in M^H$. Therefore, since H acts trivially on M^H , M^H is a G/H -module.

Definition 5.3.6. Let $H \subseteq G$ be a normal subgroup. Let M be a G -module. Then, the **inflation map**

$$\inf_{G/H}^G: H^n(G/H, M^H) \rightarrow H^n(G, M)$$

is defined as the composition

$$H^n(G/H, M^H) \xrightarrow{\text{res}_G^{G/H}} H^n(G, M^H) \rightarrow H^n(G, M).$$

Here, the first maps comes from the map $G \rightarrow G/H$, and the second is induced from the inclusion $M^H \hookrightarrow M$.

Definition 5.3.7. If $H \subseteq G$ and M is a H -module, then we define the (two) G -modules, the **induced G -module** and **co-induced G -modules**

$$\begin{aligned} \text{Ind}_H^G(M) &= \mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} M, \\ \text{CoInd}_H^G(M) &= \text{Hom}_{\mathbb{Z}[H]}(\mathbb{Z}[G], M). \end{aligned}$$

Remark 5.3.8 — The G -module action on $\text{Ind}_H^G(M)$ is

$$g' \cdot (g \otimes m) = g'g \otimes m$$

and on $\text{CoInd}_H^G(M)$, it is

$$g \cdot f(g') = f(g^{-1}g).$$

Exercise 5.3.9. Let H be a subgroup of finite index in a group G , and let M be a $\mathbb{Z}[H]$ -module. Show that there is an isomorphism of $\mathbb{Z}[G]$ -modules

$$\text{Ind}_H^G(M) \cong \text{CoInd}_H^G(M).$$

Solution: From the above remarks, we note that

$$\mathbb{Z}[G] \cong \bigoplus_{\bar{g} \in G/H} \mathbb{Z}[H] \cdot \bar{g}.$$

Since this is a finite direct sum, we know how this factors over tensor products and Hom. Let $n = [G : H]$. Then,

$$\begin{aligned} \text{Ind}_H^G(M) &\cong \left(\bigoplus_{i=1}^n \mathbb{Z}[H] \right) \otimes_{\mathbb{Z}[H]} M \cong \bigoplus_{i=1}^n M \\ \text{CoInd}_H^G(M) &\cong \text{Hom}_{\mathbb{Z}[H]} \left(\bigoplus_{i=1}^n \mathbb{Z}[H], M \right) \cong \prod_{i=1}^n M. \end{aligned}$$

These are direct sums/ products of the same size, so they coincide in **A**b. We want to show that these are also isomorphisms as G -modules.

Choose a set of representatives $g \in G$ for the left cosets $gH \in G/H$. Define $\Psi: \text{CoInd}_H^G(B) \rightarrow \text{Ind}_H^G(B)$ by

$$\Psi(\varphi) = \sum_{gH \in G/H} g \otimes \varphi(g).$$

We first show that Ψ is well defined. If g is replaced by another representative gh with $h \in H$, then

$$(gh) \otimes \varphi(gh) = gh \otimes h^{-1}\varphi(g) = g \otimes \varphi(g),$$

using the relation $xh \otimes b = x \otimes hb$ in the tensor product. Thus Ψ is well defined.

Next we check that Ψ is $\mathbb{Z}[G]$ -linear. It is clearly $\mathbb{Z}$ -linear, so it remains to check G -equivariance. Let $g' \in G$. Then

$$\Psi(g'\varphi) = \sum_{gH \in G/H} g \otimes (g'\varphi)(g) = \sum_{gH \in G/H} g \otimes \varphi((g')^{-1}g).$$

Let $u = (g')^{-1}g$. Then $g = g'u$, and since left multiplication by g' permutes the left cosets of H , the set of representatives u again runs over G/H . Hence

$$\Psi(g'\varphi) = \sum_{uH \in G/H} g'u \otimes \varphi(u) = g' \sum_{uH \in G/H} u \otimes \varphi(u) = g'\Psi(\varphi).$$

Thus Ψ is $\mathbb{Z}[G]$ -linear.

We now define its inverse $\Phi: \text{Ind}_H^G(B) \rightarrow \text{CoInd}_H^G(B)$. Since the chosen left coset representatives form a basis of the free left $\mathbb{Z}[H]$ -module $\mathbb{Z}[G]$, every element of $\text{Ind}_H^G(B)$ can be written uniquely in the form

$$\sum_{gH \in G/H} g \otimes b_g$$

with $b_g \in B$. For such an element, define $\Phi(\sum g \otimes b_g)$ to be the function $\varphi: G \rightarrow B$ determined by

$$\varphi(g) = b_g \text{ for each chosen representative } g,$$

and extended to all of G by the rule $\varphi(gh) = h^{-1}\varphi(g)$.

We check that Φ is well defined. Every $x \in G$ can be written uniquely as $x = gh$ with g a chosen representative and $h \in H$, so the above rule determines a unique value of $\varphi(x)$. Thus Φ is well defined.

We next verify that $\Phi(\sum g \otimes b_g)$ lies in $\text{CoInd}_H^G(B)$. Let $x \in G$ and $h_0 \in H$. Write $x = gh$ with g a representative and $h \in H$. Then

$$xh_0 = g(hh_0),$$

so

$$\Phi\left(\sum g \otimes b_g\right)(xh_0) = (hh_0)^{-1}b_g = h_0^{-1}h^{-1}b_g = h_0^{-1}\Phi\left(\sum g \otimes b_g\right)(x).$$

Thus $\Phi(\sum g \otimes b_g) \in \text{CoInd}_H^G(B)$.

We now check that Φ is $\mathbb{Z}[G]$ -linear. It is clearly $\mathbb{Z}$ -linear, so it suffices to check G -equivariance. Let

$$u = \sum_{gH \in G/H} g \otimes b_g$$

and $g' \in G$. Write $x \in G$ as $x = gh$ with g a representative and $h \in H$, and write $(g')^{-1}g = u_0h_1$ with u_0 a representative and $h_1 \in H$. Then

$$(g'\Phi(u))(x) = \Phi(u)((g')^{-1}x) = \Phi(u)((g')^{-1}gh) = \Phi(u)(u_0h_1h) = (h_1h)^{-1}b_{u_0}.$$

On the other hand,

$$g'u = \sum_{gH} (g'g) \otimes b_g,$$

and rewriting each $g'g$ in terms of representatives shows that the coefficient of g in $g'u$ is $h_1^{-1}b_{u_0}$. Hence

$$\Phi(g'u)(x) = h^{-1}(h_1^{-1}b_{u_0}) = (h_1h)^{-1}b_{u_0},$$

which agrees with $(g'\Phi(u))(x)$. Thus Φ is $\mathbb{Z}[G]$ -linear.

Finally, we show that Φ and Ψ are inverse maps. Let $\varphi \in \text{CoInd}_H^G(B)$. Then

$$\Psi(\varphi) = \sum_{gH \in G/H} g \otimes \varphi(g),$$

and applying Φ gives a function satisfying

$$\Phi(\Psi(\varphi))(g) = \varphi(g)$$

for each representative g , and extended by $\varphi(gh) = h^{-1}\varphi(g)$. Hence $\Phi(\Psi(\varphi)) = \varphi$.

Conversely, let

$$u = \sum_{gH \in G/H} g \otimes b_g \in \text{Ind}_H^G(B).$$

Then $\Phi(u)$ satisfies $\Phi(u)(g) = b_g$, so

$$\Psi(\Phi(u)) = \sum_{gH \in G/H} g \otimes \Phi(u)(g) = \sum_{gH \in G/H} g \otimes b_g = u.$$

Therefore Φ and Ψ are inverse $\mathbb{Z}[G]$ -module homomorphisms, and hence

$$\text{Ind}_H^G(B) \cong \text{CoInd}_H^G(B).$$

□

Theorem 5.3.10 (Shapiro's Lemma)

Let $H \subseteq G$ and M be a H -module. We have the following:

$$\begin{aligned} H_n(G, \text{Ind}_H^G(M)) &\cong H_n(H, M) \\ H^n(G, \text{CoInd}_H^G(M)) &\cong H^n(H, M) \end{aligned}$$

Proof. We prove the cohomology one. First take a projective resolution $P_\bullet \rightarrow \mathbb{Z}$ of the trivial G -module $\mathbb{Z}$. We can compute $H^n(G, \text{CoInd}_H^G(M))$, which is the n -th cohomology group of the complex

$$\text{Hom}_{\mathbb{Z}[G]}(P_\bullet, \text{CoInd}_H^G(M)).$$

We have that

$$\text{Hom}_{\mathbb{Z}[G]}(P_\bullet, \text{CoInd}_H^G(M)) = \text{Hom}_{\mathbb{Z}[G]}(P_\bullet, \text{Hom}_{\mathbb{Z}[H]}(\mathbb{Z}[G], M)) \cong \text{Hom}_{\mathbb{Z}[H]}(P_\bullet, M).$$

Then, since $P_\bullet$ is also a projective resolution of $\mathbb{Z}$ by $\mathbb{Z}[H]$ -modules, the RHS provides the cohomology $H^n(H, M)$. Note that in the second isomorphism above, we've used a special instance of the tensor-Hom adjunction.

As for the homology version, we don't need to do anything fancy. All we do is take a projective resolution of $\mathbb{Z}$ by $\mathbb{Z}[G]$ modules $P_\bullet$ then tensor with $\otimes_{\mathbb{Z}[G]} \mathbb{Z}[G] \otimes_{\mathbb{Z}[H]} M$. Then, it is clear that this provides a complex with terms $P_\bullet \otimes_{\mathbb{Z}[H]} M$, but the terms are then $H_n(H, M)$. $\square$

Remark 5.3.11 — Suppose that $[G : H] < \infty$ and M is a left $\mathbb{Z}[G]$ -module. Then the map $M \rightarrow \text{CoInd}_H^G(M)$ induces a map $H^n(G, M) \rightarrow H^n(G, \text{CoInd}_H^G(M))$. The following diagram commutes:

$$\begin{array}{ccc} H^n(G, M) & \xrightarrow{\text{Res}} & H^n(H, M) \\ \downarrow & \nearrow \cong & \\ H^n(G, \text{CoInd}_H^G(M)) & & \end{array}$$

where the isomorphism is Shapiro's lemma.

Exercise 5.3.12. Let G be a group and suppose that $H_1(G, M) = 0$ for all $\mathbb{Z}[G]$ -modules M . Prove that G is the trivial group.

What if you just assume $H_1(G, M) = 0$ for all trivial $\mathbb{Z}[G]$ -modules M ? Is that enough? (Hint: look up the definition of perfect group.)

Solution: Let M be a module with trivial G -module action. Then, we know that $H_1(G, M) \cong G^{\text{ab}} \otimes_{\mathbb{Z}} M$. Particularly, letting $M = \mathbb{Z}$, we have that $G^{\text{ab}} = 0$, so $G = [G, G]$. This is what occurs if we only assume the assumption for trivial $\mathbb{Z}[G]$ -modules M . In this case, G is a perfect group; that is, a group with $G = [G, G]$.

To finish the proof of the first part, we require Shapiro's lemma. Let $H = \langle g \rangle \subseteq G$ and $\mathbb{Z}$ be the trivial H -module. Then, $\text{Ind}_H^G(\mathbb{Z})$ is a G -module, so by assumption, $H_1(G, \text{Ind}_H^G(\mathbb{Z})) = 0$. Further, $H_1(H, \mathbb{Z}) = 0$. However, since $\mathbb{Z}$ is a trivial H -module, we get that H is cyclic, it is abelian, so we get that $H = [H, H]$, so H is trivial. Since the element g was arbitrary, it follows that G is trivial. $\square$

Theorem 5.3.13 (Inflation-Restriction Exact Sequence)

Let $H \subseteq G$ be a normal subgroup, and M be a G -module. Then, there is an exact sequence

$$0 \rightarrow H^1(G/H, M^H) \xrightarrow{\text{inf}} H^1(G, M) \xrightarrow{\text{res}} H^1(H, M).$$

Proof. Firstly, we need to show that the inflation map is injective. We use the 1-cocycles. We need to show that if $c: G/H \rightarrow M^H$ is a 1-cocycle such that $\inf(c)$ is trivial, then we need to show that c is trivial. In other words, we need to show that if the following composition (which is just the inflation map)

$$G \rightarrow G/H \xrightarrow{c} M^H \hookrightarrow M$$

is trivial then c is trivial. Thus, assume that $\inf(c): G \rightarrow M$ is equal to

$$(c(g) =) \inf(c)(g) = gm - m,$$

for some $m \in M$. Then, we want that $c(g) = gm' - m'$, for some $m' \in M^H$. Let e be the identity element. Note that since c is a 1-cocycle, $c(e) = c(e \cdot e) = ec(e) + c(e) = 2c(e)$. Thus, $c(e) = 0$. In the case of G/H , this means that $c(h) = 0$, for all $h \in H$. Because $c(g) = \inf(c)(g) = gm - m$, this implies that $hm - m = 0$ for all $h \in H$. Thus, $m \in M^H$. Thus, $c(g) = gm - m$, for some $m \in M^H$, which says that c is trivial in $H^1(G/H, M^H)$.

We next need to check exactness at $H^1(G, M)$. Let $[c] \in H^1(G, M)$. We want to see that if $[c]$ maps to 0 in $H^1(H, M)$, then $[c]$ is the inflation of a 1-cocycle from $H^1(G/H, M^H)$. By assumption $c|_H: H \rightarrow M$ is of the form $c(h) = hm - m$, for all $h \in H$, for some fixed $m \in M$. Then, we define $c': G \rightarrow M$ as $c'(g) = c(g) - (gm - m)$. By definition, $(gm - m)$ is trivial, so we have that $[c'] = [c] \in H^1(G, M)$; that is, their classes are equivalent. Now note that $c'(h) = 0$ for all $h \in H$. Moreover,

$$c'(gh) = gc'(h) + c'(g) = c'(g).$$

Therefore, c' is constant on the right cosets gH . However, H is normal, so $gH = Hg$. Consequently, $c'(hg) = c'(g)$ as well, for all $h \in H$ and $g \in G$. Thus,

$$c'(g) = c'(hg) = hc'(g) + c'(h) = hc'(g).$$

Thus, $c'(g) \in M^H$ for all $g \in G$. Since we say that c' is constant on cosets, this induces a well-defined 1-cocycle $G/H \rightarrow M^H$. For the other inclusion, it is clear that the $\text{res} \circ \inf$ is trivial. In other words, c' descends down to $\bar{c}'$:

$$\begin{array}{ccc} G & \xrightarrow{c'} & M \\ \downarrow & & \uparrow \\ G/H & \xrightarrow{\bar{c}'} & M^H \end{array}$$

This diagram tells us that $[c] = [c'] = \inf(\bar{c}')$. □

Example 5.3.14

Let $G = D_{2m}$, the Dihedral group of order $2m$. Then, let $C_m \subseteq D_{2m}$ be the subgroup of rotations. These form a normal subgroup. Thus, we have an exact sequence

$$0 \rightarrow C_m \hookrightarrow D_{2m} \twoheadrightarrow D_{2m}/C_m \cong C_2 \rightarrow 0$$

and we form the following inflation retraction sequence:

$$0 \rightarrow H^1(C_2, M^H) \rightarrow H^1(D_{2m}, M) \rightarrow H^1(C_m, M).$$

Recall now that $H^1(C_n, M) = M[n]$ for a trivial C_m -module M , where C_n is any cyclic group. Thus, there is an exact sequence as follows:

$$0 \rightarrow M[2] \rightarrow H^1(D_{2m}, M) \rightarrow M[m]$$

Note that if M is m -torsion free, then $H^1(D_{2m}, M) \cong M[2]$.

Exercise 5.3.15. Calculate $H^1(C_3, \mathbb{Q})$, where $\mathbb{Q}$ is given the trivial C_3 -action.

Solution: We have that $H^1(C_3, \mathbb{Q}) \cong \mathbb{Q}[2]$, so $H^1(C_3, \mathbb{Q}) \cong 0$. We could have also seen this from the fact that $\mathbb{Q}$ is a $\mathbb{Q}$ -vector space. $\square$

Example 5.3.16

Let $G = GL_n(k)$, where k is a field. Let $H = SL_n(k)$. Then, $G/H \cong k^\times$ via the determinant map, so H is normal in G . Then, we have an exact sequence.

$$0 \rightarrow H^1(k^\times, M) \rightarrow H^1(GL_n(K), M) \rightarrow H^1(SL_n(k), M)$$

§5.4 Applications to Group Theory

Much long before group cohomology, people were interested in groups. More specifically, how one group can be *extended* from another. That is, given H, G groups, can we find all groups E that fit into an extension?

$$1 \rightarrow H \xrightarrow{\iota} E \xrightarrow{\pi} G \rightarrow 1$$

This will be the question we will be interested now until the end. Here, we've written 1 to denote the trivial multiplicative group to emphasize that H, E, G need not be abelian. If π is a surjective group homomorphism with kernel $\iota(H)$, then $\iota(H) \subseteq E$ is normal.

Remark 5.4.1 — Note that if H, E, G are abelian, then the answer is just $\text{Ext}_{\mathbb{Z}}^1(G, H)$, so we focus on the nonabelian setting.

Definition 5.4.2. Two extensions are said to be **equivalent** if there exists a diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & H & \longrightarrow & E & \longrightarrow & G \longrightarrow 1 \\ & & \parallel & & \downarrow & & \parallel \\ 1 & \longrightarrow & H & \longrightarrow & E' & \longrightarrow & G \longrightarrow 1 \end{array}$$

Remark 5.4.3 — Of course, if we have an equivalence, that middle map will turn out to be an isomorphism

A special case occurs when $A := H$ is abelian, and **this will be the case we focus on**. Let us write

$$0 \rightarrow A \rightarrow E \xrightarrow{\pi} G \rightarrow 1.$$

Note that E can act on A by conjugation, and when A is abelian, this action is trivial: if $e \in E$ and $x \in A$, we have that

$$\pi(exe^{-1}) = \pi(e) \underbrace{\pi(x)}_{=\text{id}} \pi(e)^{-1} = \text{id}_G$$

This means that $exe^{-1} \in A$. That is, A is a normal subgroup of E . Further, A is abelian, so $exe^{-1} = x$. Thus, the conjugation action of E acting on A factors through $E/A = G$. This gives a G -module structure on A . We now want to use $H^\bullet(G, A)$ to answer this question. Explicitly, the classification will be in terms of $H^2(G, A)$.

Remark 5.4.4 — Even if we impose A, G being abelian with A having a trivial G -action, we don’t fall into the degenerate case because E is not necessarily abelian. To land in the case of $\text{Ext}_{\mathbb{Z}}^1(G, A)$, we really require the middle to be abelian too.

We first ask what the *trivial extensions* are. One might think that, as an analogue of the direct sum, the direct product may work. However, this is too naive, as the direct product will correspond to a class in $H^2(G, A_{\text{triv}})$, where A_{triv} is equipped with the trivial G -module structure. Because we want any G -module. The correct approach to this is semidirect products.

Definition 5.4.5. Let G be a group and A be a G -module. A **semidirect product** $E := A \rtimes G$ is the set $A \times G$ with the operation $(a, g) \cdot (b, h) := (a + g(b), gh)$ (where we’ve written additively in the left and multiplicatively on the right), and $g(b)$ denotes the action of g on b .

Remark 5.4.6 — One can check that this makes $A \rtimes G$ into a group with unit $(0, e)$ and inverses $(a, g)^{-1} = (-g^{-1}(a), g^{-1})$. Note that we could’ve written $g^{-1}(-a)$, but the G -action is $\mathbb{Z}$ -linear.

This provides an extension

$$0 \rightarrow A \xrightarrow{\alpha} A \rtimes G \xrightarrow{\beta} G \rightarrow 1.$$

such that $\alpha(a) := (a, e)$, so that $A \cong A \times e$ is normal in $A \rtimes G$. Further, $\beta((a, g)) = g$.

Definition 5.4.7. An extension is **split** if there exists a group homomorphism $\sigma: G \rightarrow E$ such that $\beta\sigma = \text{id}_G$. Here, β is the surjection from E to G .

Proposition 5.4.8

Let G be a group and A a G -module. Any extension

$$0 \longrightarrow A \xrightarrow{\alpha} E \xrightarrow{\beta} G \longrightarrow 1$$

is split if and only if it is equivalent to an extension

$$0 \longrightarrow A \xrightarrow{\alpha} A \rtimes G \xrightarrow{\beta} G \longrightarrow 1$$

Then, the extensions is equivalent to the semidirect product. In other words, split on the right is equivalent to having a semidirect product.

Proof. First, note that

$$0 \rightarrow A \xrightarrow{\alpha} A \rtimes G \xrightarrow{\beta} G \rightarrow 1.$$

is indeed an extension. The maps involved are indeed group homomorphisms. Define $\sigma: G \rightarrow A \rtimes G$ as $g \mapsto (0, g)$. This is indeed a group homomorphism and a section to β , so this sequence is split. Assume now that our (first) sequence above is split by some σ . We want to construct an isomorphism to the semidirect product. We define

$$\begin{aligned} \phi: E &\rightarrow A \times G \\ x &\mapsto (\alpha^{-1}(x(\sigma\beta(x))^{-1}), \beta x) \end{aligned}$$

We note that the above works because $x(\sigma\beta x)^{-1} \mapsto e_A$ under β . The claim is that this is a bijection with inverse

$$\begin{aligned} \psi: A \times G &\rightarrow E \\ (a, g) &\mapsto \alpha(a)\sigma(g) \end{aligned}$$

We have

$$\psi\phi(x) = \psi(\alpha^{-1}(x(\sigma\beta(x))^{-1}), \beta x) = x(\sigma\beta x)^{-1}\sigma\beta x = x$$

We now check that ϕ is a group homomorphism. Take $x, x' \in E$. We want to show that $\phi(xx') = \phi(x)\phi(x')$. In the second coordinate, it is clear that we have $\beta x \cdot \beta x' = \beta xx'$, since β is a group homomorphism. We move on to the first coordinate. Next, the first coordinate of $\phi(xx')$. To remind ourselves, we want to verify that α applied to the first coordinate of the semidirect product $(a, g) \cdot (b, h)$ becomes $\alpha(a) \cdot \alpha(g \cdot b) = \alpha(a)g\alpha(b)g^{-1}$. Write $g = \beta x$ and $h = \beta x'$. To further simplify, we write α of the first coordinate (which we can do due to injectivity).

$$\begin{aligned} xx'(\sigma\beta(xx'))^{-1} &= xx'(\sigma(g)\sigma(h))^{-1} \\ &= xx'\sigma(h)^{-1}\sigma(g)^{-1} \\ &= x \left(x'\sigma\beta(x')^{-1} \right) \sigma(g)^{-1} \\ &= x \left(x'\sigma\beta(x')^{-1} \right) \cdot x^{-1} \cdot \left(x(\sigma\beta x)^{-1} \right) \end{aligned}$$

Here, we note that the first bracket is the first coordinate of $\phi(x')$ and the second bracket is the first coordinate of $\phi(x)$. Then, without the second bracket, this is just the action of x on the first coordinate of $\phi(x')$. In other words, this is the first coordinate of $(g \cdot \phi(x')) \cdot \phi(x)$. By commutativity, this is the first coordinate of $\phi(x) \cdot g(\phi(x'))$. Again, this is just equal to $\alpha(\phi(x)_1 + g(\phi(x'))_1)$, where the subscript 1 means ‘the first coordinate

of’. Thus, we have the desired preservation of operations. This provides the following diagram:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & A & \longrightarrow & A \rtimes G & \longrightarrow & G \longrightarrow 1 \\
 & & \parallel & & \uparrow \phi \sim & \nearrow \exists \sigma & \parallel \\
 0 & \longrightarrow & A & \longrightarrow & E & \longrightarrow & G \longrightarrow 1
 \end{array}$$

□

Remark 5.4.9 — Note that if the left splits, we actually get a direct product (which is something stronger!)

Example 5.4.10

Let k be a field. Then, we can either take the additive group k^+ or the multiplicative group $k^\times$. Note that $k^\times$ can act on k^+ by multiplication. This makes k^+ into a $k^\times$ -module. Therefore, we can take a semidirect product $k^+ \rtimes k^\times$. This is the group of affine linear transformations, $\text{Aff}_1(k)$, where $t \mapsto (a + bt)$, for $b \neq 0$ and $t \in k$. The idea to have is the k^+ corresponds to translation and $k^\times$ corresponds to dilation.

Example 5.4.11

Consider the extension

$$1 \rightarrow \text{SL}_n(k) \rightarrow \text{GL}_n(k) \rightarrow k^\times \rightarrow 1$$

This is split because we can map any $\alpha \in k^\times$ to $\text{diag}(\alpha, 1, \dots, 1)$, providing a section to the determinant map.

Example 5.4.12

Let $A = \mathbb{Z}/n\mathbb{Z}$ and $G = C_2 = \langle 1, r \rangle$, $r^2 = 1$. We can make A into a G -module by $r \cdot a = -a$. Then, $A \rtimes G \cong D_{2n}$.

Example 5.4.13

Let Q_8 denote the quaternion group. This is the smallest group which doesn’t split. This is the group of units in the ring $R = \mathbb{Z} \oplus \mathbb{Z}i \oplus \mathbb{Z}j \oplus \mathbb{Z}k$, the Gaussian quaternions subject to the conditions

$$i^2 = j^2 = k^2 = -1, \quad ij = k = -ji.$$

Then, the units are $\{\pm 1, \pm i, \pm j, \pm k\}$. We note that all subgroups in Q_8 are normal. Let us just take $A = \{\pm 1, \pm i\} \subseteq Q_8$, which is the cyclic group of order 4. Thus, we get an extension

$$0 \rightarrow A \rightarrow Q_8 \rightarrow C_2 \rightarrow 1.$$

This is not split, for many reasons. One reason is that there is only element of order 2 in Q_8 , namely -1 , since it’s in A . Thus, $-1 \mapsto e$.

Theorem 5.4.14

Let G be a group, A be a G -module. The set of equivalence classes of extension

$$0 \rightarrow A \xrightarrow{\alpha} E \xrightarrow{\beta} G \rightarrow 1 \quad (5.1)$$

is in bijection with $H^2(G, A)$, where the induced action of G on A is the same as the G -action on the G -module A . The zero element of $H^2(G, A)$ corresponds to the class of split semidirect product with the G -module structure.

Proof. First, assign a 2-cocycle to any extension (5.1). First, choose a splitting $\sigma: G \rightarrow E$ of sets, i.e. σ is a function (set theoretic section) such that $\beta\sigma = \text{id}_G$ and $\sigma(e_G) = e_E$. If σ is a group homomorphism, then it corresponds to the split case. If not, define $\phi: G \times G \rightarrow A$ by first noting that

$$\left(\sigma(g)\sigma(h)\sigma(gh)^{-1}\right) \xrightarrow{\beta} gh(gh)^{-1} = e$$

because β is a group homomorphism. Then, we define

$$\phi(g, h) := \alpha^{-1}(\sigma(g)\sigma(h)\sigma(gh)^{-1}).$$

We note that $\phi(e, g) = \phi(g, e) = 0$, for any $g \in G$. We check that ϕ is a 2-cocycle:

$$f\phi(g, h) + \phi(f, gh) = \phi(f, g) + \phi(fg, h)$$

First, the LHS. We apply α to get that

$$\sigma(f)\sigma(g)\sigma(h)\sigma(gh)^{-1}\sigma(f)^{-1}\sigma(f)\sigma(gh)\sigma(fgh)^{-1}.$$

We note that the f in front of the $\phi(g, h)$ corresponds to the conjugation. Then, after cancellation, we have that $\sigma(f)\sigma(g)\sigma(h)\sigma(fgh)^{-1}$. Now, α of the RHS is

$$\sigma(f)\sigma(g)\sigma(fg)^{-1}\sigma(fg)\sigma(h)\sigma(fgh)^{-1},$$

and after cancellation, we get

$$\sigma(f)\sigma(g)\sigma(h)\sigma(fgh)^{-1},$$

so ϕ is a 2-cocycle.

We now need to show that this map from extension to $H^2(G, A)$ is a bijection. For this, we show that there is a bijection

$$\begin{aligned} \xi: E &\rightarrow A \times G \\ x &\mapsto (\alpha^{-1}(x(\sigma\beta x)^{-1}), \beta x). \end{aligned}$$

This is the same formula as in the previous proposition, and the same proof shows that this is a bijection. This gives a group structure on $A \times G$ as

$$(a, g) \cdot (b, h) := (a + g(b) + \phi(g, h), gh).$$

We check this directly. Write $g = \beta(x)$, $h = \beta(x')$. Then we have

$$\begin{aligned} xx' \cdot s(\beta(xx'))^{-1} &= xx's(gh)^{-1} = xx's(h)^{-1}s(g)^{-1}\alpha(\phi(g, h)) \\ &= (x\alpha(y')x^{-1}) \cdot xs(g)^{-1} \cdot \alpha(\phi(g, h)) = (x\alpha(y')x^{-1}) \cdot \alpha(y) \cdot \alpha(\phi(g, h)). \end{aligned}$$

The claim is that this makes ξ into a group homomorphism; we don't prove this here, so check yourself. We can do this by applying α and computing in E .

Conversely, given a 2-cocycle $\phi: G \times G \rightarrow A$ such that $\phi(e, g) = \phi(g, e) = 0$, for all $g \in G$ (note that we can always assume this, but proof omitted; check section 65 in [1]). Then, the formula $(a, g) \cdot (b, h) := (a + g(b) + \phi(g, h), gh)$ makes $E_\phi := A \times G$ (as a set) into a group extension

$$0 \rightarrow A \rightarrow E_\phi \rightarrow G \rightarrow 1.$$

Condition $(0, e) = (e, 0) = 0$ ensures that $(0, e)$ is the unit element. The inverse of (a, g) is

$$(-g^{-1}(a) - g^{-1}(\phi(g, g^{-1})), g^{-1}).$$

Associativity is a consequence of the 2-cocycle condition. Then A is a normal subgroup of the resulting group E_ϕ such that G acts on A by $a \mapsto g(a)$. Applying the above recipe we see that the 2-cocycle associated to E_ϕ is ϕ .

What we have done is shown that we can map between extensions E and $H^2(G, A)$. What we just need to do is show that if we quotient extensions by equivalences, we get a bijection. We can see (check) that $E \mapsto \phi \mapsto E_\phi$. Then,

$$\begin{array}{ccccc} A & \longrightarrow & E_\phi & \longrightarrow & G \\ \parallel & & \uparrow \xi & & \parallel \\ A & \longrightarrow & E & \longrightarrow & G \end{array}$$

Conversely, we can check directly from the formula $\phi \mapsto E_\phi \mapsto \phi$, we get the same 2-cocycle back.

It remains to check that extensions $E = E_\phi$ and $E' = E_\psi$ are equivalent if and only if $\phi - \psi$ is a *coboundary*, that is,

$$\phi(g, h) - \psi(g, h) = g(c(h)) - c(gh) + c(g) \quad (5.2)$$

for some function $c: G \rightarrow A$. Indeed, an equivalence of our two extensions is an isomorphism $\mu: E \rightarrow E'$ such that $\beta = \beta' \mu$. Hence μs is a set-theoretic section of β' . Any two set-theoretic sections of β' differ by a map $c: G \rightarrow A$. Thus the isomorphism $\mu: E \rightarrow E'$ sends $(a, g) \mapsto (a + c(g), g)$ which transforms the group law into a similar formula for ψ . We have

$$a + g(b) + \phi(g, h) + c(gh) = a + c(g) + g(b) + g(c(h)) + \psi(g, h),$$

which means that μ is a homomorphism precisely when (5.2) holds. $\square$

Example 5.4.15

View Q_8 again. Let $G = C_2 = \langle 1, r \rangle$, and $A = \mathbb{Z}/4\mathbb{Z}$. We give A the G -module structure by $r \cdot a = -a$. We shall classify group extensions of G by A . First, we calculate $H^2(G, A)$.

$$H^2(C_2, \mathbb{Z}/4\mathbb{Z}) \cong (\mathbb{Z}/4\mathbb{Z})^{C_2} / (1+r)(\mathbb{Z}/4\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$$

This means that there are 2-extensions here. One of them is split and one of them is not. The non-split one corresponds to Q_8 , and D_8 corresponds to the split case (the semi-direct product). Following the proof about $\phi_{Q_8}: G \times G \rightarrow A$ defined by

$$(r, r) \mapsto 2,$$

we can check that $Q_8 \cong \mathbb{Z}/4\mathbb{Z} \times C_2$ as a set but with the twisted product, which we called E_ϕ . Work out the details of this!

Solution: Let's work out the details of this. Well, first $H^2(C_2, \mathbb{Z}/4\mathbb{Z}) \cong \{0, 2\}$, where $A = \langle 1 \rangle$. We first determine our 2-cocycles. Recall that the 2-cocycle condition is

$$g\phi(h, k) - \phi(gh, k) + \phi(g, hk) - \phi(g, h) = 0.$$

We may assume that $\phi(1, 1) = \phi(1, r) = \phi(r, 1) = 0$ by normalization. Thus, we're left to check $\phi(r, r)$. The 2-cocycle condition provides then that

$$r\phi(r, r) - \phi(1, r) + \phi(r, 1) - \phi(r, r) = 0 \implies r\phi(r, r) = \phi(r, r).$$

Therefore, $\phi(r, r) \in A^G = \{0, 2\}$; we call ϕ_0 to be the map $\phi_0(r, r) = 0$ and ϕ_2 the map $\phi_2(r, r) = 2$. Now, the two extensions come from the group structure we down on the set $A \times G$. The first comes from

$$(a, g) \cdot (b, h) := (a + g(b) + \phi_0(g, h), gh),$$

which is the split one. The other is

$$(a, g) \cdot (b, h) := (a + g(b) + \phi_2(g, h), gh),$$

If we wish to find the explicit group presentations, they can be found as follows. In the first case for ϕ_0 , we have as follows:

$$(0, r) \cdot (0, r) = (0, 1), \quad x \cdot a = xax^{-1} = -a,$$

and $a^4 = 0$. Thus, ϕ_0 corresponds to the group

$$\langle a, x: a^4 = 1, \quad x^2 = 1, \quad xax^{-1} = a^{-1} \rangle \cong D_8,$$

written multiplicatively. As for the other, we have that

$$(0, r) \cdot (0, r) = (2, 1) = (1, 1) \cdot (1, 1).$$

Therefore, we have

$$\langle a, x: a^4 = 1, \quad x^2 = a^2, \quad xax^{-1} = a^{-1} \rangle \cong Q_8$$

□

Remark 5.4.16 — We note that $H^2(G, A)$ calculates the cohomology with respect to a specific G -module action A .

Example 5.4.17

Consider Q_8 , with $G = C_2$, $A = \mathbb{Z}/4\mathbb{Z}$ with the trivial action. Then, $H^2(G, A_{\text{triv}}) \cong \mathbb{Z}/2\mathbb{Z}$ again, but these correspond to C_8 and $C_2 \times C_4$. The split one is $C_2 \times C_4$.

Example 5.4.18

Note that if A is a finite abelian group of odd order, then

$$H^2(C_2, A) = 0$$

since the orders of C_2 and A are coprime. In this case, any extension of G by A is a semi-direct product $A \rtimes G$. This is the case, for example, for the dihedral group D_{2n} where n is odd: it is the unique extension of C_2 by $\mathbb{Z}/n\mathbb{Z}$ where the generator of C_2 acts on $\mathbb{Z}/n\mathbb{Z}$ as -1 . If the action of C_2 on A is trivial, then the unique extension is the direct product $A \times C_2$.

Exercise 5.4.19. Let q be a prime power. Let $n \geq 1$ be a positive integer and consider the finite field $\mathbb{F}_{q^n}$ with q^n elements. Write $\mathbb{F}_{q^n}^\times$ for the (multiplicative) group of units in $\mathbb{F}_{q^n}$.

1. Then $\mathbb{F}_q^\times$ is a subgroup of $\mathbb{F}_{q^n}^\times$ (since $\mathbb{F}_q$ is a subfield of $\mathbb{F}_{q^n}$), so we have a short exact sequence of abelian groups

$$0 \rightarrow \mathbb{F}_q^\times \rightarrow \mathbb{F}_{q^n}^\times \rightarrow \mathbb{F}_{q^n}^\times / \mathbb{F}_q^\times \rightarrow 0.$$

Show that this sequence splits if and only if $q - 1$ and $q^{n-1} + q^{n-2} + \cdots + q + 1$ are coprime.

2. Enumerate the extensions of $\mathbb{F}_{q^n}^\times / \mathbb{F}_q^\times$ by $\mathbb{F}_q^\times$ in the category of abelian groups, up to equivalence.

Solution: 1. We note that the multiplicative group of a field is cyclic. Let $m = 1 + q + \cdots + q^{n-1}$. Then, we have $\mathbb{Z}/(q-1)m\mathbb{Z} \cong \mathbb{Z}/(q-1)\mathbb{Z} \oplus \mathbb{Z}/m\mathbb{Z}$ if and only if $(q-1, m) = 1$ by the Chinese remainder theorem.

2. We're in the category of abelian groups, so we just need to classify $\text{Ext}_{\mathbb{Z}}^1(\mathbb{F}_{q^n}^\times / \mathbb{F}_q^\times, \mathbb{F}_q^\times)$. Let $n = q - 1$ and $m = q^{n+1} + \cdots + q + 1$. Then,

$$\mathbb{F}_q^\times \cong \mathbb{Z}/n\mathbb{Z}, \quad \mathbb{F}_{q^n}^\times / \mathbb{F}_q^\times \cong \mathbb{Z}/m\mathbb{Z}.$$

We know from a previous exercise that

$$\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) = (\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z}).$$

The claim is now that

$$(\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/d\mathbb{Z},$$

where $d = \gcd(n, m)$. Consider the map $\varphi: \mathbb{Z} \rightarrow (\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z})$ defined by $a \mapsto [[a]]$. Here, the first bracket is taken modulo m , and the second is modulo $n(\mathbb{Z}/m\mathbb{Z})$. This is clearly surjective, so we just find the kernel (which we claim is $d\mathbb{Z}$). Thus,

$$\phi(a) = (a \bmod m) + n(\mathbb{Z}/m\mathbb{Z}).$$

Then, $a \bmod m \in n(\mathbb{Z}/m\mathbb{Z})$ if and only if there exists some integer x such that $a \equiv nx \bmod m$. This is equivalent to saying that $a \in n\mathbb{Z} + m\mathbb{Z}$, but $n\mathbb{Z} + m\mathbb{Z} = d\mathbb{Z}$, so this proves our claim. Therefore, there are $\gcd(q-1, q^{n+1} + \cdots + q + 1)$ extensions. $\square$

Example 5.4.20

Let $G = C_3 = \langle s \rangle$, $A = C_3^2$, with C_3 -action $s \cdot (a, b) = (a, b + a)$. First, note that $H^2(C_3, C_3^2) = A^s/NA$. We have that $N = 1 + s + s^2$, so

$$(1 + s + s^2) \cdot (a, b) = (3a, 3b + 3a) = (0, 0).$$

Therefore, $NC_3 = 0$. Also, $s \cdot (a, b) = (a, b + a) = (a, b)$ if and only if $(a, b + a) = (a, b)$ if and only if $a = 0$. Thus, $H^2(C_3, C_3^2) \cong \{(0, b) : b \in C_3\} \cong C_3$. Therefore, there are three equivalence classes of extensions of C_3 by C_3^2 . We first analyze the 2-cocycles. Let $\phi: G \times G \rightarrow A$ be a 2-cocycle. We may assume that $\phi(1, 1) = \phi(1, s) = \phi(s, 1) = \phi(1, s^2) = \phi(s^2, 1) = 0$ by normalization. Recall the 2-cocycle condition:

$$g\phi(h, k) - \phi(gh, k) + \phi(g, hk) - \phi(g, h) = 0.$$

Now, before testing for some values in the cocycle condition, we reduce the problem by assuming that $\phi(s, s) = 0$. This is a valid assumption, as we will show; define $\phi(s, s) = u \in A$. Then, we can replace ϕ with some 2-cocycle ϕ' such that $\phi = \phi' + d\lambda$, for some 1-cochain λ . A 1-cochain is just some function, so define

$$\lambda(1) = \lambda(s) = 0, \quad \lambda(s^2) = u.$$

Then,

$$(d\lambda)(s, s) = s\lambda(s) - \lambda(s^2) + \lambda(s) = u.$$

Thus, $\phi'(u, u) = \phi(u, u) - u = 0$. Therefore, we may as well assume that $\phi(u, u) = 0$ originally.

Apply the 2-cocycle condition to $g = s^2$, $h = k = s$. We find that

$$s^2\phi(s, s) - \phi(1, s) + \phi(s^2, s^2) - \phi(s^2, s) = 0 \implies \phi(s^2, s^2) = \phi(s^2, s).$$

Now apply the 2-cocycle condition to $g = h = k = s$, $h = k = s^2$. We get as follows:

$$s\phi(s, s) - \phi(s^2, s) + \phi(s, s^2) - \phi(s, s) = 0 \implies \phi(s, s^2) = \phi(s^2, s) = \phi(s^2, s^2).$$

Therefore, ϕ is fully determined now; set $\phi(s^2, s^2) = c$. We just need to see what these values actually are. Apply the 2-cocycle condition with $g = s = h$, $k = s^2$:

$$s\phi(s, s^2) - \phi(s^2, s^2) + \phi(s, s) - \phi(s, s) = 0 \implies s\phi(s, s^2) = \phi(s^2, s^2) = \phi(s, s^2).$$

Therefore, we have that $c \in A^G$. Therefore, the three classes of extensions are determined by $\phi_b(s^2, s^2) = (0, b)$, for $b \in C_3$.

Remark 5.4.21 — Note that while C_3 and C_3^2 are both abelian, this did not give 9 extensions, where this number comes from the cardinality of $\text{Ext}_{\mathbb{Z}}^1(C_3, C_3^2) \cong C_3^2$. Why is this? Well, this is because now, we are not requiring E to be abelian (and so it's not some trivial action). It should be clear through this example that even

though we are considering two groups that start and end the same, there are different extensions which come from the different actions.

The Schur-Zassenhaus Theorem

This is non-examinable.

Here is a famous application of the previous theorem. We did not do this in lectures, so this is definitely not examinable at all. The question we pose is as follows: *Let E be a group and H a normal subgroup. Can we reconstruct E from H and E/H ?*

The following theorem says that the answer is (kind of) yes if E is finite and the orders of H and E/H are coprime:

Theorem 5.4.22 (Schur–Zassenhaus)

Let E be a finite group of order mn , where m and n are coprime. If H is a normal subgroup of E of order n , then

$$E \cong H \rtimes E/H.$$

Remark 5.4.23 — The theorem does not determine E uniquely, since there could be many non-isomorphic such semi-direct products. For example, if $H = C_3$ and $E/H = C_2$, then the theorem tells us that

$$E \cong C_3 \rtimes C_2.$$

There are two such groups: C_6 and $D_6 = S_3$.

Remark 5.4.24 — In our language, the theorem says that an extension

$$0 \rightarrow H \rightarrow E \rightarrow G \rightarrow 1$$

of a group G of order m by a group H of order n splits if m and n are coprime.

Remark 5.4.25 — Note that we really do need the condition that m and n are coprime. Indeed, consider the extension

$$0 \rightarrow C_2 \rightarrow C_4 \rightarrow C_2 \rightarrow 0.$$

This is not a split extension, since otherwise C_4 would contain two elements of order 2 but it only has one. So the answer to the question is no, in general.

Before we prove Schur–Zassenhaus, let me first prove a lemma:

Lemma 5.4.26 (Fratini's argument)

Let G be a finite group and H a normal subgroup of G . If S is a p -Sylow subgroup of H , then $G = N_G(S)H$.

Proof. Let $g \in G$. We want to write it as $g = nh$ where $n \in N_G(S)$ and $h \in H$. Consider $g^{-1}Sg$. Then this is a subgroup of H , because H is normal, and has the same size as S , so it must also be a p -Sylow subgroup of H . But all p -Sylow subgroups of H are conjugate to one another, i.e. there exists an $h \in H$ such that

$$g^{-1}Sg = h^{-1}Sh.$$

So $S = gh^{-1}Shg^{-1}$, i.e. $gh^{-1} \in N_G(S)$. Therefore $g \in N_G(S)H$. $\square$

Proof of Schur–Zassenhaus (Theorem 5.4.22). It suffices to show that E contains a subgroup of order m , since then such a subgroup will be isomorphic to G under $E \rightarrow G$. Indeed, if G' is a subgroup of order m then $G' \cap H$ is trivial since m and n are coprime, $|G' \cap H|$ divides both $|G'| = m$ and $|H| = n$ by Lagrange's theorem. We shall prove this by induction on the order n of H . So suppose $|H| = n > 1$ and the theorem is true for all subgroups of smaller order.

Step 1: Without loss of generality, we may assume that H is a minimal normal subgroup. Indeed, suppose that H is not a minimal normal subgroup. Then there exists another normal subgroup

$$1 \neq H_0 \trianglelefteq E \quad \text{with} \quad H_0 \leq H.$$

Consider $H/H_0 \trianglelefteq E/H_0$. Then

$$|H/H_0| \text{ divides } |H| = n,$$

so $|H/H_0|$ and $|E/H| = m$ are coprime. So by the induction hypothesis there exists a subgroup

$$\overline{H} \leq E/H_0$$

such that $|\overline{H}| = m$. Let $\hat{H} \leq E$ be such that $\hat{H}/H_0 = \overline{H}$. Then

$$|\hat{H}/H_0| = |\overline{H}| = m$$

and

$$|H_0| \text{ divides } |H| = n,$$

so $|H_0|$ and $|H/H_0|$ are coprime. So by the induction hypothesis there exists an $H' \leq \hat{H}$ such that

$$|H'| = |\hat{H}/H_0| = m.$$

Now let p be a prime dividing $|H| = n$.

Step 2: Without loss of generality, we may assume that H is a p -group. Indeed, let S be a p -Sylow subgroup of H . By Frattini's argument,

$$E = N_E(S)H.$$

So

$$E/H \cong N_E(S)/(H \cap N_E(S)).$$

Since $|H \cap N_E(S)|$ divides $|H| = n$ and $|N_E(S)/H \cap N_E(S)| = |E/H| = m$, we have that

$$|N_E(S)/H \cap N_E(S)|$$

and

$$|H \cap N_E(S)|$$

are coprime. Since $S \trianglelefteq H$ and H is a minimal normal subgroup of E , we must have that S is not a normal subgroup of E . So $S \neq H$. Since $S \trianglelefteq N_E(S)$, hence

$$|N_E(S)| < |E|.$$

So by the induction hypothesis there exists an $H' \leq N_E(S)$ such that

$$|H'| = |N_E(S)/H \cap N_E(S)| = |E/H| = m.$$

Step 3: By steps 1 and 2 we have reduced to proving the claim when H is a p -group and a minimal normal subgroup of E . I claim that this makes H abelian. Indeed, since H is a p -group its centre $Z(H)$ is non-trivial. But $Z(H)$ is a normal subgroup of E . Indeed, for all $g \in E$ we have

$$gZ(H)g^{-1} = Z(gHg^{-1}) = Z(H),$$

so $Z(H)$ is a non-trivial subgroup of H and a normal subgroup of E . Since H is a minimal normal subgroup of E we must have $H = Z(H)$, i.e. H is abelian.

Step 4: We have reduced to the case that H is abelian. We have seen that extensions of the form

$$0 \rightarrow H \rightarrow E \rightarrow G \rightarrow 1$$

are classified by $H^2(G, H)$. But this group is trivial since G and H have coprime orders, so all such extensions split. $\square$

Remark 5.4.27 — Note that the proof actually proved that if H is a normal subgroup of a finite group G such that $|H|$ and $|G/H|$ are coprime, then there exists a subgroup K of G with

$$|K| = |G/H|.$$

6 Abelian Categories, Limits, and Colimits

This is the mastery content of the course. This should be read alongside sections 2.6, 3.1, 3.5 and A.5 of [1].

§6.1 Abelian Categories

Definition 6.1.1. A category $\mathcal{C}$ is called **additive** if for each $C, D \in \text{ob}(\mathcal{C})$ the set $\text{Hom}_{\mathcal{C}}(C, D)$ carries the structure of an abelian group and the following axioms hold:

- The composition law is linear (i.e. a homomorphism) in each argument; that is, given a diagram

$$A \xrightarrow{f} B \begin{array}{c} \xrightarrow{g} \\ \xrightarrow{g'} \end{array} C \xrightarrow{h} D$$

we have $h(g + g')f = hgf + hg'f$.

- The **zero object** exists; that is, there exists an object 0 such that, for each $C \in \mathcal{C}$, $\text{Hom}_{\mathcal{C}}(0, C)$ and $\text{Hom}_{\mathcal{C}}(C, 0)$ are 1-element sets.
- $\mathcal{C}$ admits finite products and finite coproducts.

Remark 6.1.2 — One can actually show that the existence of the zero object follows from the other two axioms.

Exercise 6.1.3. Show that finite products and finite coproducts coincide in any additive category.

Solution: First consider the product $C := \prod_{i \in I} C_i$, where the $C_i \in \text{ob}(\mathcal{C})$ is indexed by a finite indexing set I . Then, we have the projections $p_i: C \rightarrow C_i$. Note that we have maps $\iota_i: C_i \rightarrow C$ such that $\iota_i p_i = \text{id}_{C_i}$ and $\iota_i p_j = 0$ for $i \neq j$. Using these, we may construct the identity on C as $\sum_{i \in I} \iota_i p_i = \text{id}_C$. We show that with this ι_i , C satisfies the universal property of the coproduct. Let $X \in \text{ob}(\mathcal{C})$ be an arbitrary and let $f_i: C_i \rightarrow X$ be a collection of maps. We want to find a unique $f: C \rightarrow X$ such that $f_i = f \circ \iota_i$. We define the map f as

$$f((c_i)_{i \in I}) = \sum_{i \in I} f_i(p_i((c_i)_{i \in I})).$$

Then, we see that

$$f(\iota_i(c_i)) = \sum_{j \in I} f_j(p_j(\iota_i(c_i))) = f_i(p_i(\iota_i(c_i))) = f(c_i).$$

We now need to show that this is unique. Let $g: C \rightarrow X$ be another map which carries the universal property. That is, $g \circ \iota_i = f_i$. Then, we have as follows:

$$g((c_i)_{i \in I}) = g\left(\sum_{i \in I} \iota_i(c_i)\right) \stackrel{*}{=} \sum_{i \in I} g(\iota_i(c_i)) = \sum_{i \in I} f_i(c_i) = \sum_{i \in I} f_i(p_i((c_i)_{i \in I})) = f((c_i)_{i \in I}).$$

At $*$, we are using that the composition law is linear in each argument. Thus, we've shown uniqueness and we are done. Now, I can cheat and say that the dual statement holds by reversing all arrows, but I will do this formally (soon). $\square$

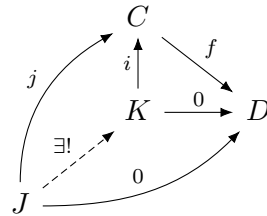
Definition 6.1.4. A functor $F: \mathcal{C} \rightarrow \mathcal{C}'$ between additive categories is **additive** if

$$F: \text{Hom}_{\mathcal{C}}(C, D) \longrightarrow \text{Hom}_{\mathcal{C}'}(F(C), F(D))$$

is a homomorphism of abelian groups for each $C, D \in \text{ob}(\mathcal{C})$.

Remark 6.1.5 — Let $\mathcal{C}$ be an additive category. Then for any two objects C and D , there is a distinguished element in $\text{Hom}_{\mathcal{C}}(C, D)$, namely the composition $C \rightarrow 0 \rightarrow D$. We will call this map the zero map and denote it by 0 ; no confusion should result.

Definition 6.1.6. Let $f: C \rightarrow D$ be a morphism in an additive category $\mathcal{C}$. A **kernel** of f is an object $K \in \text{ob}(\mathcal{C})$ and a morphism $i: K \rightarrow C$ such that $f \circ i = 0$ and such that every morphism $j: J \rightarrow C$ such that $f \circ j = 0$ factors uniquely through i .

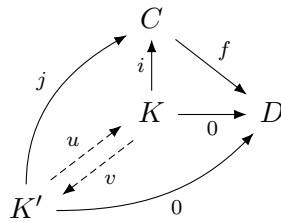


That is, there exists a unique morphism $j': J \rightarrow K$ such that $j = i \circ j'$.

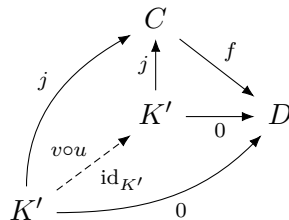
Exercise 6.1.7. Suppose that a kernel $i: K \rightarrow C$ of $f: C \rightarrow D$ exists.

1. Show that the kernel is unique up to unique isomorphism.
2. Show that the morphism i is monic.

Solution: 1. Suppose we have two kernels, (K, i) and (K', j) . Consider the following diagram:



Here, note that the two arrows going in between K' and K come from the universal property on K and K' . Therefore, we have that $j = j \circ v \circ u$. We want to show that $u \circ v$ to get that K and K' are isomorphic objects with i and j different up to isomorphism. For this, we consider the following diagram:

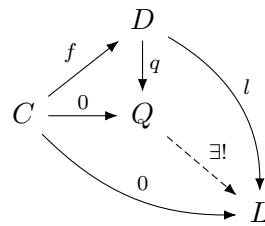


Note that $\text{id}_{K'}$ appears because clearly $j = j \circ \text{id}_{K'}$. Therefore, by uniqueness, it follows that $v \circ u = \text{id}_{K'}$, so we have that the two kernels are isomorphic.

2. This follows by universality. Suppose we have two maps $f_1, f_2: X \rightarrow K$ such that $i \circ f_1 = i \circ f_2$. Then, by the universal property of the kernel, the map $X \rightarrow K$ must be unique, so $f_1 = f_2$. $\square$

Remark 6.1.8 — Since kernels are unique up to unique isomorphism if they exist, we generally refer to the kernel of a morphism $f: C \rightarrow D$, and write it as $\ker(f) \rightarrow C$.

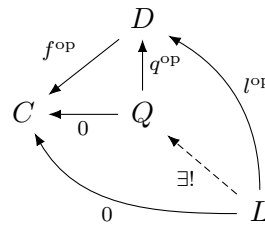
Definition 6.1.9. Let $f: C \rightarrow D$ be a morphism in an additive category $\mathcal{C}$. A **cokernel** of f is an object $Q \in \text{ob}(\mathcal{C})$ and a morphism $q: D \rightarrow Q$ such that $q \circ f = 0$ and such that every morphism $l: D \rightarrow L$ such that $l \circ f = 0$ factors uniquely through q .



That is, there exists a unique morphism $l': Q \rightarrow L$ such that $l = l' \circ q$.

Exercise 6.1.10. Show that the cokernel of a morphism $f: C \rightarrow D$ in $\mathcal{C}$ is the kernel of the corresponding morphism in the opposite category $\mathcal{C}^{\text{op}}$.

Solution: In $\mathcal{C}^{\text{op}}$, we reverse all the arrows, which give the following diagram for the cokernel of a morphism $f: C \rightarrow D$:



It is clear that this is the appropriate universal diagram for the kernel of $f: C \rightarrow D$ then. $\square$

Exercise 6.1.11. Suppose that a cokernel $q: D \rightarrow Q$ of $f: C \rightarrow D$ exists.

1. Show that the cokernel is unique up to unique isomorphism.
2. Show that the morphism q is epic.
3. Show that, for any $A \in \text{ob}(\mathcal{C})$, the sequence of abelian groups

$$0 \longrightarrow \text{Hom}_{\mathcal{C}}(Q, A) \longrightarrow \text{Hom}_{\mathcal{C}}(D, A) \longrightarrow \text{Hom}_{\mathcal{C}}(C, A)$$

is exact.

Solution: 1 and 2 follows by the dual statements of kernel.

3. First, we have the complex

$$C \xrightarrow{f} D \xrightarrow{q} Q \rightarrow 0.$$

Since $q \circ f = 0$, this is actually a complex. We note that we have the complex

$$0 \longrightarrow \operatorname{Hom}_{\mathcal{C}}(Q, A) \xrightarrow{- \circ q} \operatorname{Hom}_{\mathcal{C}}(D, A) \xrightarrow{- \circ f} \operatorname{Hom}_{\mathcal{C}}(C, A),$$

and we want to show this is exact. First, we show exactness of the map $p^* = - \circ p$. Suppose we have some $g \in \operatorname{Hom}_{\mathcal{C}}(Q, A)$ such that $g \circ q = 0$. Then, since q is epic, it is right cancellative, so $g = 0$. Next, suppose that $g \in \operatorname{Hom}_{\mathcal{C}}(D, A)$ such that $g \circ f = 0$. We want to show that there exists some $h \in \operatorname{Hom}_{\mathcal{C}}(Q, A)$ such that $h \circ q = g$. This h exists uniquely by the universal property of the cokernel. Thus, we have exactness. $\square$

Remark 6.1.12 — Since cokernels are unique up to unique isomorphism if they exist, we generally refer to the cokernel of a morphism $f: C \rightarrow D$, and write it as $D \rightarrow \operatorname{coker}(f)$.

Exercise 6.1.13. Let $\mathcal{C}$ be an additive category with $A, B \in \operatorname{ob} \mathcal{C}$ and $f \in \operatorname{Hom}_{\mathcal{C}}(A, B)$. Then, we have the following:

1. f is monic if and only if $\ker f = 0$.
2. f is epic if and only if $\operatorname{coker} f = 0$.

Solution: 1. First assume f is monic. Let $(\ker f, i)$ be our kernel. Then, $fi = 0$. Since f is monic, it follows that $i = 0$, the zero map. Now, since the kernel is a monic, this implies that $\ker f = 0$. To see this, we use a rather tacky argument. Since i is the zero map, we have that

$$i \circ \operatorname{id}_{\ker f} = i \circ i.$$

Thus, using that i is monic, this implies that $\operatorname{id}_{\ker f} = i = 0$, so in fact, $\ker f$ must be zero then. For the converse, suppose the $\ker f = 0$. Explicitly, we mean that $(0, 0)$ is the kernel. Suppose that $C \in \mathcal{C}$ is another object and we have that $fg = fh$, for $g, h \in \operatorname{Hom}_{\mathcal{C}}(C, A)$. Then, we have that $f(g - h) = 0$. Thus, $g - h$ factors through the kernel, so we have $g - h = k \circ 0$, where $k: C \rightarrow \ker f = 0$. Thus, $g - h = 0$, so $g = h$ as desired.

2. This is similar to the previous; or I can cheat and just say this is true by the dual statement of 1. $\square$

Definition 6.1.14. An **abelian category** $\mathcal{C}$ is an additive category such that the following axioms hold.

- Kernels and cokernels exist.
- If $f \in \operatorname{Hom}_{\mathcal{C}}(A, B)$ such that $\ker(f) = 0$, then f is the kernel of $B \rightarrow \operatorname{coker}(f)$. If $\operatorname{coker}(f) = 0$, then f is the cokernel of $\ker(f) \rightarrow A$. If $\ker(f) = 0$ and $\operatorname{coker}(f) = 0$, then f is an isomorphism.

Remark 6.1.15 — We note that this second axiom can be rephrased using the exercise above: every monic (respectively epic) in $\mathcal{C}$ is the kernel (respectively cokernel) of its cokernel (respectively kernel).

Example 6.1.16

An example of an additive category which is not an abelian category is $\text{Free}(\mathbb{Z})$, the category of free $\mathbb{Z}$ -modules. The maps in $\text{Free}(\mathbb{Z})$ are just matrices, since we are looking for maps from $\mathbb{Z}^m \rightarrow \mathbb{Z}^n$. However, cokernels do not exist. For instance, we have

$$\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z},$$

so $\mathbb{Z}/2\mathbb{Z}$ is a cokernel, but this isn't a free $\mathbb{Z}$ -module.

Example 6.1.17

The category of abelian groups is an abelian category; here morphisms are group homomorphisms.

Example 6.1.18

The category of R -modules, where R is a ring; here morphisms are homomorphisms of R -modules.

Example 6.1.19

The category of vector spaces over k , where k is a field; here morphisms are k -linear maps.

Example 6.1.20

The category of complexes of R -modules; here morphisms are morphisms of complexes.

Remark 6.1.21 — Note that the category of groups is not an abelian category (in fact not even an additive category). Can you see why? It is a little bit subtle. Note that group homomorphisms do have kernels: they are the quotient of the codomain by normal closure of the image, so that's not really the problem. One way to see it is to notice that finite products do not coincide with finite coproducts.

Definition 6.1.22. If $f: A \rightarrow B$ is a morphism in an abelian category $\mathcal{C}$, we define the **image** $\text{Im}(f)$ as the kernel of the cokernel of f .

Remark 6.1.23 — From the definition of the kernel we immediately obtain that there is a unique morphism $A \rightarrow \text{Im}(f)$ such that f is the composition $A \rightarrow \text{Im}(f) \rightarrow B$.

Definition 6.1.24. A sequence of morphisms in an abelian category $\mathcal{C}$ is called **exact** if the kernel of each map equals the image of the previous map.

Definition 6.1.25. A (covariant) additive functor $F: \mathcal{C} \rightarrow \mathcal{C}'$ between abelian categories is called **left exact** if for any short exact sequence in $\mathcal{C}$

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

we have an exact sequence in $\mathcal{C}'$

$$0 \longrightarrow F(A) \longrightarrow F(B) \longrightarrow F(C).$$

F is called **right exact** if the following sequence is exact in $\mathcal{C}'$:

$$F(A) \longrightarrow F(B) \longrightarrow F(C) \longrightarrow 0.$$

Further, we say that a functor is **exact** if it is both left exact and right exact.

Remark 6.1.26 — Thus, an exact functor is an additive functor that preserves short exact sequences.

Theorem 6.1.27

Let $L: \mathcal{C} \rightarrow \mathcal{D}$ and $R: \mathcal{C} \rightarrow \mathcal{D}$ be an adjunction of additive functors. Then, L is right exact and R is left exact.

Proof. Let $\tau: \text{Hom}_{\mathcal{D}}(L(A), B) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(A, R(B))$ be our natural isomorphism. Suppose we have that $0 \rightarrow B' \rightarrow B \rightarrow B'' \rightarrow 0$ is exact in $\mathcal{D}$. By naturality of τ , we have, for every A in $\text{ob } \mathcal{C}$, a commutative diagram as follows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Hom}_{\mathcal{D}}(L(A), B') & \longrightarrow & \text{Hom}_{\mathcal{D}}(L(A), B) & \longrightarrow & \text{Hom}_{\mathcal{D}}(L(A), B'') \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & \text{Hom}_{\mathcal{C}}(A, R(B')) & \longrightarrow & \text{Hom}_{\mathcal{C}}(A, R(B)) & \longrightarrow & \text{Hom}_{\mathcal{C}}(A, R(B'')) \end{array}$$

The top row is exact because $\text{Hom}_{\mathcal{D}}(L(A), -)$ is left exact. Thus, the bottom row is exact for all A . The Yoneda lemma tells us then that

$$0 \rightarrow R(B') \rightarrow R(B) \rightarrow R(B'')$$

must be exact. Therefore, every right adjoint R is left exact. In particular, $L^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}^{\text{op}}$, which is a right adjoint, is left exact; that is L is right exact. $\square$

§6.2 Colimits in Abelian Categories

Let $\mathcal{A}$ be an abelian category. If you want, you can focus on the case $\mathcal{A} = \mathbf{mod}\text{-}R$ for some ring R (but the definition works the same for any abelian category). Let $\mathcal{I}$ be another category.

Definition 6.2.1. A **colimit** of a functor $\mathcal{F}: \mathcal{I} \rightarrow \mathcal{A}$ is an object $L \in \text{ob } (\mathcal{A})$ together with morphisms

$$\alpha_i: \mathcal{F}(i) \rightarrow L \quad \text{for all } i \in \text{ob } (\mathcal{I})$$

such that for every morphism $f: i \rightarrow j$ in $\mathcal{I}$ the diagram

$$\begin{array}{ccc} \mathcal{F}(i) & \xrightarrow{\alpha_i} & L \\ F(f) \downarrow & \nearrow \alpha_j & \\ \mathcal{F}(j) & & \end{array}$$

commutes, and (L, α) is universal with this property. That is, if L' and $\alpha'_i: \mathcal{F}(i) \rightarrow L'$ satisfy

$$\begin{array}{ccc} \mathcal{F}(i) & \xrightarrow{\alpha'_i} & L' \\ F(f) \downarrow & \nearrow \alpha'_j & \\ \mathcal{F}(j) & & \end{array}$$

for all $f: i \rightarrow j$, then there exists a unique morphism $h: L \rightarrow L'$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{F}(i) & \xrightarrow{\mathcal{F}(f)} & \mathcal{F}(j) \\ \searrow \alpha_i & & \nearrow \alpha_j \\ & L & \\ \swarrow \alpha'_i & \downarrow \exists! h & \searrow \alpha'_j \\ & L' & \end{array}$$

Remark 6.2.2 — The universal property implies that any two colimits of F are isomorphic via a distinguished isomorphism, and we therefore denote any such colimit by $\text{colim}_{i \in \mathcal{I}} \mathcal{F}(i)$.

Example 6.2.3

If $\mathcal{I}$ is a set with no non-trivial morphisms (no maps between i and j , for $i \neq j$), then

$$\text{colim}_{i \in \mathcal{I}} \mathcal{F}(i) \cong \coprod_{i \in \mathcal{I}} \mathcal{F}(i)$$

in $\mathcal{A}$.

Definition 6.2.4. An abelian category $\mathcal{A}$ is called **cocomplete** if the coproduct $\coprod_{i \in I} A_i$ exists for every set I for every choice of objects $A_i \in \text{ob}(\mathcal{A})$, $i \in I$.

Example 6.2.5

Many familiar categories like the category of sets, rings, or left R -modules are cocomplete.

Definition 6.2.6. A **small category** $\mathcal{C}$ is a category such that $\text{ob}(\mathcal{C})$ is a set.

Lemma 6.2.7

Let $\mathcal{I}$ be a small category. Then, $\mathcal{A}$ be a cocomplete abelian category if and only if $\text{colim}_{i \in \mathcal{I}} \mathcal{F}(i)$ exists for all functors $\mathcal{F}: \mathcal{I} \rightarrow \mathcal{A}$.

Proof. First note that if the colimit exists for all functors, then $\mathcal{A}$ is immediately co-complete because the coproduct is an instance of the colimit, but with no transition maps.

Now for the forward implication. $\mathcal{I}$ is a small category, then $\text{ob}(\mathcal{I})$ is a set. Thus, since $\mathcal{A}$ is cocomplete, $\coprod_{i \in \mathcal{I}} \mathcal{F}(i)$ exists. Let $\mathcal{F}: \mathcal{I} \rightarrow \mathcal{A}$ be a functor then. We have two obvious maps

$$\iota_i, \iota_j \circ \mathcal{F}(i): \coprod_{f: i \rightarrow j} \mathcal{F}(i) \rightarrow \coprod_{i \in \mathcal{I}} \mathcal{F}(i),$$

where the ι_i, ι_j are inclusions. Because we want these two maps to be equal (and the category is abelian), we can take $K := \text{coker}(\iota_i - \iota_j \circ \mathcal{F}(f))$ and show that $K = \text{colim}_{i \in \mathcal{I}} \mathcal{F}(i)$. We just need to show that this K satisfies the universal property of the colimit. If we suppress the notation of the collection of maps $\iota_i - \iota_j \circ \mathcal{F}(f)$ into one map d (so $d|_{\mathcal{F}(f)} = \iota_i - \iota_j \circ \mathcal{F}(f)$), we have the following diagram:

$$\begin{array}{ccc} & \coprod_{i \in \mathcal{I}} \mathcal{F}(i) & \\ & \downarrow q & \\ \coprod_{f: i \rightarrow j} \mathcal{F}(i) & \xrightarrow{0} & K \end{array}$$

$\nearrow d$

To check the universal property, we must also construct maps $\alpha_i: \mathcal{F}(i) \rightarrow \text{colim}_{i \in \mathcal{I}} \mathcal{F}(i)$ that has the desired compatibility. We have the following diagram:

$$\begin{array}{ccc} \mathcal{F}(i) & \xrightarrow{\mathcal{F}(f)} & \mathcal{F}(j) \\ & \searrow \iota_i \quad \swarrow \iota_j & \\ & \coprod_{i \in \mathcal{I}} \mathcal{F}(i) & \\ & \downarrow q & \\ & K & \end{array}$$

$\swarrow \alpha_i \quad \searrow \alpha_j$

Here, the maps $\alpha_i := q \circ \iota_i$. Thus, it is clear that $\alpha_i = \alpha_j \circ \mathcal{F}(f)$, since

$$\alpha_i = q \circ \iota_i = q \circ \iota_j \circ \mathcal{F}(f) = \alpha_j \circ \mathcal{F}(f).$$

We now need to verify this has the universal property as desired. Suppose X is some object in $\mathcal{A}$ such that the following diagram holds:

$$\begin{array}{ccc} \mathcal{F}(i) & \xrightarrow{\mathcal{F}(f)} & \mathcal{F}(j) \\ & \searrow \iota_i \quad \swarrow \iota_j & \\ & \coprod_{i \in \mathcal{I}} \mathcal{F}(i) & \\ & \downarrow q & \\ & K & \\ & \swarrow \alpha_i \quad \searrow \alpha_j & \\ & X & \end{array}$$

$\swarrow \beta_i \quad \searrow \beta_j$

By the universal property of the coproduct, we note that there exists a map $p: \coprod_{i \in \mathcal{I}} \mathcal{F}(i) \rightarrow X$ with $\beta_i = p \circ \iota_i$. Then, we have that

$$p(\iota_i - \iota_j \circ \mathcal{F}(f)) = \beta_i - \beta_j \circ \mathcal{F}(f) = 0,$$

so it must factor uniquely through the cokernel by the universal property of the cokernel. This provides the desired universal property.

The converse is clear, since a coproduct is a small category with no relation in between them. Thus, we just need to specify the objects via $\mathcal{F}$. $\square$

§6.2.i Filtered Colimits

We now introduce a special instance of colimits, which are filtered colimits. These are useful because they commute with Tor , as in Corollary 6.2.23.

Definition 6.2.8. A non-empty category $\mathcal{I}$ is called **filtered** if:

- For every $i, j \in \text{ob}(\mathcal{I})$ there are arrows $i \rightarrow k$ and $j \rightarrow k$ for some $k \in \text{ob}(\mathcal{I})$.
- For every two morphisms $u, v: i \rightarrow j$, there is a morphism $w: j \rightarrow k$ such that $wu = wv$.

Definition 6.2.9. A **filtered colimit** in $\mathcal{A}$ is the colimit of a functor

$$\mathcal{F}: \mathcal{I} \rightarrow \mathcal{A}$$

in which $\mathcal{I}$ is a filtered category.

Definition 6.2.10. Let $(I, \leq)$ be a partially ordered set (poset) such that for all $i, j \in I$, there exists $k \in I$ with $i \leq k$ and $j \leq k$. Such a poset is called filtered.

Definition 6.2.11. A poset I determines a category whose objects are elements of I and for any two elements $i, j \in I$, the morphism set $\text{Hom}_{\mathcal{I}}(i, j)$ has a unique element if $i \leq j$ and is empty otherwise.

Example 6.2.12

A filtered poset I determines a filtered category $\mathcal{I}$, and given a functor $\mathcal{F}$ from $\mathcal{I}$ to an abelian category $\mathcal{A}$, we refer to $\text{colim}_{i \in \mathcal{I}} \mathcal{F}(i)$ as the **direct limit** of $\mathcal{F}(i)$. This is often written as

$$\lim_{\rightarrow} \mathcal{F}(i),$$

with the filtered poset I being implicit in the notation and $\mathcal{F}(i)$ being the image of $i \in I$ under F .

Example 6.2.13

The set of natural numbers $I = \mathbb{N}$ forms a poset with relations

$$1 \leq 2 \leq 3 \leq \dots$$

Let $\mathcal{F}$ be the functor from the corresponding category $\mathcal{I}$ to the category of rings such that $\mathcal{F}(i) = \mathbb{Z}/2^i\mathbb{Z}$ and if $i \leq j$, the unique morphism in $\mathcal{I}$ gives multiplication by 2^{j-i} . Then one can show that

$$\lim_{\rightarrow} \mathcal{F}(i) = \lim_{\rightarrow} \mathbb{Z}/2^n\mathbb{Z} = \mathbb{Z}[\frac{1}{2}]/\mathbb{Z}.$$

Exercise 6.2.14. More generally, show that

$$\varinjlim \mathbb{Z}/p^n\mathbb{Z} = \mathbb{Z}[\frac{1}{p}]/\mathbb{Z},$$

as rings. This is called the **Prüfer group**.

Solution: We just show that the Prüfer group enjoys the universal property of the direct limit. We note that $f_{nm}: \mathbb{Z}/p^n\mathbb{Z} \rightarrow \mathbb{Z}/p^m\mathbb{Z}$ if $n < m$ and this map is multiplication by p^{m-n} and taking the class. We define the maps $\iota_n: \mathbb{Z}/p^n\mathbb{Z}$ as

$$\iota([a]) = \frac{a}{p^n} + \mathbb{Z}.$$

This is clearly a well-defined morphism of rings. Commutativity of the diagram is then clear. Now let R be another ring has the same commutativity with maps $\varphi_n: \mathbb{Z}/p^n\mathbb{Z} \rightarrow R$. We define $\varphi: \mathbb{Z}[\frac{1}{p}]/\mathbb{Z} \rightarrow R$ as the map which maps

$$\frac{a}{p^n} + \mathbb{Z} \mapsto \varphi_n([a]).$$

This is a well-defined morphism of rings. This map satisfies the desired commutativity, but we now must show uniqueness, so let $\psi: \mathbb{Z}[\frac{1}{p}]/\mathbb{Z} \rightarrow R$ be another map satisfying the commutativity. Then, $\psi \circ \iota_n = \varphi_n$. Therefore,

$$\psi\left(\frac{a}{p^n} + \mathbb{Z}\right) = \psi(\iota_n([a])) = \varphi_n([a]) = \varphi\left(\frac{a}{p^n} + \mathbb{Z}\right).$$

□

Example 6.2.15

Similarly, we have

$$\mathbb{Q}/\mathbb{Z} = \varinjlim \mathbb{Z}/n\mathbb{Z},$$

where the limit is taken in the category of abelian groups, and the maps $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ are the obvious homomorphisms.

Exercise 6.2.16. Prove that any abelian group G is isomorphic to the direct limit of its finitely generated subgroups. *Hint: Define the poset I as follows: I is the set of all finitely generated subgroups of G , i.e. for every $i \in I$, there is a corresponding finitely generated subgroup $H_i \subset G$. For $i, j \in I$, we set $i \leq j$ if and only if $H_i \subset H_j$. Then I is a filtered poset. Consider the functor $\mathcal{F}$ from the corresponding category $\mathcal{I}$ to the category of abelian groups such that $\mathcal{F}(i) = H_i$ and that $\mathcal{F}(i \rightarrow j)$ is the inclusion map $H_i \rightarrow H_j$. Then you only need to show that*

$$\varinjlim H_i = G.$$

Solution: We follow the hint as suggested. We first need to define a filtered poset. We let I to be the set of all finitely generated subgroups of G ; that is, for every $i \in I$, we let i correspond to $H_i \subset G$, where $H_i \subset G$ is finitely generated. On I , we endow it with the partial order $\leq$ so that $i \leq j$ if and only if $H_i \subset H_j$. It is clear then that I is a poset. Further, it is filtered because for any two objects i, j , we may find some bigger k such that $H_i, H_j \subset H_k$. Such a H_k certainly exists because we can just let H_k be the group generated by the union of generators for H_i and H_j . Further, the condition on the morphisms

is trivially true since there is at most one morphism between any two objects in I . Thus, I is a filtered poset, and coproducts exist in the category $\mathbf{Ab}$, so colimits/ direct limits exist.

Now, let $\mathcal{F}: I \rightarrow \mathbf{Ab}$ be the functor which establishes this correspondence; that is, $\mathcal{F}(i) = H_i$ and $\mathcal{F}(i \rightarrow j) = \iota_{ij}: H_i \hookrightarrow H_j$. Now, suppose that $H := \operatorname{colim}_{i \in I} H_i$. Then, we note that G has the inclusions $g_i: H_i \hookrightarrow G$ such that we have the desired compatibility $g_i = g_j \iota_{ij}$. Then, by the universal property of the colimit, we have a unique map $\phi: H \rightarrow G$ such that $g_i = \phi \circ h_i$, where $h_i: H_i \rightarrow H$ are the maps that come with H . We want to show that ϕ is an isomorphism. Then, since g_i is an injection, ϕ is an injection; particularly, if $\phi(x) = 0$, then $x = h_i(a)$, for some a in a finitely generated subgroup H_i . Therefore, $0 = \phi(h_i(a)) = g_i(a)$, but g_i is injective, so $a = 0$. To show that ϕ is a surjection, consider some element $g \in G$. Then, let $H_i = \langle g \rangle$. Then, g is in the image of g_i , so it must be in the image of ϕ as well. In other words, ϕ is surjective. Thus, ϕ is isomorphic and we have that

$$\lim_{\rightarrow} H_i \cong G.$$

□

Definition 6.2.17. Let $\mathcal{C}$ be any category and $\mathcal{A}$ be an abelian category. The **functor category** $[\mathcal{C}, \mathcal{A}]$ is the abelian category whose objects are functors $F: \mathcal{C} \rightarrow \mathcal{A}$ and maps are natural transformations.

Remark 6.2.18 — Let $\mathcal{I}$ be a fixed category, $\mathcal{A}$ be another category. Then, we note that colim is a functor from $[\mathcal{I}, \mathcal{A}]$ to $\mathcal{A}$, when the colimit exists for all functors $F: \mathcal{I} \rightarrow \mathcal{A}$.

Proof. We prove this now. First, consider the $\operatorname{id}_F: F \Rightarrow F$. We have that

$$\operatorname{colim}_{i \in I}(\operatorname{id}_F) = \operatorname{id}_F,$$

by the universal property of the colimit. Now, let $F, G, H: \mathcal{I} \rightarrow \mathcal{A}$ be functors and $\alpha: F \Rightarrow G$, $\beta: G \Rightarrow H$ be natural transformations. We denote the colimits of F, G, H as $(F^c, f_i), (G^c, g_i), (H^c, h_i)$. Then, note that by the universal property of colimits, there exist maps

$$\operatorname{colim}_{i \in I}(\alpha): \operatorname{colim}_{i \in I} F(i) \rightarrow \operatorname{colim}_{i \in I} G(i)$$

and

$$\operatorname{colim}_{i \in I}(\beta): \operatorname{colim}_{i \in I} G(i) \rightarrow \operatorname{colim}_{i \in I} H(i).$$

We want to show that

$$\operatorname{colim}_{i \in I}(\beta \circ \alpha) = \operatorname{colim}_{i \in I}(\beta) \circ \operatorname{colim}_{i \in I}(\alpha),$$

since the universal property ensures the uniqueness of such a map. □

Definition 6.2.19. Let $\mathcal{I}$ be a fixed category. The **diagonal functor** Δ from every category $\mathcal{A}$ to the functor category $[\mathcal{I}, \mathcal{A}]$ is defined as follows. If $A \in \mathcal{A}$, then ΔA is the constant functor $(\Delta A)_i = A$, for all i .

Exercise 6.2.20. Show that colim is left adjoint to Δ . Therefore, conclude that colim is a right exact functor when $\mathcal{A}$ is abelian.

Solution: In other words, we need to show that there is a natural bijection

$$\tau_{F,A}: \text{Hom}_{\mathcal{A}}(\text{colim}_{i \in \mathcal{I}} F(i), A) \xrightarrow{\sim} \text{Hom}_{[\mathcal{I}, \mathcal{A}]}(F, \Delta(A)).$$

If $\phi \in \text{Hom}_{\mathcal{A}}(\text{colim}_{i \in \mathcal{I}} F(i), A)$, we define $\tau_{F,A}(\phi) := \phi \circ \iota_i$, where the ι_i are the maps $\iota_i: F(i) \rightarrow \text{colim}_{i \in \mathcal{I}} F(i)$ which provide the desired commutativity. First, we check that this is indeed a natural transformation. Let $u: F(i) \rightarrow F(j)$ and $\text{id}_A: A \rightarrow A$. Then, if $\psi \in \text{Hom}_{\mathcal{A}}(\text{colim}_{i \in \mathcal{I}} F(i), A)$, we have that

$$\psi \circ \iota_j \circ F(u) = \text{id}_A \circ \psi \circ \iota_j = \psi \circ \iota_i,$$

which is clearly true by the commutativity $\iota_j \circ F(u) = \iota_i$.

Next, must check that this provides the naturality as desired. Let G be another functor in $[\mathcal{I}, \mathcal{A}]$, and suppose we have a natural transformation $\theta: F \Rightarrow G$. Further, let B be in $\mathcal{A}$ and $f: A \rightarrow B$ be a morphism in $\text{Hom}_{\mathcal{A}}(A, B)$. Then, suppose that $\varphi \in \text{Hom}_{\mathcal{A}}(\text{colim}_{i \in \mathcal{I}} G(i), A)$. Then, we have that

$$\varphi \circ \text{colim}_{i \in \mathcal{I}}(\theta) \circ \iota_i = \varphi \circ \iota_i \circ \theta_i,$$

using the commutativity $\text{colim}_{i \in \mathcal{I}}(\theta) \circ \iota_i = \iota_i \circ \theta(i)$. Similarly, we have that

$$f \circ \psi \circ \iota_i = \Delta(f) \circ \psi \circ \iota_i,$$

since $\Delta(f) = f \circ \text{id}_A = f$. Thus, we have the desired naturalities.

Finally, we show that these are bijections as well. For this, we construct an inverse to $\tau_{F,A}$. We define $\eta_{F,A}: \text{Hom}_{[\mathcal{I}, \mathcal{A}]}(F, \Delta(A)) \rightarrow \text{Hom}_{\mathcal{A}}(\text{colim}_{i \in \mathcal{I}} F(i), A)$ to send a natural transformation $\eta \in \text{Hom}_{[\mathcal{I}, \mathcal{A}]}(F, \Delta(A))$ to the unique map $\tilde{\eta}: \text{colim}_{i \in \mathcal{I}} F(i) \rightarrow A$ such that $\tilde{\eta} \circ \iota_i = \eta_i$. This exists by the universal property of the colimit. These maps are indeed inverses, as one can check:

$$\tau_{F,A} \circ \eta_{F,A}(\xi_i) = \tau_{F,A}(\tilde{\xi}) = \tilde{\xi} \circ \iota_i = \xi_i$$

and

$$\eta_{F,A} \circ \tau_{F,A}(\xi) = \eta_{F,A}(\xi \circ \iota_i) = \xi,$$

so we are done.

To state that colim is right exact, this follows from the fact that colim is a left adjunction to the diagonal functor; thus, right exact. $\square$

§6.2.ii R -mod

We now focus in on the category of left R -modules for a ring R , which we denote as $R\text{-mod}$. One key result to keep in mind is that in $R\text{-mod}$, (arbitrary) coproducts and direct sums coincide. The following is a key result (Theorem 2.6.15 in [1]), which requires Lemma 6.2.22 (placed right after).

Theorem 6.2.21

Filtered colimits of R -modules are exact, when considered as functors from the category of functors $[\mathcal{I}, R\text{-mod}]$ to $R\text{-mod}$. Here $\mathcal{I}$ is the indexing category for the colimit, which is a filtered category.

Proof. Let $\mathcal{A} := R\text{-}\mathbf{mod}$ for ease of notation. We have seen in the exercise above that $\text{colim}_{i \in \mathcal{I}}$ is already right exact, so we just need to show that we have exactness at the left. In other words, we must show that if $t: A \rightarrow B$ is monic in $[\mathcal{I}, \mathcal{A}]$ (each $t_i: A_i \rightarrow B_i$ is monic), then $\text{colim}_{i \in \mathcal{I}} A_i \rightarrow \text{colim}_{i \in \mathcal{I}} B_i$ is monic in $\mathcal{A}$. Let $a \in \text{colim}_{i \in \mathcal{I}} A_i$ such that $\text{colim}_{i \in \mathcal{I}} t(a) = 0$ in $\text{colim}_{i \in \mathcal{I}} B_i$. Then, a is in the image of some $a_i \in A_i$ by Lemma 6.2.22. Thus, $t_i(a_i) \in B_i$ vanishes in $\text{colim}_{i \in \mathcal{I}} B_i$, so by Lemma 6.2.22 again, there exists some map $\varphi: i \rightarrow j$ such that

$$0 = \varphi(t_i(a_i)) = t_j(\varphi(a_i))$$

in B_j . Since t_j is monic, $\varphi(a_i) = 0$ in A_j . Thus, $a = 0$ in $\text{colim}_{i \in \mathcal{I}} A_i$. $\square$

Lemma 6.2.22

Let $\mathcal{I}$ be a filtered category and $A: \mathcal{I} \rightarrow R\text{-}\mathbf{mod}$ a functor. Then

- (i) Every element $a \in \text{colim}_{i \in \mathcal{I}} A_i$ is the image of some element $a_i \in A_i$ (for some $i \in \mathcal{I}$) under the canonical map

$$A_i \rightarrow \text{colim}_{i \in \mathcal{I}} A_i.$$

- (ii) For every i , the kernel of the canonical map

$$A_i \rightarrow \text{colim}_{i \in \mathcal{I}} A_i$$

is the union of the kernels of the maps

$$\varphi: A_i \rightarrow A_j,$$

where $\varphi: i \rightarrow j$ is a map in $\mathcal{I}$.

Proof. We shall use the explicit construction of $\text{colim}_{i \in \mathcal{I}} A_i$. Let $\lambda_i: A_i \rightarrow \bigoplus_{i \in \mathcal{I}} A_i$ be the canonical maps. Every element a of $\text{colim}_{i \in \mathcal{I}} A_i$ is the image of

$$\sum_{j \in J} \lambda_j(a_j)$$

for some finite set $J = \{i_1, \dots, i_n\}$. There is an upper bound i in $\mathcal{I}$ for $\{i_1, \dots, i_n\}$; using the maps $A_j \rightarrow A_i$ we can represent each a_j as an element in A_i and take a_i to be their sum. Evidently, a is the image of a_i , so the first statement holds.

Now suppose that $a_i \in A_i$ vanishes in $\text{colim}_{i \in \mathcal{I}} A_i$. Then there are $\varphi_{jk}: j \rightarrow k$ in $\mathcal{I}$ and $a_j \in A_j$ so that

$$\lambda_i(a_i) = \sum \lambda_k(\varphi_{jk}(a_j)) - \lambda_j(a_j)$$

is in $\bigoplus_{i \in \mathcal{I}} A_i$. Choose an upper bound t in $\mathcal{I}$ for all the i, j, k in this expression. Adding

$$\lambda_t(\varphi_{it}(a_i)) - \lambda_i(a_i)$$

to both sides we may assume that $i = t$. Adding zero terms of the form

$$[\lambda_t \varphi_{jt}(a_j) - \lambda_k \varphi_{jk}(a_j)] + [\lambda_t \varphi_{jt}(-a_j) - \lambda_k \varphi_{jk}(-a_j)],$$

we can assume that all the k 's are t . If any φ_{jt} are parallel arrows in $\mathcal{I}$, then by changing t we can equalize them. Therefore we have

$$\lambda_t(a_t) = \lambda_t\left(\sum \varphi_{jt}(a_j)\right) = \sum \lambda_j(a_j)$$

with all the j 's distinct and none equal to t . Since the λ_j are injections, all the a_j must be zero. Hence

$$\varphi_{it}(a_i) = a_t = 0,$$

that is, $a_i \in \ker(\varphi_{it})$. □

This is Corollary 2.6.16 in [1].

Corollary 6.2.23

Let $\mathcal{A} = R\text{-}\mathbf{mod}$ and let $\mathcal{B}$ be an abelian category. If $F: \mathcal{A} \rightarrow \mathcal{B}$ is a left adjoint to a functor from $\mathcal{B}$ to $\mathcal{A}$, then for every functor $A: \mathcal{I} \rightarrow \mathcal{A}$ when $\mathcal{I}$ is a filtered category, we have

$$L_*F(\operatorname{colim}_{i \in \mathcal{I}} A(i)) = \operatorname{colim}_{i \in \mathcal{I}} L_*F(A(i)).$$

Proof. Filtered colimits are exact in $R\text{-}\mathbf{mod}$, so they commute homology. □

Note that left adjoint functors are right-exact. As a special case of the above Corollary, we have the following.

Corollary 6.2.24

Given a filtered category $\mathcal{I}$, a functor $A: \mathcal{I} \rightarrow R\text{-}\mathbf{mod}$, and a right R -module B , we have

$$\operatorname{Tor}_k^R(B, \operatorname{colim}_{i \in \mathcal{I}} A(i)) \cong \operatorname{colim}_{i \in \mathcal{I}} \operatorname{Tor}_k^R(B, A(i))$$

for all $k \geq 0$.

Proof. This is a special instance of the above, applied to $B \otimes_R -$. □

Exercise 6.2.25. Compute $\operatorname{Tor}_k^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}[\frac{1}{p}]/\mathbb{Z})$.

Solution: First note that the Prüfer group is the colimit of the sequence

$$\dots \xleftarrow{\times p} \mathbb{Z}/p^n\mathbb{Z} \xleftarrow{\times p} \dots \xleftarrow{\times p} \mathbb{Z}/p^2\mathbb{Z} \xleftarrow{\times p} \mathbb{Z}/p\mathbb{Z}.$$

Well, we have that

$$\operatorname{Tor}_k^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}[\frac{1}{p}]/\mathbb{Z}) \cong \varinjlim \operatorname{Tor}_k^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p^n\mathbb{Z}),$$

and we have that

$$\operatorname{Tor}_k^{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}/p^n\mathbb{Z}) = \begin{cases} (\mathbb{Z}/p^n\mathbb{Z})/(p \cdot \mathbb{Z}/p^n\mathbb{Z}) & k = 0 \\ (\mathbb{Z}/p^n\mathbb{Z})[p] & k = 1 \\ 0 & k \geq 2. \end{cases}$$

Now, while the groups above for Tor are both $\mathbb{Z}/p\mathbb{Z}$, they mean different things with the maps. First, $k = 0$. Here, the multiplication by p maps become 0, since we are quotienting by p at each stage. Therefore,

$$\varinjlim \text{Tor}_0^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p^n\mathbb{Z}) \cong \varinjlim \mathbb{Z}/p\mathbb{Z} = 0.$$

As for $k = 1$, this is the group generated by the class $[p^{n-1}]$ in $\mathbb{Z}/p^n\mathbb{Z}$. This gets mapped to the class $[p^n]$ in $\mathbb{Z}/p^{n+1}\mathbb{Z}$, so is an isomorphism. Thus,

$$\varinjlim \text{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p^n\mathbb{Z}) \cong \varinjlim \mathbb{Z}/p\mathbb{Z} \cong \mathbb{Z}/p\mathbb{Z}.$$

□

§6.3 Limits

There is also the dual notion of limit. That is, the limit of a functor $\mathcal{F}: \mathcal{I} \rightarrow \mathcal{A}$ is the colimit of the corresponding functor $\mathcal{F}^{\text{op}}: \mathcal{I}^{\text{op}} \rightarrow \mathcal{A}^{\text{op}}$. We write the formal definition out for completeness.

Definition 6.3.1. A **limit** of a functor $\mathcal{F}: \mathcal{I} \rightarrow \mathcal{A}$ is an object $L \in \text{ob}(\mathcal{A})$ together with morphisms

$$\alpha_i: L \rightarrow \mathcal{F}(i) \quad \text{for all } i \in \text{ob}(\mathcal{I})$$

such that for every morphism $f: j \rightarrow i$ in $\mathcal{I}$, the diagram

$$\begin{array}{ccc} & \mathcal{F}(i) & \xleftarrow{\alpha_i} L \\ \mathcal{F}(f) \uparrow & & \searrow \alpha_j \\ & \mathcal{F}(j) & \end{array}$$

commutes, and (L, α) is universal with this property. In other words, if there is another object L' with a collection of maps $\alpha'_i: L' \rightarrow \mathcal{F}(i)$ satisfying the same commutativity diagram as above, then there exists a unique morphism $h: L' \rightarrow L$ such that the following diagram commutes:

$$\begin{array}{ccccc} & & \mathcal{F}(f) & & \\ & & \xleftarrow{\quad} & & \xrightarrow{\quad} \\ \mathcal{F}(i) & & & & \mathcal{F}(j) \\ & \nwarrow \alpha_i & & \nearrow \alpha_j & \\ & L & & & \\ & \uparrow \exists! h & & & \\ & L' & & & \\ & \nwarrow \alpha'_i & & \nearrow \alpha'_j & \end{array}$$

Remark 6.3.2 — One can show that $\lim$ is right adjoint to the diagonal functor Δ . Therefore, $\lim$ is left exact.

Example 6.3.3

Let $\mathcal{I} = \mathbb{N}$, and $\mathcal{F}: \mathcal{I} \rightarrow \mathbf{Ab}$ be the map $\mathcal{F}(i) = \mathbb{Z}/p^i\mathbb{Z}$, with a map between $\mathcal{F}(j) \rightarrow \mathcal{F}(i)$ if $j > i$ defined as the canonical mod p^i maps. Then,

$$\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}.$$

§6.3.i Countable Towers

We now introduce a particularly common limit. This is when the index category $\mathcal{I}$ is the opposite category of the category appearing in Example 6.2.13 (so objects are natural numbers but the maps go ‘down’). We shall work in this setting now.

Notation 6.3.4. Let $\mathcal{I}$ be the opposite category of Example 6.2.13. Let $\mathcal{A}$ be $\mathbf{Ab}$. We shall call the objects of $\mathbf{Ab}^{\mathcal{I}}$ countable **towers** of abelian groups; they have the form

$$\{A_i\}: \quad \cdots \rightarrow A_2 \rightarrow A_1 \rightarrow A_0.$$

Remark 6.3.5 (Generalizing Towers) — This construction generalizes from $\mathbf{Ab}$ to other abelian category $\mathcal{A}$ which satisfy the following axiom, introduced by Grothendieck:

$$\mathcal{A} \text{ is complete, and the product of any set of surjections is a surjection.} \quad (6.1)$$

The reason for this is that if $\mathcal{I}$ is a discrete set, $\mathcal{A}^{\mathcal{I}}$ is the product category $\prod_{i \in \mathcal{I}} \mathcal{A}$ of indexed families of objects $\{A_i\}$ in $\mathcal{A}$. For $\{A_i\}$ in $\mathcal{A}^{\mathcal{I}}$, $\varprojlim_{i \in \mathcal{I}} A_i$ is the product $\prod A_i$. Axiom (6.1) states that the left exact functor $\prod$ from $\mathcal{A}^{\mathcal{I}}$ to $\mathcal{A}$ is exact for all discrete $\mathcal{I}$. Axiom (6.1) fails ($\prod_{i=1}^{\infty}$ is not exact) for some important abelian categories, such as $\text{Sheaves}(X)$. On the other hand, axiom (6.1) is satisfied by many abelian categories in which objects have underlying sets, such as $\mathbf{Ab}$, $R\text{-mod}$. Therefore, we can think of whatever we define from now to hold for $R\text{-mod}$ as well.

Definition 6.3.6. Given a tower $\{A_i\}$ in $\mathbf{Ab}$, define the map

$$\Delta: \prod_{i=0}^{\infty} A_i \rightarrow \prod_{i=0}^{\infty} A_i$$

by the element-theoretic formula

$$\Delta(\dots, a_i, \dots, a_0) = (\dots, a_i - \bar{a}_{i+1}, \dots, a_1 - \bar{a}_2, a_0 - \bar{a}_1),$$

where $\bar{a}_{i+1}$ denotes the image of $a_{i+1} \in A_{i+1}$ in A_i . We define $\varprojlim^1 A_i$ to be the cokernel of Δ , so that $\varprojlim^1$ is a functor from $\mathbf{Ab}^{\mathcal{I}}$ to $\mathbf{Ab}$. We also set $\varprojlim^n A_i = 0$ for $n \neq 0, 1$.

Exercise 6.3.7. Show that $\varprojlim A_i = \ker \Delta$. Therefore, we also set $\varprojlim^0 A_i = \varprojlim A_i$.

Solution: We just need to show that $\varprojlim A_i$ satisfies the universal property of the kernel for Δ . We note that by definition, $(a_i)_i \in \varprojlim A_i$ if $a_i = \bar{a}_{i+1}$. Thus, we define $\pi: \varprojlim A_i \rightarrow \prod_{i=0}^{\infty} A_i$ as $\pi(a) = (\pi_i(a))_i$, where π_i are the projections that come with the projective limit. Then, it is clear that $\Delta(\pi(a)) = 0$. Now, suppose that there is some other $j: J \rightarrow \prod_{i=0}^{\infty} A_i$ such that $\Delta j = 0$. Then, let p_i be the projection from the product $\prod_{i=0}^{\infty} A_i \rightarrow A_i$. Then, Δj implies that $p_i j = \bar{p}_{i+1} j$. Therefore, by the universal property of the limit, we find a unique map from J to $\varprojlim A_i$, so we are done. $\square$

Example 6.3.8

We have seen the profinite integers $\widehat{\mathbb{Z}}$. In fact, we have

$$\widehat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n\mathbb{Z}$$

where the limit is taken in the category of abelian groups, and the maps $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z}$ are the obvious homomorphisms.

Lemma 6.3.9

The functors $\{\varprojlim^n\}$ form a long exact sequence.

Proof. If

$$0 \rightarrow \{A_i\} \rightarrow \{B_i\} \rightarrow \{C_i\} \rightarrow 0$$

is a short exact sequence of towers, apply the Snake Lemma to

$$\begin{array}{ccccccc} 0 & \longrightarrow & \prod A_i & \longrightarrow & \prod B_i & \longrightarrow & \prod C_i \longrightarrow 0 \\ & & \Delta \downarrow & & \Delta \downarrow & & \Delta \downarrow \\ 0 & \longrightarrow & \prod A_i & \longrightarrow & \prod B_i & \longrightarrow & \prod C_i \longrightarrow 0 \end{array}$$

to get the requisite natural long exact sequence. $\square$

Corollary 6.3.10

$\varprojlim$ is a left exact functor.

Proof. By the lemma above, the terms are

$$0 \rightarrow \ker \Delta \rightarrow \ker \Delta \rightarrow \ker \Delta \rightarrow \operatorname{coker} \Delta \rightarrow \operatorname{coker} \Delta \rightarrow \dots$$

Since $\ker \Delta = \varprojlim A_i$ and $\operatorname{coker} \Delta := \varprojlim^1 A_i$ (analogously for the others), we find that

$$0 \rightarrow \varprojlim A_i \rightarrow \varprojlim B_i \rightarrow \varprojlim C_i \rightarrow \varprojlim^1 A_i \rightarrow \varprojlim^1 B_i \rightarrow \dots$$

Therefore, we have left-exactness. $\square$

Lemma 6.3.11

If all the maps $A_{i+1} \rightarrow A_i$ are surjective, then $\varprojlim^1 A_i = 0$. Moreover, $\varprojlim A_i \neq 0$ (unless every $A_i = 0$), because each of the natural projections $\varprojlim A_i \rightarrow A_j$ are surjective.

Proof. We just need to show that if all $A_{i+1} \rightarrow A_i$ are surjective, then so is Δ . Given elements $b_i \in A_i$ for all i , and any $a_0 \in A_0$, we inductively choose $a_{i+1} \in A_{i+1}$ to be a lift of $a_i - b_i \in A_i$. By surjectivity, these lifts exist. The map Δ sends $(\dots, a_1, a_0)$ to $(\dots, b_1, b_0)$, so Δ is surjective and $\operatorname{coker}(\Delta) = 0$. If all the $b_i = 0$, then we've constructed a lift such that

$$(\dots, a_1, a_0) \in \ker(\Delta) = \varprojlim A_i.$$

$\square$

Corollary 6.3.12

We have

$$\varprojlim^1 A_i = \text{coker } \Delta \cong (R^1 \varprojlim)(A_i)$$

and

$$R^n \varprojlim A_i = 0 = \varprojlim^n A_i$$

for $n \neq 0, 1$.

Proof. Note that we just wish to show that $R^n \varprojlim A_i = \varprojlim^n A_i$, for all n . For this, we must show that $\varprojlim^1$ vanishes on injectives, and that $R^n \varprojlim A_i = 0$ for $n \geq 2$. The vanishing on injectives is explained below. Now, we can construct a product of injective towers, in which all the maps in each injective tower is surjective, so surjective in the product as well. Therefore, $\varprojlim^1$ vanishes on these injective towers. The vanishing of $R^n \varprojlim = 0$ is then immediate from the long exact sequence of $\varprojlim^n$. $\square$

Definition 6.3.13. A tower $\{A_i\}$ of abelian groups satisfies the **Mittag-Leffler condition** if for each k there exists a $j \geq k$ such that the image of $A_i \rightarrow A_k$ equals the image of $A_j \rightarrow A_k$ for all $i \geq j$. That is, the images of the A_i in A_k satisfy the descending chain condition:

$$\text{Im}(A_k \rightarrow A_k) \supseteq \text{Im}(A_{k+1} \rightarrow A_k) \supseteq \cdots \supseteq \text{Im}(A_j \rightarrow A_k) = \text{Im}(A_{j+1} \rightarrow A_k) = \cdots$$

We say that $\{A_i\}$ satisfies the **trivial Mittag-Leffler condition** if for each k there exists a $j > k$ such that the map $A_j \rightarrow A_k$ is zero.

Example 6.3.14

For example, the Mittag-Leffler condition is satisfied if all the maps

$$A_{i+1} \rightarrow A_i$$

in the tower $\{A_i\}$ are onto.

Proposition 6.3.15

If $\{A_i\}$ satisfies the Mittag-Leffler condition, then

$$\varprojlim^1 A_i = 0.$$

Proof. If $\{A_i\}$ satisfies the trivial Mittag-Leffler condition, and $b_i \in A_i$ are given, set

$$a_k = b_k + \bar{b}_{k+1} + \cdots + \bar{b}_{j-1},$$

where $\bar{b}_i$ denotes the image of b_i in A_k . Note that $\bar{b}_i = 0$ for $i \geq j$. Then Δ maps

$$(\dots, a_1, a_0)$$

to

$$(\dots, b_1, b_0).$$

Thus Δ is onto and

$$\varprojlim^1 A_i = 0$$

when $\{A_i\}$ satisfies the trivial Mittag-Leffler condition.

In the general case, let $B_k \subseteq A_k$ be the image of $A_i \rightarrow A_k$ for large i . The maps

$$B_{k+1} \rightarrow B_k$$

are all onto, so

$$\varprojlim^1 B_k = 0.$$

The tower $\{A_k/B_k\}$ satisfies the trivial Mittag-Leffler condition, so

$$\varprojlim^1 A_k/B_k = 0.$$

From the short exact sequence

$$0 \rightarrow \{B_i\} \rightarrow \{A_i\} \rightarrow \{A_i/B_i\} \rightarrow 0$$

of towers, we see that

$$\varprojlim^1 A_i = 0$$

as claimed. □

Exercise 6.3.16. Show that

$$\varprojlim^1 A_i = 0$$

if $\{A_i\}$ is a tower of finite abelian groups, or a tower of finite-dimensional vector spaces over a field.

Solution: The claim is that these both satisfy the Mittag-Leffler condition. Let k be arbitrary. Then, A_k is a finite abelian group. Therefore, the image of $A_j \rightarrow A_k$ is a finite abelian group as well. As j increases, we form a descending chain

$$\text{Im}(A_k \rightarrow A_k) \supseteq \text{Im}(A_{k+1} \rightarrow A_k) \supseteq \text{Im}(A_{k+2} \rightarrow A_k) \supseteq \dots$$

Since this is a descending chain of finite objects, this must stabilize, so it satisfies the Mittag-Leffler condition. As for finite dimensional vector spaces, we essentially get the same stabilization due to the dimension being finite. Therefore, the above theorem finishes off the desired result. □

Example 6.3.17

Here, we present an example of a tower of abelian groups not satisfying the Mittag-Leffler condition. Here, we let $A_n = \mathbb{Z}$ with transition maps $A_{n+1} \xrightarrow{\times p} A_n$ (for p a prime number), so the tower looks like this:

$$\dots \xrightarrow{\times p} \mathbb{Z} \xrightarrow{\times p} \dots \xrightarrow{\times p} \mathbb{Z} \xrightarrow{\times p} \mathbb{Z}$$

This is not Mittag-Leffler. Moreover, $\varprojlim^1 A_\bullet = \mathbb{Z}_p/\mathbb{Z} \neq 0$.

Proof. It follows that $\text{Im}(A_{k+r} \rightarrow A_k) = p^r \mathbb{Z}$. Therefore, we have a chain of images

$$\supseteq p^r \supseteq \cdots \supseteq p^3 \mathbb{Z} \supseteq p^2 \mathbb{Z} \supseteq p \mathbb{Z} \supseteq \mathbb{Z}.$$

This is clearly not stable, so doesn’t satisfy the condition. Furthermore, we can exhibit that $\varprojlim^1 A_i \neq 0$. For this, we first show that $\varprojlim A_i = 0$. For this, we use the fact that $\varprojlim A_i = \ker \Delta$, where $\Delta: \prod_{i=0}^{\infty} A_i \rightarrow \prod_{i=0}^{\infty} A_i$ is as defined above:

$$\Delta(\dots, a_i, \dots, a_0) = (\dots, a_i - a_{i+1}^-, \dots, a_1 - a_2^-, a_0 - a_1^-).$$

We note that then a_0 is some integer which must satisfy the following conditions:

$$a_0 = pa_1 = p^2 a_2 = p^3 a_3 = \cdots = p^r a_r = \cdots, \quad \forall r > 0.$$

This isn’t possible, because there is no integer infinitely divisible by p -powers. Thus, $\varprojlim A_i = 0$. In order to compute this, we want to use the long exact sequence induced by $\varprojlim^n$. For this, we observe that $A_\bullet$ sits inside a short exact sequence of complexes:

$$0 \rightarrow A_\bullet \rightarrow B_\bullet \rightarrow C_\bullet \rightarrow 0$$

Here,

$$B_\bullet: \quad \cdots \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \xrightarrow{\text{id}} \cdots \xrightarrow{\text{id}} \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z}$$

and

$$C_\bullet: \quad \cdots \mathbb{Z}/p^n \mathbb{Z} \rightarrow \mathbb{Z}/p^{n-1} \mathbb{Z} \rightarrow \cdots \rightarrow \mathbb{Z}/p^2 \mathbb{Z} \rightarrow \mathbb{Z}/p \mathbb{Z}$$

The long exact sequence we get is therefore

$$0 \rightarrow \varprojlim A_\bullet \rightarrow \varprojlim B_\bullet \rightarrow \varprojlim C_\bullet \rightarrow \varprojlim^1 A_\bullet \rightarrow \varprojlim^1 B_\bullet \rightarrow \cdots$$

Now, we make some observations. First of all $\varprojlim A_\bullet = 0$ as computed above. Further, it is clear that $\varprojlim B_\bullet = \mathbb{Z}$ and that $\varprojlim C_\bullet = \mathbb{Z}_p$. It is also immediate that $B_\bullet$ satisfies the Mittag-Leffler condition because we only have the identity maps and all components are the same. Therefore, the above long exact sequence truncates to the following:

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}_p \rightarrow \varprojlim^1 A_\bullet \rightarrow 0.$$

In other words, we have a s.e.s. here, and so

$$\varprojlim^1 A_\bullet \cong \mathbb{Z}_p / \mathbb{Z}.$$

□

Remark 6.3.18 — As for some motivation as to why one would think of the towers $B_\bullet$ and $C_\bullet$, this comes from observing this:

$$\begin{array}{ccccccc} & & \vdots & & \vdots & & \vdots \\ 0 & \longrightarrow & p^2 \mathbb{Z} & \hookrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z}/p^2 \mathbb{Z} \longrightarrow 0 \\ & & \downarrow \hookrightarrow & & \parallel & & \downarrow \\ 0 & \longrightarrow & p \mathbb{Z} & \hookrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z}/p \mathbb{Z} \longrightarrow 0 \\ & & \downarrow \hookrightarrow & & \parallel & & \downarrow \\ 0 & \longrightarrow & \mathbb{Z} & \xlongequal{\quad} & \mathbb{Z} & \longrightarrow & 0 \longrightarrow 0 \end{array}$$

§6.4 The Big Theorem of Limits and Colimits

This is really the big theorem which makes the final results in Section 6.2.ii work.

Theorem 6.4.1 (Adjoint and Limits)

Let $L: \mathcal{A} \rightarrow \mathcal{B}$ be left adjoint to a functor $R: \mathcal{B} \rightarrow \mathcal{A}$, where $\mathcal{A}$ and $\mathcal{B}$ are arbitrary categories. Then:

1. L preserves all colimits (coproducts, direct limits, cokernels, etc.). That is, if $A: I \rightarrow \mathcal{A}$ has a colimit, then so does $LA: I \rightarrow \mathcal{B}$, and

$$L(\operatorname{colim}_{i \in I} A_i) = \operatorname{colim}_{i \in I} L(A_i).$$

2. R preserves all limits (products, inverse limits, kernels, etc.). That is, if $B: I \rightarrow \mathcal{B}$ has a limit, then so does $RB: I \rightarrow \mathcal{A}$, and

$$R\left(\lim_{i \in I} B_i\right) = \lim_{i \in I} R(B_i).$$

Proof. Check MacLane. □

§6.5 Exercises

Example 6.5.1

$\mathbb{Z}[\frac{1}{p}] \cong \operatorname{colim}_{i \in I} F(i)$, where $I = \mathbb{N}$ with respect to the ordering $i \rightarrow j$ if $j < i$, with the maps $\mathbb{Z} \xrightarrow{\times p} \mathbb{Z}$:

$$\mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \xrightarrow{\times p} \dots \xrightarrow{\times p} \mathbb{Z} \xrightarrow{\times p} \dots$$

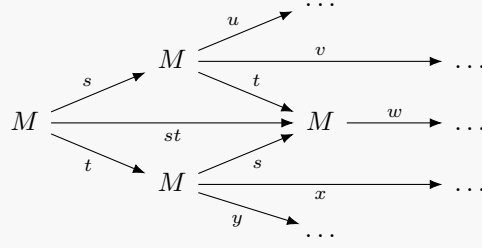
Exercise 6.5.2. Show that a colimit of torsion groups is torsion.

Solution: Let A_i be torsion groups and $A \cong \operatorname{colim}_{i \in I} A_i$. □

Exercise 6.5.3. Let R be a commutative ring and let $S \subseteq R$ be a multiplicatively closed set. Let M be an R -module. Let $\mathcal{C}_S$ be the category whose objects are elements of S and with morphisms $s \rightarrow t$ also given by elements of $u \in S$ such that $t = us$. Define a functor $F: \mathcal{C}_S \rightarrow R\text{-mod}$ by $F(s) = M$ for all s and

$$F\left(s \xrightarrow{u} t\right) = \left(M \xrightarrow{u} M\right),$$

where the map is multiplication by u . This looks like the following:



Show that this is a filtered colimit, and show that its colimit is isomorphic to $S^{-1}M$.

$$\varinjlim(F) \cong S^{-1}M.$$

Hint: To show that $\varinjlim(F) \cong S^{-1}M$ you need to write down lots of maps from the diagram to $S^{-1}M$, and show that $S^{-1}M$ has the universal property of the colimit.

Solution: We first show this is filtered colimit; for this, we must show that $\mathcal{C}_S$ is filtered. Let $s, t \in \mathcal{C}_S$. Then, we note that there are arrows $s \xrightarrow{t} st$, $t \xrightarrow{s} st$. Next, let $s, t \in \mathcal{C}_s$ such that there are two maps from $s \rightarrow t$, given by u, v . This means that $t = us = vs$. We want to find a map given by w such that $wu = wv$ as maps. This means that $wus = wvs$ as elements, so taking $w = 1$ is good, since we know that $us = vs = t$. As F is a functor from $\mathcal{C}_S \rightarrow R\text{-mod}$. Now, let $\phi_s: M \rightarrow S^{-1}M$ be given by $m \mapsto s^{-1}m$. Then, we have the following diagram:

$$\begin{array}{ccc} F(s) & \xrightarrow{F(u)} & F(t) \\ & \searrow \phi_s & \swarrow \phi_t \\ & S^{-1}M & \end{array}$$

Then, in this diagram, we have that $\phi_s(m) = \frac{m}{s}$. Going the other way, we have

$$\phi_t(F(u)(m)) = \phi_t(um) = \frac{um}{t} = \frac{um}{us} = \frac{m}{s}.$$

Therefore, each diagram of the form above commutes. Now, let N be another R -module with a collection of maps ψ_s (for $s \in \mathcal{C}_s$) such that $\psi_t = \psi_s \circ F(u)$. Then, there exists a unique map $\Phi: S^{-1}M \rightarrow N$ defined by $\frac{m}{s} \mapsto \psi_s$. We note that such a map is well defined because if $\frac{m}{s} = \frac{m'}{s'}$, then there exists some nonzero element $u \in S$ such that $u(ms' - m's) = 0$. Then, compatibility of the diagram tells us that

$$\psi_{us'}(um') = \psi_{s'}(m'), \quad \phi_{us}(um) = \phi_s(m).$$

Then,

$$\Phi\left(\frac{m'}{s'}\right) = \psi_{s'}(m') = \psi_{us'}(um') = \psi_{us's}(um's).$$

What we have done in the above is just go to bigger indices and compare. Then,

$$\psi_{uss'}(um's) = \psi_{uss'}(ums'),$$

where the second equality is justified by $um's = ums'$. By using a similar chain of equalities, we find that

$$\Phi\left(\frac{m'}{s'}\right) = \Phi\left(\frac{m}{s}\right).$$

Since $S^{-1}M$ satisfies the universal property of the colimit, we are done. $\square$

Exercise 6.5.4. Let $k[x]$ be a polynomial ring in one variable. Consider the following diagram in the category of $k[x]$ -modules:

$$\cdots \rightarrow k[x]/(x^n) \rightarrow k[x]/(x^{n-1}) \rightarrow \cdots \rightarrow k[x]/(x^2) \rightarrow k[x]/(x).$$

Show that the limit of this diagram, i.e. the inverse or projective limit, is the power series ring $k[[x]]$.

Solution: The index category in concern here is the opposite of natural numbers, so we can describe this concretely. Particularly, our indexing category as follows:

$$1 \geq 2 \geq 3 \geq \cdots,$$

and we define our functor F to map $i \mapsto k[x]/(x^i)$. First, we have the diagram

$$\begin{array}{ccc} k[x]/(x^n) & \xrightarrow{\text{mod } x^m} & k[x]/(x^m) \\ & \nwarrow \text{mod } x^m \quad \nearrow \text{mod } x^n & \\ & k[[x]] & \end{array}$$

Then, if $f(x) := \sum_{i=0}^{\infty} a_i x^i \in k[[x]]$, then $f(x) \text{ mod } x^m = \sum_{i=0}^{m-1} a_i x^i$, then applying $\text{mod } x^n$, we get $\sum_{i=0}^{n-1} a_i x^i$, which coincides with applying $\text{mod } x^n$ to $f(x)$ in the first place. Therefore, this diagram commutes. Now, if we have any other $k[x]$ -module M , with a family of compatible maps, say given by π_m , then we can define a unique map $M \rightarrow k[[x]]$ which maps some $\tau \in M$ to the unique $\sum_{i=0}^{\infty} a_i x^i$ such that taking $\text{mod } x^n$, it coincides with $\pi_n(\tau)$, for all n . Therefore, the formal power series ring satisfies the universal property, so we are done. $\square$

Exercise 6.5.5. Let $k[x]$ be a polynomial ring in one variable. Let $k(x)$ be the ring of rational functions in x . Consider $k(x)$ as a module over $k[x]$. Explain why it is impossible to express $k(x)$ as the limit of a diagram

$$\cdots \rightarrow A_n \rightarrow A_{n-1} \rightarrow \cdots \rightarrow A_2 \rightarrow A_1$$

where the A_i are finitely generated $k[x]$ -modules, like we did for $k[[x]]$ in the previous question. *Hint: $k(x)$ is a divisible $k[x]$ -module.*

Solution: Note that $k(x)$ is a divisible $k[x]$ -module, as mentioned in the hint. This implies that for any $f \in k[x]$, the multiplication by f map is surjective from $k(x) \rightarrow k(x)$. Suppose, for the sake of contradiction, that $k[[x]] \cong \varprojlim A_i$, where each A_i is a finitely generated $k[x]$ -module. Then, since we have a map from $k(x) \rightarrow k(x)$ given by multiplication by f , compatibility of the diagram implies that this multiplication by f map descends to the finite levels. In other words, this is what we have:

$$\begin{array}{ccc} k(x) & & \\ \downarrow f & & \\ k(x) & & \\ \swarrow \pi_n \quad \searrow \pi_m & & \\ \text{Im}(\pi_n) & \xrightarrow{\phi_{nm}} & \text{Im}(\pi_m) \\ \cap & & \cap \\ A_n & \xrightarrow{\phi_{nm}} & A_m \end{array}$$

Let us define $B_n := \text{Im}(\pi_n)$; note that as $B_n \subset \text{Im}(\pi_n)$ and $k[x]$ is Noetherian, B_n is finitely generated. Then, multiplication by f induces a surjection from $B_n \rightarrow B_n$. If we take some $b_n \in B_n$, then it comes from some $g \in k[x]$, so $\pi_n(g) = b_n$. Further, f is a surjection, so $g = f \cdot h$, where $h \in k[x]$. Therefore, we follow this chain of equalities to deduce that

$$b_n = f\pi_n(h).$$

Here, $f \in k[x]$, so we can factor it out of π_n , and $\pi_n(h) \in B_n$ since the same map π_n exists at the top of the diagram. Therefore, f is a surjective map from $B_n \rightarrow B_n$. This means that B_n is a divisible $k[x]$ -module, since f was chosen to be arbitrary. Therefore, B_n is a divisible finitely generated module. Now, if f were a nonunit in $k[x]$, then (f) is contained in some maximal ideal $\mathfrak{m}$. Considering the localization of B_n at $\mathfrak{m}$, we then find that $(f)(B_n)_{\mathfrak{m}} = (B_n)_{\mathfrak{m}}$. Then, Nakayama’s lemma implies that $(B_n)_{\mathfrak{m}} = 0$. Now, because f was an arbitrary nonunit, $(B_n)_{\mathfrak{m}} = 0$ for all maximal ideals $\mathfrak{m}$. This implies that $B_n = 0$, which is a contradiction. $\square$

Exercise 6.5.6. If A is an abelian group, then the set $\text{Hom}_{\mathbb{Z}}(A, A)$ of group homomorphisms $A \rightarrow A$ is a ring: the group structure is given by the sum of homomorphisms, and the ring multiplication is composition. We prove this in this exercise.

Proposition 6.5.7

There is a ring isomorphism

$$\text{Hom}_{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \cong \widehat{\mathbb{Z}}.$$

The ring $\widehat{\mathbb{Z}}$ is called the **profinite integers** (or ‘zed hat’ if you prefer). It is defined as the following subring of $\prod_{n=1}^{\infty} \mathbb{Z}/n\mathbb{Z}$:

$$\widehat{\mathbb{Z}} := \left\{ (a_1, a_2, a_3, \dots) \in \prod_{n=1}^{\infty} \mathbb{Z}/n\mathbb{Z} : a_n \equiv a_m \pmod{m} \text{ for all } m, n \text{ with } m \mid n \right\}.$$

There is a canonical injective ring homomorphism $\mathbb{Z} \hookrightarrow \widehat{\mathbb{Z}}$ given by

$$n \mapsto (n \bmod 1, n \bmod 2, n \bmod 3, \dots),$$

but it should be reasonably clear that this is far from being surjective.

1. Write down a non-trivial element of $\widehat{\mathbb{Z}}/\mathbb{Z}$. (In fact, it turns out that $\widehat{\mathbb{Z}}/\mathbb{Z}$ is a $\mathbb{Q}$ -vector space isomorphic to $\mathbb{R}$.)
2. Show that

$$\widehat{\mathbb{Z}} = \varprojlim \mathbb{Z}/n\mathbb{Z} \quad \text{and} \quad \mathbb{Q}/\mathbb{Z} = \varinjlim \mathbb{Z}/n\mathbb{Z}.$$

Use the fact that

$$\text{Hom}_{\mathbb{Z}}\left(\varinjlim A_i, B\right) \cong \varprojlim \text{Hom}_{\mathbb{Z}}(A_i, B).$$

and the fact that

$$\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong A[n]$$

to prove the proposition.

3. Alternatively, prove the proposition with the following steps: Given $a = (a_1, a_2, \dots) \in \widehat{\mathbb{Z}}$, define a map

$$\phi_a : \mathbb{Q}/\mathbb{Z} \rightarrow \mathbb{Q}/\mathbb{Z}, \quad m/n \mapsto a_n m/n.$$

Show that this defines a well-defined ring homomorphism

$$\widehat{\mathbb{Z}} \rightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}), \quad a \mapsto \phi_a.$$

4. Given a $\psi \in \text{Hom}_{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, \mathbb{Q}/\mathbb{Z})$ define

$$a_\psi := (\psi(1), 2\psi(1/2), 3\psi(1/3), \dots) \in \prod_{n=1}^{\infty} \mathbb{Z}/n\mathbb{Z}.$$

Show that this defines a well-defined ring homomorphism

$$\text{Hom}_{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \rightarrow \hat{\mathbb{Z}}, \quad \psi \mapsto a_\psi.$$

Check that it is inverse to $a \mapsto \phi_a$.

5. Using the short exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$$

or otherwise, show that

$$\text{Hom}_{\mathbb{Z}}(\mathbb{Q}, \mathbb{Q}/\mathbb{Z}) \cong \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Note that as a vector space this is isomorphic to $\mathbb{R}$.

Solution: 1. Consider the element which is 1 at prime components, but 0 everywhere else. This is not an integer, since if it were, then it would be congruent to 1 modulo every p .

2. We first define our index category. We let its objects be $\mathbb{N}$, but we define an arrow from $n \rightarrow m$ if $m \mid n$. We show this is filtered first. Let n, m be two positive integers. Then, there is an arrow from n, m to $\text{gcd}(n, m)$. Further, if we have two arrows $n \rightrightarrows m$, then we can just take $m \rightarrow m$, since an arrow implies $m \mid n$. Thus, our index category is filtered, and we now just define functors on it. We let $F(n) = \mathbb{Z}/n\mathbb{Z}$ and $F(n \rightarrow m)$ be the mod m map. We shall show that $\hat{\mathbb{Z}}$ has the universal property of the colimit. Given some sequence $(a_n)_n$, we define the map $\pi_m: \hat{\mathbb{Z}} \rightarrow \mathbb{Z}/m\mathbb{Z}$ as the map $(a_n)_n \mapsto a_m$, where it just extracts the m -th index. Then, it is clear that π_m is compatible. $F(n \rightarrow m)$. Let M be an abelian group, with a collection of maps φ_n such that $\varphi_n(\omega) \bmod m \equiv \varphi_m(\omega)$, if $m \mid n$. Here, $\omega \in M$ (sorry for the notation, I wanted to reserve alphabets for integers). Then, we can define a map from $\varphi: M \rightarrow \hat{\mathbb{Z}}$ send ω to the sequence such that it respects this compatibility. That is, we define $\varphi(\omega) = (\varphi_n(\omega))_n$. This is also unique. If $\psi: M \rightarrow \hat{\mathbb{Z}}$ is any other map, then it better satisfy $\pi_n(\psi(\omega)) = \varphi_n(\omega)$. Therefore, we have as follows:

$$\psi(\omega) = (\pi_n(\psi(\omega)))_n = (\varphi_n(\omega))_n = \varphi(\omega).$$

Thus, $\hat{\mathbb{Z}}$ has the universal property of the colimit.

As for the second one, we just reverse the arrows of our indexing category. Explicitly, $m \rightarrow n$ if $m \mid n$, but we define $F(m \rightarrow n)$ to be the canonical inclusion of $\mathbb{Z}/m\mathbb{Z} \hookrightarrow \mathbb{Z}/n\mathbb{Z}$. We claim that $\mathbb{Q}/\mathbb{Z}$ possesses the universal property of the limit. Let $x \in \mathbb{Q}/\mathbb{Z}$; then, it is necessarily of some order n . We then define $\iota_n: \mathbb{Z}/n\mathbb{Z} \hookrightarrow \mathbb{Q}/\mathbb{Z}$ as $[x] \mapsto \frac{x}{n} + \mathbb{Z}$. We also define the maps between $\tau_{mn}: \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ as $[x] \mapsto [\frac{n}{m}x]$. Then, starting with $[x] \in \mathbb{Z}/m\mathbb{Z}$, we find that $\iota_m([x]) = \frac{x}{m} + \mathbb{Z}$, and going the other way, we have $[x] \mapsto [\frac{n}{m}x] \mapsto \frac{nx}{mx} + \mathbb{Z} = \frac{x}{m} + \mathbb{Z}$. Therefore, these commute. Now, suppose we have some other abelian group M with a collection of maps φ_n such that $\varphi_m = \varphi_n \circ \tau_{mn}$. Then, we can define a map $\varphi: \mathbb{Q}/\mathbb{Z} \rightarrow M$ defined by mapping $\frac{x}{n} \mapsto \varphi_n([x])$. We have the desired commutativity. We claim this is also unique. Let $\psi: \mathbb{Q}/\mathbb{Z} \rightarrow M$ also satisfy $\psi \circ \iota_n = \varphi_n$. Let $\frac{x}{n} + \mathbb{Z}$ be an arbitrary element in $\mathbb{Q}/\mathbb{Z}$. Then, we have that

$$\psi\left(\frac{x}{n} + \mathbb{Z}\right) = \psi \circ \iota_n([x]) = \varphi_n([x]) = \varphi\left(\frac{x}{n} + \mathbb{Z}\right).$$

Therefore, $\psi = \varphi$, so we have uniqueness. Then, using the fact that $\text{Hom}_{\mathbb{Z}}(\varinjlim A_i, B) \cong \varprojlim \text{Hom}_{\mathbb{Z}}(A_i, B)$, we have that

$$\text{Hom}_{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \cong \varprojlim \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \cong \varprojlim (\mathbb{Q}/\mathbb{Z})[n] \cong \varinjlim \mathbb{Z}/n\mathbb{Z} = \hat{\mathbb{Z}}.$$

3. While we already have the isomorphism from the above, we shall just do this. If we have $a, b \in \mathbb{Z}$, then clearly $\phi_a \phi_b = \phi_{ab}$; it is also clear that $\phi_a + \phi_b = \phi_{a+b}$. It also maps 1 to $\phi_1 = \text{id}_{\mathbb{Q}/\mathbb{Z}}$ and 0 to $\phi_0 = 0$.

4. This is clearly a morphism of rings, by looking component wise. We shall note that indeed, $\text{id}_{\mathbb{Q}/\mathbb{Z}} \mapsto 1$ and $0 \mapsto 0$. We just check that these are inverses to each other. First, we check that $\phi_{a_\psi} = \psi$. Applying $\frac{m}{n}$, the LHS gets mapped to

$$\frac{m}{n} \cdot (a_\psi)_n = \frac{m}{n} \cdot n\psi\left(\frac{1}{n}\right) = m \cdot \psi\left(\frac{1}{n}\right) = \psi\left(\frac{m}{n}\right).$$

Next, we want to show that $a = a_{\phi_a}$. We look at the n -th component of the RHS.

$$(a_{\phi_a})_n = n\varphi_a\left(\frac{1}{n}\right) = na_n \cdot \frac{1}{n} = a_n.$$

Thus, we have verified these are inverses.

5. We apply $\text{Hom}_{\mathbb{Z}}(-, \mathbb{Q}/\mathbb{Z})$, which is contravariant and left exact, and $\hat{\mathbb{Z}} \otimes_{\mathbb{Z}} -$, which is covariant and right exact. Then, we attain two new exact sequences

$$\hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Z} \longrightarrow \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

$$\text{Hom}_{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \longrightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Q}, \mathbb{Q}/\mathbb{Z}) \longrightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Q}/\mathbb{Z}) \longrightarrow 0$$

However, we actually get exactness at the left (in the diagram) as well. Since $\hat{\mathbb{Z}}$ is torsion free and $\mathbb{Z}$ is a PID, we get exactness at the left to be preserved. Also, since $\mathbb{Q}/\mathbb{Z}$ is injective, exactness extends at the right (in the diagram, to the left). Also, using the isomorphisms we know, this becomes the following:

$$0 \longrightarrow \hat{\mathbb{Z}} \longrightarrow \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

$$0 \longrightarrow \hat{\mathbb{Z}} \longrightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Q}, \mathbb{Q}/\mathbb{Z}) \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

We now wish to show we can connect these via isomorphisms at the left and right, for then the five lemma finishes it off. At the left, we can just take the identity. At the right, we want to show that $\hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}$. This goes as follows. First, we have that

$$\hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} \cong \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \varinjlim \mathbb{Z}/n\mathbb{Z} \cong \varinjlim (\hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}).$$

The second isomorphism needs some justification, but we shall leave this to the below. Then,

$$\varinjlim (\hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}) \cong \varinjlim \hat{\mathbb{Z}}/n\hat{\mathbb{Z}} \cong \varinjlim \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Q}/\mathbb{Z}.$$

The first isomorphism follows because for any abelian group M , $M \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong M/nM$. As for the second, this is just the first isomorphism theorem applied to $\hat{\mathbb{Z}} \rightarrow \mathbb{Z}/n\mathbb{Z}$. Connecting, we have that $\hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} \cong \mathbb{Q}/\mathbb{Z}$. Therefore, through this chain of isomorphisms,

we have the following diagram.

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \hat{\mathbb{Z}} & \longrightarrow & \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q} & \longrightarrow & \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} \longrightarrow 0 \\
 & & \parallel & & & & \downarrow \sim \\
 0 & \longrightarrow & \hat{\mathbb{Z}} & \longrightarrow & \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Q}, \mathbb{Q}/\mathbb{Z}) & \longrightarrow & \mathbb{Q}/\mathbb{Z} \longrightarrow 0
 \end{array}$$

Therefore, by the five lemma (or 2-out-of-3 lemma/ snake lemma), we are done. $\square$

Exercise 6.5.8. Prove that

$$\mathrm{Hom}_{\mathbb{Z}}\left(\varinjlim A_i, B\right) \cong \varprojlim \mathrm{Hom}_{\mathbb{Z}}(A_i, B).$$

Solution: We note that $\mathrm{Hom}_{\mathbb{Z}}(-, B): \mathbf{Ab}^{\mathrm{op}} \rightarrow \mathbf{Ab}$ is covariant and is right adjoint to the tensor product. Therefore, it commutes with limits. However, a limit in $\mathbf{Ab}^{\mathrm{op}}$ is a colimit in $\mathbf{Ab}^{\mathrm{op}}$, so we arrive at the desired isomorphism. $\square$

Exercise 6.5.9. Let M be an R -module and (N_i) be a collection of R -modules over some index category I . Then,

$$\mathrm{colim}_{i \in I} (M \otimes_R N_i) \cong M \otimes_R \mathrm{colim}_{i \in I} N_i.$$

Thus, deduce that $\varinjlim (\hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}) \cong \hat{\mathbb{Z}} \otimes_{\mathbb{Z}} \varinjlim \mathbb{Z}/n\mathbb{Z}$.

Solution: $M \otimes_R -$ is a left adjoint to $\mathrm{Hom}_R(M, -)$. Therefore, it commutes with colimits. We are done. $\square$

Exercise 6.5.10. Show that $\hat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ as rings.

Proof. For this, we use the isomorphism $\hat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z}$. As we will be commuting many limits, we've added the subscript to make it obvious which limit does what. CRT tells us

$$\varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \varprojlim_n \left(\prod_p \mathbb{Z}/p^{\nu_p(n)}\mathbb{Z} \right)$$

Here, the outer limit is taken with respect to projections, for $m \mid n$. Now, we note that the product is a limit with no maps between the index category. Therefore the the product is a right adjoint to the diagonal Δ -functor, so it preserves limits. In particular, we have that

$$\varprojlim_n \left(\prod_p \mathbb{Z}/p^{\nu_p(n)}\mathbb{Z} \right) \cong \prod_p \left(\varprojlim_n \mathbb{Z}/p^{\nu_p(n)}\mathbb{Z} \right).$$

If we can show the isomorphism below, then we are done, because then $\hat{\mathbb{Z}}/\mathbb{Z} \cong \prod_p \mathbb{Z}_p$.

$$\varprojlim_n \mathbb{Z}/p^{\nu_p(n)}\mathbb{Z} \cong \varprojlim_{\ell} \mathbb{Z}/p^{\ell}\mathbb{Z}.$$

Note that on the left, this is taken with respect to projections for $m \mid n$, and the right is taken over all natural ℓ . For this, we construct maps from both sides which compose to the identity. Consider $\varphi: \varprojlim_n \mathbb{Z}/p^{\nu_p(n)}\mathbb{Z} \rightarrow \varprojlim_{\ell} \mathbb{Z}/p^{\ell}\mathbb{Z}$ defined by

$$(x_n + p^{\nu_p(n)}\mathbb{Z})_n \longmapsto (x_n + p^{\ell}\mathbb{Z})_{\ell},$$

where $\nu_p(n) = \ell$. We note that this is well defined, because $p^r \mid p^{r+1}$, and so

$$x_{p^{r+1}} + p^{\nu_p(p^r)}\mathbb{Z} = x_{p^r} + p^{\nu_p(p^r)}\mathbb{Z}.$$

We note that this is also clearly a morphism of rings. Conversely, we may define the map $\psi: \varprojlim_{\ell} \mathbb{Z}/p^{\ell}\mathbb{Z} \rightarrow \varprojlim_n \mathbb{Z}/p^{\nu_p(n)}\mathbb{Z}$ by mapping $(x_{\ell} + p^{\ell}\mathbb{Z})_{\ell}$ to the element $(y_n + p^{\nu_p(n)}\mathbb{Z})_n$, where

$$\begin{aligned} y_n &:= x_{\ell}, & \nu_p(n) &= \ell \geq 1, \\ y_n &:= 0, & \nu_p(n) &= 0. \end{aligned}$$

This is also a well-defined ring morphism. If $m \mid n$ and $p \mid m$, then $\nu_p(m) \leq \nu_p(n)$. Therefore,

$$y_n = x_{\nu_p(n)} \equiv x_{\nu_p(m)} = y_m \pmod{p^{\nu_p(m)}}.$$

If $m \mid n$ and $p \nmid m$, then

$$y_n \equiv 0 = y_m \pmod{1}.$$

Thus, we have well-definedness. Note that directly above, since we're taking modulo 1 (or modulo $\mathbb{Z}$), it doesn't matter what y_n is, since it is an integer.

We conclude by checking that they are inverses to each other. It should be immediate that

$$\varphi \circ \psi \left((x_{\ell} + p^{\ell}\mathbb{Z})_{\ell} \right) = (x_{\ell} + p^{\ell}\mathbb{Z})_{\ell}.$$

Therefore, $\varphi \circ \psi = \text{id}$. Conversely,

$$\varphi \circ \psi \left((x_n + p^{\nu_p(n)}\mathbb{Z})_n \right) = (x_n + p^{\nu_p(n)}\mathbb{Z})_n,$$

which is again immediate from the formulae. Therefore, we are done. $\square$

Exercise 6.5.11. Compute $\text{Tor}_k^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}[\frac{1}{p}]\mathbb{Z})$.

Solution: We first show that $\mathbb{Z}[\frac{1}{p}] \cong \varinjlim \mathbb{Z}$, with transition maps as follows:

$$\mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \xrightarrow{\times p} \dots \xrightarrow{\times p} \mathbb{Z} \xrightarrow{\times p} \dots$$

Now let X be an abelian group with maps $\varphi_n: \mathbb{Z}^n \rightarrow X$ which satisfy the commutativity of the diagram. Then, we can construct the map $\varphi: \mathbb{Z}[\frac{1}{p}] \rightarrow X$ as the map $\frac{a}{p^n} \mapsto \varphi_n(a)$, where $a \in \mathbb{Z}^n$. This clearly commutes, so we consider another map $\psi: \mathbb{Z}[\frac{1}{p}] \rightarrow X$ which satisfies $\psi \circ \iota_n = \varphi_n$. We have the following:

$$\psi\left(\frac{a}{p^n}\right) = \psi \circ \iota_i(a) = \varphi_n(a) = \varphi \circ \iota_n(a) = \varphi\left(\frac{a}{p^n}\right).$$

Thus, they are unique.

Thus, we have that

$$\text{Tor}_k^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}[\frac{1}{p}]\mathbb{Z}) \cong \varinjlim \text{Tor}_k^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}).$$

Now, we have that

$$\mathrm{Tor}_k^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}) = \begin{cases} \mathbb{Z}/p\mathbb{Z} & k = 0, \\ \mathbb{Z}[p] & k = 1, \\ 0 & k \geq 2. \end{cases}$$

For $k = 1$, we have $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}) = 0$. Also, for $k = 0$, we note that the transition maps simply become the zero maps. Therefore,

$$\mathrm{Tor}_k^{\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}[\frac{1}{p}]\mathbb{Z}) = 0, \forall k.$$

□

Exercise 6.5.12. Let k be a field, and let $\mathcal{I}$ be the poset of finite subsets $S \subseteq \mathbb{N}$, ordered by inclusion. For each $S \in \mathcal{I}$, define $F(S) = k[x_i : i \in S]$, for $S \subseteq T$ and let $F(S) \rightarrow F(T)$ be the natural inclusion of polynomial rings. Prove that

$$\varinjlim_{S \in \mathcal{I}} k[x_i : i \in S] \cong k[x_1, x_2, x_3, \dots].$$

Explain why the analogous statement for power series is false; that is, explain why

$$\varinjlim_{S \in \mathcal{I}} k[[x_i : i \in S]] \not\cong k[[x_1, x_2, x_3, \dots]]$$

Instead, show that

$$k[[x_1, x_2, \dots]] \cong \varprojlim_n k[x_1, x_2, \dots] / (x_1, x_2, \dots)^n.$$

Solution: We shall denote by $k[S]$ to mean $k[x_i : i \in S]$ for brevity. We first note that $\mathcal{I}$ forms a filtered poset. For any two sets S, T , we may take the union $S \cup T$ to get an inclusion $F(S), F(T) \rightarrow F(S \cup T)$. The other axiom stating that if we have two arrows from $S \rightarrow T$, then there is some $T \rightarrow V$ such that their composition coincide is immediate, by taking $V = T$. Therefore, we may take the colimit. We show that $k[x_1, \dots]$ possesses the universal property for colimits now. Well, the canonical inclusions from $k[S] \rightarrow k[x_1, \dots]$ allows for the commutativity condition to hold. If there exists another ring X such that it comes with a collection of maps satisfying the commutativity with the transition maps, we can clearly find a unique map from $k[x_1, \dots] \rightarrow X$ (we won't verify this here as this goes analogously as to usual).

The same construction can't hold for $k[[x_1, \dots, x_n]]$ because elements such as $\sum_{i=1}^{\infty} x_i$ are not contained in any finite $k[[S]]$.

Conversely, we shall show that $k[[x_1, \dots]]$ carries the universal property for the limit $\varprojlim_n k[x_1, x_2, \dots] / (x_1, x_2, \dots)^n$. For this, we use the natural modulo $\deg n$ maps. It is clear these satisfy the commutativity condition. Now, if X is some ring which satisfies the commutativity condition with the transition maps, then we can find a unique map from X to $k[[x_1, \dots]]$ which maps an element in X to the formal power series in $k[[x_1, \dots]]$ which is equal to the modulo n power series. □

Exercise 6.5.13. Let R be a commutative ring and S be the set of non-zero divisors. That

is if $s \in S$, then there exists no $r \in R$ such that $rs = 0$. Show that

$$M_{\text{tors}} \cong \varinjlim_{s \in S} M[s].$$

Solution: We first note that S forms a filtered category. We order $s \rightarrow t$ if $s \mid t$. If $s, t \in S$, then clearly $s \rightarrow st$ and $t \rightarrow st$. Further, if there are two arrows $s \rightrightarrows t$, then taking $t \rightarrow t$ is enough, since these two compositions must coincide. Now, we show that M_{tors} possesses the universal property. First, we note that if $s \mid t$, then there is an inclusion $M[s] \hookrightarrow M[t]$ because if $m \in M[s]$, then $sm = 0$, so $tm = 0$. Now, we construct the maps $\phi_s: M[s] \rightarrow M_{\text{tors}}$. We define $\phi_s(m) \mapsto m$. The commutativity condition is immediate. Now suppose X is some other R -module which satisfies the commutativity of transition maps. Then, we construct a unique map from $M_{\text{tors}} \rightarrow X$ such that $m \in M_{\text{tors}}$ maps to $\iota_s(m)$, where ι_s are the maps that come with X . Here, $m \in M_{\text{tors}}$ has some $sm = 0$, which is this s . The uniqueness is then clear. If we have some other map respecting the commutativity, they must be equal. $\square$

Bibliography

- [1] Charles A. Weibel. *An Introduction to Homological Algebra*. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 1994.